



RTL Design Sherpa

RTL AMBA Monitor Subsystem Module Reference — Rev 0.90

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1 AMBA4 Event Code Reference

This is the per-protocol event-code reference for the AMBA4 family — AXI4 / AXI4-Lite, APB4, and AXI4-Stream. The 8-bit `event_code` field of every monbus packet sourced by an AMBA4 monitor takes its value from one of the enums defined in `rtl/amba/includes/monitor_amba4_pkg.sv` and listed below. For the universal packet layout, `protocol` / `packet_type` enums, and helper functions see `monitor_package_spec.md`.

Each **event-code** enum is 8 bits wide. Slots 8'h0–8'hE are reserved for predefined codes (with `_RESERVED_*` placeholders absorbing unused indices) and slot 8'hF is always the `*_USER_DEFINED` escape hatch.

One exception: `axi_perfhist_select_t` is **4 bits**, not 8. It is not an event code — it is the `event_code[7:4]` *select nibble* that says which histogram a PerfHist packet carries, so its `AXI_PERFHIST_SEL_USER = 4'hF` escape hatch sits in a 4-bit space.

1.1 Design Notes

This is a package, not a module. It has no ports and no clock; it declares the types, enums and parameters that the monitor family shares, and exists so that a field width or event encoding has exactly one definition. Changing a width here changes every module that imports it, which is the point.

Compile order matters. A package must be analysed before anything that imports it. The filelists under `rtl/amba/filelists/` place the `includes/` packages first for that reason; hand-listing sources in a different order is one of the ways a build breaks confusingly (`vault/handbook/design/filelists.md`).

No test of its own, and none expected. A package has no behaviour to simulate. It is verified by every module that imports it failing to elaborate if a declaration is wrong.

1.1.1 AXI4 Event Codes

Nine categories cover the AXI4 / AXI4-Lite monitor surface: error, timeout, completion, threshold, performance, address-match, debug, and the two windowed-performance categories perf-window (`axi_perfwin_code_t`, `packet_type = PktTypePerfWin = 4'hD`) and perf-histogram (`axi_perfhist_select_t`, `packet_type = PktTypePerfHist = 4'hE`); see the RTL package for the full perf-window / perf-histogram value tables. The `protocol` field for every code in this section is `PROTOCOL_AXI (4'h0)`.

1.1.2 axi_error_code_t

Packet context: `packet_type = PktTypeError`, `protocol = PROTOCOL_AXI`

Value	Name	Description
8'h0	<code>AXI_ERR_RESP_SLVERR</code>	Slave error response
8'h1	<code>AXI_ERR_RESP_DECERR</code>	Decode error response
8'h2	<code>AXI_ERR_DATA_ORPHAN</code>	Data without command
8'h3	<code>AXI_ERR_RESP_ORPHAN</code>	Response without

Value	Name	Description
		transaction
8'h4	AXI_ERR_PROTOCOL	Protocol violation
8'h5	AXI_ERR_BURST_LENGTH	Invalid burst length
8'h6	AXI_ERR_BURST_SIZE	Invalid burst size
8'h7	AXI_ERR_BURST_TYPE	Invalid burst type
8'h8	AXI_ERR_ID_COLLISION	ID collision detected
8'h9	AXI_ERR_WRITE_BEFORE_ADDR	Write data before address
8'hA	AXI_ERR_RESP_BEFORE_DATA	Response before data complete
8'hB	AXI_ERR_LAST_MISSING	Missing LAST signal
8'hC	AXI_ERR_STROBE_ERROR	Write strobe error
8'hD	AXI_ERR_ADDR_RANGE	Address-range violation (from axi_monitor_addr_check)
8'hE	AXI_ERR_EVENT_DROPPED	monitor-lite: events lost to monbus backpressure; event_data is the count
8'hF	AXI_ERR_USER_DEFINED	User-defined error

1.1.3 axi_timeout_code_t

Packet context: packet_type = PktTypeTimeout, protocol = PROTOCOL_AXI

Value	Name	Description
8'h0	AXI_TIMEOUT_CMD	Command/Address timeout
8'h1	AXI_TIMEOUT_DATA	Data timeout
8'h2	AXI_TIMEOUT_RESP	Response timeout
8'h3	AXI_TIMEOUT_HANDSHAKE	Handshake timeout
8'h4	AXI_TIMEOUT_BURST	Burst completion timeout
8'h5	AXI_TIMEOUT_EXCLUS	Exclusive access

Value	Name	Description
	IVE	timeout
8'h6–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	AXI_TIMEOUT_USER_DEFINED	User-defined timeout

1.1.4 axi_completion_code_t

Packet context: packet_type = PktTypeCompletion, protocol = PROTOCOL_AXI

Value	Name	Description
8'h0	AXI_COMPL_TRANS_COMPLETE	Transaction completed successfully
8'h1	AXI_COMPL_READ_COMPLETE	Read transaction complete
8'h2	AXI_COMPL_WRITE_COMPLETE	Write transaction complete
8'h3	AXI_COMPL_BURST_COMPLETE	Burst transaction complete
8'h4	AXI_COMPL_EXCLUSIVE_OK	Exclusive access completed
8'h5	AXI_COMPL_EXCLUSIVE_FAIL	Exclusive access failed
8'h6	AXI_COMPL_ATOMIC_OK	Atomic operation completed
8'h7	AXI_COMPL_ATOMIC_FAIL	Atomic operation failed
8'h8–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	AXI_COMPL_USER_DEFINED	User-defined completion

1.1.5 axi_threshold_code_t

Packet context: packet_type = PktTypeThreshold, protocol = PROTOCOL_AXI

Value	Name	Description
8'h0	AXI_THRESH_ACTIVE_COUNT	Active transaction count threshold
8'h1	AXI_THRESH_LATENCY	Latency threshold

Value	Name	Description
8'h2	AXI_THRESH_ERROR_RATE	Error rate threshold
8'h3	AXI_THRESH_THROUGHPUT	Throughput threshold
8'h4	AXI_THRESH_QUEUE_DEPTH	Queue depth threshold
8'h5	AXI_THRESH_BANDWIDTH	Bandwidth utilization threshold
8'h6	AXI_THRESH_OUTSTANDING	Outstanding transactions threshold
8'h7	AXI_THRESH_BURST_SIZE	Average burst size threshold
8'h8–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	AXI_THRESH_USER_DEFINED	User-defined threshold

1.1.6 axi_performance_code_t

Packet context: packet_type = PktTypePerf, protocol = PROTOCOL_AXI

Value	Name	Description
8'h0	AXI_PERF_ADDR_LATENCY	Address phase latency
8'h1	AXI_PERF_DATA_LATENCY	Data phase latency
8'h2	AXI_PERF_RESP_LATENCY	Response phase latency
8'h3	AXI_PERF_TOTAL_LATENCY	Total transaction latency
8'h4	AXI_PERF_THROUGHPUT	Transaction throughput
8'h5	AXI_PERF_ERROR_RATE	Error rate
8'h6	AXI_PERF_ACTIVE_COUNT	Current active transaction count
8'h7	AXI_PERF_COMPLETED_COUNT	Total completed transaction count
8'h8	AXI_PERF_ERROR_COUNT	Total error transaction count
8'h9	AXI_PERF_BANDWIDTH_UTIL	Bandwidth utilization
8'hA	AXI_PERF_QUEUE_DEPTH	Average queue depth

Value	Name	Description
8'hB	AXI_PERF_BURST_EFFICIENCY	Burst efficiency metric
8'hC–8'hD	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hE	AXI_PERF_READ_WRITE_RATIO	Read/Write transaction ratio
8'hF	AXI_PERF_USER_DEFINED	User-defined performance

1.1.7 axi_addr_match_code_t

Packet context: packet_type = PktTypeAddrMatch, protocol = PROTOCOL_AXI

Value	Name	Description
8'h0	AXI_ADDR_EXACT_MATCH	Exact address match
8'h1	AXI_ADDR_RANGE_MATCH	Address within range
8'h2	AXI_ADDR_MASK_MATCH	Address mask match
8'h3	AXI_ADDR_PATTERN_MATCH	Address pattern match
8'h4	AXI_ADDR_SEQUENTIAL	Sequential address access
8'h5	AXI_ADDR_STRIDE_MATCH	Stride pattern match
8'h6	AXI_ADDR_HOTSPOT	Address hotspot detected
8'h7	AXI_ADDR_CONFLICT	Address conflict detected
8'h8–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	AXI_ADDR_USER_DEFINED	User-defined address match

1.1.8 axi_debug_code_t

Packet context: packet_type = PktTypeDebug, protocol = PROTOCOL_AXI

Value	Name	Description
8'h0	AXI_DEBUG_STATE_CHANGE	AXI state machine change
8'h1	AXI_DEBUG_PIPELINE_STALL	Pipeline stall event
8'h2	AXI_DEBUG_BACKPRESSURE	Backpressure event
8'h3	AXI_DEBUG_OUTSTANDING	Outstanding transaction count
8'h4	AXI_DEBUG_REORDER_BUFFER	Reorder buffer status
8'h5	AXI_DEBUG_ID_ALLOCATION	Transaction ID allocation
8'h6	AXI_DEBUG_QOS_ESCALATION	QoS escalation event
8'h7	AXI_DEBUG_HANDSHAKE	Handshake timing event
8'h8	AXI_DEBUG_QUEUE_STATUS	Queue status change
8'h9	AXI_DEBUG_COUNTER	Counter snapshot
8'hA	AXI_DEBUG_FIFO_STATUS	FIFO status
8'hB–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	AXI_DEBUG_USER_DEFINED	User-defined debug

1.1.9 APB4 Event Codes

Six categories cover the APB4 monitor surface: error, timeout, completion, threshold, performance, and debug. The protocol field for every code in this section is `PROTOCOL_APB` (4'h2).

1.1.10 apb_error_code_t

Packet context: `packet_type = PktTypeError`, `protocol = PROTOCOL_APB`

Value	Name	Description
8'h0	APB_ERR_PSLVERR	Peripheral slave error
8'h1	APB_ERR_SETUP_VIOLATION	Setup phase protocol violation
8'h2	APB_ERR_ACCESS_VIOLATION	Access phase protocol violation

Value	Name	Description
8'h3	APB_ERR_STROBE_ERROR	Write strobe error
8'h4	APB_ERR_ADDR_DECODE	Address decode error
8'h5	APB_ERR_PROT_VIOLATION	Protection violation (PPROT)
8'h6	APB_ERR_ENABLE_ERROR	Enable phase error
8'h7	APB_ERR_READY_ERROR	PREADY protocol error
8'h8	APB_ERR_ADDR_RANGE	Address-range violation (from apb_monitor_addr_check)
8'h9–8'hE	(reserved)	Reserved for future use
8'hF	APB_ERR_USER_DEFINED	User-defined error

1.1.11 apb_timeout_code_t

Packet context: packet_type = PktTypeTimeout, protocol = PROTOCOL_APB

Value	Name	Description
8'h0	APB_TIMEOUT_SETUP	Setup phase timeout
8'h1	APB_TIMEOUT_ACCESS	Access phase timeout
8'h2	APB_TIMEOUT_ENABLE	Enable phase timeout (PREADY stuck)
8'h3	APB_TIMEOUT_PREADY_STUCK	PREADY stuck low
8'h4	APB_TIMEOUT_TRANSFER	Overall transfer timeout
8'h5–8'hE	(reserved)	Reserved for future use
8'hF	APB_TIMEOUT_USER_DEFINED	User-defined timeout

1.1.12 apb_completion_code_t

Packet context: packet_type = PktTypeCompletion, protocol = PROTOCOL_APB

Value	Name	Description
8'h0	APB_COMPL_TRANS_COMPLETE	Transaction completed
8'h1	APB_COMPL_READ_COM	Read transaction

Value	Name	Description
	PLETE	complete
8'h2	APB_COMPL_WRITE_COMPLETE	Write transaction complete
8'h3–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	APB_COMPL_USER_DEFINED	User-defined completion

1.1.13 apb_threshold_code_t

Packet context: packet_type = PktTypeThreshold, protocol = PROTOCOL_APB

Value	Name	Description
8'h0	APB_THRESH_LATENCY	Transaction latency threshold
8'h1	APB_THRESH_ERROR_RATE	Error rate threshold
8'h2	APB_THRESH_ACTIVE_COUNT	Active transaction count threshold
8'h3	APB_THRESH_THROUGHPUT	Throughput threshold
8'h4–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	APB_THRESH_USER_DEFINED	User-defined threshold

1.1.14 apb_performance_code_t

Packet context: packet_type = PktTypePerf, protocol = PROTOCOL_APB

Value	Name	Description
8'h0	APB_PERF_READ_LATENCY	Read transaction latency
8'h1	APB_PERF_WRITE_LATENCY	Write transaction latency
8'h2	APB_PERF_THROUGHPUT	Transaction throughput
8'h3	APB_PERF_ERROR_RATE	Error rate
8'h4	APB_PERF_ACTIVE_COUNT	Active transaction

Value	Name	Description
	UNT	count
8'h5	APB_PERF_COMPLETED_COUNT	Completed transaction count
8'h6–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	APB_PERF_USER_DEFINED	User-defined performance

1.1.15 apb_debug_code_t

Packet context: packet_type = PktTypeDebug, protocol = PROTOCOL_APB

Value	Name	Description
8'h0	APB_DEBUG_STATE_CHANGE	APB state changed
8'h1	APB_DEBUG_SETUP_PHASE	Setup phase event
8'h2	APB_DEBUG_ACCESS_PHASE	Access phase event
8'h3	APB_DEBUG_ENABLE_PHASE	Enable phase event
8'h4	APB_DEBUG_PSEL_TRACE	PSEL trace
8'h5	APB_DEBUG_PENABLE_TRACE	PENABLE trace
8'h6	APB_DEBUG_PREADY_TRACE	PREADY trace
8'h7	APB_DEBUG_PPROT_TRACE	PPROT trace
8'h8	APB_DEBUG_PSTRB_TRACE	PSTRB trace
8'h9–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	APB_DEBUG_USER_DEFINED	User-defined debug

1.1.16 AXIS4 Event Codes

Six categories cover the AXI4-Stream monitor surface: error, timeout, completion, credit, channel, and stream. The protocol field for every code in this section is PROTOCOL_AXIS (4'h1).

1.1.17 axis_error_code_t

Packet context: packet_type = PktTypeError, protocol = PROTOCOL_AXIS

Value	Name	Description
8'h0	AXIS_ERR_PROTOCOL	Protocol violation
8'h1	AXIS_ERR_READY_TIMING	TREADY timing violation
8'h2	AXIS_ERR_VALID_TIMING	TVALID timing violation
8'h3	AXIS_ERR_LAST_MISSING	Missing TLAST signal
8'h4	AXIS_ERR_LAST_ORPHAN	Orphaned TLAST signal
8'h5	AXIS_ERR_STRB_INVALID	Invalid TSTRB pattern
8'h6	AXIS_ERR_KEEP_INVALID	Invalid TKEEP pattern
8'h7	AXIS_ERR_DATA_ALIGNMENT	Data alignment error
8'h8	AXIS_ERR_ID_VIOLATION	TID violation
8'h9	AXIS_ERR_DEST_VIOLATION	TDEST violation
8'hA	AXIS_ERR_USER_VIOLATION	TUSER violation
8'hB–8'hE	(reserved)	Reserved for future use
8'hF	AXIS_ERR_USER_DEFINED	User-defined error

1.1.18 axis_timeout_code_t

Packet context: packet_type = PktTypeTimeout, protocol = PROTOCOL_AXIS

Value	Name	Description
8'h0	AXIS_TIMEOUT_HANDSHAKE	TVALID/TREADY handshake timeout
8'h1	AXIS_TIMEOUT_STREAM	Stream completion

Value	Name	Description
		timeout
8'h2	AXIS_TIMEOUT_PACKET	Packet timeout (TLAST)
8'h3	AXIS_TIMEOUT_BACKPRESSURE	Backpressure timeout
8'h4	AXIS_TIMEOUT_BUFFER	Buffer drain timeout
8'h5	AXIS_TIMEOUT_STALL	Stream stall timeout
8'h6–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	AXIS_TIMEOUT_USER_DEFINED	User-defined timeout

1.1.19 axis_completion_code_t

Packet context: packet_type = PktTypeCompletion, protocol = PROTOCOL_AXIS

Value	Name	Description
8'h0	AXIS_COMPL_STREAM_END	Stream completed (TLAST)
8'h1	AXIS_COMPL_PACKET_SENT	Packet sent successfully
8'h2	AXIS_COMPL_TRANSFER	Data transfer completed
8'h3	AXIS_COMPL_BURST_END	Burst completed
8'h4	AXIS_COMPL_HANDSHAKE	Successful handshake
8'h5–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	AXIS_COMPL_USER_DEFINED	User-defined completion

1.1.20 axis_credit_code_t

Packet context: packet_type = PktTypeCredit, protocol = PROTOCOL_AXIS

Value	Name	Description
8'h0	AXIS_CREDIT_READY_ASSERT	TREADY asserted
8'h1	AXIS_CREDIT_READY_DEASSERT	TREADY deasserted

Value	Name	Description
	SSERT	
8'h2	AXIS_CREDIT_BUFFER_AVAILABLE	Buffer space available
8'h3	AXIS_CREDIT_BUFFER_FULL	Buffer full
8'h4	AXIS_CREDIT_FLOW_CONTROL	Flow control event
8'h5	AXIS_CREDIT_BACKPRESSURE	Backpressure applied
8'h6	AXIS_CREDIT_THROUGHPUT	Throughput event
8'h7	AXIS_CREDIT EFFICIENCY	Transfer efficiency
8'h8–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	AXIS_CREDIT_USER_DEFINED	User-defined credit event

1.1.21 axis_channel_code_t

Packet context: packet_type = PktTypeChannel, protocol = PROTOCOL_AXIS

Value	Name	Description
8'h0	AXIS_CHAN_CONNECT	Channel connected
8'h1	AXIS_CHAN_DISCONNECT	Channel disconnected
8'h2	AXIS_CHAN_STALL	Channel stalled
8'h3	AXIS_CHAN_RESUME	Channel resumed
8'h4	AXIS_CHAN_CONGESTION	Channel congestion
8'h5	AXIS_CHAN_ID_CHANGE	TID change detected
8'h6	AXIS_CHAN_DEST_CHANGE	TDEST change detected
8'h7	AXIS_CHAN_CONFIG_CHANGE	Channel configuration change
8'h8–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>

Value	Name	Description
8'hF	AXIS_CHAN_USER_DEFINED	User-defined channel event

1.1.22 axis_stream_code_t

Packet context: packet_type = PktTypeStream, protocol = PROTOCOL_AXIS

Value	Name	Description
8'h0	AXIS_STREAM_START	Stream started
8'h1	AXIS_STREAM_END	Stream ended (TLAST)
8'h2	AXIS_STREAM_PAUSE	Stream paused
8'h3	AXIS_STREAM_RESUME	Stream resumed
8'h4	AXIS_STREAM_OVERFLOW	Stream buffer overflow
8'h5	AXIS_STREAM_UNDERFLOW	Stream buffer underflow
8'h6	AXIS_STREAM_TRANSFER	Active data transfer
8'h7	AXIS_STREAM_IDLE	Stream idle
8'h8–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	AXIS_STREAM_USER_DEFINED	User-defined stream event

1.1.23 Unified Event Code Union

monitor_amba4_pkg also exports a unified_event_code_t packed union that overlays the **8-bit event-code enums** in this document onto a single 8-bit field, plus a raw view for direct 8-bit access. The two perf-window enums (axi_perfwid_code_t, axi_perfhist_select_t) are **not** union members and have no create_* helper — pack those through the raw view. Helper functions create_axi*_event, create_apb4*_event, and create_axis*_event construct the union from a typed enum value — use these from any wrapper that needs to publish a protocol-tagged event_code without manual bit packing. ## Related Modules - **monitor_package_spec.md** — Universal types, packet layout, helper functions. - **monitor_amba5_pkg.md** — AXI5 / APB5 / AXIS5 extended event codes. - **monitor_arbiter_pkg.md** — ARB and CORE event codes. - **The Monitor System** — architecture and capabilities: how packets are produced, filtered, transported and captured, and how to instrument a block that is not a bus protocol.

1.2 Timing Characteristics

This module is **purely combinational** – it contains no `always_ff` and no latch, so it holds no state and adds no clock cycles. Its outputs settle a propagation delay after its inputs, and it introduces no latency into a pipeline that instantiates it.

Timing closure is therefore a question of the surrounding logic’s slack, not of this module’s cycle count. No synthesis figures are quoted; none have been measured.

1.3 Testing

No dedicated testbench for this module. It has no `val/**/test_monitor_amba4_pkg.py`. It is exercised indirectly, through the tests of modules that instantiate it (directly or further up):

- `apb4_monitor` – `val/**/test_apb4_monitor.py`
- `apb5_monitor` – `val/**/test_apb5_monitor.py`
- `apb_monitor_addr_check` – `val/**/test_apb_monitor_addr_check.py`
- `arbiter_monbus_common` – `val/**/test_arbiter_monbus_common.py`
- `axi4_master_rd_mon` – `val/**/test_axi4_master_rd_mon.py`

Indirect coverage exercises this module only in the configurations those parents elaborate. A parameter or mode no parent uses is untested.

Treat any behaviour described on this page as unverified by simulation.

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2 AMBA5 Extended Event Code Reference

This is the per-protocol event-code reference for the AMBA5 family — AXI5, APB5, and AXIS5. The enums here **extend** the AMBA4 set with the new event categories introduced by the AMBA5 specifications: atomic operations and tracing on AXI5; wake-up signaling, parity, and user signals on APB5; wake-up, parity, and CRC on AXIS5. For the AMBA4 baseline see

monitor_amba4_pkg.md; for the universal packet layout and helper functions see monitor_package_spec.md.

Source: rtl/amba/includes/monitor_amba5_pkg.sv. Each enum is 8 bits wide; slot 8'hF is the *_USER_DEFINED escape hatch. AMBA5 codes reuse the AMBA4 protocol values (PROTOCOL_AXI / PROTOCOL_APB / PROTOCOL_AXIS); the distinguishing field is the chosen event_code enum and the packet_type context the producer uses.

2.1 Design Notes

This is a package, not a module. It has no ports and no clock; it declares the types, enums and parameters that the monitor family shares, and exists so that a field width or event encoding has exactly one definition. Changing a width here changes every module that imports it, which is the point.

Compile order matters. A package must be analysed before anything that imports it. The filelists under rtl/amba/filelists/ place the includes/ packages first for that reason; hand-listing sources in a different order is one of the ways a build breaks confusingly (vault/handbook/design/filelists.md).

No test of its own, and none expected. A package has no behaviour to simulate. It is verified by every module that imports it failing to elaborate if a declaration is wrong.

2.1.1 AXI5 Extended Event Codes

Two new categories on top of the AXI4 baseline: atomic-operation events and QoS / trace events.

2.1.2 axi5_atomic_code_t

Packet context: extends axi_completion_code_t semantics (packet_type = PktTypeCompletion, protocol = PROTOCOL_AXI) — used to report which AXI5 atomic primitive completed.

Value	Name	Description
8'h0	AXI5_ATOMIC_LOAD	Atomic load operation
8'h1	AXI5_ATOMIC_SWAP	Atomic swap operation
8'h2	AXI5_ATOMIC_COMPARE	Atomic compare operation
8'h3	AXI5_ATOMIC_ADD	Atomic add

Value	Name	Description
		operation
8'h4	AXI5_ATOMIC_CLR	Atomic clear operation
8'h5	AXI5_ATOMIC_XOR	Atomic XOR operation
8'h6	AXI5_ATOMIC_SET	Atomic set operation
8'h7	AXI5_ATOMIC_SMAX	Atomic signed max
8'h8	AXI5_ATOMIC_SMIN	Atomic signed min
8'h9	AXI5_ATOMIC_UMAX	Atomic unsigned max
8'hA	AXI5_ATOMIC_UMIN	Atomic unsigned min
8'hB–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	AXI5_ATOMIC_USER_DEFINED	User-defined atomic

2.1.3 axi5_trace_code_t

Packet context: packet_type = PktTypeDebug (trace / QoS / poison / MPAM events), protocol = PROTOCOL_AXI.

Value	Name	Description
8'h0	AXI5_TRACE_START	Trace session start
8'h1	AXI5_TRACE_END	Trace session end
8'h2	AXI5_TRACE_DATA	Trace data packet
8'h3	AXI5_QOS_ESCALATION	QoS level escalation
8'h4	AXI5_QOS_DEESCALATION	QoS level de-escalation
8'h5	AXI5_POISON_DETECTED	Poison bit detected
8'h6	AXI5_LOOP_DETECTED	Loop detection triggered
8'h7	AXI5_MPAM_EVENT	MPAM partition event

Value	Name	Description
8'h8–8'hE	(reserved)	Reserved for future use
8'hF	AXI5_USER_DEFINED	User-defined

2.1.4 APB5 Extended Event Codes

Three new categories on top of the APB4 baseline: wake-up signaling (PWAKEUP), parity (PPARITY / PRDATAPARITY / PREADYPARITY / PSLVERRPARITY), and user signals (PUSER / PSUSER).

2.1.5 apb5_wakeup_code_t

Packet context: packet_type = PktTypeDebug (or producer-defined), protocol = PROTOCOL_APB — APB5 wake-up / sleep transitions.

Value	Name	Description
8'h0	APB5_WAKEUP_REQUEST	PWAKEUP asserted by slave
8'h1	APB5_WAKEUP_ACKNOWLEDGED	Wake-up acknowledged
8'h2	APB5_WAKEUP_TIMEOUT	Wake-up request timeout
8'h3	APB5_WAKEUP_REJECTED	Wake-up request rejected
8'h4	APB5_SLEEP_REQUEST	Sleep mode request
8'h5	APB5_SLEEP_ENTERED	Entered sleep mode
8'h6–8'hE	(reserved)	Reserved for future use
8'hF	APB5_WAKEUP_USER_DEFINED	User-defined wake-up

2.1.6 apb5_parity_code_t

Packet context: packet_type = PktTypeError (parity errors) or PktTypeDebug (status), protocol = PROTOCOL_APB.

Value	Name	Description
8'h0	APB5_PARITY_PWDATA_ERROR	PWDATA parity error (PPARITY)

Value	Name	Description
8'h1	APB5_PARITY_PRDATA_ERROR	PRDATA parity error (PRDATAPARITY)
8'h2	APB5_PARITY_PREADY_ERROR	PREADY parity error (PREADYPARITY)
8'h3	APB5_PARITY_PSLVERR_ERROR	PSLVERR parity error (PSLVERRPARITY)
8'h4	APB5_PARITY_CORRECTED	Parity error corrected
8'h5	APB5_PARITY_UNCORRECTED	Parity error uncorrectable
8'h6	APB5_PARITY_CHECK_DISABLED	Parity checking disabled
8'h7	APB5_PARITY_CHECK_ENABLED	Parity checking enabled
8'h8–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	APB5_PARITY_USER_DEFINED	User-defined parity

2.1.7 apb5_user_code_t

Packet context: packet_type = PktTypeDebug, protocol = PROTOCOL_APB — APB5 PUSER / PSUSER side-band signal events.

Value	Name	Description
8'h0	APB5_USER_PUSER_VALID	PUSER valid on master
8'h1	APB5_USER_PSUSER_VALID	PSUSER valid on slave
8'h2	APB5_USER_SIGNAL_MISMATCH	User signal mismatch
8'h3–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	APB5_USER_USER_DEFINED	User-defined

2.1.8 AXIS5 Extended Event Codes

Three new categories on top of the AXIS4 baseline: wake-up (TWAKEUP), parity (TPARITY), and CRC (TCRC_ERROR).

2.1.9 axis5_wakeup_code_t

Packet context: packet_type = PktTypeDebug (or producer-defined), protocol = PROTOCOL_AXIS — AXIS5 wake-up / sleep transitions.

Value	Name	Description
8'h0	AXIS5_WAKEUP_REQUEST	TWAKEUP asserted
8'h1	AXIS5_WAKEUP_ACKNOWLEDGED	Wake-up acknowledged
8'h2	AXIS5_WAKEUP_TIMEOUT	Wake-up timeout
8'h3	AXIS5_WAKEUP_ACTIVE	Wake-up active, data pending
8'h4	AXIS5_SLEEP_ENTERING	Entering sleep mode
8'h5	AXIS5_SLEEP_EXITING	Exiting sleep mode
8'h6–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	AXIS5_WAKEUP_USER_DEFINED	User-defined wake-up

2.1.10 axis5_parity_code_t

Packet context: packet_type = PktTypeError (parity errors) or PktTypeDebug (status), protocol = PROTOCOL_AXIS.

Value	Name	Description
8'h0	AXIS5_PARITY_TDATA_ERROR	TDATA parity error (TPARITY)
8'h1	AXIS5_PARITY_CORRECTED	Parity error corrected
8'h2	AXIS5_PARITY_UNCORRECTED	Parity error uncorrectable
8'h3	AXIS5_PARITY_CHECK_DISABLED	Parity checking disabled
8'h4	AXIS5_PARITY_CHECK_ENABLED	Parity checking enabled
8'h5–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	AXIS5_PARITY_USER_DEFINED	User-defined parity

2.1.11 axis5_crc_code_t

Packet context: packet_type = PktTypeError (CRC failures) or PktTypeDebug (status),
protocol = PROTOCOL_AXIS — optional CRC feature in AXIS5.

Value	Name	Description
8'h0	AXIS5_CRC_VALID	CRC check passed
8'h1	AXIS5_CRC_ERROR	CRC error detected (TCRC_ERROR)
8'h2	AXIS5_CRC_COMPUTED	CRC computed and sent
8'h3	AXIS5_CRC_DISABLED	CRC checking disabled
8'h4	AXIS5_CRC_ENABLED	CRC checking enabled
8'h5–8'hE	(reserved)	Reserved for future use
8'hF	AXIS5_CRC_USER_DEFINE D	User-defined CRC

2.1.12 Unified Event Code Union

monitor_amba5_pkg exports an amba5_event_code_t packed union that overlays the eight AMBA5 enums above (plus a raw 8-bit view) onto a single field. Helper functions create_axi5_atomic_event, create_axi5_trace_event, create_apb5_wakeup_event, create_apb5_parity_event, create_apb5_user_event, create_axis5_wakeup_event, create_axis5_parity_event, and create_axis5_crc_event construct the union from a typed enum value. ## Related Modules - **monitor_package_spec.md** — Universal types, packet layout, helper functions. - **monitor_amba4_pkg.md** — AXI4 / APB4 / AXIS4 baseline event codes. - **monitor_arbiter_pkg.md** — ARB and CORE event codes. - **The Monitor System** — architecture and capabilities: how packets are produced, filtered, transported and captured, and how to instrument a block that is not a bus protocol.

2.2 Timing Characteristics

This module is **purely combinational** – it contains no always_ff and no latch, so it holds no state and adds no clock cycles. Its outputs settle a propagation delay after its inputs, and it introduces no latency into a pipeline that instantiates it.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

2.3 Testing

No dedicated testbench for this module. It has no `val/**/test_monitor_amba5_pkg.py`. It is exercised indirectly, through the tests of modules that instantiate it (directly or further up):

- `apb5_monitor – val/**/test_apb5_monitor.py`
- `axi4_master_rd_mon – val/**/test_axi4_master_rd_mon.py`
- `axi4_master_rd_mon_cg – val/**/test_axi4_master_rd_mon_cg.py`
- `axi4_master_wr_mon – val/**/test_axi4_master_wr_mon.py`
- `axi4_master_wr_mon_cg – val/**/test_axi4_master_wr_mon_cg.py`

Indirect coverage exercises this module only in the configurations those parents elaborate. A parameter or mode no parent uses is untested.

Treat any behaviour described on this page as unverified by simulation.

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3 ARB and CORE Event Code Reference

This is the per-protocol event-code reference for the two non-AMBA monbus protocols: **ARB** (arbiter monitoring — fairness, starvation, weighted round-robin, ACK protocol) and **CORE** (internal accelerator / DMA-engine monitoring — descriptor fetch, credit cycles, sub-engine state). Both sets follow the same packet layout as the AMBA monitors; only the protocol and event_code fields differ. For the universal packet layout and helper functions see `monitor_package_spec.md`.

Source: `rtl/amba/includes/monitor_arbiter_pkg.sv`. Each enum is 8 bits wide; slot 8'hF is the *_USER_DEFINED escape hatch. ARB codes carry `PROTOCOL_ARB` (4'h3); CORE codes carry `PROTOCOL_CORE` (4'h4).

3.1 Design Notes

This is a package, not a module. It has no ports and no clock; it declares the types, enums and parameters that the monitor family shares, and exists so that a field width or event encoding has

exactly one definition. Changing a width here changes every module that imports it, which is the point.

Compile order matters. A package must be analysed before anything that imports it. The filelists under `rtl/amba/filelists/` place the `includes/` packages first for that reason; hand-listing sources in a different order is one of the ways a build breaks confusingly (`vault/handbook/design/filelists.md`).

No test of its own, and none expected. A package has no behaviour to simulate. It is verified by every module that imports it failing to elaborate if a declaration is wrong.

3.1.1 ARB Event Codes

Six categories cover the arbiter monitor surface: error, timeout, completion, threshold, performance, and debug. The protocol field for every code in this section is `PROTOCOL_ARB` (4'h3).

3.1.2 `arb_error_code_t`

Packet context: `packet_type = PktTypeError, protocol = PROTOCOL_ARB`

Value	Name	Description
8'h0	ARB_ERR_STARVATION	Client request starvation
8'h1	ARB_ERR_ACK_TIMEOUT	Grant ACK timeout
8'h2	ARB_ERR_PROTOCOL_VIOLATION	ACK protocol violation
8'h3	ARB_ERR_CREDIT_VIOLATION	Credit system violation
8'h4	ARB_ERR_FAIRNESS_VIOLATION	Weighted fairness violation
8'h5	ARB_ERR_WEIGHT_UNDERFLOW	Weight credit underflow
8'h6	ARB_ERR_CONCURRENT_GRANTS	Multiple simultaneous grants
8'h7	ARB_ERR_INVALID_GRANT_ID	Invalid grant ID detected
8'h8	ARB_ERR_ORPHAN_ACK	ACK without pending grant
8'h9	ARB_ERR_GRANT_OVERLAP	Overlapping grant periods
8'hA	ARB_ERR_MASK_ERROR	Round-robin mask error

Value	Name	Description
8'hB	ARB_ERR_STATE_MACHINE	FSM state error
8'hC	ARB_ERR_CONFIGURATION	Invalid configuration
8'hD–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	ARB_ERR_USER_DEFINED	User-defined error

3.1.3 arb_timeout_code_t

Packet context: packet_type = PktTypeTimeout, protocol = PROTOCOL_ARB

Value	Name	Description
8'h0	ARB_TIMEOUT_GRANT_ACK	Grant ACK timeout
8'h1	ARB_TIMEOUT_REQUEST_HOLD	Request held too long
8'h2	ARB_TIMEOUT_WEIGHT_UPDATE	Weight update timeout
8'h3	ARB_TIMEOUT_BLOCK_RELEASE	Block release timeout
8'h4	ARB_TIMEOUT_CREDIT_UPDATE	Credit update timeout
8'h5	ARB_TIMEOUT_STATE_CHANGE	State machine timeout
8'h6–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	ARB_TIMEOUT_USER_DEFINED	User-defined timeout

3.1.4 arb_completion_code_t

Packet context: packet_type = PktTypeCompletion, protocol = PROTOCOL_ARB

Value	Name	Description
8'h0	ARB_COMPL_GRANT_ISSUED	Grant successfully issued
8'h1	ARB_COMPL_ACK_RECEIVED	ACK successfully received

Value	Name	Description
8'h2	ARB_COMPL_TRANSACTION	Complete transaction (grant+ack)
8'h3	ARB_COMPL_WEIGHT_UPDATE	Weight update completed
8'h4	ARB_COMPL_CREDIT_CYCLE	Credit cycle completed
8'h5	ARB_COMPL_FAIRNESS_PERIOD	Fairness analysis period
8'h6	ARB_COMPL_BLOCK_PERIOD	Block period completed
8'h7–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	ARB_COMPL_USER_DEFINED	User-defined completion

3.1.5 arb_threshold_code_t

Packet context: packet_type = PktTypeThreshold, protocol = PROTOCOL_ARB

Value	Name	Description
8'h0	ARB_THRESH_REQUEST_LATENCY	Request-to-grant latency threshold
8'h1	ARB_THRESH_ACK_LATENCY	Grant-to-ACK latency threshold
8'h2	ARB_THRESH_FAIRNESS_DEVIATION	Fairness deviation threshold
8'h3	ARB_THRESH_ACTIVE_REQUESTS	Active request count threshold
8'h4	ARB_THRESH_GRANT_RATE	Grant rate threshold
8'h5	ARB_THRESH EFFICIENCY	Grant efficiency threshold
8'h6	ARB_THRESH_CREDIT_LOW	Low credit threshold
8'h7	ARB_THRESH_WEIGHT_IMBALANCE	Weight imbalance threshold
8'h8	ARB_THRESH_STARVATION_TIME	Starvation time threshold
8'h9–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>

Value	Name	Description
8'hF	ARB_THRESH_USER_DEFINED	User-defined threshold

3.1.6 arb_performance_code_t

Packet context: packet_type = PktTypePerf, protocol = PROTOCOL_ARB

Value	Name	Description
8'h0	ARB_PERF_GRANT_ISSUED	Grant issued event
8'h1	ARB_PERF_ACK_RECEIVED	ACK received event
8'h2	ARB_PERF_GRANT_EFFICIENCY	Grant completion efficiency
8'h3	ARB_PERF_FAIRNESS_METRIC	Fairness compliance metric
8'h4	ARB_PERF_THROUGHPUT	Arbitration throughput
8'h5	ARB_PERF_LATENCY_AVG	Average latency measurement
8'h6	ARB_PERF_WEIGHT_COMPLIANCE	Weight compliance metric
8'h7	ARB_PERF_CREDIT_UTILIZATION	Credit utilization efficiency
8'h8	ARB_PERF_CLIENT_ACTIVITY	Per-client activity metric
8'h9	ARB_PERF_STARVATION_COUNT	Starvation event count
8'hA	ARB_PERF_BLOCK_EFFICIENCY	Block/unblock efficiency
8'hB–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	ARB_PERF_USER_DEFINED	User-defined performance

3.1.7 arb_debug_code_t

Packet context: packet_type = PktTypeDebug, protocol = PROTOCOL_ARB

Value	Name	Description
8'h0	ARB_DEBUG_STATE_CHANGE	Arbiter state machine change

Value	Name	Description
8'h1	ARB_DEBUG_MASK_UPDATE	Round-robin mask update
8'h2	ARB_DEBUG_WEIGHT_CHANGE	Weight configuration change
8'h3	ARB_DEBUG_CREDIT_UPDATE	Credit level update
8'h4	ARB_DEBUG_CLIENT_MASK	Client enable/disable mask
8'h5	ARB_DEBUG_PRIORITY_CHANGE	Priority level change
8'h6	ARB_DEBUG_BLOCK_EVENT	Block/unblock event
8'h7	ARB_DEBUG_QUEUE_STATUSES	Request queue status
8'h8	ARB_DEBUG_COUNTER_SNAPSHOT	Counter values snapshot
8'h9	ARB_DEBUG_FIFO_STATUS	FIFO status change
8'hA	ARB_DEBUG_FAIRNESS_STATE	Fairness tracking state
8'hB	ARB_DEBUG_ACK_STATE	ACK protocol state
8'hC–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	ARB_DEBUG_USER_DEFINED	User-defined debug

3.1.8 CORE Event Codes

Six categories cover the core/accelerator monitor surface: error, timeout, completion, threshold, performance, and debug. The protocol field for every code in this section is `PROTOCOL_CORE` (4'h4). These codes are used by custom DMA / accelerator subsystems (descriptor engines, control read/write engines, credit-based flow control).

3.1.9 core_error_code_t

Packet context: `packet_type = PktTypeError, protocol = PROTOCOL_CORE`

Value	Name	Description
8'h0	CORE_ERR_DESCRIPTOR_MALFORMED	Missing magic number (0x900dc0de)
8'h1	CORE_ERR_DESCRIPTOR_BAD_ADDR	Invalid descriptor address

Value	Name	Description
8'h2	CORE_ERR_DATA_BAD_ADDR	Invalid data address (fetch or runtime)
8'h3	CORE_ERR_FLAG_COMPARISON	Flag mask/compare mismatch
8'h4	CORE_ERR_CREDIT_UNDERFLOW	Credit system violation
8'h5	CORE_ERR_STATE_MACHINE	Invalid FSM state transition
8'h6	CORE_ERR_DESCRIPTOR_ENGINE	Descriptor engine FSM error
8'h7	CORE_ERR_FLAG_ENGINE	Flag engine FSM error (legacy — pre-ctrlrd)
8'h8	CORE_ERR_CTRLWR_ENGINE	Control write engine FSM error
8'h9	CORE_ERR_DATA_ENGINE	Data engine error
8'hA	CORE_ERR_CHANNEL_INVALIDID	Invalid channel ID
8'hB	CORE_ERR_CONTROL_VIOLATION	Control register violation
8'hC	CORE_ERR_OVERFLOW	Overflow
8'hD	CORE_ERR_CTRLRD_MAX_RETRIES	Control read max retries exceeded
8'hE	CORE_ERR_CTRLRD_ENGINE	Control read engine FSM error
8'hF	CORE_ERR_USER_DEFINED	User-defined error

3.1.10 core_timeout_code_t

Packet context: packet_type = PktTypeTimeout, protocol = PROTOCOL_CORE

Value	Name	Description
8'h0	CORE_TIMEOUT_DESCRIPTOR_FETCH	Descriptor fetch timeout
8'h1	CORE_TIMEOUT_CTRLRD_RETRY	Control read retry timeout
8'h2	CORE_TIMEOUT_CTRLWR_WRITE	Control write timeout

Value	Name	Description
8'h3	CORE_TIMEOUT_DATA_TRANSFER	Data transfer timeout
8'h4	CORE_TIMEOUT_CREDIT_WAIT	Credit wait timeout
8'h5	CORE_TIMEOUT_CONTROL_WAIT	Control enable wait timeout
8'h6	CORE_TIMEOUT_ENGINE_RESPONSE	Sub-engine response timeout
8'h7	CORE_TIMEOUT_STATE_TRANSITION	FSM state transition timeout
8'h8–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	CORE_TIMEOUT_USER_DEFINED	User-defined timeout

3.1.11 core_completion_code_t

Packet context: packet_type = PktTypeCompletion, protocol = PROTOCOL_CORE

Value	Name	Description
8'h0	CORE_COMPL_DESCRIPTOR_LOADED	Descriptor successfully loaded
8'h1	CORE_COMPL_DESCRIPTOR_CHAIN	Descriptor chain completed
8'h2	CORE_COMPL_CTRLRD_COMPLETED	Control read completed (with match)
8'h3	CORE_COMPL_CTRLWR_COMPLETED	Control write completed
8'h4	CORE_COMPL_DATA_TRANSFER	Data transfer completed
8'h5	CORE_COMPL_CREDIT_CYCLE	Credit cycle completed
8'h6	CORE_COMPL_CHANNEL_COMPLETE	Channel processing complete
8'h7	CORE_COMPL_ENGINE_READY	Sub-engine ready
8'h8–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	CORE_COMPL_USER_DEFINED	User-defined completion

3.1.12 core_threshold_code_t

Packet context: packet_type = PktTypeThreshold, protocol = PROTOCOL_CORE

Value	Name	Description
8'h0	CORE_THRESH_DESCRIPTOR_QUEUE	Descriptor queue depth threshold
8'h1	CORE_THRESH_CREDIT_LOW	Credit low threshold
8'h2	CORE_THRESH_FLAG_RETRY_COUNT	Flag retry count threshold
8'h3	CORE_THRESH_LATENCY	Processing latency threshold
8'h4	CORE_THRESH_ERROR_RATE	Error rate threshold
8'h5	CORE_THRESH_THROUGHPUT	Throughput threshold
8'h6	CORE_THRESH_ACTIVE_CHANNELS	Active channel count threshold
8'h7	CORE_THRESH_PROGRAM_LATENCY	Program write latency threshold
8'h8	CORE_THRESH_DATA_RATE	Data transfer rate threshold
8'h9–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	CORE_THRESH_USER_DEFINED	User-defined threshold

3.1.13 core_performance_code_t

Packet context: packet_type = PktTypePerf, protocol = PROTOCOL_CORE

Value	Name	Description
8'h0	CORE_PERF_END_OF_DATA	Stream continuation signal
8'h1	CORE_PERF_END_OF_STREAM	Stream termination signal
8'h2	CORE_PERF_ENTERING_IDLE	FSM returning to idle
8'h3	CORE_PERF_CREDIT_INCR	Credit added by software

Value	Name	Description
8'h4	CORE_PERF_CREDIT_EXHAUSTED	Credit blocking execution
8'h5	CORE_PERF_STATE_TRANSITION	FSM state change
8'h6	CORE_PERF_DESCRIPTOR_ACTIVE	Data processing started
8'h7	CORE_PERF_CTRLRD_RETRY	Control read retry attempt
8'h8	CORE_PERF_CHANNEL_ENABLE	Channel enabled by software
8'h9	CORE_PERF_CHANNEL_DISABLE	Channel disabled by software
8'hA	CORE_PERF_CREDIT_UTILIZATION	Credit utilization metric
8'hB	CORE_PERF_PROCESSING_LATENCY	Total processing latency
8'hC	CORE_PERF_QUEUE_DEPTH	Current queue depth
8'hD–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	CORE_PERF_USER_DEFINED	User-defined performance

3.1.14 core_debug_code_t

Packet context: packet_type = PktTypeDebug, protocol = PROTOCOL_CORE

Value	Name	Description
8'h0	CORE_DEBUG_FSM_STATE_CHANGE	Descriptor FSM state change
8'h1	CORE_DEBUG_DESCRIPTOR_CONTENT	Descriptor content trace
8'h2	CORE_DEBUG_CTRLRD_ENGINE_STATE	Control read engine state trace
8'h3	CORE_DEBUG_CTRLWR_ENGINE_STATE	Control write engine state trace
8'h4	CORE_DEBUG_CREDIT_OPERATION	Credit system operation

Value	Name	Description
8'h5	CORE_DEBUG_CONTROL_REGISTER	Control register access
8'h6	CORE_DEBUG_ENGINE_HANDSHAKE	Engine handshake trace
8'h7	CORE_DEBUG_QUEUE_STATUS	Queue status change
8'h8	CORE_DEBUG_COUNTER_SNAPSHOT	Counter values snapshot
8'h9	CORE_DEBUG_ADDRESS_TRACE	Address progression trace
8'hA	CORE_DEBUG_PAYLOAD_TRACE	Payload content trace
8'hB–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	CORE_DEBUG_USER_DEFINED	User-defined debug

3.1.15 Unified Event Code Union (arb_core_event_code_t)

monitor_arbiter_pkg exports a single arb_core_event_code_t packed union that overlays all twelve enums (six ARB + six CORE) plus a raw 8-bit view onto a single field. Use this when a wrapper carries event codes from either protocol on a shared bus — the protocol field of the monbus packet disambiguates which enum interpretation applies.

```
typedef union packed {
    arb_error_code_t      arb_error;
    arb_timeout_code_t    arb_timeout;
    arb_completion_code_t arb_completion;
    arb_threshold_code_t  arb_threshold;
    arb_performance_code_t arb_performance;
    arb_debug_code_t      arb_debug;

    core_error_code_t      core_error;
    core_timeout_code_t    core_timeout;
    core_completion_code_t core_completion;
    core_threshold_code_t  core_threshold;
    core_performance_code_t core_performance;
    core_debug_code_t      core_debug;

    logic [7:0] raw;
} arb_core_event_code_t;
```

Helper constructors create_arb_error_event, create_arb_timeout_event, create_arb_completion_event, create_arb_threshold_event, create_arb_performance_event, and create_arb_debug_event build the union from a typed

ARB enum value. (Equivalent CORE constructors are not exported by the package — code producing CORE codes assigns the union field directly.) ## Related Modules - **monitor_package_spec.md** — Universal types, packet layout, helper functions. - **monitor_amba4_pkg.md** — AXI4 / APB4 / AXIS4 event codes. - **monitor_amba5_pkg.md** — AXI5 / APB5 / AXIS5 extended event codes. - **The Monitor System** — architecture and capabilities: how packets are produced, filtered, transported and captured, and how to instrument a block that is not a bus protocol.

3.2 Timing Characteristics

This module is **purely combinational** – it contains no `always_ff` and no latch, so it holds no state and adds no clock cycles. Its outputs settle a propagation delay after its inputs, and it introduces no latency into a pipeline that instantiates it.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

3.3 Testing

No dedicated testbench for this module. It has no `val/**/test_monitor_arbiter_pkg.py`. It is exercised indirectly, through the tests of modules that instantiate it (directly or further up):

- `arbiter_monbus_common` – `val/**/test_arbiter_monbus_common.py`
- `axi4_master_rd_mon` – `val/**/test_axi4_master_rd_mon.py`
- `axi4_master_rd_mon_cg` – `val/**/test_axi4_master_rd_mon_cg.py`
- `axi4_master_wr_mon` – `val/**/test_axi4_master_wr_mon.py`
- `axi4_master_wr_mon_cg` – `val/**/test_axi4_master_wr_mon_cg.py`

Indirect coverage exercises this module only in the configurations those parents elaborate. A parameter or mode no parent uses is untested.

Treat any behaviour described on this page as unverified by simulation.

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4 Monitor Package Specification

This is the canonical reference for the monitor-bus packet format and the universal types shared by every protocol monitor in the AMBA subsystem. For per-protocol event-code enums see the three sibling docs:

- `monitor_amba4_pkg.md` — AXI4 / APB4 / AXIS4 event codes
 - `monitor_amba5_pkg.md` — AXI5 / APB5 / AXIS5 extended codes
 - `monitor_arbiter_pkg.md` — ARB and CORE event codes
-

4.1 Package Organization

The RTL splits monitor type definitions across four SystemVerilog packages:

Package	File	Purpose
<code>monitor_common_pkg</code>	<code>rtl/amba/includes/ monitor_common_pkg.sv</code>	Universal types: packet structure, timestamp, protocol enum, packet-type enum, helper functions. This doc.
<code>monitor_amba4_pkg</code>	<code>rtl/amba/includes/ monitor_amba4_pkg.sv</code>	AXI4 / APB4 / AXIS4 event-code enums (error / timeout / completion / threshold / perf / addr-match / debug).
<code>monitor_amba5_pkg</code>	<code>rtl/amba/includes/ monitor_amba5_pkg.sv</code>	AXI5 / APB5 / AXIS5 extended event-code enums (atomic, trace, wakeup, parity, user).
<code>monitor_arbiter_pkg</code>	<code>rtl/amba/includes/ monitor_arbiter_pkg.s v</code>	ARB and CORE event codes for arbiter monitors and custom subsystems.

A wrapper `monitor_pkg` re-exports all four for back-compatibility — code that imports `monitor_pkg::*`; continues to compile unchanged.

4.2 The Monbus Record

A monbus emission is two paired wires carried atomically through every level of the transport: a **128-bit packet** (`monbus_packet`) plus a **64-bit side-band timestamp** (`monbus_timestamp`). 192 bits total.

Wire	Bits	Width	Field	Owner	Notes
<code>monbus_timestamp</code>	[63:0]	64	timestamp	system	Free-running counter from <code>monbus_group_core</code> , sampled at emission. Consumers treat it as an opaque ordering key.
<code>monbus_packet</code>	[127:124]	4	packet_type	enum (fixed)	See Packet Type Enum .
<code>monbus_packet</code>	[123:109]	15	reserved	(slack)	Currently emitted as zero.
<code>monbus_packet</code>	[108:105]	4	protocol	enum (fixed)	See Protocol Enum .
<code>monbus_packet</code>	[104:97]	8	event_code	enum (fixed)	Protocol-specific; see sibling docs.
<code>monbus_packet</code>	[96:88]	9	channel_id	designer	AXI ID or channel index.
<code>monbus_packet</code>	[87:72]	16	agent_id	designer	Designer-allocated

Wire	Bits	Width	Field	Owner	Notes
monbus_packet	[71:64]	8	unit_id	designer	agent ID. Designer-allocated unit ID.
monbus_packet	[63:0]	64	event_data	payload	Layout depends on (packet_type, event_code) — see event_data Payload Conventions .

4.2.1 SystemVerilog Type Aliases

```
localparam int MONBUS_PKT_WIDTH = 128;
localparam int MONBUS_TS_WIDTH  = 64;
```

```
typedef logic [MONBUS_PKT_WIDTH-1:0] monitor_packet_t;
typedef logic [MONBUS_TS_WIDTH-1:0]  monbus_timestamp_t;
```

The widths are localparams in `monitor_common_pkg`, **not** per-module parameters — there is exactly one packet format and one timestamp format across the codebase. Any consumer that declares a bus or FIFO holding a packet must reference these constants, not hard-code 128 / 64.

4.3 Protocol Enum

The 4-bit protocol field identifies which protocol family produced the event. The enum is defined in `monitor_common_pkg`:

Value	Name	Description
4'h0	PROTOCOL_AXI	AXI / AXI-Lite / AXI5
4'h1	PROTOCOL_AXIS	AXI4-Stream
4'h2	PROTOCOL_APB	Advanced Peripheral Bus (APB4 / APB5)
4'h3	PROTOCOL_ARB	Arbiter-specific events

Value	Name	Description
4'h4	PROTOCOL_CORE	Core / custom subsystem events
4'h5	PROTOCOL_WB	Wishbone B4 events (wb4_monitor; event codes in monitor_wb4_pkg, not re-exported by monitor_pkg)
4'h5–4'hF	(reserved)	Forward-compat slack — 16 protocols max.

```

typedef enum logic [3:0] {
    PROTOCOL_AXI    = 4'h0,
    PROTOCOL_AXIS   = 4'h1,
    PROTOCOL_APB    = 4'h2,
    PROTOCOL_ARB    = 4'h3,
    PROTOCOL_CORE   = 4'h4,
    PROTOCOL_WB     = 4'h5
} protocol_type_t;

```

4.4 Packet Type Enum

The 4-bit packet_type field classifies the event independent of protocol. The 16 codes are defined as localparams in monitor_common_pkg:

Value	Name	Description
4'h0	PktTypeError	Protocol violation, SLVERR/DECERR, orphan, range violation
4'h1	PktTypeCompletion	Transaction completed successfully
4'h2	PktTypeThreshold	Threshold crossed (latency, queue depth, etc.)
4'h3	PktTypeTimeout	Timeout detected on command / data / response phase

Value	Name	Description
4'h4	PktTypePerf	Performance metric event
4'h5	PktTypeCredit	Credit status (AXIS)
4'h6	PktTypeChannel	Channel status (AXIS)
4'h7	PktTypeStream	Stream event (AXIS — start/end/abort)
4'h8	PktTypeAddrMatch	Address-match watchpoint (AXI). Produced by axi_monitor_addr_check on a DEBUG-flavored range hit (event code AXI_ADDR_RANGE_MATCH 8'h01), gated by cfg_debug_enable.
4'h9	PktTypeAPB	APB-specific event
4'hA–4'hC	(reserved)	Forward-compat slack
4'hD	PktTypePerfWin	Windowed-performance event (AXI perf window close)
4'hE	PktTypePerfHist	Latency-histogram bucket event (AXI)
4'hF	PktTypeDebug	Debug / trace event
localparam logic [3:0]	PktTypeError	= 4'h0;
localparam logic [3:0]	PktTypeCompletion	= 4'h1;
localparam logic [3:0]	PktTypeThreshold	= 4'h2;
localparam logic [3:0]	PktTypeTimeout	= 4'h3;
localparam logic [3:0]	PktTypePerf	= 4'h4;
localparam logic [3:0]	PktTypeCredit	= 4'h5;
localparam logic [3:0]	PktTypeChannel	= 4'h6;
localparam logic [3:0]	PktTypeStream	= 4'h7;
localparam logic [3:0]	PktTypeAddrMatch	= 4'h8;
localparam logic [3:0]	PktTypeAPB	= 4'h9;
localparam logic [3:0]	PktTypePerfWin	= 4'hD;
localparam logic [3:0]	PktTypePerfHist	= 4'hE;
localparam logic [3:0]	PktTypeDebug	= 4'hF;

4.5 event_data Payload Conventions

The 64-bit event_data field is interpreted in the context of (packet_type, event_code). There is no universal layout — each producer defines its own packing. The table below lists the most common shapes:

Producer	packet_type	event_code	event_data layout
axi_monitor_addr_check	Error	AXI_ERR_ADDR_RANGE (8'h0D)	[63:60] = 4'hF no-range sentinel (this packet is the allowlist MISS, so no range matched), [59:0] = the offending unmatched address
apb_monitor_addr_check	Error	APB_ERR_ADDR_RANGE (8'h08)	[63:60] = range_index (4b), [59] = is_read, [58:0] = address
axi_monitor_reporter	Error / Timeout / Completion	various	Full 64-bit address, or zero-extended ID / latency / counter
axi_monitor_reporter	Perf	AXI_PERF_*	Zero-extended counter value (completed/error counts, latencies)
axi_monitor_reporter	Threshold	AXI_THRESH_ACTIVE_COUNT	Zero-extended active-transaction count
arbiter_monbus_common	Error / Perf	ARB_*	Arbiter-specific payload (winning client, weight, etc.)

Notable design rule: **direction is not embedded in event_data for AXI addr_check**. Each AXI monitor instance watches a single direction (AR or AW) set at build time via the IS_READ

parameter; consumers recover direction from the emitting monitor's (unit_id, agent_id) tuple. APB addr_check is the exception — one APB monitor sees both reads and writes on the same channel, so it carries an explicit is_read flag.

4.6 Helper Functions

Field accessors and a packet builder are declared in monitor_common_pkg:

```
function automatic logic [3:0]      get_packet_type
(monitor_packet_t pkt); return pkt[127:124]; endfunction
function automatic logic [14:0]     get_reserved
(monitor_packet_t pkt); return pkt[123:109]; endfunction
function automatic protocol_type_t get_protocol_type
(monitor_packet_t pkt); return protocol_type_t'(pkt[108:105]);
endfunction
function automatic logic [7:0]      get_event_code
(monitor_packet_t pkt); return pkt[104:97]; endfunction
function automatic logic [8:0]      get_channel_id
(monitor_packet_t pkt); return pkt[96:88]; endfunction
function automatic logic [15:0]     get_agent_id
(monitor_packet_t pkt); return pkt[87:72]; endfunction
function automatic logic [7:0]      get_unit_id
(monitor_packet_t pkt); return pkt[71:64]; endfunction
function automatic logic [63:0]     get_event_data
(monitor_packet_t pkt); return pkt[63:0]; endfunction

function automatic monitor_packet_t create_monitor_packet(
    logic [3:0]      packet_type,
    protocol_type_t protocol,
    logic [7:0]      event_code,
    logic [8:0]      channel_id,
    logic [7:0]      unit_id,
    logic [15:0]     agent_id,
    logic [63:0]     event_data
);
    return {packet_type, 15'h0, protocol, event_code,
            channel_id, agent_id, unit_id, event_data};
endfunction
```

create_monitor_packet zeroes the reserved field; do not rely on its bits.

4.6.1 Construction Example

```
// AXI master-side wrapper emits a range-violation error packet
monitor_packet_t pkt = create_monitor_packet(
    .packet_type ( PktTypeError
    .protocol    ( PROTOCOL_AXI
```



```

        .event_code ( AXI_ERR_ADDR_RANGE                                ), // 8'h0D
        .channel_id ( 9'(cmd_id)                                       ),
        .unit_id    ( UNIT_ID                                           ), // build-
time
        .agent_id   ( AGENT_ID                                          ), // build-
time
        // MISS packet: the nibble is the no-range sentinel, never a range
index.
        .event_data ( {4'hF, 60'(cmd_addr)}                             )
    );

```

The wrapper drives pkt on monbus_packet and the sampled i_mon_time on monbus_timestamp in the same cycle.

4.7 Side-Band Timestamp

The timestamp lives outside the packet so the packet layout is fixed at 128 bits and create_monitor_packet doesn't need a timestamp argument.

```
typedef logic [MONBUS_TS_WIDTH-1:0] monbus_timestamp_t; // 64 bits
```

Stage	Wire	Notes
Source	i_mon_time	A free-running counter generated in monbus_group_core and broadcast to every wrapper via the shared mon_time_w net.
Sampling	addr_pkt_timestamp / monbus_timestamp	Each producer (addr_check, reporter, debug) samples i_mon_time on the cycle its packet asserts valid.
Transport	monbus_arbiter	The arbiter carries (packet, timestamp) atomically through a 192-bit skid so consumers never see a mismatched pair.
Drain	monbus_group_core	Emits a 3-beat record on

Stage	Wire	Notes
		m_mon_axil for bulk capture: beat 0 = {tag[3:0], source_ts[59:0]} (tag = 4'h0 raw; non-zero tags reserved for a future on-the-wire compression encoder), beat 1 = packet[127:64], beat 2 = packet[63:0]. Single-record reads via s_mon_axil use the same slice order for IRQ-driven consumption.

Consumers (host scripts, parsers, the bin/TBClasses/monbus decoder) treat the timestamp as an opaque ordering key — its semantics (cycle counter, microsecond, PTP time) are deployment-specific, and nothing in the RTL depends on which you choose. The tradeoffs between those choices used to be written up in the MonitorSystem whitepaper, removed in 2026-07 as superseded; they are not currently documented anywhere.

4.8 Per-Protocol Event Code Packages

Each event-code enum is 8 bits wide and lives in the protocol-specific sibling package. The pattern is:

```
event_code = enum value from
              monitor_{amba4, amba5, arbiter}_pkg::
{protocol}_{category}_code_t
```

Categories per protocol (each ~16-entry enum):

Package	Protocol	Categories
monitor_amba4_pkg	AXI4	error, timeout, completion, threshold, performance, addr_match, debug, perfwin, perfhist (nine)
monitor_amba4_pkg	APB4	error, timeout,

Package	Protocol	Categories
monitor_amba4_pkg	AXIS4	completion, threshold, performance, debug error, timeout, completion, credit, channel, stream
monitor_amba5_pkg	AXI5	atomic, trace
monitor_amba5_pkg	APB5	wakeup, parity, user
monitor_amba5_pkg	AXIS5	wakeup, parity, CRC
monitor_arbiter_pkg	ARB	error, timeout, completion, threshold, performance, debug
monitor_arbiter_pkg	CORE	error, timeout, completion, threshold, performance, debug

Full enum-value tables are in the sibling docs.

4.9 Backward Compatibility

The aggregating wrapper `monitor_pkg` re-exports every type from the four split packages, so existing code that does:

```
import monitor_pkg::*;
```

continues to compile. New code should import the specific package it needs:

```
// Universal types
import monitor_common_pkg::*;

// Plus the protocol you actually use:
import monitor_amba4_pkg::*;
```

This keeps namespace pollution down and makes the dependency graph explicit at the file level.

4.10 Related Modules

- [The Monitor System](#) — architecture and capabilities: drain paths, capture strategies (bulk / compressed / counting), and the aggregation topology.
 - `../monitor/axi_monitor_base.md` — Core monitor that emits packets.
 - `../monitor/axi_monitor_reporter.md` — Packet formatting logic (where `create_monitor_packet` is invoked).
 - `../monitor/axi_monitor_addr_check.md` — Address-range checker, canonical example of structured `event_data`.
 - `../monitor/monbus_arbiter.md` — 192-bit atomic packet+timestamp arbitration.
-

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5 Wishbone B4 Event Code Reference

The per-protocol event-code reference for Wishbone B4. The 8-bit `event_code` field of every monbus packet sourced by `wb4_monitor` takes its value from one of the enums declared in `rtl/amba/includes/monitor_wb4_pkg.sv` and listed below. For the universal packet layout, the protocol and `packet_type` enums and the helper functions, see `monitor_package_spec.md`.

Each event-code enum is 8 bits. Slots `8'h0–8'hE` hold predefined codes, with `_RESERVED_*` placeholders absorbing the unused indices, and slot `8'hF` is the `*_USER_DEFINED` escape hatch, matching the AMBA4 and AMBA5 packages.

5.1 Design Notes

This is a package, not a module. No ports, no clock. It declares the event encodings the Wishbone monitor emits so that each one has exactly one definition.

Deliberately NOT re-exported by `monitor_pkg`. The aggregating wrapper re-exports the AMBA4, AMBA5 and arbiter packages, but not this one. Only `wb4_monitor`'s filelist includes it, so adding Wishbone to the monitor family changed no existing consumer's compile closure. If a future block needs both, include the package directly rather than widening the aggregator.

PROTOCOL_WB lives elsewhere. The protocol id 4'h5 is an additive entry in monitor_common_pkg's protocol_type_t, alongside AXI, AXIS, APB, ARB and CORE, because the protocol enum is universal. Only the event codes are Wishbone-specific and live here.

Address-range violations share APB's code. apb_monitor_addr_check is reused by wb4_monitor through its PROTOCOL parameter. The checker emits event code 8'h08, which is APB_ERR_ADDR_RANGE in the AMBA4 package and WB_ERR_ADDR_RANGE here; the two agree by construction so one checker can serve both protocols.

No test of its own, and none expected. A package has no behaviour to simulate. It is verified by wb4_monitor failing to elaborate if a declaration is wrong, and by val/amba/test_wb4_monitor.py decoding the codes through TBClasses.monbus, whose Python enums mirror this file.

5.1.1 wb_error_code_t

Packet context: packet_type = PktTypeError, protocol = PROTOCOL_WB

Value	Name	Description
8'h0	WB_ERR_ERR	Slave terminated the transfer with ERR
8'h1	WB_ERR_ORPHAN_RSP	A response arrived with no request outstanding
8'h2	WB_ERR_TRACK_LOST	Request accepted with no free tracking slot; the transfer is untracked and its response will read as an orphan
8'h3–8'h7	(reserved)	Reserved for future use
8'h8	WB_ERR_ADDR_RANGE	Address-range violation (from apb_monitor_addr_check, tagged PROTOCOL_WB)
8'h9–8'hE	(reserved)	Reserved for future use
8'hF	WB_ERR_USER_DEFINED	User-defined error

5.1.2 wb_timeout_code_t

Packet context: packet_type = PktTypeTimeout, protocol = PROTOCOL_WB

Value	Name	Description
8'h0	WB_TIMEOUT_CMD	A request sat on cmd_valid without cmd_ready for cfg_cmd_timeout_cnt clocks
8'h1	WB_TIMEOUT_RSP	The oldest open transfer went unterminated for cfg_rsp_timeout_cnt clocks
8'h2–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	WB_TIMEOUT_USER_DEFINED	User-defined timeout

Each fires once: the command timeout re-arms when the request is taken or withdrawn, the response timeout is flagged per queue entry.

5.1.3 wb_completion_code_t

Packet context: packet_type = PktTypeCompletion, protocol = PROTOCOL_WB

Value	Name	Description
8'h0	WB_COMPL_ACK	Terminated ACK (direction in aux_data)
8'h1	WB_COMPL_READ	Read terminated ACK
8'h2	WB_COMPL_WRITE	Write terminated ACK
8'h3	WB_COMPL_RTY	Terminated RTY. A completion, not an error: the FUB decides whether to retry
8'h4–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	WB_COMPL_USER_DEFINED	User-defined completion

RTY sits here rather than in the error enum on purpose. A retry is the slave asking for the transfer again, not a failure, and wb4_retry may absorb it before the FUB ever sees it.

5.1.4 wb_perf_code_t

Packet context: packet_type = PktTypePerf, protocol = PROTOCOL_WB

Value	Name	Description
8'h0	WB_PERF_READ_LATENCY	A read's latency crossed <code>cfg_latency_threshold</code>
8'h1	WB_PERF_WRITE_LATENCY	A write's latency crossed <code>cfg_latency_threshold</code>
8'h2–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	WB_PERF_USER_DEFINED	User-defined performance event

5.1.5 wb_debug_code_t

Packet context: `packet_type = PktTypeDebug`, `protocol = PROTOCOL_WB`

Value	Name	Description
8'h0	WB_DEBUG_QUEUE_ACTIVE	The tracking queue went from empty to non-empty
8'h1	WB_DEBUG_QUEUE_IDLE	The tracking queue drained
8'h2–8'hE	<i>(reserved)</i>	<i>Reserved for future use</i>
8'hF	WB_DEBUG_USER_DEFINED	User-defined debug event

5.2 Related

- `wb4_monitor` - the only module that imports this package
- `monitor_package_spec.md` - packet layout and the protocol enum, including `PROTOCOL_WB` (4'h5)
- `monitor_amba4_pkg.md` - the AMBA4 equivalent, whose 8'h08 address-range code this package matches

6 apb_monitor_addr_check

Module: `apb_monitor_addr_check.sv` **Location:** `rtl/amba/monitor/` **Status:** Production Ready

6.1 Overview

A configurable N-range address-violation filter for the APB monitor pipeline — the APB mirror of `axi_monitor_addr_check`. It watches the `cmd_valid/cmd_ready` handshake the `apb4_monitor` already snoops, and when an accepted command's `paddr` falls inside any of N configured `[low, high]` inclusive ranges it emits a `PktTypeError` MonBus packet with event code `APB_ERR_ADDR_RANGE (8'h08)`.

APB peripherals tend to grow reserved or protected address windows that software must never touch, and flagging those accesses is the whole point of this block. It adds no logic to the APB datapath itself: it snoops the command handshake the monitor already observes, compares the accepted address against the programmed ranges, and reports which range was hit, the offending address, and whether the access was a read or a write.

- Up to N configurable inclusive `[low, high]` address ranges (default 4, up to 16 usable via the 4-bit range index)
- Per-range enable plus a global `cfg_addr_check_enable`
- Emits a standard MonBus error packet with the offending address, range index, and direction
- Preserves an `is_read` bit (APB has no separate AR/AW channels, so direction must be carried explicitly)
- Per-range pending mask latches hits and drains them one packet at a time under backpressure; repeat hits on one range coalesce newest-wins (see Operation)
- Side-band timestamp captured from the broadcast free-running counter on emission
- Exact-match support by setting a range's `low == high`

Typical uses: flagging accesses to reserved/protected APB register windows, security or safety guard-banding of peripheral address maps, debug tripwires on unexpected address regions during bring-up, and exact-address watchpoints (`low == high`). In short: programmable, per-range address-violation reporting on the MonBus that mirrors the AXI variant while carrying the APB-specific read/write direction bit.

6.2 Parameters

Parameter	Type	Default	Description
<code>N_ADDR_RANGES</code>	<code>int</code>	4	Number of configurable <code>[low, high]</code>

Parameter	Type	Default	Description
ADDR_WIDTH	int	32	APB address width
UNIT_ID	logic [7:0]	8'h00	Unit id stamped into emitted packets
AGENT_ID	logic [15:0]	16'h0000	Agent id stamped into emitted packets
PROTOCOL	logic [3:0]	PROTOCOL_APB	Protocol tag on the emitted packet. wb4_monitor reuses the checker on its command queue with PROTOCOL_WB; the event code 8'h08 is *_ERR_ADDR_RANGE in both packages
M	int	ADDR_WIDTH	Internal address-width alias

6.3 Ports

6.3.1 Clock and Reset

Port	Direction	Width	Description
clk	input	1	Clock
aresetn	input	1	Active-low asynchronous reset

6.3.2 Timestamp

Port	Direction	Width	Description
i_mon_time	input	monbus_timestamp_t	Free-running counter broadcast by the monbus_group family; sampled on emission

6.3.3 Snooped APB Command Stream

Port	Direction	Width	Description
cmd_paddr	input	M	Command address (APB paddr)
cmd_pwrite	input	1	Command direction (1 = write, 0 = read)
cmd_valid	input	1	Command valid (from the monitor's snoop of the APB handshake)
cmd_ready	input	1	Command ready; valid && ready && enable marks an accepted command

6.3.4 Range Configuration

Port	Direction	Width	Description
cfg_addr_check_enable	input	1	Global enable for the checker (also gates output valid)
cfg_addr_range_enable	input	N_ADDR_RANGES	Per-range enable mask

Port	Direction	Width	Description
cfg_addr_range_low	input	N_ADDR_RANGES × M	Per-range inclusive lower bound
cfg_addr_range_high	input	N_ADDR_RANGES × M	Per-range inclusive upper bound

6.3.5 Outgoing MonBus Packet

Port	Direction	Width	Description
addr_pkt_valid	output	1	Packet valid (a pending violation is ready to emit)
addr_pkt_ready	input	1	Downstream ready; valid && ready accepts the packet
addr_pkt_data	output	monitor_packet_t	Assembled MonBus error packet
addr_pkt_timestamp	output	monbus_timestamp_t	Timestamp side-band (i_mon_time)

6.4 Functional Description

6.4.1 Combinational Range Hits

A command “fires” when `cmd_valid` && `cmd_ready` && `cfg_addr_check_enable`. For each range `i`, a one-hot hit is asserted when the range is enabled, the command fires, and `cmd_paddr` lies within `[cfg_addr_range_low[i], cfg_addr_range_high[i]]` inclusive. Setting a range’s low and high equal yields exact-match (watchpoint) semantics.

6.4.2 Pending Mask and Latched Snapshot

Each range has a pending bit. On a hit, the range’s pending bit sets and its address and direction are latched (`r_lat_addr[i] = cmd_paddr`, `r_lat_is_read[i] = !cmd_pwrite`). This decouples detection from emission so a range with a pending violation keeps its slot while the

MonBus is backpressured. It does NOT preserve every violation: there is one latch per range, so repeated hits on the SAME range coalesce and the newest address wins. Two hits on range 0 during a stall yield one packet carrying the second address; the first address is not reported. This is the same lossy-but-honest behaviour the AXI sibling documents for its DEBUG ranges. Distinct ranges do not coalesce with each other. A first-match priority encoder over `r_pending` selects the range to emit (`emit_oh / emit_idx`); `addr_pkt_valid` asserts when any range is pending and the checker is enabled. When the packet is accepted (`addr_pkt_valid && addr_pkt_ready`), the emitted range's pending bit clears. If a fresh hit and an accept collide on the same range in one cycle, the hit wins (the pending bit stays set), so a back-to-back violation is not dropped.

6.4.3 Packet Encoding

The emitted packet is the standard 128-bit MonBus format with a 64-bit `event_data` field, assembled via `create_monitor_packet`:

- **packet_type** = `PktTypeError (4'h0)`
- **protocol** = `PROTOCOL` parameter: `PROTOCOL_APB (4'h2)` by default, `PROTOCOL_WB (4'h5)` inside `wb4_monitor`
- **event_code** = `APB_ERR_ADDR_RANGE (8'h08)`
- **channel_id** = 0 (APB has no ID concept)
- **unit_id** = `UNIT_ID`, **agent_id** = `AGENT_ID`
- **event_data[63:60]** = range index (4 bits → up to 16 ranges)
- **event_data[59]** = `is_read` (1 = read, 0 = write)
- **event_data[58:0]** = offending `cmd_paddr` (zero-padded if `M < 59`, or its low 59 bits if `M >= 59`)

6.4.4 The is_read Carve-Out

The AXI variant drops the direction bit because AR and AW are separate channels — direction is implied by which monitor (read vs write) emitted the packet. APB has no such split: the same monitor sees both directions, so this block preserves an explicit `is_read` bit (carved out of the 60-bit address slot) so consumers can disambiguate read from write.

6.4.5 Timestamp Side-Band

Exactly as in the AXI variant, `i_mon_time` (the free-running counter broadcast by the `monbus_group` family) is driven straight out on `addr_pkt_timestamp`, so the consuming group can stamp the violation with a bus-consistent time.

6.5 Timing Characteristics

This module is **sequential**: it contains clocked logic (via `always_ff` or the repository's `ALWAYS_FF_RST` macro) and therefore holds state. Outputs driven from those blocks are registered and appear one clock after the inputs that produced them.

Per-path cycle counts are not enumerated here; read the block that drives the signal you care about. No synthesis frequency or area figures are quoted – none have been measured against a target device.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

6.6 Usage Examples

```
// Guard two reserved APB windows; report violations on the MonBus.
apb_monitor_addr_check #(
    .N_ADDR_RANGES    (4),
    .ADDR_WIDTH        (32),
    .UNIT_ID           (8'h10),
    .AGENT_ID           (16'h0001)
) u_apb_addr_check (
    .clk                (pclk),
    .aresetn            (presetn),
    .i_mon_time         (mon_time),           // from the monbus
group

    // Snooped APB command handshake (from apb4_monitor)
    .cmd_paddr          (apb_paddr),
    .cmd_pwrite         (apb_pwrite),
    .cmd_valid          (cmd_valid),
    .cmd_ready          (cmd_ready),

    // Range config (range 0 = a window, range 1 = an exact
watchpoint).
    //
    // cfg_addr_range_low/high are PACKED 2D arrays, so an assignment
pattern
    // fills left-to-right starting at the HIGHEST index -- exactly
like a
    // concatenation. The leftmost item is range 3, the rightmost is
range 0.
    // Getting this backwards silently programs the wrong ranges: the
enable
    // mask lights up ranges 0-1 while the intended bounds sit on 3-2.
    //
        { r3,      r2,      r1,
```

```

r0
    }
    .cfg_addr_check_enable (1'b1),
    .cfg_addr_range_enable (4'b0011),
    .cfg_addr_range_low    ('{32'h0, 32'h0, 32'h0000_2000,
32'h0000_1000}),
    .cfg_addr_range_high   ('{32'h0, 32'h0, 32'h0000_2000,
32'h0000_1FFF}),

    // MonBus output (connect to the group's error-drain path)
    .addr_pkt_valid        (addr_pkt_valid),
    .addr_pkt_ready        (addr_pkt_ready),
    .addr_pkt_data          (addr_pkt_data),
    .addr_pkt_timestamp     (addr_pkt_timestamp)
);

```

6.7 Design Notes

6.7.1 Mirror of the AXI Variant

The block shares `axi_monitor_addr_check`'s *structure* – timestamp handling, the pending-mask/priority-encoder emission path, and the range-index/address packet layout – but it is **not** a behavioural mirror. The differences are load-bearing:

	<code>apb_monitor_addr_check</code>	<code>axi_monitor_addr_check</code>
Range meaning	Blocklist — an in-range hit raises <code>APB_ERR_ADDR_RANGE</code>	Allowlist — a miss (outside every enabled error range) raises <code>AXI_ERR_ADDR_RANGE</code>
Packet classes	Error only	Error <i>and</i> <code>PktTypeAddrMatch</code>
Per-range flavor	none	<code>ADDR_RANGE_IS_ERROR</code> selects <code>DEBUG</code> vs <code>ERROR</code> per range
Enable gating	<code>cfg_addr_check_enable</code>	plus <code>cfg_debug_enable</code> / <code>cfg_error_enable</code> per path
Channel id	hardwired <code>9'h0</code> (APB has no ID)	from <code>cmd_id</code>
<code>is_read</code>	preserved in the payload	dropped (recovered from <code>IS_READ</code> + unit/agent id)

The polarity inversion is the one that bites. Programming the APB ranges as if they were an allowlist — the AXI convention — flags every access *inside* your intended-legal window as a violation and lets everything outside it pass silently. On APB, a configured range is a region you want to be told about.

6.7.2 Backpressure Safety

Detection latches into `r_pending` and emission drains one range per accepted packet, so a pending RANGE survives downstream backpressure – though repeat hits on that range coalesce to the newest address, so the count of packets is not the count of violations. The hit-versus-accept collision rule keeps the pending bit set when a new hit lands on a range being drained the same cycle.

A hit landing on a range whose beat is already being presented goes to a per-range **shadow slot** rather than overwriting the latched payload, and is installed when that beat is accepted. Rewriting a beat already on the bus would change the payload under a held valid, which the MonBus valid/ready contract forbids; the shadow keeps newest-wins for every beat except the one on the wire. The AXI sibling carries the same guard for the same reason.

6.7.3 Address Field Width

For `ADDR_WIDTH >= 59` the low 59 bits of the address are carried; for narrower buses the address is zero-extended into the 59-bit payload slot. The upper four `event_data` bits are always the range index.

6.7.4 Exact-Match Watchpoints

Program `cfg_addr_range_low[i] == cfg_addr_range_high[i]` to turn a range into a single-address watchpoint.

6.8 Related Modules

Used by: the `apb4_monitor` pipeline (address-range violation reporting).

Uses: `monitor_common_pkg` (`PktTypeError`, `PROTOCOL_APB`, `monbus_timestamp_t`, `create_monitor_packet`); `monitor_amba4_pkg` (`APB_ERR_ADDR_RANGE` event code); `reset_defs.svh` (`ALWAYS_FF_RST` / `RST_ASSERTED` reset macros).

See also: `axi_monitor_addr_check.sv` (AXI-side equivalent, no `is_read` bit); `apb4_monitor.sv` (the APB monitor this checker plugs into).

6.9 Testing

val/amba/test_apb_monitor_addr_check.py exercises this module. It collects 1 parameter cases at the default REG_LEVEL.

```
source env_python
pytest val/amba/test_apb_monitor_addr_check.py -v
```

6.10 References

- RTL: rtl/amba/monitor/apb_monitor_addr_check.sv
 - Packet format: docs/markdown/rtl-amba/includes/monitor_package_spec.md
 - Architecture: docs/markdown/rtl-amba/shared/README.md
 - Design Guide: docs/markdown/rtl-amba/index.md
-

Last Updated: 2026-07-15

6.11 Navigation

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7 arbiter_monbus_common

Module: arbiter_monbus_common.sv **Location:** rtl/amba/monitor/ **Category:** Core Infrastructure **Status:** Production Ready

7.1 Overview

arbiter_monbus_common is the shared **telemetry block** that watches an arbiter and reports on it. It is instantiated inside arbiter_rr_pwm_monbus and arbiter_wrr_pwm_monbus as u_monitor.

It performs **no arbitration of its own**. Every arbiter-facing signal (request, grant_valid, grant, grant_id, grant_ack, block_arb) is an **input**: the module snoops the decisions an

external arbiter has already made, measures them, and emits 128-bit monitor packets about what it saw. There is no request/grant output, no data mux, and no client stream inputs.

Looking for the N:1 monitor-bus merge? That is a different module: `monbus_arbiter`, which takes N `monbus_valid`/packet streams and arbitrates them onto one. This page has described that module in the past; it does not describe this one.

What it does, in five verbs:

1. **Observe:** sample the arbiter's request/grant/ACK activity without perturbing it
2. **Detect:** identify starvation, ACK timeouts, protocol violations, and threshold crossings
3. **Measure:** maintain fairness-deviation and grant-efficiency metrics
4. **Report:** format findings as `monitor_packet_t` records and queue them for the monitor bus
5. **Expose:** drive silicon-debug status outputs (`debug_*`) for direct observation

Feature summary:

- **Snoops any arbiter** — RR or WRR, with or without the grant-ACK handshake
- **Per-client ACK timeout tracking** with a configurable threshold
- **Protocol violation detection** — multiple simultaneous grants, spurious ACKs, grant without request
- **Starvation detection** with per-client timing
- **Fairness deviation** measured against configured client weights
- **Grant efficiency** tracking (grants issued vs. completed)
- **128-bit monitor packets** emitted through an internal FIFO, plus a 64-bit side-band timestamp

7.2 Parameters

Parameter	Type	Default	Description
CLIENTS	int	4	Width of the snooped arbiter's client vectors (request,

Parameter	Type	Default	Description
			grant, grant_ack). Nothing is arbitrated here – this sizes what is observed
WAIT_GNT_ACK	int	0	1 = require a grant-ACK handshake
WEIGHTED_MODE	int	0	Dead — declared but referenced by no logic. Fairness analysis always uses the cfg_max_thresh weights, and this module performs no arbitration for a mode switch to modulate. Setting it changes nothing
MON_AGENT_ID	logic [15:0]	16'h0010	Monitor agent identifier (16-bit)
MON_UNIT_ID	logic [7:0]	8'h00	Monitor unit identifier (8-bit)
MON_FIFO_DEPTH	int	8	Monitor packet FIFO depth
MON_FIFO_ALMOST_MARGIN	int	1	Almost-full margin for the monitor FIFO
FAIRNESS_REPORT_CYCLES	int	256	How often the fairness deviation is recomputed (cycles). Not a sliding window: r_grant_counters

Parameter	Type	Default	Description
			and r_total_grants are cleared on reset alone, so the deviation is always computed from grants accumulated since reset. MIN_GRANTS_FOR_F AIRNESS is likewise compared against the lifetime total. After a traffic- phase or weight change the figure converges toward the new distribution rather than snapping to it
MIN_GRANTS_FOR_F AIRNESS	int	100	Minimum grants before a fairness report is valid
DEFAULT_ACK_TIME OUT	int	64	Dead — declared and referenced by no logic. The effective threshold is whatever is driven on the cfg_mon_ack_time out_thresh port (arbiter_rr_pwm_ monbus hardwires 16'h40;

Parameter	Type	Default	Description
			arbiter_wrr_pwm_monbus passes its own port through). Overriding this parameter changes nothing

MAX_LEVELS (default 16) is an independent parameter, not a derived one: it sets the per-client weight resolution and therefore the width of the `cfg_max_thresh` port ($CXMTW = CLIENTS * \lceil \log_2(MAX_LEVELS) \rceil$). `arbiter_wrr_pwm_monbus` passes its own MAX_LEVELS down. Change it and the port width changes with it.

$N(\lceil \log_2(CLIENTS) \rceil)$, `MON_FIFO_COUNT_WIDTH`, `MAX_LEVELS_WIDTH` and the weight-vector widths are derived from the parameters above. They are declared with `parameter` rather than `localparam`, so they are overridable in principle — don't, since they must stay consistent with what they are derived from.

7.2.1 Derived Parameters (do not override)

These are declared as `parameter` so the elaborator can compute them, not so callers can set them. Each defaults to an expression over the parameters above; overriding one desynchronises it from its source and the design fails to elaborate or silently mis-sizes a bus. Set the parameters they are derived FROM and leave these alone.

Derived parameter	Default expression
MFCW	MON_FIFO_COUNT_WIDTH
MTW	MAX_LEVELS_WIDTH

7.3 Ports

All ports, from `rtl/amba/monitor/arbiter_monbus_common.sv`. Every arbiter-facing signal is an INPUT: this module snoops and never drives the arbiter.

Clock and reset

Port	Dir	Width	Description
clk	In	1	Clock. Note this module uses <code>clk/rst_n</code> , not the AMBA

Port	Dir	Width	Description
rst_n	In	1	aclk/aresetn Active-low reset

Snooped arbiter interface (inputs only)

Port	Dir	Width	Description
request	In	CLIENTS	Client request vector
grant_valid	In	1	Grant is valid this cycle
grant	In	CLIENTS	One-hot grant vector
grant_id	In	N	Binary-encoded grant ID
grant_ack	In	CLIENTS	Per-client grant acknowledge
block_arb	In	1	Arbiter is blocked (optional; tie 0 if unused)

Configuration

Port	Dir	Width	Description
cfg_mon_enable	In	1	Master enable for the monitor
cfg_mon_pkt_type_enable	In	16	Per-packet-type enable mask
cfg_mon_latency_thresh	In	16	Grant-latency threshold
cfg_mon_starvation_thresh	In	16	Per-client starvation threshold
cfg_mon_fairness_thresh	In	16	Fairness-deviation threshold
cfg_mon_active_thresh	In	16	Active-client-count threshold

Port	Dir	Width	Description
cfg_mon_ack_time_out_thresh	In	16	ACK timeout threshold
cfg_mon_efficiency_thresh	In	16	Grant-efficiency threshold
cfg_mon_sample_period	In	8	Sampling period for the periodic metrics
cfg_max_thresh	In	CXMTW	Packed per-client weight/threshold vector
i_mon_time	In	monbus_timestamp_t	Shared timestamp input

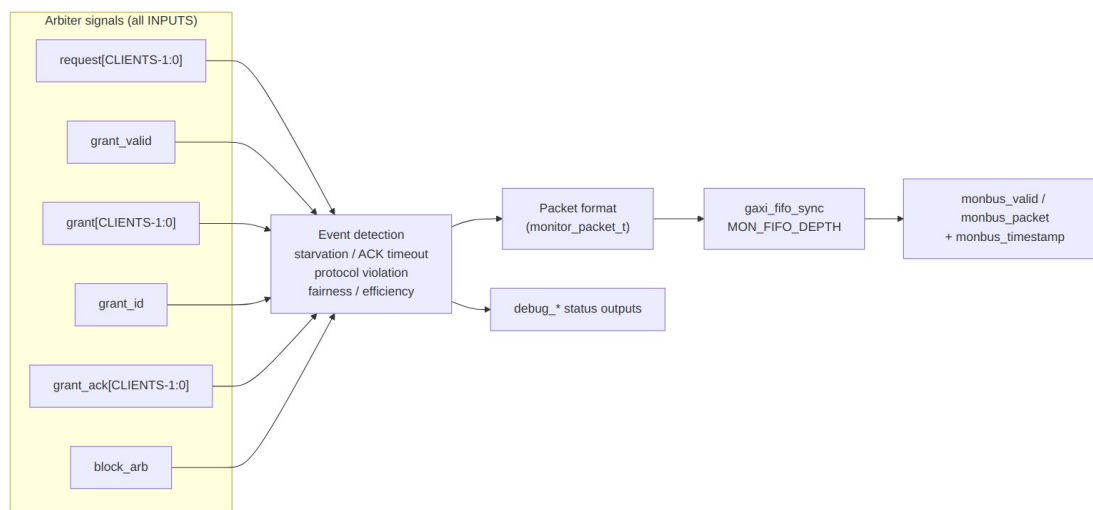
Monitor bus output

Port	Dir	Width	Description
monbus_valid	Out	1	Packet valid
monbus_ready	In	1	Downstream accepts the packet
monbus_packet	Out	monitor_packet_t	128-bit monitor packet
monbus_timestamp	Out	monbus_timestamp_t	Side-band sampled time

Debug status outputs

Port	Dir	Width	Description
debug_fifo_count	Out	$\$clog2(MON_FIFO_DEPTH)+1$	Occupancy of the packet FIFO
debug_packet_count	Out	16	Packets emitted
debug_ack_timeout	Out	CLIENTS	Per-client ACK timeout status
debug_protocol_violations	Out	16	Protocol-violation count
debug_grant_efficiency	Out	16	Grant efficiency,

Port	Dir	Width	Description
debug_client_starvation	Out	CLIENTS	percent Per-client starvation flags
debug_fairness_deviation	Out	16	Fairness-deviation metric
debug_monitor_state	Out	3	Monitor internal state



Functional Description

The single monitor-bus output carries this module’s **own** event packets. It is not a merge of upstream streams — there are none.

This module is instantiated automatically within the higher-level monitor modules (arbiter_rr_pwm_monbus, arbiter_wrr_pwm_monbus); you configure its behavior through the top-level monitor parameters rather than instantiating it yourself. See the individual monitor pages for configuration examples.

7.4 Timing Characteristics

Metric	Value	Notes
Latency	1-2 cycles	Typical processing delay

Metric	Value	Notes
Throughput	1 operation/cycle	Maximum rate

7.5 Usage Examples

Every parameter and port below is taken from the module declaration.

```
arbiter_monbus_common #(
    .CLIENTS                (4),
    .WAIT_GNT_ACK           (0),
    .WEIGHTED_MODE          (0),
    .MON_AGENT_ID           (16'h0010),
    .MON_UNIT_ID            (8'h00),
    .MON_FIFO_DEPTH         (8),
    .MON_FIFO_ALMOST_MARGIN(1),
    .FAIRNESS_REPORT_CYCLES(256),
    .MIN_GRANTS_FOR_FAIRNESS(100),
    .DEFAULT_ACK_TIMEOUT    (64)
) u_arbiter_monbus_common (
    .clk                     (clk),
    .rst_n                  (rst_n),
    .cfg_max_thresh          (cfg_max_thresh),
    .request                 (request),
    .grant_valid             (grant_valid),
    .grant                  (grant),
    .grant_id               (grant_id),
    .grant_ack              (grant_ack),
    .block_arb              (block_arb),
    .cfg_mon_enable         (cfg_mon_enable),
    .cfg_mon_pkt_type_enable(cfg_mon_pkt_type_enable),
    .cfg_mon_latency_thresh(cfg_mon_latency_thresh),
    .cfg_mon_starvation_thresh(cfg_mon_starvation_thresh),
    .cfg_mon_fairness_thresh(cfg_mon_fairness_thresh),
    .cfg_mon_active_thresh (cfg_mon_active_thresh),
    .cfg_mon_ack_timeout_thresh(cfg_mon_ack_timeout_thresh),
    .cfg_mon_efficiency_thresh(cfg_mon_efficiency_thresh),
    .cfg_mon_sample_period (cfg_mon_sample_period),
    .i_mon_time             (i_mon_time),
    .monbus_valid           (monbus_valid),
    .monbus_ready           (monbus_ready),
    .monbus_packet          (monbus_packet),
    .monbus_timestamp       (monbus_timestamp),
    .debug_fifo_count       (debug_fifo_count),
    .debug_packet_count     (debug_packet_count),
    .debug_ack_timeout      (debug_ack_timeout),
    .debug_protocol_violations(debug_protocol_violations),

```



```
.debug_grant_efficiency(debug_grant_efficiency),  
.debug_client_starvation(debug_client_starvation),  
.debug_fairness_deviation(debug_fairness_deviation),  
.debug_monitor_state    (debug_monitor_state)  
);
```

7.6 Design Notes

This module snoops and never drives. Every arbiter-facing port is an input; it observes request/grant/ACK activity and cannot perturb the arbitration it is measuring. That is the property that lets it be dropped into an existing arbiter without changing behaviour.

It uses `clk/rst_n`, not the AMBA `aclk/aresetn`. An easy mis-wire in a file where every neighbour uses the AMBA names.

The compliance model belongs to the DV framework, not here. Round-robin order, starvation and fairness verdicts are computed in CocoTBFramework’s `ArbiterCompliance`; several historical “violations” were defects in that model rather than in any arbiter (COMMON-016 through COMMON-019).

7.7 Related Modules

Used by: `arbiter_rr_pwm_monbus`, `arbiter_wrr_pwm_monbus` (as `u_monitor`).

See also: [axi_monitor_reporter](#)

7.8 Testing

Coverage targets:

- Functional correctness of core logic
- Boundary conditions (min/max values)
- Error handling and recovery
- Interface protocol compliance

See: `val/amba/test_arbiter_monbus_common.py` for verification tests

7.9 References

- **Monitor Architecture:** docs/markdown/rtl-amba/overview.md
 - **Monitor Configuration Guide:** [Monitor Base Configuration](#)
 - **Packet Format Specification:**
docs/markdown/rtl-amba/includes/monitor_package_spec.md
-

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8 arbiter_rr_pwm_monbus

Module: arbiter_rr_pwm_monbus.sv **Location:** rtl/amba/monitor/ **Status:** Production Ready

8.1 Overview

Three blocks in one wrapper: a fair round-robin arbiter, a PWM generator that gates it, and the arbiter_monbus_common telemetry block watching the result. The internal sizing is fixed on purpose — 16-bit PWM resolution, 16-entry monitor FIFO, 256-cycle fairness reporting — so every instance behaves identically and there are fewer knobs to get wrong.

Use it where multiple masters need equal-priority access and you also want periodic arbitration windows: the PWM output drives the arbiter's block_arb input directly, so you get precise time-division control, while the monitor bus gives you real-time visibility into arbitration behavior, request latencies, and problems like starvation or protocol violations. The standardized internal configurations (PWM width, FIFO depth, fairness intervals) keep behavior consistent across instantiations, simplifying integration and reducing parameter proliferation.

- Fair round-robin arbitration across N clients
- PWM-based arbiter blocking for periodic control
- Standardized 16-bit PWM resolution
- Comprehensive monitoring via arbiter_monbus_common
- Optional ACK protocol support
- Fixed internal sizing for consistency (16-entry FIFO, 256-cycle fairness reporting)

- Configurable client count (1-64)
- Real-time performance and fairness tracking

8.2 Parameters

Parameter	Type	Default	Description
CLIENTS	int	4	Number of arbitration clients (1-64)
WAIT_GNT_ACK	int	0	ACK protocol enable (0=disable, 1=enable)
MON_AGENT_ID	logic [15:0]	16'h0010	Monitor agent identifier (16-bit per monitor bus protocol)
MON_UNIT_ID	logic [7:0]	8'h00	Monitor unit identifier (8-bit per monitor bus protocol)
USE_MONITOR	bit	1	Synthesis-time monitor enable. When 0, omits monbus telemetry block (arbiter and PWM remain intact). When 1, full monitoring.

8.2.1 Fixed Internal Configurations (Not User-Configurable)

Configuration	Value	Rationale
PWM_WIDTH	16 bits	Adequate resolution for most use cases
MON_FIFO_DEPTH	16 entries	Optimal for monitoring scenarios

Configuration	Value	Rationale
MON_FIFO_ALMOST_MARGIN	2 entries	Safety margin for FIFO
FAIRNESS_REPORT_CYCLES	256 cycles	Power-of-2 efficient interval
MIN_GRANTS_FOR_FAIRNESS	64 grants	Statistical significance threshold

8.3 Ports

8.3.1 Clock and Reset

Port	Direction	Width	Description
clk	input	1	Clock signal
rst_n	input	1	Active-low asynchronous reset

8.3.2 Arbiter Interface

Port	Direction	Width	Description
request	input	CLIENTS	Client request signals
grant_valid	output	1	Grant is valid this cycle
grant	output	CLIENTS	One-hot grant vector
grant_id	output	$\$clog2(CLIENTS)$	Binary encoded grant ID
grant_ack	input	CLIENTS	Grant acknowledge signals (ACK mode only)

8.3.3 PWM Configuration

Port	Direction	Width	Description
cfg_pwm_sync_rst_n	input	1	PWM synchronous reset (active-low)
cfg_pwm_start	input	1	Start PWM generation
cfg_pwm_duty	input	16	Duty cycle value (0 to period-1)
cfg_pwm_period	input	16	Period value (1 to 65535)
cfg_pwm_repeat_count	input	16	Number of periods to generate (0=infinite)
cfg_pwm_sts_done	output	1	PWM sequence complete status
pwm_out	output	1	PWM output (controls arbiter blocking)

8.3.4 Monitor Configuration

Port	Direction	Width	Description
cfg_mon_enable	input	1	Global monitor enable
cfg_mon_pkt_type_enable	input	16	Packet type enable mask
cfg_mon_latency	input	16	Latency threshold (cycles)
cfg_mon_starvation	input	16	Starvation threshold (cycles)
cfg_mon_fairness	input	16	Fairness deviation threshold (percent)
cfg_mon_active	input	16	Active request

Port	Direction	Width	Description
cfg_mon_period	input	8	count threshold Sample period for metrics

8.3.5 Monitor Bus Output

Port	Direction	Width	Description
monbus_valid	output	1	Monitor bus packet valid
monbus_ready	input	1	Monitor bus ready (from downstream)
monbus_packet	output	128	monitor_packet_t event packet
monbus_timestamp	output	64	Side-band timestamp sampled at packet emission, travelling with the packet

8.3.6 Monitor Time Input

Port	Direction	Width	Description
i_mon_time	input	64	Free-running monitor-time broadcast from the monbus*_group family. Must be connected — leaving it open floats the timestamp every emitted packet carries

8.3.7 Debug Outputs

Port	Direction	Width	Description
debug_fifo_count	output	\$clog2(17)	Monitor FIFO fill level (0-16)
debug_packet_count	output	16	Total packets generated

8.4 Functional Description

The module integrates three key components.

8.4.1 Round-Robin Arbiter

Provides fair arbitration using the `arbiter_round_robin` module: - Equal priority for all clients - Sequential rotation through active requests - Optional ACK protocol (`WAIT_GNT_ACK=1`) - `Block_arb` input for external flow control

8.4.2 PWM Generator

Controls arbiter availability through periodic blocking: - 16-bit resolution (65536 discrete levels) - Configurable duty cycle and period - Repeat count for finite sequences - PWM output directly drives arbiter `block_arb` input

Duty Cycle Calculation: - `pwm_out = 1` (arbiter blocked) for `cfg_pwm_duty` cycles - `pwm_out = 0` (arbiter active) for `cfg_pwm_period - cfg_pwm_duty` cycles

Example: 25% arbiter availability:

```
cfg_pwm_period = 16'd100;      // 100-cycle period
cfg_pwm_duty   = 16'd75;       // Block for 75 cycles
// Result: arbiter active for 25 cycles, blocked for 75 cycles
```

8.4.3 Monitor Bus Common

Comprehensive monitoring via `arbiter_monbus_common`: - Default client weights = 1 (equal priority) - Tracks grant distribution, latencies, starvation - Fixed internal configurations ensure consistency - Event packets include request latency, starvation detection, active request count

8.4.4 Operation Flow

1. Clients assert request signals
2. PWM output modulates arbiter availability (`pwm_out` -> `block_arb`)
3. When arbiter active (`pwm_out=0`), round-robin arbiter grants one request
4. Monitor tracks arbitration events (grants, timeouts, fairness)

5. Event packets generated to monitor bus output
 6. If ACK mode enabled, arbiter waits for grant_ack before next grant
-

8.5 Timing Characteristics

This module is **sequential**: the AST shows clocked processes, so it holds state. Outputs driven from those blocks are registered and appear one clock after the inputs that produced them.

Per-path cycle counts are not enumerated here; read the block that drives the signal you care about. No synthesis frequency or area figures are quoted – none have been measured against a target device.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

8.6 Usage Examples

```
// 8-client round-robin arbiter with 30% availability window
arbiter_rr_pwm_monbus #(
    .CLIENTS      (8),
    .WAIT_GNT_ACK (0),           // No ACK protocol
    .MON_AGENT_ID (16'h0015),
    .MON_UNIT_ID  (8'h03)
) u_rr_arb (
    .clk           (clk),
    .rst_n         (rst_n),

    // Arbiter interface
    .request       (client_requests),
    .grant_valid   (grant_valid),
    .grant         (grant_onehot),
    .grant_id      (grant_binary),
    .grant_ack     (8'h00), // Not used (ACK disabled)

    // PWM configuration - 30% availability
    .cfg_pwm_sync_rst_n (1'b1),
    .cfg_pwm_start      (pwm_start),
    .cfg_pwm_duty       (16'd70), // Block for 70 cycles
    .cfg_pwm_period     (16'd100), // Out of 100 cycles total
    .cfg_pwm_repeat_count (16'd0), // Infinite repeat
    .cfg_pwm_sts_done   (pwm_done),
    .pwm_out            (pwm_signal),
```



```

    // Monitor configuration
    .cfg_mon_enable          (1'b1),
    .cfg_mon_pkt_type_enable (16'h000F), // Error, timeout,
completion, threshold
    .cfg_mon_latency         (16'd50),    // 50-cycle latency warning
    .cfg_mon_starvation      (16'd200),    // 200-cycle starvation
threshold
    .cfg_mon_fairness        (16'd15),    // 15% fairness deviation
    .cfg_mon_active          (16'd7),      // Warn if 7+ active
requests
    .cfg_mon_period          (8'd128),

    // Monitor bus output
    .monbus_valid            (mon_valid),
    .monbus_ready            (mon_ready),
    .monbus_packet           (mon_packet),
    .monbus_timestamp        (mon_timestamp),

    // Monitor time broadcast (from the monbus group) -- not optional
    .i_mon_time              (mon_time),

    // Debug
    .debug_fifo_count        (mon_fifo_level),
    .debug_packet_count       (mon_pkt_count)
);

// Start PWM generation
initial begin
    pwm_start = 1'b0;
    @(posedge clk);
    pwm_start = 1'b1;
    @(posedge clk);
    pwm_start = 1'b0;
end

```

8.7 Design Notes

8.7.1 PWM Control Strategy

The PWM output directly controls arbiter blocking: - **pwm_out = 1**: Arbiter blocked (no grants issued) - **pwm_out = 0**: Arbiter active (can issue grants)

This allows precise timing control for: - Time-division multiple access (TDMA) scheduling - Bandwidth reservation windows - Periodic maintenance operations - Energy-efficient arbitration (idle periods)

8.7.2 Fixed Configuration Benefits

The standardized internal configurations provide: - **Consistency**: All instances behave identically - **Predictability**: Known FIFO depths, fairness intervals - **Simplified Integration**: Fewer parameters to configure - **Verified Operation**: Tested configurations

If different internal sizes are needed, create a new variant rather than parameterizing.

8.7.3 Fairness in Round-Robin Mode

All clients have equal priority (weight = 1 in monitoring logic). Fairness deviation events are generated if any client receives significantly more or fewer grants than expected. The measurement is **cumulative since reset**, not a sliding window: FAIRNESS_REPORT_CYCLES (256) only paces how often the deviation is recomputed from the lifetime grant counters, which are cleared on reset alone.

Expected: Each client receives $(1/\text{CLIENTS}) * 100\%$ of grants Threshold: `cfg_mon_fairness` parameter (percent)

8.7.4 Monitor Bus Integration

See `arbiter_monbus_common.md` for complete monitoring details. Key points: - Uses `PROTOCOL_ARB` (3'b011) event encoding - Fixed 16-entry FIFO absorbs bursts, but does **not** guarantee zero loss: the event registers are overwritten every cycle and the FIFO write is not held when full, so a one-cycle event (grant, completion) that fires while the FIFO is full is dropped. Persistent conditions (starvation, thresholds) re-present until accepted - Per-client latency and starvation tracking - Grant efficiency monitoring

8.8 Related Modules

Used by: system arbitration hierarchies, multi-master interconnects, TDMA scheduling systems.

Uses: `arbiter_round_robin.sv` (core RR arbiter); `pwm.sv` (PWM generator); `arbiter_monbus_common.sv` (monitoring infrastructure).

See also: `arbiter_wrr_pwm_monbus.sv` (weighted variant); `monbus_arbiter.sv` (aggregates monitor bus streams).

8.9 Testing

`val/amba/test_arbiter_rr_pwm_monbus.py` exercises this module. It collects 8 parameter cases at the default `REG_LEVEL`.

```
source env_python
pytest val/amba/test_arbiter_rr_pwm_monbus.py -v
```

8.10 References

- Internal: docs/markdown/rtl-amba/index.md (AMBA subsystem requirements)
 - Internal: docs/markdown/rtl-amba/arbiter_monbus_common.md (monitoring details)
 - RTL: rtl/amba/monitor/arbiter_rr_pwm_monbus.sv
 - Tests: val/amba/test_arbiter_rr_pwm_monbus.py
-

Last Updated: 2025-10-24

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9 arbiter_wrr_pwm_monbus

Module: arbiter_wrr_pwm_monbus.sv **Location:** rtl/amba/monitor/ **Status:** Production Ready

9.1 Overview

The weighted sibling of arbiter_rr_pwm_monbus: priority-based arbitration with configurable client weights, the same PWM flow control, and the same arbiter_monbus_common telemetry with standardized internal configurations. Higher-weight clients receive proportionally more grants while lower-weight clients still get guaranteed service — no starvation — and the monitoring infrastructure tracks whether the actual grant distribution matches the configured weights and reports fairness violations when it drifts.

The PWM integration adds temporal control of arbiter availability on top: TDMA windows or bandwidth reservation, with priority-based selection inside each active window.

- Weighted round-robin arbitration with configurable priorities
- Per-client weight thresholds (1 to MAX_LEVELS - 1; the field is $\lceil \log_2(\text{MAX_LEVELS}) \rceil$ bits, so MAX_LEVELS itself is not representable — see Design Notes)
- PWM-based arbiter blocking for periodic control
- Standardized 16-bit PWM resolution
- Comprehensive fairness deviation monitoring
- Optional ACK protocol support
- Fixed internal sizing (16-entry FIFO, 256-cycle fairness reporting)
- Enhanced debug outputs for priority tracking

9.2 Parameters

Parameter	Type	Default	Description
MAX_LEVELS	int	16	Maximum weight levels per client (1-64)
CLIENTS	int	4	Number of arbitration clients (1-64)
WAIT_GNT_ACK	int	0	ACK protocol enable (0=disable, 1=enable)
MON_AGENT_ID	logic [15:0]	16'h0010	Monitor agent identifier (16-bit per monitor bus protocol)
MON_UNIT_ID	logic [7:0]	8'h00	Monitor unit identifier (8-bit per monitor bus protocol)
USE_MONITOR	bit	1	Synthesis-time monitor enable. When 0, omits monbus telemetry

Parameter	Type	Default	Description
			block (arbiter and PWM remain intact). When 1, full monitoring.

9.2.1 Fixed Internal Configurations (Not User-Configurable)

Configuration	Value	Rationale
PWM_WIDTH	16 bits	Adequate resolution for most use cases
MON_FIFO_DEPTH	16 entries	Optimal for monitoring scenarios
MON_FIFO_ALMOST_MARGIN	2 entries	Safety margin for FIFO
FAIRNESS_REPORT_CYCLES	256 cycles	Power-of-2 efficient interval
MIN_GRANTS_FOR_FAIRNESS	64 grants	Statistical significance threshold

9.2.2 Derived Parameters (do not override)

These are declared as parameter so the elaborator can compute them, not so callers can set them. Each defaults to an expression over the parameters above; overriding one desynchronises it from its source and the design fails to elaborate or silently mis-sizes a bus. Set the parameters they are derived FROM and leave these alone.

Derived parameter	Default expression
MAX_LEVELS_WIDTH	$\$clog2(\text{MAX_LEVELS})$
N	$\$clog2(\text{CLIENTS})$
CXMTW	$\text{CLIENTS} * \text{MAX_LEVELS_WIDTH}$

9.3 Ports

9.3.1 Clock and Reset

Port	Direction	Width	Description
clk	input	1	Clock signal
rst_n	input	1	Active-low

Port	Direction	Width	Description
			asynchronous reset

9.3.2 Arbiter Configuration and Interface

Port	Direction	Width	Description
cfg_arb_max_thresh	input	CLIENTS * \$clog2(MAX_LEVELS)	Per-client weight thresholds
request	input	CLIENTS	Client request signals
grant_valid	output	1	Grant is valid this cycle
grant	output	CLIENTS	One-hot grant vector
grant_id	output	\$clog2(CLIENTS)	Binary encoded grant ID
grant_ack	input	CLIENTS	Grant acknowledge signals (ACK mode only)

9.3.3 PWM Configuration

Port	Direction	Width	Description
cfg_pwm_sync_rst_n	input	1	PWM synchronous reset (active-low)
cfg_pwm_start	input	1	Start PWM generation
cfg_pwm_duty	input	16	Duty cycle value (0 to period-1)
cfg_pwm_period	input	16	Period value (1 to 65535)

Port	Direction	Width	Description
cfg_pwm_repeats_count	input	16	Number of periods (0=infinite)
cfg_pwm_status_done	output	1	PWM sequence complete status
pwm_out	output	1	PWM output (controls arbiter blocking)

9.3.4 Monitor Configuration

Port	Direction	Width	Description
cfg_mon_enable	input	1	Global monitor enable
cfg_mon_pkt_type_enable	input	16	Packet type enable mask
cfg_mon_latency_thresh	input	16	Latency threshold (cycles)
cfg_mon_starvation_thresh	input	16	Starvation threshold (cycles)
cfg_mon_fairness_thresh	input	16	Fairness deviation threshold (percent)
cfg_mon_active_thresh	input	16	Active request count threshold
cfg_mon_ack_timeout_thresh	input	16	ACK timeout threshold (cycles)
cfg_mon_efficiency_thresh	input	16	Grant efficiency threshold (percent)
cfg_mon_sample_period	input	8	Sample period for metrics

9.3.5 Monitor Bus Output

Port	Direction	Width	Description
monbus_valid	output	1	Monitor bus packet valid
monbus_ready	input	1	Monitor bus ready (from downstream)
monbus_packet	output	128	monitor_packet_t event packet
monbus_timestamp	output	64	Side-band timestamp sampled at packet emission, travelling with the packet

9.3.6 Monitor Time Input

Port	Direction	Width	Description
i_mon_time	input	64	Free-running monitor-time broadcast from the monbus_*_group family. Must be connected — leaving it open floats the timestamp every emitted packet carries

9.3.7 Enhanced Debug Outputs

Port	Direction	Width	Description
debug_fifo_count	output	$\lceil \log_2(17) \rceil$	Monitor FIFO fill level
debug_packet_count	output	16	Total packets

Port	Direction	Width	Description
nt			generated
debug_ack_timeout	output	CLIENTS	Per-client ACK timeout status
debug_protocol_violations	output	16	Protocol violation count
debug_grant_efficiency	output	16	Grant efficiency percentage
debug_client_starvation	output	CLIENTS	Per-client starvation flags
debug_fairness_deviation	output	16	Maximum fairness deviation
debug_monitor_state	output	3	Monitor internal state

9.4 Functional Description

The module integrates four key components.

9.4.1 Weighted Round-Robin Arbiter

Provides priority-based arbitration using `arbiter_round_robin_weighted`: - Per-client weight configuration via `cfg_arb_max_thresh` - Higher weight = higher priority - Fair rotation within same weight level - Prevents starvation of low-priority clients - Optional ACK protocol support

Weight Encoding: `cfg_arb_max_thresh` is a concatenated vector:

```
cfg_arb_max_thresh = {weight[CLIENTS-1], ..., weight[1], weight[0]}
```

Where each weight is $\lceil \log_2(\text{MAX_LEVELS}) \rceil$ bits wide.

Arbitration Algorithm (`arbiter_round_robin_weighted`, credit-based): 1. Each client's weight loads a per-client **credit counter** on replenish (`w_credit_counter[i] = w_client_weight[i]`) 2. A client is eligible while it has credit left (`w_has_crd[i] = (r_credit_counter[i] > 0) && ...`) 3. Selection among *eligible* clients is **plain round-robin** — no comparison of weights picks a winner 4. Granting spends a credit; when all requesting clients are out of credit the counters replenish and the cycle repeats

Weight therefore controls **how many grants** a client gets per replenishment round, not **which** client wins a given contest. A high-weight client does not lock out low-weight ones — they interleave, which is what makes the share figures below achievable.

9.4.2 PWM Generator

Controls arbiter availability (same as RR variant): - 16-bit resolution - Configurable duty cycle and period - PWM output directly drives arbiter block_arb

9.4.3 Monitor Bus Common

Comprehensive monitoring with enhanced fairness tracking: - **WEIGHTED_MODE** is a **dead parameter** in `arbiter_monbus_common` — declared and never referenced. Fairness analysis is weight-aware unconditionally (it always uses `cfg_max_thresh`); setting or clearing this changes nothing - Compares actual vs expected grant distribution based on configured weights - Detects fairness violations when distribution deviates from weight ratios

Fairness Calculation Example:

4 clients with weights [8, 4, 2, 1] (total weight = 15)

Expected distribution:

Client 0: $(8/15) * 100 = 53.3\%$

Client 1: $(4/15) * 100 = 26.7\%$

Client 2: $(2/15) * 100 = 13.3\%$

Client 3: $(1/15) * 100 = 6.7\%$

Actual distribution is measured over **all grants since reset**, not over a

sliding window: ``r_grant_counters`` and ``r_total_grants`` in ``arbiter_monbus_common`` are cleared only on reset.

``FAIRNESS_REPORT_CYCLES``

(256) paces how often the deviation is **recomputed** from those lifetime

totals, and ``MIN_GRANTS_FOR_FAIRNESS`` is likewise compared against the lifetime

total. After a traffic-phase or weight change the reported deviation therefore

still reflects all prior history; it converges toward the new distribution

rather than snapping to it.

Deviation = $|actual\% - expected\%|$ for each client.

Event generated if $max\ deviation > cfg_mon_fairness_thresh$.

9.4.4 Operation Flow

1. Configure per-client weights via `cfg_arb_max_thresh`
2. Clients assert request signals
3. PWM output modulates arbiter availability
4. When active, the weighted arbiter grants a **credit-eligible** requester, chosen round-robin among those with credit remaining — weight budgets

how many grants a client gets per replenishment round, it does not rank clients

5. Monitor tracks grants and calculates fairness metrics
 6. Fairness violations reported if distribution deviates from weights
-

9.5 Timing Characteristics

This module is **sequential**: the AST shows clocked processes, so it holds state. Outputs driven from those blocks are registered and appear one clock after the inputs that produced them.

Per-path cycle counts are not enumerated here; read the block that drives the signal you care about. No synthesis frequency or area figures are quoted – none have been measured against a target device.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

9.6 Usage Examples

```
// 4-client weighted arbiter with priorities [8, 4, 2, 1]
localparam int MAX_LEVELS = 16;
localparam int CLIENTS = 4;
localparam int WEIGHT_WIDTH = $clog2(MAX_LEVELS); // 4 bits

// Configure client weights
logic [WEIGHT_WIDTH-1:0] client_weights [CLIENTS];
assign client_weights[0] = 4'd8; // Highest priority
assign client_weights[1] = 4'd4;
assign client_weights[2] = 4'd2;
assign client_weights[3] = 4'd1; // Lowest priority

wire [CLIENTS*WEIGHT_WIDTH-1:0] cfg_weights = {
    client_weights[3], client_weights[2],
    client_weights[1], client_weights[0]
};

arbiter_wrr_pwm_monbus #(
    .MAX_LEVELS      (MAX_LEVELS),
    .CLIENTS         (CLIENTS),
    .WAIT_GNT_ACK    (1), // Enable ACK protocol
    .MON_AGENT_ID    (16'h0018),
    .MON_UNIT_ID     (8'h04)
```

```

) u_wrr_arb (
    .clk                (clk),
    .rst_n              (rst_n),

    // Arbiter configuration
    .cfg_arb_max_thresh (cfg_weights),
    .request             (client_requests),
    .grant_valid        (grant_valid),
    .grant              (grant_onehot),
    .grant_id           (grant_binary),
    .grant_ack          (grant_ack),

    // PWM configuration - 40% availability
    .cfg_pwm_sync_rst_n (1'b1),
    .cfg_pwm_start      (pwm_start),
    .cfg_pwm_duty       (16'd60),    // Block for 60 cycles
    .cfg_pwm_period     (16'd100),   // Out of 100 total
    .cfg_pwm_repeat_count (16'd0),   // Infinite
    .cfg_pwm_sts_done   (pwm_done),
    .pwm_out            (pwm_signal),

    // Monitor configuration
    .cfg_mon_enable     (1'b1),
    .cfg_mon_pkt_type_enable (16'h003F),
    .cfg_mon_latency_thresh (16'd100),
    .cfg_mon_starvation_thresh (16'd500),
    .cfg_mon_fairness_thresh (16'd10), // 10% deviation threshold
    .cfg_mon_active_thresh (16'd4),
    .cfg_mon_ack_timeout_thresh (16'd200),
    .cfg_mon_efficiency_thresh (16'd75),
    .cfg_mon_sample_period (8'd100),

    // Monitor bus output
    .monbus_valid       (mon_valid),
    .monbus_ready       (mon_ready),
    .monbus_packet      (mon_packet),
    .monbus_timestamp   (mon_timestamp),

    // Monitor time broadcast (from the monbus group) -- not optional
    .i_mon_time         (mon_time),

    // Enhanced debug outputs
    .debug_fifo_count   (mon_fifo_level),
    .debug_packet_count (mon_pkt_count),
    .debug_ack_timeout  (mon_ack_timeouts),
    .debug_protocol_violations (mon_violations),
    .debug_grant_efficiency (mon_efficiency),

```

```

        .debug_client_starvation (mon_starvation),
        .debug_fairness_deviation (mon_fairness_dev),
        .debug_monitor_state      (mon_state)
    );

```

9.7 Design Notes

9.7.1 Weight Configuration

The `cfg_arb_max_thresh` parameter requires careful configuration:

Valid Range: Each client weight must be 1 to (MAX_LEVELS-1) **Zero Weights:** Zero-weight clients are effectively disabled **Dynamic Updates:** Weights can be changed runtime, but may cause temporary fairness violations during transition

Example Calculation (4 clients, MAX_LEVELS=16):

```

localparam WEIGHT_WIDTH = 4; // $clog2(16) = 4
logic [WEIGHT_WIDTH-1:0] w0 = 4'd12; // 75% of max
logic [WEIGHT_WIDTH-1:0] w1 = 4'd6;  // 37.5% of max
logic [WEIGHT_WIDTH-1:0] w2 = 4'd3;  // 18.75% of max
logic [WEIGHT_WIDTH-1:0] w3 = 4'd1;  // 6.25% of max

assign cfg_arb_max_thresh = {w3, w2, w1, w0}; // LSB is client 0

```

9.7.2 Fairness Deviation Threshold

The `cfg_mon_fairness_thresh` parameter should account for statistical variation:

Recommended Values: - 5% for strict fairness enforcement - 10% for typical systems (allows reasonable variation) - 20% for relaxed monitoring

Factors Affecting Deviation: - Short measurement windows (MIN_GRANTS_FOR_FAIRNESS) - Bursty request patterns - Weight ratio extremes (e.g., 16:1 vs 2:1) - PWM blocking periods

9.7.3 Enhanced Debug Outputs

This module exposes complete debug outputs from `arbiter_monbus_common`:

- **debug_client_starvation:** Bit vector showing starved clients
 - Use to identify clients not receiving service
 - Indicates potential weight configuration issues
- **debug_fairness_deviation:** Maximum deviation percentage
 - Real-time fairness metric without consuming monitor bus bandwidth

- Connect to system debug registers for runtime visibility
- **debug_grant_efficiency**: Grant completion ratio
 - Low efficiency may indicate ACK timeout issues or downstream congestion

9.7.4 ACK Mode with Weighted Arbitration

When WAIT_GNT_ACK=1, the arbiter waits for grant_ack before issuing next grant. This can affect fairness:

Impact on Fairness: - Slow-ACK clients may receive fewer grants than their weight warrants - ACK timeout tracking helps identify problematic clients - Consider increasing `cfg_mon_ack_timeout_thresh` for clients with known slower response

9.8 Related Modules

Used by: priority-based interconnect arbiters, QoS-aware memory controllers, multi-tier scheduling hierarchies.

Uses: `arbiter_round_robin_weighted.sv` (core WRR arbiter); `pwm.sv` (PWM generator); `arbiter_monbus_common.sv` (monitoring; note WEIGHTED_MODE is inert).

See also: `arbiter_rr_pwm_monbus.sv` (equal-priority variant); `monbus_arbiter.sv` (aggregates monitor bus streams).

9.9 Testing

`val/amba/test_arbiter_wrr_pwm_monbus.py` exercises this module. It collects 8 parameter cases at the default REG_LEVEL.

```
source env_python
pytest val/amba/test_arbiter_wrr_pwm_monbus.py -v
```

9.10 References

- Internal: `docs/markdown/rtl-amba/index.md` (AMBA subsystem requirements)
- Internal: `docs/markdown/rtl-amba/arbiter_monbus_common.md` (monitoring details)

- RTL: rtl/amba/monitor/arbiter_wrr_pwm_monbus.sv
 - Tests: val/amba/test_arbiter_wrr_pwm_monbus.py
-

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10 axi_monitor_addr_check

Module: axi_monitor_addr_check.sv **Location:** rtl/amba/monitor/ **Category:** Core Infrastructure **Status:** Production Ready (Formally Verified)

10.1 Overview

axi_monitor_addr_check is a parallel address-range comparator instantiated within AXI monitor wrappers. It observes command-phase handshakes (AR/AW) and classifies each accepted address against N user-configured inclusive ranges [low, high]. Each range carries a **flavor** (ADDR_RANGE_IS_ERROR[i]) that selects one of two independent report paths:

- **DEBUG range** (ADDR_RANGE_IS_ERROR[i] = 0): a hit emits a PktTypeAddrMatch (4'h8) packet with event code AXI_ADDR_RANGE_MATCH = 8'h01, gated by cfg_debug_enable.
- **ERROR range** (ADDR_RANGE_IS_ERROR[i] = 1): the enabled ERROR ranges form an **allowlist**; a command whose address is in NONE of them emits a PktTypeError (4'h0) packet with event code AXI_ERR_ADDR_RANGE = 8'h0D, gated by cfg_error_enable.

DEBUG and ERROR ranges are evaluated independently, so a single command may produce both a match and a miss; two pending slots hold each and the output stream serialises them.

This is a **shared infrastructure module** used internally by AXI4/AXIL4/AXI5 monitor wrappers when parameterized with N_ADDR_RANGES > 0. It is not typically instantiated directly by users but is critical for address-space validation (allowlist enforcement) and targeted debug tracing. In practice it buys you four things:

1. **Allowlist enforcement:** ERROR ranges declare the *expected* address space; any access outside it raises an Error/ADDR_RANGE packet.
2. **Targeted debug tracing:** DEBUG ranges watch specific regions; a hit emits an AddrMatch packet without implying an error.
3. **Independent debug/error regions:** the two flavors can cover different address sets, so “watch this region” and “everything must stay inside that region” are configured separately.
4. **Design Verification:** verify address constraints in functional simulation and formal proof.

The ERROR/allowlist path is the checker built into the monitor whenever `N_ADDR_RANGES > 0`, **independent of the error reporter cone** (`ENABLE_ERROR_LOGIC`). It is therefore the most controllable way to **deliberately inject an error** for coverage: point an enabled ERROR range at a region the legitimate traffic never touches, and every access becomes an allowlist miss that emits `PktTypeError/AXI_ERR_ADDR_RANGE`. Because an error is a fault condition, this injection is expected to hang the traffic that provoked it – see [Healthy classes vs fault classes](#) for why that stall is the point, not a defect.

Feature summary:

- **Parallel Range Comparators:** `N` independent [low, high] inclusive range checkers
 - **Two flavors per range:** `ADDR_RANGE_IS_ERROR` selects DEBUG (match) vs ERROR (allowlist-miss) behavior per range; default all-0 leaves the ERROR/miss path inert (feature unused by default)
 - **Independent path enables:** `cfg_debug_enable` gates AddrMatch packets, `cfg_error_enable` gates Error packets; `cfg_addr_check_enable` is the master gate
 - **Zero-Area Synthesis:** `N_ADDR_RANGES = 0` omits this module entirely (no gates, no regs) – done by the **parent**, not here: `axi_monitor_base` wraps the instance in `if (N_ADDR_RANGES > 0) gen_addr_check` and ties the packet stream off in `gen_no_addr_check`
 - **Monbus Integration:** AddrMatch packets (8'h8/8'h01) and Error packets (8'h0/8'h0D)
 - **Coalescing:** per-range latched address for MATCH; a single latched slot for MISS; one packet per cycle (MISS has emit priority)
 - **Formal Verification:** all properties proven (prove + cover PASS)
-

10.2 Parameters

Parameter	Type	Default	Description
N_ADDR_RANGES	int	4	Number of address-range comparators (≥ 1). Max 16 (4-bit range index).
ADDR_WIDTH	int	32	Address bus width (matches AXI_ADDR_WIDTH of parent monitor). Must be ≤ 60 (fits the event_data address field).
ID_WIDTH	int	6	Transaction ID width (clipped to 9 bits when copied into the packet's channel_id).
UNIT_ID	logic [7:0]	8'h00	8-bit unit identifier in monitor packets.
AGENT_ID	logic [15:0]	16'h0000	16-bit agent identifier in monitor packets.
IS_READ	bit	1	Build-time flag: 1 if this monitor watches reads (AR), 0 for writes (AW). Drives direction recovery at the consumer (see Event Encoding).

Parameter	Type	Default	Description
ADDR_RANGE_IS_ERROR	logic [N_ADDR_RANGE S-1:0]	'0	Per-range flavor: bit i = 0 → range i is a DEBUG range (hit → AddrMatch); bit i = 1 → range i is an ERROR range (allowlist; miss → Error/ADDR_RANGE). Default all-0 keeps the ERROR/miss path inert.

10.2.1 Derived Parameters (do not override)

These are declared as parameter so the elaborator can compute them, not so callers can set them. Each defaults to an expression over the parameters above; overriding one desynchronises it from its source and the design fails to elaborate or silently mis-sizes a bus. Set the parameters they are derived FROM and leave these alone.

Derived parameter	Default expression
M	ADDR_WIDTH
IW	ID_WIDTH

10.3 Ports

10.3.1 Clock and Reset

Port	Direction	Width	Description
clk	Input	1	AXI clock
aresetn	Input	1	AXI active-low reset

10.3.2 Side-Band Timestamp

Port	Direction	Width	Description
i_mon_time	Input	64	Free-running counter from

Port	Direction	Width	Description
			monbus group, broadcast to every wrapper via the shared mon_time_w net. Sampled when addr_pkt_valid asserts and driven out on addr_pkt_timestamp.

10.3.3 Command Interface (Snoop)

Port	Direction	Width	Description
cmd_valid	Input	1	Command valid (AR or AW handshake)
cmd_ready	Input	1	Command ready (slave accepting)
cmd_addr	Input	ADDR_WIDTH	Command address (araddr or awaddr)
cmd_id	Input	ID_WIDTH	Command transaction ID (arid or awid)

10.3.4 Configuration Inputs

Port	Direction	Width	Description
cfg_addr_check_enable	Input	1	Master enable for all comparators. 0 = no packets generated.
cfg_debug_enable	Input	1	Enables the

Port	Direction	Width	Description
			MATCH path: DEBUG-range hits emit AddrMatch packets.
cfg_error_enable	Input	1	Enables the MISS path: an address outside the ERROR allowlist emits an Error/ADDR_RAN GE packet.
cfg_addr_range_enable[N_ADDR_RANGES-1:0]	Input	N	Per-range enable bits. 1 = range active, 0 = range disabled.
cfg_addr_range_low[N_ADDR_RANGES-1:0] [ADDR_WIDTH-1:0]	Input	$N \times \text{ADDR_WIDTH}$	Inclusive low bound for each range.
cfg_addr_range_high[N_ADDR_RANGES-1:0] [ADDR_WIDTH-1:0]	Input	$N \times \text{ADDR_WIDTH}$	Inclusive high bound for each range.

10.3.5 Monitor Bus Output

Port	Direction	Width	Description
addr_pkt_valid	Output	1	Address-check packet valid (AddrMatch or Error)
addr_pkt_ready	Input	1	Downstream ready to accept packet
addr_pkt_data	Output	128	Monitor packet (monitor_packet_ t, 128-bit format)
addr_pkt_timesta	Output	64	Sampled

Port	Direction	Width	Description
mp			i_mon_time paired atomically with addr_pkt_data

10.4 Functional Description

10.4.1 Internal Architecture

The module instantiates N parallel comparators and splits them by flavor:

1. **Per-Range Comparators:** Each range i computes $\text{raw_hit}[i] = \text{cfg_addr_range_enable}[i] \ \&\& \ (\text{cmd_addr} \geq \text{cfg_addr_range_low}[i]) \ \&\& \ (\text{cmd_addr} \leq \text{cfg_addr_range_high}[i])$.
2. **Flavor split:**
 - $\text{debug_hit}[i] = \text{raw_hit}[i] \ \&\& \ \text{!ADDR_RANGE_IS_ERROR}[i]$ — a hit in a DEBUG range.
 - $\text{err_hit} = |(\text{raw_hit} \ \& \ \text{ADDR_RANGE_IS_ERROR})$ — the address is inside some enabled ERROR range (allowed).
 - $\text{err_ranges_exist} = |(\text{cfg_addr_range_enable} \ \& \ \text{ADDR_RANGE_IS_ERROR})$.
3. **Per-command events** (qualified by $\text{cmd_fire} = \text{cmd_valid} \ \&\& \ \text{cmd_ready} \ \&\& \ \text{cfg_addr_check_enable}$):
 - $\text{match_set}[i] = \text{cmd_fire} \ \&\& \ \text{cfg_debug_enable} \ \&\& \ \text{debug_hit}[i] \rightarrow \text{MATCH}$ (AddrMatch packet).
 - $\text{miss_set} = \text{cmd_fire} \ \&\& \ \text{cfg_error_enable} \ \&\& \ \text{err_ranges_exist} \ \&\& \ \text{!err_hit} \rightarrow \text{MISS}$ (Error packet).
4. **Pending + emit:** MATCH events coalesce per DEBUG range ($\text{r_match_pending}[i]$, latched address, latest-win); MISS events coalesce into a single slot (r_miss_pending). One packet drains per cycle — **MISS has priority**, then the lowest-index pending DEBUG range. Direction (read vs. write) is not embedded — the build-time `IS_READ` parameter fixes the channel, and consumers recover direction from (`UNIT_ID`, `AGENT_ID`).

Because the two flavors are independent, a command that hits a DEBUG range and is also outside the ERROR allowlist sets BOTH pending slots; the two packets emit on successive cycles.

10.4.2 Event Encoding

Two packet types share the 128-bit `monitor_packet_t` layout; only `packet_type`, `event_code`, and the range-index nibble differ:

Field	MATCH (debug hit)	MISS (error allowlist)
[127:124] Packet Type	4'h8 PktTypeAddrMatch	4'h0 PktTypeError
[108:105] Protocol	4'h0 PROTOCOL_AXI	4'h0 PROTOCOL_AXI
[104:97] Event Code	8'h01 AXI_ADDR_RANGE_MAT CH	8'h0D AXI_ERR_ADDR_RANGE
[96:88] Channel ID	cmd_id clipped/zero- extended to 9 bits	same
[87:72] Agent ID	from AGENT_ID	from AGENT_ID
[71:64] Unit ID	from UNIT_ID	from UNIT_ID
[63:60] Range Index	matching DEBUG range (0..N-1)	4'hF (no-range sentinel)
[59:0] Address	full cmd_addr, zero- padded	full cmd_addr, zero- padded

is_read flag dropped: Earlier revisions reserved a bit in `event_data` for read-vs-write direction. The 128-bit layout drops it because each AXI monitor instance watches a single direction (set at build time by `IS_READ`); consumers recover direction from (`UNIT_ID`, `AGENT_ID`). Note: `apb_monitor_addr_check` still carries `is_read` since a single APB monitor sees both directions on the same channel.

Side-band timestamp: `addr_pkt_timestamp` carries the sampled `i_mon_time` paired atomically with the packet through the arbiter and into `monbus_group`.

10.5 Timing Characteristics

This module is **sequential**: it contains clocked logic (via `always_ff` or the repository's `ALWAYS_FF_RST` macro) and therefore holds state. Outputs driven from those blocks are registered and appear one clock after the inputs that produced them.

Per-path cycle counts are not enumerated here; read the block that drives the signal you care about. No synthesis frequency or area figures are quoted – none have been measured against a target device.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

10.6 Usage Examples

Every parameter and port below is read from the module declaration.

```
axi_monitor_addr_check #(
    .N_ADDR_RANGES      (4),
    .ADDR_WIDTH          (32),
    .ID_WIDTH            (6),
    .UNIT_ID             (8'h00),
    .AGENT_ID            (16'h0000),
    .IS_READ              (1'b1),
    .ADDR_RANGE_IS_ERROR ('0')
) u_axi_monitor_addr_check (
    .clk                  (clk),
    .aresetn              (aresetn),
    .i_mon_time           (i_mon_time),
    .cmd_addr             (cmd_addr),
    .cmd_id               (cmd_id),
    .cmd_valid            (cmd_valid),
    .cmd_ready            (cmd_ready),
    .cfg_addr_check_enable (cfg_addr_check_enable),
    .cfg_debug_enable     (cfg_debug_enable),
    .cfg_error_enable     (cfg_error_enable),
    .cfg_addr_range_enable (cfg_addr_range_enable),
    .cfg_addr_range_low   (cfg_addr_range_low),
    .cfg_addr_range_high  (cfg_addr_range_high),
    .addr_pkt_valid       (addr_pkt_valid),
    .addr_pkt_ready       (addr_pkt_ready),
    .addr_pkt_data        (addr_pkt_data),
    .addr_pkt_timestamp   (addr_pkt_timestamp)
);
```

10.7 Design Notes

10.7.1 Filtering Integration

The checker produces two packet classes, each filtered by the parent monitor's existing drop masks (axi_monitor_filtered / monbus_group_core):

- **MATCH** (PktTypeAddrMatch, event 0x01) — dropped by `cfg_axi_addr_mask[1]`.

- **MISS** (PktTypeError, event 0x0D AXI_ERR_ADDR_RANGE) — dropped by `cfg_axi_error_mask[13]`.

No new mask wiring is needed — both event codes already have reserved bits in the per-class 16-bit masks.

Example: suppress the error/allowlist packets while keeping debug matches:

```
.cfg_axi_error_mask(16'h2000) // bit 13 high = drop ADDR_RANGE error packets
```

10.7.2 Instantiation Pattern

The module is instantiated **inside** AXI monitor wrappers (`axi4_master_wr_mon`, etc.) by the monitor architecture. Users do not instantiate this module directly but configure it via wrapper parameters.

10.7.3 Synthesis Behavior

- **N_ADDR_RANGES = 0:** handled by the **instantiating parent**. `axi_monitor_base` generates the instance only under `N_ADDR_RANGES > 0`; its `gen_no_addr_check` branch drives `addr_pkt_valid = 0` and a zeroed packet, so from the monitor's arbiter the stream is constant 0 and the checker costs nothing.

This module itself has **no N == 0 guard** and must not be instantiated directly with 0 – its parameter is documented `>= 1`. At 0 the per-range vectors degenerate (logic `[-1:0]`), the range loops are empty so the enable terms are never driven, and `addr_pkt_valid` would be undriven (X in simulation) rather than a clean constant 0.

- **N_ADDR_RANGES > 0:** Full comparator logic synthesized. Area scales with N.

10.7.4 Downstream FIFO

The monbus output should be fed into a standard FIFO (e.g., `gaxi_fifo_sync`) to prevent backpressure stalls:

In practice the `addr_check` output is merged with the reporter's main packet stream by an arbiter (`monbus_arbiter`) that carries packet+timestamp atomically through a 192-bit skid. The arbiter's downstream FIFO sits at the monbus group boundary, sized for the aggregate of all per-wrapper streams. A standalone FIFO on the `addr_check` output is normally not needed.

10.8 Related Modules

- **`axi_monitor_base`** — Core monitor infrastructure that instantiates this module

- [axi_monitor_filtered](#) — packet-type and event-code filtering (sibling to `addr_check`)
- [axi4_master_wr_mon](#) — Wrapper that uses this module (`N_ADDR_RANGES` parameter)

10.9 Testing

All properties proven via formal verification (see `formal/amba/axi_monitor_addr_check/formal_axi_monitor_addr_check.sv`):

Property	Description	Status
P1: Reset quiet	After reset, <code>addr_pkt_valid</code> is deasserted.	PASS
P2: Master gate	When <code>cfg_addr_check_enable = 0</code> , <code>addr_pkt_valid</code> stays low.	PASS
P3: Packet class	An emitted packet is <code>PROTOCOL_AXI</code> and is either <code>MATCH</code> (<code>AddrMatch/0x01</code>) or <code>MISS</code> (<code>Error/0x0D</code>) — nothing else.	PASS
P4: MATCH validity	A <code>MATCH</code> packet's <code>range_index</code> points to an enabled DEBUG range (<code>ADDR_RANGE_IS_ERROR=0</code>) and the address lies within its [<code>low</code> , <code>high</code>].	PASS
P5: MISS validity	A <code>MISS</code> packet carries the <code>0xF</code> sentinel, at least one ERROR range is enabled, and the address lies in no enabled ERROR range.	PASS
P6: Sticky valid	Once <code>addr_pkt_valid</code>	PASS

Property	Description	Status
cover	asserts it stays asserted until the consumer accepts. Both emission and the emit+handshake are reachable (non-vacuous).	PASS

10.10 References

- **Monitor Architecture:** docs/markdown/rtl-amba/overview.md
- **Monitor Configuration Guide:** [Monitor Base Configuration](#)
- **Packet Format Specification:** docs/markdown/rtl-amba/includes/monitor_package_spec.md
- **Formal Verification:** formal/amba/axi_monitor_addr_check/

10.11 Navigation

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11 AXI Monitor Base

Module: axi_monitor_base.sv **Location:** rtl/amba/monitor/ **Category:** Core Infrastructure
Status: Production Ready

11.1 Overview

The axi_monitor_base module provides the core transaction tracking and event reporting behind every AXI/AXIL monitor.

This is a **shared infrastructure module** used internally by the AXI/AXIL monitors. You won't instantiate it directly — the wrappers do that — but it's the piece the whole monitor architecture hangs off, so it's worth understanding before you touch any wrapper.

Key features:

- Transaction-based tracking for AXI and AXI-Lite protocols
- Out-of-order transaction handling with ID-based tracking
- Data-before-address support (slave-side scenarios)
- 128-bit standardized monitor bus packet output, plus a 64-bit side-band timestamp
- Configurable performance metrics tracking
- Timeout detection and threshold monitoring
- Per-transaction state-change debug packets (via `ENABLE_DEBUG_LOGIC + cfg_debug_enable`; the verbosity-level/mask interface is dead — see the parameter notes)

Five jobs land in this module:

1. **Transaction Tracking:** maintains state for all outstanding AXI/AXIL transactions
 2. **Event Detection:** identifies protocol errors, timeouts, threshold violations
 3. **Packet Generation:** creates standardized 128-bit `monitor_packet_t` records paired with a 64-bit side-band timestamp
 4. **Flow Control:** manages backpressure and transaction-table exhaustion
 5. **Performance Metrics:** optional latency and throughput tracking
-

11.2 Parameters

11.2.1 Identity and Sizing

Parameter	Type	Default	Description
<code>UNIT_ID</code>	logic [7:0]	8'h09	8-bit unit identifier in monitor packets
<code>AGENT_ID</code>	logic [15:0]	16'h0063	16-bit agent identifier in monitor packets
<code>MAX_TRANSACTIONS</code>	int	16	Maximum outstanding transactions in the CAM
<code>ADDR_WIDTH</code>	int	32	Address bus width

Parameter	Type	Default	Description
ID_WIDTH	int	8	Transaction ID width (0 for AXIL). Hard maximum 8 — bus_transaction_t.id is an 8-bit field, so a wider key would disagree with the payload and mis-attribute transactions; axi_monitor_transactions_mgr refuses ID_WIDTH > 8 at elaboration with an \$error
ADDR_BITS_IN_PKT	int	38	Inert. Intended as the number of address LSBs carried in a packet, but the derived ADDR_BITS is never referenced: bus_transaction_t.addr is 32 bits, so packets carry the low 32 address bits regardless. With ADDR_WIDTH > 32 the upper bits are lost at table-allocation time
IS_READ	bit	1	1=read monitor (R data channel, AR

Parameter	Type	Default	Description
			bursts), 0=write monitor (W data channel, AW bursts)
IS_AXI	bit	1	1=AXI protocol, 0=AXI-Lite
INTR_FIFO_DEPTH	int	8	Depth of the reporter's outgoing interrupt/event FIFO
DEBUG_FIFO_DEPTH	int	8	Dead. No debug-trace FIFO is instantiated; referenced by no logic
N_ADDR_RANGES	int	0	Number of address-range comparators. 0 removes the axi_monitor_addr_check block entirely (zero area)
ADDR_RANGE_IS_ERROR	logic [N_ADDR_RANGE-1:0]	'0	Per-range flavor forwarded to axi_monitor_addr_check: bit i = 0 -> DEBUG range (hit -> AddrMatch, cfg_debug_enable); bit i = 1 -> ERROR range (allowlist miss -> Error/ADDR_RANGE,

Parameter	Type	Default	Description
			cfg_error_enable) . Default all-0.

11.2.2 Master Switches

Parameter	Type	Default	Description
ENABLE_PERF_PACKETS	bit	0	Master switch for the reporter's legacy perf-rollup packets. Setting it also defaults ENABLE_PERF_LOGIC on (see Performance Monitoring)
ENABLE_DEBUG_MODULE	bit	0	Inert / reserved. The debug-trace sub-module it was meant to switch on does not exist in this design; the parameter is kept only because the wrapper family plumbs it. The live debug path is ENABLE_DEBUG_LOGIC + cfg_debug_enable (the reporter's state-change emitter)

11.2.3 Transaction-Table Shaping

These size and shape the transaction table. Defaults reproduce the classic single-bank, age-ranked behaviour exactly.

Parameter	Type	Default	Description
USE_WDATA_ORDER_Q	bit	0	Write monitors only. Recover AXI4's WID-less W-beat ordering with an AWID FIFO (push on the AW handshake, pop on W-last) instead of an $O(N^2)$ oldest-select over the whole table. Required for banking on a write monitor
NUM_BANKS	int	1	Split the table into banks by low ID bits so the two $O(N^2)$ structures (oldest-select, rank update) are restricted to same-bank pairs. Must be a power of 2 and divide MAX_TRANSACTIONS

Sizing rule — this one bites. Every transaction sharing an ID lands in the *same* bank, so per-ID concurrency is capped by the **bank** depth, not by MAX_TRANSACTIONS: $\text{MAX_TRANSACTIONS} / \text{NUM_BANKS} \geq (\text{IDs per bank}) \times (\text{outstanding per ID})$. Undersize it and entries are refused rather than mis-tracked — which surfaces three layers up as missing packets.

NUM_BANKS > 1 on a write monitor requires USE_WDATA_ORDER_Q = 1. The combination without it refuses to elaborate (\$error in axi_monitor_trans_mgr): the WID-less fallback is a state predicate spanning every bank, so the same-bank oldest-select returns one winner

per bank and a single W beat advances one transaction in each. There is no correct fallback, so the build is refused rather than silently double-counting.

11.2.4 ID-Range Filter

Lets several monitors snoop one ID-multiplexed bus, each owning a slice, so no single table has to hold the whole concurrency. Default OFF — bit-identical to an unfiltered build.

Parameter	Type	Default	Description
ID_FILTER_ENABLE	bit	0	Enable the filter
ID_MATCH_BASE	int	0	First ID owned by this instance
ID_MATCH_COUNT	int	0	How many IDs; 0 means all (no filter)
ADDR_FILTER_ENABLE	bit	0	Address-range packet filter, forwarded to axi_monitor_transactions_mgr. 0 leaves it inert

The ID window can also be set **at runtime**, which matters because ID_MATCH_BASE/ID_MATCH_COUNT are elaboration constants — retargeting which master an instance watches otherwise means a rebuild:

Port	Description
cfg_id_filter_enable	High: use the cfg_id_* window below. Low: use the parameters, bit-identical to a build without this feature
cfg_id_match_base	First ID owned, runtime
cfg_id_match_count	How many; 0 means all, matching the parameter rule so a zeroed register block does not silently filter everything away

AXI-Lite has no transaction IDs, so the axil4*_mon wrappers tie these off rather than exposing them — a filter keyed on a field the protocol lacks has nothing to match against.

The **address** filter is driven the same way. ADDR_FILTER_ENABLE only builds the logic; without these three ports driven it stays inert, which is the failure a reader hits when the parameter is the only thing documented:

Port	Width	Description
cfg_addr_filter_enable	1	High: suppress packets for transactions outside the window. Low: filter inert, regardless of the parameter
cfg_addr_filter_low	ADDR_WIDTH	Window base, inclusive
cfg_addr_filter_high	ADDR_WIDTH	Window limit, inclusive

The filter is live only when ADDR_FILTER_ENABLE && cfg_addr_filter_enable. The verdict is latched per table entry at ALLOCATION (filtered_mask), not re-evaluated at emission, so a transaction is judged once on its command address and its data and response beats follow that verdict — see axi_monitor_trans_mgr.

This gates the monitor’s **observation** inputs only, never the datapath the wrapper drives, so a filtered instance stays transparent on the bus. All three channels filter on the same range deliberately: filtering the command alone would leave data/resp for other IDs arriving unmatched, and the unmatched path allocates orphans — the table would fill with other channels’ traffic, which is the problem the filter exists to avoid.

11.2.5 Timer LUT Sizing (CFI_*)

Forwarded to axi_monitor_timer, which divides aclk down to a 1 us tick. The divisor is the clock frequency in MHz, so a table built for this design’s clock gives an exact tick.

Parameter	Type	Default	Description
CFI_MIN_FREQ_MHZ	int	100	Lowest frequency in the LUT
CFI_MAX_FREQ_MHZ	int	100	Highest frequency in the LUT
CFI_NUM_FREQ_ENTRIES	int	16	LUT entries. Note it does not size this module’s cfg_freq_sel, which is a fixed 4-bit port here — only axi_monitor_time

Parameter	Type	Default	Description
CFI_FREQ_STRATEGY	int	0	r derives its select width from this 0 = LINEAR spacing, 1 = POW2

The defaults set **every entry to 100 MHz**, so the tick is exactly 1 us at 100 MHz regardless of `cfg_freq_sel`. Override to a real MIN..MAX range only if the design changes `aclk` at runtime.

11.2.6 Synthesis-Cone Enables (ENABLE_*_LOGIC)

Each detection cone can be compiled out to save area. These gate the **logic**, not just packet emission — a disabled cone synthesizes away entirely. Defaults keep the classic cones on and perf/debug off.

Parameter	Type	Default	Effect when 0
ENABLE_ERROR_LOGIC	bit	1	Drop the error-detection cone (orphans, response errors)
ENABLE_TIMEOUT_LOGIC	bit	1	Drop the timeout cone and the <code>axi_monitor_timeout</code> instance
ENABLE_COMPL_LOGIC	bit	1	Drop the completion cone
ENABLE_THRESHOLD_LOGIC	bit	1	Drop the threshold cone (latency / active-count thresholds)
ENABLE_PERF_LOGIC	bit	= ENABLE_PERF_PACKETS	Drop the reporter's legacy perf-rollup cone (<code>axi_monitor_reporter_perf</code> : two 16-bit lifetime counters + the 5-state emit FSM). It does not gate the

Parameter	Type	Default	Effect when 0
			perfmon measurement window or its bucket counters — those are unconditional (see below)
ENABLE_DEBUG_LOG_IC	bit	0	Drop the debug cone

Removed: the former CAM_PIPELINE / TRANS_CAM_PIPELINE parameters no longer exist. The transaction CAM is now **always pipelined** (one extra cycle of active_count latency). The block_ready margin is derived from monitor_common_pkg::cmd_entry_reserve() — see [Flow Control and the Saturation-Recovery Contract](#). Overshoot past MAX_TRANSACTIONS is structurally impossible regardless of the margin: the CAM allocates from its exact combinational free vector.

11.2.7 Derived Parameters (do not override)

These are declared as parameter so the elaborator can compute them, not so callers can set them. Each defaults to an expression over the parameters above; overriding one desynchronises it from its source and the design fails to elaborate or silently mis-sizes a bus. Set the parameters they are derived FROM and leave these alone.

Derived parameter	Default expression
IW	ID_WIDTH

11.3 Ports

11.3.1 Command Phase Interface (AW/AR)

Port	Direction	Width	Description
cmd_addr	Input	ADDR_WIDTH	Command address value
cmd_id	Input	ID_WIDTH	Transaction ID
cmd_len	Input	8	Burst length (AXI only)

Port	Direction	Width	Description
cmd_size	Input	3	Burst size (AXI only)
cmd_burst	Input	2	Burst type (AXI only)
cmd_valid	Input	1	Command valid
cmd_ready	Input	1	Command ready

11.3.2 Data Channel Interface (R/W)

Port	Direction	Width	Description
data_id	Input	ID_WIDTH	Data transaction ID
data_last	Input	1	Last data beat indicator
data_resp	Input	2	Response code (OKAY/EXOKA Y/SLVERR/DEC ERR)
data_valid	Input	1	Data valid
data_ready	Input	1	Data ready

11.3.3 Response Channel Interface (B)

Port	Direction	Width	Description
resp_id	Input	ID_WIDTH	Response transaction ID
resp_code	Input	2	Response code
resp_valid	Input	1	Response valid
resp_ready	Input	1	Response ready

11.3.4 Configuration Interface

Port	Direction	Width	Description
clear	Input	1	Synchronous

Port	Direction	Width	Description
			clear — passes through to axi_monitor_transactions_mgr to empty the transaction CAM and zero the active-count pipeline atomically, without a full aresetn. Pulse one cycle while the monitor is idle.
cfg_freq_sel	Input	4	Frequency selection for timeout scaling
cfg_addr_cnt	Input	16	Address phase timeout, in microseconds (1 us tick)
cfg_data_cnt	Input	16	Data phase timeout, in microseconds
cfg_resp_cnt	Input	16	Response phase timeout, in microseconds
cfg_error_enable	Input	1	Enable error event packets
cfg_compl_enable	Input	1	Enable completion packets
cfg_threshold_enable	Input	1	Enable threshold packets
cfg_timeout_enab	Input	1	Enable timeout

Port	Direction	Width	Description
le			packets
cfg_perf_enable	Input	1	Enable performance metric packets
cfg_debug_enable	Input	1	Enable debug/trace packets (feeds the debug reporter sub-block)
cfg_active_trans_threshold	Input	16	Active-transaction count that triggers a threshold packet
cfg_latency_threshold	Input	32	Latency value that triggers a threshold packet
cfg_debug_level	Input	4	Dead — referenced by no logic (interface to the absent debug sub-module)
cfg_debug_mask	Input	16	Dead — referenced by no logic

11.3.5 Address-Range Checker Interface

Active only when `N_ADDR_RANGES > 0`; otherwise these inputs are ignored and the `axi_monitor_addr_check` block is not synthesized. Address-range violation packets are the lowest-priority source on the monitor bus (reporter > debug > addr_check).

Port	Direction	Width	Description
cfg_addr_check_enable	Input	1	Master enable for the address-range checker
cfg_addr_range_enable	Input	<code>N_ADDR_RANGES</code>	Per-range enable bit vector

Port	Direction	Width	Description
cfg_addr_range_low	Input	N_ADDR_RANGES x ADDR_WIDTH	Per-range low (inclusive) address bounds
cfg_addr_range_high	Input	N_ADDR_RANGES x ADDR_WIDTH	Per-range high (inclusive) address bounds

11.3.6 Side-Band Timestamp Input

Port	Direction	Width	Description
i_mon_time	Input	64	Free-running counter from the monbus_group family (any wrapper), sampled at packet emission

11.3.7 Monitor Bus Output

Port	Direction	Width	Description
monbus_valid	Output	1	Monitor packet valid
monbus_ready	Input	1	Monitor packet ready (from downstream)
monbus_packet	Output	128	Standardized monitor_packet_t
monbus_timestamp	Output	64	monbus_timestamp_t paired atomically with monbus_packet
block_ready	Output	1	Flow control: 1 = upstream may proceed, 0 = transaction-table occupancy is at the blocking

Port	Direction	Width	Description
			threshold. Derived from active_count vs MAX_TRANSACTIONS ; the reporter's INTR_FIFO_DEPTH has no path to it. See Flow Control and the Saturation- Recovery Contract
busy	Output	1	Monitor is busy indicator (active_count > 0)
active_count	Output	8	Number of live transaction-table entries (registered pop-count of CAM occupancy; lags true occupancy by 1 cycle)
perf_completed_c ount	Output	16	Lifetime completion count from axi_monitor_repo rter_perf – packets actually emitted, not window-scoped. Reads 0 when ENABLE_PERF_LOGI C = 0
perf_error_count	Output	16	Lifetime error/timeout count from the same block; 0

Port	Direction	Width	Description
			when ENABLE_PERF_LOGI C = 0

The base module multiplexes three internal sources onto the monbus output — reporter packets (highest priority), debug packets, and addr_check error packets — driving monbus_timestamp from the source's sampled timestamp (i_mon_time for reporter/debug; addr_pkt_timestamp from the addr_check submodule).

11.3.8 Performance-Window Control Inputs

These configure the measurement-window state machine. Tie the selectors to a reserved code (e.g. 3'b111) and the pulses/force-close low at instances that don't use perfmon. See [Performance Monitoring](#).

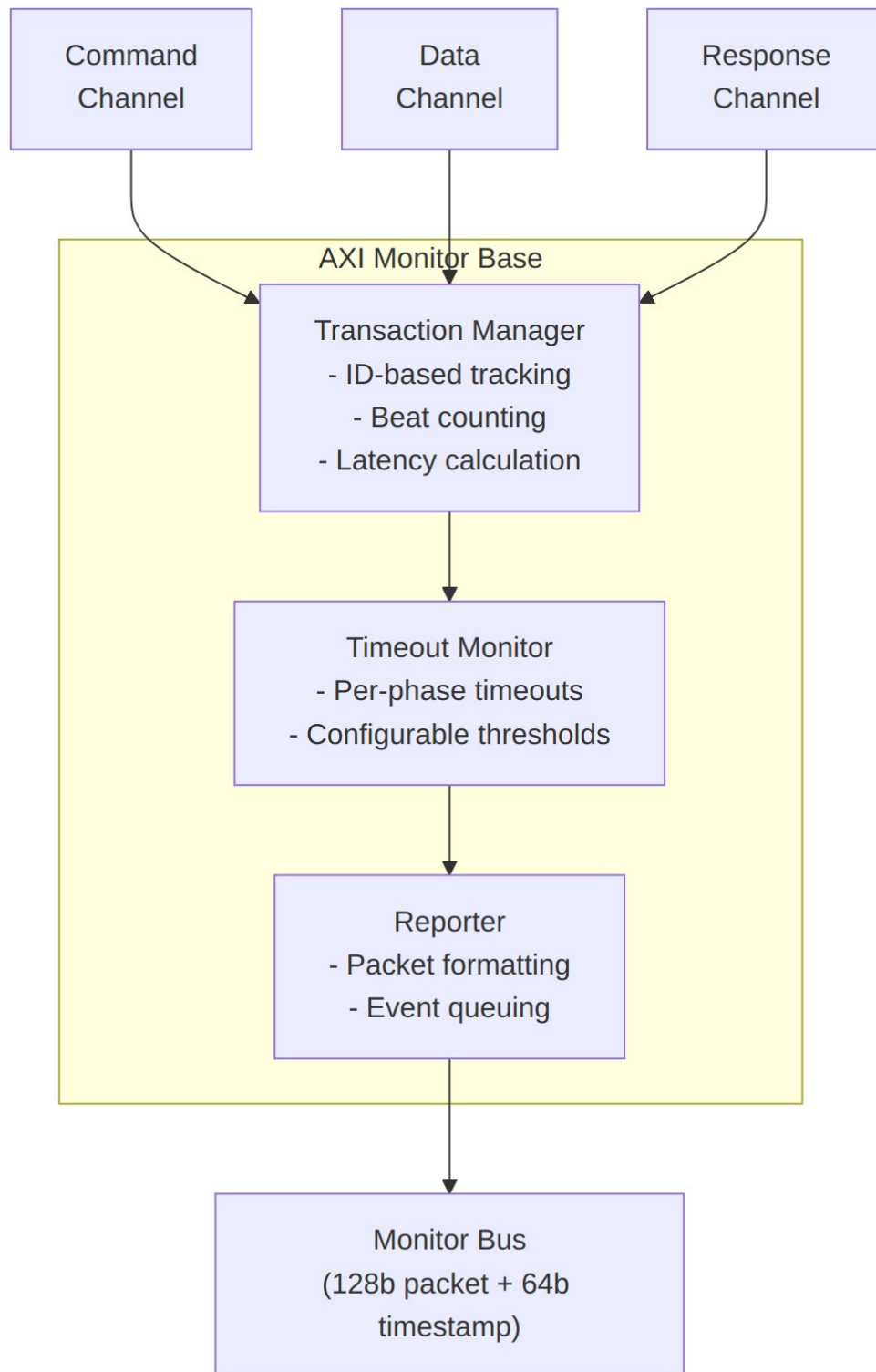
Port	Direction	Width	Description
cfg_start_event_sel	Input	3	Window start event source select
cfg_end_event_sel	Input	3	Window end event source select
cfg_start_trigger	Input	1	Software/engine pulse: open the measurement window
cfg_end_trigger	Input	1	Software/engine pulse: close the measurement window
cfg_window_force_close	Input	1	Software override: force the window closed immediately

11.3.9 Performance-Window Status and Counter Outputs

All counters accumulate only while `window_active=1`, reset at window-start, and hold their values through `WIN_CLOSING`/`WIN_IDLE` so the integrating block can sample a completed window's totals after `window_active` deasserts.

Port	Direction	Width	Description
<code>window_active</code>	Output	1	High while a measurement window is open
<code>window_cycles</code>	Output	32	Cycles elapsed inside the current window
<code>perf_prod_cycles</code>	Output	32	<code>data_valid</code> && <code>data_ready</code> cycles (productive beats)
<code>perf_bp_cycles</code>	Output	32	<code>data_valid</code> && ! <code>data_ready</code> cycles (back-pressure)
<code>perf_starv_cycles</code>	Output	32	! <code>data_valid</code> && <code>data_ready</code> cycles (starvation)
<code>perf_idle_cycles</code>	Output	32	! <code>data_valid</code> && ! <code>data_ready</code> cycles (idle)
<code>perf_beat_count</code>	Output	32	Data beats transferred (= <code>perf_prod_cycles</code> , 1 beat/cycle)
<code>perf_byte_count</code>	Output	64	Bytes transferred = beats x (1 << latched <code>AXSIZE</code>)
<code>perf_burst_count</code>	Output	32	Address-phase handshakes inside the window (AR for reads, AW for writes)

11.4 Functional Description



Architecture

The module coordinates four sub-components:

1. **Transaction Manager:** tracks transactions, manages the table
2. **Timeout Monitor:** detects stuck transactions
3. **Performance Tracker:** optional latency/throughput metrics
4. **Reporter:** generates standardized packets

11.4.1 Flow Control and the Saturation-Recovery Contract

This is the canonical statement of the monitor's flow-control behavior. The wrapper pages (`axi4*_mon`, `axi5*_mon`, `axil4*_mon`) [link here](#).

`block_ready` is a positive-enable flow-control output: 1 = the upstream handshake may proceed, 0 = stall. It is driven **solely by transaction-table occupancy** (`active_count` vs `MAX_TRANSACTIONS`); the reporter's packet FIFO (`INTR_FIFO_DEPTH`) has no path to it. In the `*_mon` wrappers, `block_ready` is an internal net ANDed into the upstream-facing ready signal — it is **not** a port on the wrappers.

The contract (single source of truth: `monitor_common_pkg::cmd_entry_reserve()` in `rtl/amba/includes/monitor_common_pkg.sv`):

- **Command-entry cap.** The transaction manager caps command-originated entries (allocated by AW/AR activity) at `MAX_TRANSACTIONS - cmd_entry_reserve(MAX_TRANSACTIONS)`. The reserve is 4 for `MAX_TRANSACTIONS >= 16` and 0 below. Orphan entries (data/response beats with no matching command) may still use every slot commands are not entitled to; orphans always drain via error reporting, so they cannot pin occupancy.
- **Reopen threshold strictly above the cap.** `block_ready` re-asserts at `active_count < MAX_TRANSACTIONS - (cmd_entry_reserve - 1)`. Because this threshold sits **strictly above** the command cap, a saturated table always drains back below the reopen point: even if every command entry is permanently in flight, occupancy cannot reach the threshold once the orphan entries drain. Blocking therefore **throttles, never deadlocks**.
- **The blocking margin covers all three allocators.** `active_count` is a registered pop-count that lags true occupancy by one cycle, and three independent allocators (address, data, response) can each take a slot in that stale cycle. The derived margin (`cmd_entry_reserve - 1 = 3`) covers all three, so a command can never be admitted against stale occupancy and then find no free slot. The reserve of 4 is the only value that satisfies both

this and the reopen constraint above simultaneously; with the earlier reserve of 2 the margin derived to 1, and commands admitted in the stale cycle went untracked while their un-backpressureable data beats were silently discarded (measured as an observer-vs-in-core burst-count gap of 4096 vs 3073). `val/amba/test_axi_mon_block_ready.py` asserts no command is admitted without an allocation, on every wrapper.

- **Overshoot is impossible.** The CAM allocates from its exact combinational free vector, so the table can never hold more than `MAX_TRANSACTIONS` entries. A command that handshakes while the cap is reached is simply not tracked (lossy-but-honest degrade), never stalled forever.
- **Small tables trade recovery for capacity.** Tables with `MAX_TRANSACTIONS < 16` get `cmd_entry_reserve = 0` and keep the full legacy allocation (with the legacy flat margin of 3 on `block_ready`). Small tables cannot spare slots — even one reserved slot at `MAX=8` measurably starves oversubscribed same-ID tracking — so they give up the recovery guarantee in exchange for full tracking capacity.

Both `axi_monitor_trans_mgr` (the cap) and this module (the `block_ready` margin) derive their constants from the same package function, so the two cannot drift. The contract is enforced by in-RTL formal properties in `axi_monitor_trans_mgr.sv` (`ap_cmd_entry_cap`, `ap_no_reopened_complete`, `mutation-checked`) and by a 100-seed deliberately-undersized-table stream sweep (`test_stream_core_mon_backpressure` in the `STREAM` project area, plus `val/amba/test_axi_monitor_trans_mgr.py::phase_saturation_recovers`).

History: before commit `cb29e226`, a stray non-last data beat could poison a terminal entry into an unclosable state, occupancy ratcheted to `MAX`, and the old flat `MAX-3` margin placed the reopen threshold exactly AT the saturation point — `block_ready` latched low forever and the monitored datapath wedged (the `stream_core` multi-channel hang). Commit `95c9490a` extended the fix to runtime-disabled packet classes (see the auto-retire notes in [axi_monitor_reporter](#)).

Sizing rule for shared masters: a monitor watching a bus shared by multiple channels/requesters must size `MAX_TRANSACTIONS` to cover the SUM of outstanding transactions — `NUM_CHANNELS × per-channel outstanding`, plus margin (the orphan reserve and the 1-cycle `active_count` lag). Sizing to the per-channel limit alone makes the monitor throttle the shared master from two channels up; this exact mistake shipped in `stream_core` (fixed in `95c9490a` by making the monitor table parametric on channel count).

Verilator note: transaction tables deeper than 64 require raising `--unroll-count` (default 64) in simulation builds, or the per-slot generate loops fail with `BLKLOOPINIT` errors. The stream sim builds use `--unroll-count 256`.

11.4.2 Performance Monitoring

`axi_monitor_base` is the **canonical home** of the monitor's performance subsystem. Every AXI/AXIL wrapper (`axi4*_mon`, `axil4*_mon`, and the `axi_monitor_filtered` wrapper) forwards these ports straight through to this core. Because the base is generic, the data channel it measures is chosen by `IS_READ`: the **R** channel and **AR** bursts for read monitors, the **W** channel and **AW** bursts for write monitors.

The perfmon logic is **always instantiated**: the measurement-window state machine and the bank of data-channel utilization and throughput counters are plain `always_ff` blocks in this module with no generate guard, so they cost their area in every build. `ENABLE_PERF_LOGIC` gates only the reporter's legacy perf-rollup sub-block (`axi_monitor_reporter_perf`), which is a different thing: two 16-bit lifetime counters and the FSM that emits `PktTypePerf` rollup packets. Building with `ENABLE_PERF_PACKETS=0` therefore removes the rollup packets, not the window. All window counters accumulate **only while a window is open** and hold their values between windows so the host can read a completed window's totals.

11.4.2.1 The Measurement Window

A window is opened by a **start event** and closed by an **end event**. Event sources are chosen by the 3-bit `cfg_start_event_sel` / `cfg_end_event_sel` selectors:

Code	Start event	End event
3'b000	<code>cfg_start_trigger</code> pulse (software/CSR)	<code>cfg_end_trigger</code> pulse
3'b001	first command handshake (<code>cmd_valid</code> && <code>cmd_ready</code>)	last data (reads: <code>data_last</code> beat; writes: response handshake)
3'b010	<code>cfg_perf_enable</code> rising edge	<code>cfg_perf_enable</code> falling edge
3'b011	first data handshake (<code>data_valid</code> && <code>data_ready</code>)	<code>window_cycles</code> saturation
3'b100	<code>cfg_start_trigger</code> pulse (external-trigger convention)	<code>cfg_end_trigger</code> pulse
others	never fires (reserved)	never fires (reserved)

`cfg_window_force_close` is a software override that closes an open window immediately regardless of the end-event selector.

The state machine has three states:

- **WIN_IDLE** — waiting for a start event; window_active=0.
- **WIN_ACTIVE** — window open; window_active=1; window_cycles free-runs (starting at 1 so the total is inclusive of the first cycle). All counters accumulate.
- **WIN_CLOSING** — one-cycle hold before re-arming; counters are frozen and stable for sampling.

11.4.2.2 Utilization Buckets (data channel)

Every cycle inside the window is classified by the data channel's valid/ready into exactly one of four mutually-exclusive buckets. The four buckets sum to window_cycles - 1 by construction (the start cycle seeds window_cycles to 1 while the buckets reset to 0), so utilization = perf_prod_cycles / (window_cycles - 1) — **until a counter saturates**. window_cycles freezes at 32'hFFFF_FFFE and each bucket sticks at 32'hFFFF_FFFF rather than wrapping, so on a window longer than ~2³² cycles (~43 s at 100 MHz) the identity and this formula both break. A reader seeing 32'hFFFF_FFFF, or a bucket sum below window_cycles - 1, is looking at an overflowed window.

Counter	Condition	Meaning
perf_prod_cycles	valid && ready	productive beat transferred
perf_bp_cycles	valid && !ready	back-pressure (data offered, sink not ready)
perf_starv_cycles	!valid && ready	starvation (sink ready, no data)
perf_idle_cycles	!valid && !ready	idle

11.4.2.3 Throughput Counters

Counter	Width	Meaning
perf_beat_count	32	data beats transferred (= perf_prod_cycles, 1 beat/cycle)
perf_byte_count	64	bytes transferred = beats x (1 << latched AXSIZE). AXSIZE is captured (cmd_size) at the most recent address-phase handshake and assumed constant within a burst

Counter	Width	Meaning
perf_burst_count	32	per the AXI4 mandate. 64-bit width prevents wrap on long windows at wide buses address-phase handshakes inside the window (AR for read monitors, AW for write monitors)

The integrator computes average burst length as $\text{perf_beat_count} / \text{perf_burst_count}$.

Note: the perfmon counters are exposed on the module interface for the integrating block to sample; the reporter's PerfWin packet emission is staged separately (RFC Stage B+). The four-bucket model follows DMA_UTILIZATION_MEASUREMENT.md Section 3.

11.5 Timing Characteristics

Metric	Value	Notes
Latency	2-3 cycles	Event detection to packet output
Throughput	1 packet per 2 cycles	The reporter's registered output stage cannot load a new packet on the same cycle the previous one is accepted, so sustained rate is at most one packet every other cycle
Table Lookup	1 cycle	ID-based transaction lookup

11.6 Usage Examples

Not typically instantiated directly by users. Use the high-level monitors instead:

```
// User instantiates this:
axi4_master_rd_mon #(...) u_mon (...);

// Which internally uses:
// - axi4_master_rd (frontend)
// - axi_monitor_base (this module)
// - axi_monitor_filtered
```

11.7 Design Notes

11.7.1 Transaction Table Sizing

Design Scenario	MAX_TRANSACTIONS	Rationale
AXI4 Master (Burst)	16-32	Support multiple outstanding bursts
AXI4 Slave (Burst)	16-32	Handle out-of-order responses
AXI4-Lite Master	4-8	Single-beat, limited concurrency
AXI4-Lite Slave	4-8	Simple protocol, fewer outstanding
Shared master (multi-channel)	NUM_CHANNELS x per-channel outstanding + margin	Per-channel limit alone throttles the shared bus — see the sizing rule in Flow Control and the Saturation-Recovery Contract

Tables of 16 or more slots reserve `cmd_entry_reserve(MAX) = 4` slots (the saturation-recovery guarantee plus a blocking margin wide enough for all three same-cycle allocators); tables below 16 keep full legacy allocation. Tables deeper than 64 need `--unroll-count` raised above Verilator's default of 64 in sim builds.

11.7.2 Timeout Configuration

- **cfg_addr_cnt**: command phase timeout (AR/AW)

- **cfg_data_cnt**: data phase timeout (R/W)
- **cfg_resp_cnt**: response phase timeout (B)

Typical values:

- Burst traffic: `cfg_*_cnt` = 4-8 (aggressive timeout)
 - Memory controllers: `cfg_*_cnt` = 10-15 (allow latency)
 - External interfaces: a large `cfg_*_cnt` (the field is 16 bits, so up to 65535 us ~ 65 ms; 0xFFFF is effectively “never time out”)
-

11.8 Related Modules

- [axi_monitor_filtered](#)
- [axi_monitor_trans_mgr](#)
- [axi_monitor_reporter](#)
- [axi_monitor_timeout](#)

Used by:

- **axi4_master_rd_mon**
- **axi4_master_wr_mon**
- **axi4_slave_rd_mon**
- **axi4_slave_wr_mon**
- **axil4_master_rd_mon**
- **axil4_master_wr_mon**
- **axil4_slave_rd_mon**
- **axil4_slave_wr_mon**

See also:

- **Monitor Architecture:** [docs/markdown/rtl-amba/overview.md](#)
 - **Monitor Configuration Guide:** [Monitor Base Configuration](#)
 - **Packet Format Specification:**
[docs/markdown/rtl-amba/includes/monitor_package_spec.md](#)
-

11.9 Testing

Key test scenarios:

1. **Transaction Table Management:**
 - Fill table to MAX_TRANSACTIONS
 - Verify table exhaustion handling
 - Test ID reuse after completion
 2. **Out-of-Order Transactions:**
 - Issue multiple transactions with different IDs
 - Complete in random order
 - Verify correct ID matching
 3. **Timeout Detection:**
 - Initiate transaction without completion
 - Verify timeout event at configured threshold
 - Check timeout packet contains correct ID
 4. **Performance Metrics:**
 - Enable ENABLE_PERF_PACKETS
 - Verify latency calculations
 - Check throughput tracking
-

11.10 Navigation

- [Back to Shared Infrastructure Index](#)
- [Back to rtl-amba Index](#)
- [Back to Main Documentation Index](#)

12 AXI Monitor Filtered

Module: axi_monitor_filtered.sv **Location:** rtl/amba/monitor/ **Category:** Core Infrastructure **Status:** Production Ready

12.1 Overview

axi_monitor_filtered is a **filtering wrapper around axi_monitor_base**. It instantiates one axi_monitor_base internally, taps the same AXI/AXIL command, data, and response channels, and then applies a configurable drop filter to the 128-bit monitor packets the base emits before forwarding them downstream. It does **not** take a monitor-bus input stream — it produces the monitor bus from the channels it observes.

This is a **shared infrastructure module** used internally by the AXI/AXIL monitors. You won't instantiate it directly, but it's the piece that keeps monitor-bus traffic manageable, so it's worth understanding.

Key features:

- **Wraps axi_monitor_base:** all transaction tracking, timeout, threshold, and perfmon logic lives in the base; this module only filters its output stream.
- **Level 1 — packet-type drop mask** (cfg_axi_pkt_mask): drop entire packet types by index.
- **Level 2 — error select** (cfg_axi_err_select): reserved for cross-routing; in the AXI wrapper it is used only for **configuration-conflict validation** (see cfg_conflict_error), not applied to the stream.
- **Level 3 — per-event-code masking:** one 16-bit drop mask per packet type (cfg_axi_error_mask, cfg_axi_timeout_mask, cfg_axi_compl_mask, cfg_axi_thresh_mask, cfg_axi_perf_mask, cfg_axi_addr_mask, cfg_axi_debug_mask) indexed by the packet's event code.
- **Configuration conflict detection:** cfg_conflict_error flags overlapping cfg_axi_pkt_mask / cfg_axi_err_select bits.
- **Bypass mode:** ENABLE_FILTERING=0 passes every packet straight through.
- **Optional pipeline stage:** ADD_PIPELINE_STAGE=1 registers the filtered output for timing closure.

Four reasons this block exists:

1. **Traffic Management:** reduces monitor bus congestion by dropping unwanted packet types and event codes at the source.
 2. **Granular Control:** supports per-packet-type (Level 1) and per-event-code (Level 3) drop masking.
 3. **Configuration Validation:** flags conflicting mask settings via cfg_conflict_error.
 4. **Protocol Isolation:** drops any non-AXI-protocol packet (protocol field != 4'h0), which should never occur inside an AXI monitor.
-

12.2 Parameters

Parameter	Type	Default	Description
UNIT_ID / AGENT_ID	logic	8'h01 / 16'h000A	Identity bits stamped into

Parameter	Type	Default	Description
MAX_TRANSACTIONS	int	16	emitted monitor packets. Outstanding transaction table depth (passed to axi_monitor_base -> axi_monitor_transactions_mgr).
ADDR_WIDTH / ID_WIDTH	int	32 / 8	Address and AXI ID widths. ID_WIDTH has a hard maximum of 8 (the transaction-table id field is 8 bits; wider is refused at elaboration with an \$error). ADDR_WIDTH > 32 is allowed but truncates the address carried in packets to the low 32 bits.
IS_READ / IS_AXI	bit	1 / 1	Direction (read vs write) and protocol family (AXI4 vs AXIL).
ENABLE_PERF_PACKETS	bit	1	Emit perf packets onto the monitor bus.
ENABLE_DEBUG_MODULE	bit	0	Inert / reserved – the debug-trace sub-module it names does not

Parameter	Type	Default	Description
Reporter sub-block enables			exist; forwarded to axi_monitor_base only because the wrapper family plumbs it. Real debug packets come from ENABLE_DEBUG_LOG IC + cfg_debug_enable .
			Each gates one of the six reporter sub-blocks (axi_monitor_reporter_*) at elaboration time. Pass-through to axi_monitor_base .
	ENABLE_ERROR_LOG IC	bit 1	Error reporter (orphans, resp errors).
	ENABLE_TIMEOUT_LOG IC	bit 1	Timeout reporter (addr/data/resp timers).
	ENABLE_COMPL_LOG IC	bit 1	Completion reporter.
	ENABLE_THRESHOLD_LOGIC	bit 1	Threshold reporter (latency / active-count thresholds).
ENABLE_PERF_LOGIC	bit	ENABLE_PERF_PACKETS	Gates axi_monitor_reporter_perf only:

Parameter	Type	Default	Description
			two 16-bit lifetime counters plus the 5-state emit FSM. It does not gate the Stage B window counters below — those are unconditional always_ff blocks in axi_monitor_base and are live in every build.
ENABLE_DEBUG_LOG IC	bit	0	Debug reporter.
ENABLE_FILTERING	bit	1	Master enable for filtering: two active drop levels (packet type, then event code). Level 2 is reserved and routes nothing
ADD_PIPELINE_STAGE	bit	0	Add register stage for timing.
N_ADDR_RANGES	int	0	Number of address-range comparators in the axi_monitor_addr_check sub-block (0 = the comparator block is not synthesised at all).
ADDR_RANGE_IS_ERROR	logic [N_ADDR_RANGE	'0	Per-range flavor forwarded to the

Parameter	Type	Default	Description
	S-1:0]		checker: 0 = DEBUG (hit -> AddrMatch), 1 = ERROR (allowlist miss -> Error/ADDR_RAN GE). Default all-0.

12.2.1 Forwarded to axi_monitor_base (pass-through)

These are declared here and passed straight down; the base module's page is the authority on what each one does. They are listed so this wrapper's own page is a complete inventory.

Parameter	Type	Default	Description
USE_WDATA_ORDER_Q	bit	0	Write monitors: recover WID-less W-beat order with an AWID FIFO. Required when NUM_BANKS > 1 on a write monitor — the combination without it refuses to elaborate
NUM_BANKS	int	1	Bank the transaction table by low ID bits. Power of 2, must divide MAX_TRANSACTIONS ; per-ID concurrency is capped by the bank depth
ID_FILTER_ENABLE	bit	0	Track only a slice of the IDs on a shared bus

Parameter	Type	Default	Description
ID_MATCH_BASE	int	0	First ID owned by this instance
ID_MATCH_COUNT	int	0	How many IDs; 0 = all (no filter)
ADDR_FILTER_ENABLE	bit	0	Address-range packet filter. Builds the logic only; the <code>cfg_addr_filter_*</code> ports below arm it
CFI_MIN_FREQ_MHZ	int	100	Timer LUT lower bound (MHz)
CFI_MAX_FREQ_MHZ	int	100	Timer LUT upper bound (MHz)
CFI_NUM_FREQ_ENTRIES	int	16	Timer LUT entries. Does not size this module's <code>cfg_freq_sel</code> , which is a fixed 4-bit port here
CFI_FREQ_STRATEGY	int	0	0 = LINEAR, 1 = POW2

See [Transaction-Table Shaping, ID-Range Filter and Timer LUT Sizing in `axi\_monitor\_base`](#) for the sizing rules and the elaboration error.

12.2.2 Filter configuration and status (pass-through)

Also declared here and passed straight down. The two filters are inert unless their `cfg_*_enable` is driven, whatever the parameters say:

Port	Dir	Width	Description
<code>cfg_id_filter_enable</code>	input	1	Use the runtime ID window instead of <code>ID_MATCH_BASE/COUNT</code>

Port	Dir	Width	Description
cfg_id_match_base	input	ID_WIDTH	First ID owned, runtime
cfg_id_match_count	input	ID_WIDTH+1	How many; 0 = all
cfg_addr_filter_enable	input	1	Arm the address-range packet filter
cfg_addr_filter_low	input	ADDR_WIDTH	Window base, inclusive
cfg_addr_filter_high	input	ADDR_WIDTH	Window limit, inclusive
block_ready	output	1	Table near capacity: stop admitting new commands
busy	output	1	At least one transaction is being tracked
active_count	output	8	Registered count of occupied table entries

block_ready, busy and active_count are real outputs and must be connected or explicitly left open; they are how an integrator sees the monitor's own backpressure and occupancy.

12.3 Ports

12.3.1 Observed AXI/AXIL Channels (inputs to the wrapped base)

This module has **no monitor-bus input stream**. It taps the same command, data, and response channels as axi_monitor_base and passes them through unchanged. See axi_monitor_base for full descriptions.

Group	Ports
Command (AW/AR)	cmd_addr, cmd_id, cmd_len, cmd_size, cmd_burst, cmd_valid, cmd_ready
Data (W/R)	data_id, data_last, data_resp,

Group	Ports
Response (B)	data_valid, data_ready resp_id, resp_code, resp_valid, resp_ready
Timer / enables / thresholds	cfg_freq_sel, cfg_addr_cnt, cfg_data_cnt, cfg_resp_cnt, cfg_error_enable, cfg_compl_enable, cfg_threshold_enable, cfg_timeout_enable, cfg_perf_enable, cfg_debug_enable, cfg_debug_level, cfg_debug_mask, cfg_active_trans_threshold, cfg_latency_threshold
Address-range checker (when N_ADDR_RANGES > 0)	cfg_addr_check_enable, cfg_addr_range_enable, cfg_addr_range_low, cfg_addr_range_high
Side-band time	i_mon_time (monbus_timestamp_t)

All of the above are passed through to the internal axi_monitor_base verbatim.

12.3.2 Monitor Bus Output (filtered)

Port	Direction	Width	Description
monbus_valid	Output	1	Filtered packet valid (base valid AND not dropped)
monbus_ready	Input	1	Downstream ready
monbus_packet	Output	128	Filtered monitor_packet_t (unmodified; only passed or dropped)
monbus_timestamp	Output	64	monbus_timestamp_t paired atomically with monbus_packet

12.3.3 Reset and Synchronous Control

Port	Direction	Width	Description
aclk / aresetn	Input	1 / 1	Standard clock and active-low async reset.
clear	Input	1	Synchronous clear — passes through to axi_monitor_base -> axi_monitor_transactions_mgr to empty the transaction CAM and zero the active-count pipeline without aresetn. Pulse one cycle while idle.

12.3.4 Filter Configuration

All filter masks use **drop semantics**: a set bit drops the corresponding packet type or event code. All masks are 16-bit.

Port	Direction	Width	Description
cfg_axi_pkt_mask	Input	16	Level 1 packet-type drop mask. Bit pkt_type set -> drop all packets of that type
cfg_axi_err_select	Input	16	Level 2 error-select. Reserved for cross-routing; in the AXI wrapper it is consumed only by the conflict check,

Port	Direction	Width	Description
			not applied to the stream
cfg_axi_error_mask	Input	16	Level 3 per-event-code drop mask for Error packets
cfg_axi_timeout_mask	Input	16	Level 3 drop mask for Timeout packets
cfg_axi_compl_mask	Input	16	Level 3 drop mask for Completion packets
cfg_axi_thresh_mask	Input	16	Level 3 drop mask for Threshold packets
cfg_axi_perf_mask	Input	16	Level 3 drop mask for Performance packets
cfg_axi_addr_mask	Input	16	Level 3 drop mask for Address-Match packets
cfg_axi_debug_mask	Input	16	Level 3 drop mask for Debug packets

The Level 3 masks are indexed by the packet's event code (low nibble). For each packet the wrapper selects the mask matching its packet type, then drops the packet if `mask[event_code[3:0]]` is set.

12.3.5 Configuration Status Output

Port	Direction	Width	Description
cfg_conflict_error	Output	1	Asserted when <code>cfg_axi_pkt_mask</code> & <code>cfg_axi_err_select</code> is non-zero (overlapping type is both dropped)

Port	Direction	Width	Description
			and error-selected)

12.3.6 Performance Window Control (Stage A of perfmon RFC)

Port	Direction	Width	Description
cfg_start_event_sel	Input	3	Event selector that opens the perf window (e.g. command-handshake, first beat, software pulse).
cfg_end_event_sel	Input	3	Event selector that closes the perf window.
cfg_start_trigger	Input	1	Software-driven open pulse (combines with cfg_start_event_sel).
cfg_end_trigger	Input	1	Software-driven close pulse.
cfg_window_force_close	Input	1	Synchronous software override: forces the measurement window closed on the next clock edge. Perf totals are held , not dropped – the counters keep their values through WIN_CLOSING/WIN_IDLE so a host can

Port	Direction	Width	Description
			read the forced window, and reset only at the next window start.
window_active	Output	1	The perf window is currently open.
window_cycles	Output	32	Number of cycles the current window has been open.

Tie `cfg_start_event_sel/cfg_end_event_sel` to 3'b111 and the triggers + `cfg_window_force_close` to 1'b0 at instances that don't use perfmon.

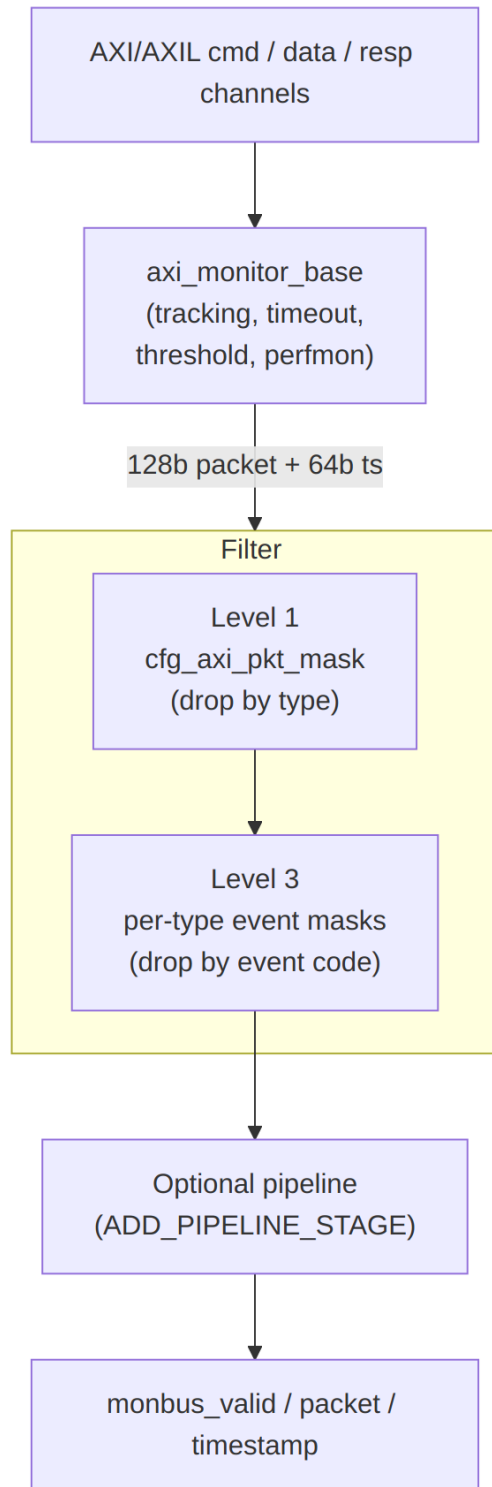
12.3.7 Performance Counters (Stage B of perfmon RFC)

Port	Direction	Width	Description
perf_prod_cycles	Output	32	Productive cycles (valid + ready both high).
perf_bp_cycles	Output	32	Back-pressure cycles (valid high, ready low).
perf_starv_cycles	Output	32	Starvation cycles (valid low, ready high).
perf_idle_cycles	Output	32	Idle cycles (both low).
perf_beat_count	Output	32	Beat handshakes accumulated this window.
perf_byte_count	Output	64	Bytes transferred this window: productive beats x (1 << latched cmd_size). cmd_len is not

Port	Direction	Width	Description
perf_burst_count	Output	32	involved. Bursts (command handshakes) this window.
perf_completed_count	Output	16	Lifetime completion count from axi_monitor_reporter_perf (not window-scoped). Reads 0 when ENABLE_PERF_LOGIC = 0
perf_error_count	Output	16	Lifetime error/timeout count from the same block; 0 when ENABLE_PERF_LOGIC = 0

These counters latch on each window_active open and freeze on close; software reads them after the close edge.

12.4 Functional Description



Architecture

The base monitor generates every packet; the filter then drops unwanted ones. A dropped packet is acknowledged back to the base (`base_monbus_ready`) so the base never stalls on packets that will be discarded. Level 2 (`cfg_axi_err_select`) is a validation-only input in this wrapper and is not shown in the drop path.

12.5 Timing Characteristics

Metric	Value	Notes
Filtering Latency	0-1 cycles	Combinatorial (0) or registered (1)
Throughput	1 packet per 2 cycles	Limited by the reporter's registered output stage; the filter itself introduces no additional backpressure

12.6 Usage Examples

This module is instantiated automatically within higher-level monitor modules — users configure behavior through top-level monitor parameters.

12.6.1 Filter Strategy

`cfg_axi_pkt_mask` uses **drop semantics** — a bit set at index `pkt_type` drops that type. Leave a bit 0 to keep the type. Indices follow `monitor_common_pkg` (`PktTypeError`, `PktTypeCompletion`, `PktTypeThreshold`, `PktTypeTimeout`, `PktTypePerf`, `PktTypeAddrMatch`, `PktTypeDebug`). Tie the per-event Level 3 masks to `16'h0000` unless you need to suppress specific event codes.

Pass everything (no filtering):

```
.cfg_axi_pkt_mask    (16'h0000), // drop nothing
// or set ENABLE_FILTERING = 0 at elaboration for full bypass
```

Suppress performance packets (typical functional run):

```
.cfg_axi_pkt_mask    (16'h0000 | (16'h1 << PktTypePerf)), // drop only Perf
```

Performance-only capture (drop completions to cut traffic):

```
.cfg_axi_pkt_mask    ((16'h1 <= PktTypeCompletion) |  
                    (16'h1 <= PktTypeTimeout)),           // keep  
Error + Perf
```

Suppress one Error event code (Level 3):

```
.cfg_axi_pkt_mask    (16'h0000),                          // keep all types  
.cfg_axi_error_mask (16'h1 <= ERR_EVENT_CODE),           // drop just that  
error event
```

12.7 Design Notes

Two active filter levels, not three. Packet-type masks (cfg_axi_pkt_mask), then per-event-code masks. cfg_axi_err_select is RESERVED and routes nothing: this module uses it in exactly one expression, cfg_conflict_error = |(cfg_axi_pkt_mask & cfg_axi_err_select), so its only effect is to raise that flag on an overlap. Its own port comment says “unused in this context”.

A mask bit set means DROP. The sense catches people out; a mask of all ones emits nothing.

12.8 Related Modules

- [axi_monitor_base](#)

Used by:

- axi4_master_rd_mon
- axi4_master_wr_mon
- axi4_slave_rd_mon
- axi4_slave_wr_mon

See also:

- **Monitor Architecture:** docs/markdown/rtl-amba/overview.md
 - **Monitor Configuration Guide:** [Monitor Base Configuration](#)
 - **Packet Format Specification:**
docs/markdown/rtl-amba/includes/monitor_package_spec.md
-

12.9 Testing

Key test scenarios:

1. **Level 1 masking:**
 - Generate all packet types from the base
 - Set individual `cfg_axi_pkt_mask` bits and verify only the masked types are dropped
 2. **Level 3 masking:**
 - Set a per-type event mask (e.g. `cfg_axi_error_mask`) and verify only packets whose event code matches the set bit are dropped
 3. **Configuration conflict:**
 - Set the same bit in both `cfg_axi_pkt_mask` and `cfg_axi_err_select` and verify `cfg_conflict_error` asserts
 4. **Packet integrity:**
 - Verify passed packets are bit-identical to the base output (filter only drops, never mutates)
 - Verify a dropped packet is acked to the base so it never stalls
 - Exercise `ADD_PIPELINE_STAGE=1` and confirm one-cycle latency with correct backpressure
-

12.10 Navigation

- [Back to Shared Infrastructure Index](#)
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13 AXI Monitor Lite

Module: `axi_monitor_lite.sv` **Location:** `rtl/amba/monitor/axi_monitor_lite.sv`
Category: Monitor Infrastructure **Status:** Production Ready (amba/monitor-lite TASK-001, 2026-09-25)

13.1 Overview

`axi_monitor_lite` is the AXI transaction monitor rebuilt for gate count and timing: three quarters of what `axi_monitor_base` reports, for about a fifth of its LUTs. It sits behind the same `axi4/axi5/axil4/axil5_{master,slave}_{rd,wr}_mon` wrappers as their `_monlite` siblings (`axi4_slave_rd_monlite` and fifteen more), and emits the same 128-bit `monitor_packet_t`

with the same UNIT_ID/AGENT_ID on the same monbus handshake – the arbiter, the group, the tally and the host tooling cannot tell which monitor produced a packet.

What it reports:

- **Error** packets: a SLVERR/DECERR response, a data or response beat with no owning transaction (orphan), a burst whose LAST came early or never came, a write burst arriving ahead of a second address, and EVENT_DROPPED (see below) – each with the transaction’s ID in `channel_id` and its address in `event_data`, exactly as the full monitor packs them.
- **Completion** packets on a read’s last beat or a write’s B, with the address and, in `event_data[63:48]`, the latency in cycles from the command handshake.
- **Timeout** packets when a transaction makes no progress for `cfg_timeout_cnt` microseconds, naming the phase (CMD for a stalled command channel, DATA, RESP).
- **Threshold** packets when the number of outstanding transactions crosses `cfg_active_trans_threshold` upward.

What it does not do, by design: performance packets and windows (`axi_bus_meter` is the perf path), debug state-change packets, the address-range checker and report-time address filter, the ID-range filter, the latency threshold, three independent per-phase timers, and the `block_ready` admission stall. The lite never touches the traffic it watches.

13.2 Why it is small

The full monitor tracks every transaction as a ~285-bit record and re-derives everything about it every cycle: three CAM lookup ports, an oldest-first attribution done with age ranks that every entry recomputes on any free, a per-entry state machine that a reporter then scans, a second copy of the table in the reporter, three 16-bit timers per entry and an 8-deep packet FIFO. Measured on `bridge_1x2_rd_mon` (Artix-7 100T -1, default preset, 16 slots) that is 3,249 LUTs and 1,628 FFs per read monitor, 2,370 of the LUTs in the per-slot cones.

The lite keeps one 90-bit entry per transaction and computes each event at the handshake that causes it:

Mechanism	Full monitor	Lite
Allocation	three-port CAM with free-slot encode and hit suppression	free-slot priority encode on the command handshake
R/B attribution to an	ID match, oldest-first by	the head of that ID’s

Mechanism	Full monitor	Lite
entry	dense age ranks ($O(N^2)$ age matrix)	linked list (head, tail and a 3-bit next pointer per slot); a one-hot match, no age compare
W attribution (AXI4 W beats carry no ID)	age ranks over all live entries	an AW-order FIFO of slot indices: the head owns the beat
Timeout	three 16-bit timers per entry, all counting	one microsecond stamp per entry (a copy of a shared tick counter), one rotating subtract-and-compare
Latency	three 32-bit timestamps per entry	one 16-bit cycle stamp per entry, one shared subtractor at completion
Event detection	reporter scans the table with priority encoders	computed on the handshake, no scan; the cycle's events are registered, and the packet is picked and formatted from flops the next cycle
Output	8-deep FIFO of 85-bit entries plus a table copy	OUT_DEPTH-deep queue (default 4) of 66-bit entries (type, code, id, latency, address) in an unreset array; the constant fields are added at the output
Backpressure	admission stall (block_ready)	drop and count, reported as Error/EVENT_DROPPED
Address-range checker	axi_monitor_addr_check, N ranges, muxed onto the monbus	the same module, optional (N_ADDR_RANGES), its packets muxed onto the

Mechanism	Full monitor	Lite
		monbus with a presented-packet hold (Sean, 2026-09-26: STREAM's data ports use it)

An entry holds stamps, not counters: the only per-slot arithmetic is an 8-bit beat decrement, kept local so the beat that lands reads one flag (“was that the last expected”) instead of an 8-bit value through a mux. The two 16-bit subtractors (latency, timeout age) sit after the read mux and are shared by every slot, and one id/address mux serves every packet class.

Five builds got here, each measured in the same fixture. The first had a serial running-max loop for “oldest matching entry” (89 LUT levels). The second had a 16-bit incremter, a 16-bit subtractor and an 8-bit decremter in every slot (10,120 LUTs, 38 levels at sixteen slots). The third moved the arithmetic behind the read mux but ordered by an allocation sequence stamp, which both wraps for a long-lived entry and puts a subtract in front of every tournament compare, and it still formatted and queued the packet in the same cycle as the attribution (6,339 LUTs for the bridge, 23 levels). The fourth ordered by dense rank and registered the events before the pick (5,575 LUTs, 12 levels, 0.6 ns short); its hierarchy report put 346 of the lite’s 838 LUTs in a generic skid buffer, and its critical path through the rank tournament. The fifth, described here, replaced the tournament with per-ID linked lists and the skid with a four-entry unreset queue.

13.3 Measured

Same fixture, same flow: bridge_1x2_rd regenerated with mon_preset = "error_only" (the full monitor, 16 slots, error+timeout+compl+threshold cones) and with mon_preset = "lite" (this block, 8 slots), synthesized and routed out of context through projects/components/bridge/fpga/ on 2026-09-25, Vivado 2025.1. (Since 2026-09-26 the bridge generator builds the lite on every monitored port, so bridge_1x2_rd_mon regenerated today is a lite bridge too; the full-monitor column is the 2026-09-25 baseline.) Per-instance numbers are the hierarchical utilization of one read monitor inside cpu_rd_adapter.

	Full monitor	Lite	Lite / full
One read monitor, LUTs	3,249	677	21%
One read monitor, FFs	1,628	831	51%
Whole bridge (3 monitors + arbiter + group), LUTs,	12,625	5,339	42%

	Full monitor	Lite	Lite / full
Artix-7 100T - 1			
Whole bridge, FFs	7,433	5,270	71%
Bridge WNS at 10 ns, Artix- 7 100T -1	-0.303 ns	-0.160 ns	
Bridge WNS at 6.667 ns, Kintex-7 325T -2	+0.212 ns	+1.092 ns	
Bridge without monitors, LUTs / FFs	826 / 767		

The flops are the table: eight 90-bit entries, the event register and the four-entry queue. The full monitor's 16-slot CAM is 800 FFs on its own, so halving the slots does most of the flop saving; the LUT saving is the design.

Neither bridge meets 10 ns on the Artix-7, and in neither case is the path in a monitor: the full bridge's worst path is inside `monitor_trans_cam`, the lite bridge's inside `monbus_axil4_axil4_group` (the address-window compare chain `s1_beats_to_limit`, 11 CARRY4 in a row), a block both bridges share. No path through `axi_monitor_lite` is among the twenty worst; all of them have at least 0.85 ns of slack at 10 ns. The group's chain is filed as its own item.

13.4 Parameters

Parameter	Type	Default	Description
UNIT_ID	logic [7:0]	8'h09	Unit id in every packet
AGENT_ID	logic [15:0]	16'h0063	Agent id in every packet
MAX_TRANSACTIONS	int	8	Table slots. A command that

Parameter	Type	Default	Description
			finds none free is counted (refused_count), not tracked; its beats then report as orphans
ADDR_WIDTH	int	32	Address width stored per entry and reported
ID_WIDTH	int	8	Transaction ID width; 0 for AXI-Lite (one bit is kept)
IS_READ	bit	1	1 = AR/R monitor, 0 = AW/W/B monitor
IS_AXI	bit	1	0 = AXI-Lite: no ID compare
TS_WIDTH	int	16	Cycle stamp width: ordering and latency (latency saturates at 2 ¹⁶ cycles)
AGE_WIDTH	int	16	Microsecond age width for the timeout (cfg_timeout_cnt is 16 bits)
OUT_DEPTH	int	4	Output queue depth in 66-bit entries; a power of two
N_ADDR_RANGES	int	0	Address-range checker windows (axi_monitor_add

Parameter	Type	Default	Description
ADDR_RANGE_IS_ER ROR	logic [N-1:0]	'0	r_check, the full monitor's); 0 = not built, zero area Per range: 1 = a MISS is an Error/ADDR_RANGE packet, 0 = a HIT is an AddrMatch packet
CFI_*		as axi_monitor_time r	Frequency-invariant microsecond tick (counter_freq_invariant)

13.5 Ports

The command/data/response taps and the `cfg_*` names are the subset of `axi_monitor_base`'s that the lite implements; the wrappers connect the same signals to both. `cfg_timeout_cnt` is one threshold for every phase (the wrappers feed it their `cfg_timeout_cycles`, microseconds, 0 = never). `cfg_axi_pkt_mask[type]` drops a packet type at the source.

Outputs beyond the monbus: `active_count` (live entries), `busy`, `perf_completed_count` / `perf_error_count` (16-bit saturating, the wrapper's `transaction_count` / `error_count`), `dropped_count` (events lost to backpressure since reset or clear) and `refused_count` (commands that found no slot).

13.6 Functional Description

13.6.1 Attribution

A read beat belongs to the oldest live entry whose ID matches RID; AXI returns same-ID responses in issue order, so oldest-matching is exact and the 16-bit allocation stamp orders it. A write beat belongs to the entry at the head of the AW-order FIFO, because AXI4 W beats carry no ID and follow AW issue order; the FIFO pops on WLAST. A B belongs to the oldest matching entry that has finished its data phase.

A write burst that arrives before its AW (legal on a slave-side monitor) is buffered as a beat count and applied when the AW allocates; a second early burst before that AW is an Error/WRITE_BEFORE_ADDR and is dropped.

13.6.2 Errors, once per transaction

A response error, an early or missing LAST, an orphan beat: each is reported the cycle it happens. An entry that has reported an error reports no more for that transaction and yields no completion; it still frees on its last beat or B, so the table never leaks on error.

13.6.3 Timeout

Each entry carries a microsecond age cleared on every beat. One rotating pointer compares one entry per cycle against `cfg_timeout_cnt`, so a stuck transaction is reported within `MAX_TRANSACTION`s cycles of crossing – a rounding error against a microsecond tick – and reported once. A command channel that holds VALID without READY for the same threshold reports Timeout/CMD.

13.6.4 Drop and count

Events go into the output queue. If the queue is full when an event fires, or two events fire in one cycle (error > timeout > completion > threshold picks the one that goes), the rest are dropped and counted. The next time the queue has room and nothing else wants it, one Error/EVENT_DROPPED packet carries the count and the counter restarts. The consumer therefore always knows how many events it did not see.

13.7 Verification

- `val/amba/monitor-lite/test_axi_monitor_lite.py` through `axi4_slave_{rd,wr}_monlite`: exact packets for singles and bursts (address, id, non-zero latency), one RESP_SLVERR and no completion for an out-of-range access, one timeout naming the stuck phase, one active-count threshold, and a held monbus that drops events and then reports the count. 8 cells at full.
- `val/amba/monitor-lite/test_<wrapper>_monlite.py`, sixteen tests derived from the `_mon` wrapper tests on the existing monitor TB classes: the same scenarios with the `_monlite` DUT. 105 cells at full with the above.
- `formal/amba/axi_monitor_lite`: under unconstrained taps, `active_count` never exceeds the table, `clear` empties it, and an offered packet is held unchanged until taken; covers reach a completion, an error, a timeout, a threshold and a full table (the proof is not vacuous).
- The bridge's generated monitor stress tests run unchanged on `bridge_1x2_rd_lite_mon` (`mon_preset = "lite"`).

13.8 Related

`axi_monitor_base` (the full monitor), the sixteen `*_monlite` wrappers (one per `*_mon`, e.g. `axi4_slave_rd_monlite`), `monbus_arbiter` and `monbus_group` (unchanged consumers), `vault/Tasks/amba/amba/monitor-lite TASK-001` (the review that sized this).

14 AXI Monitor Reporter — Completion Cone

Module: `axi_monitor_reporter_compl.sv` **Location:** `rtl/amba/monitor/` **Status:** Production Ready

14.1 Overview

The AXI Monitor Reporter Completion Cone is the completion-packet detection sub-block of `axi_monitor_reporter`. It scans the outstanding-transaction table for slots that have reached the `TRANS_COMPLETE` state but have not yet been reported, priority-encodes the first such slot, and emits the fields for a `PktTypeCompletion` MonBus packet. It was split out of the monolithic reporter so integrators who do not need completion packets can drop it with `ENABLE_COMPL_LOGIC=0` and pay zero LUT cost for the per-slot scan and priority encoder. The block is purely combinational.

Key features:

- Scans the shared transaction table for unreported `TRANS_COMPLETE` slots
- First-match priority encoder selects one slot per cycle
- Emits `PktTypeCompletion` packet fields with event code `EVT_TRANS_COMPLETE`
- Runtime-gated by `cfg_compl_enable`; compile-time removable via the parent's `ENABLE_COMPL_LOGIC`
- Reports the selected slot index (`sel_idx`) back to the top reporter for `event_reported` feedback
- Pure combinational — all state (table, `event_reported`) lives in the top reporter

The transaction manager tracks every outstanding AXI transaction and marks a slot `TRANS_COMPLETE` when it finishes cleanly. Someone must turn those completions into MonBus packets exactly once each. This cone does the detect-and-select half of that job: it finds the completed-but-unreported slots, picks the first, and hands the packet fields plus the chosen slot index up to the top reporter, which pushes the packet into the shared MonBus FIFO and sets the slot's `event_reported` bit so it is not emitted again. Factoring it into its own module means the

(non-trivial) MAX_TRANSACTIONS-wide scan and encoder synthesizes only when completion reporting is actually wanted.

Use cases:

- Emitting one completion packet per successfully-finished AXI transaction
- Completion-tracking test/verification configurations
- Transaction-throughput accounting on the MonBus

Key benefit: isolates the completion scan/encoder so it can be compiled out (ENABLE_COMPL_LOGIC=0) for zero area when completion packets are not needed.

14.2 Parameters

Parameter	Type	Default	Description
MAX_TRANSACTIONS	int	16	Depth of the shared transaction table scanned
IDX_W	int	$\lceil \log_2(\text{MAX_TRANSACTIONS}) \rceil$	Width of the selected-slot index

14.3 Ports

14.3.1 Inputs (shared monitor state)

Port	Direction	Width	Description
trans_table	Input	bus_transaction_t x MAX_TRANSACTIONS	The outstanding-transaction table
event_reported	Input	MAX_TRANSACTIONS	Per-slot “already reported” mask (from the top reporter)
cfg_compl_enable	Input	1	Runtime enable for completion reporting

14.3.2 Outputs (packet fields to the top reporter)

Port	Direction	Width	Description
pkt_valid	Output	1	A completed-unreported slot was found this cycle
pkt_type	Output	4	Packet type = PktTypeCompletion
pkt_event_code	Output	8	Event code = EVT_TRANS_COMPLETE
pkt_channel	Output	9	Channel id {3'b0, slot.channel[5:0]}
pkt_data	Output	64	Slot address, zero-padded to 64 bits
sel_idx	Output	IDX_W	Index of the selected slot (for event_reported feedback)

14.4 Functional Description

14.4.1 Completion Scan

For every slot in `trans_table`, the slot qualifies as a completion event when it is `valid`, its `event_reported` bit is clear, its `state == TRANS_COMPLETE`, and `cfg_compl_enable` is set. All qualifying slots are collected into a one-hot-per-bit `w_events` vector.

14.4.2 First-Match Selection

A second loop scans `w_events` from index 0 upward and latches the first set bit into `w_sel`, asserting `w_has_event`. The `WIDTHTRUNC` lint is locally waived where the loop index is narrowed to `IDX_W`. This gives strict low-index-first priority so exactly one completion is emitted per cycle even when several slots complete together; the rest are picked up on subsequent cycles once the winner is marked reported.

14.4.3 Emitted Fields

- `pkt_valid = w_has_event`
- `sel_idx = w_sel` (tells the top reporter which slot to mark reported)
- `pkt_type = PktTypeCompletion`
- `pkt_event_code = EVT_TRANS_COMPLETE`
- `pkt_channel = {3'b0, trans_table[w_sel].channel[5:0]}`
- `pkt_data = pad_address(trans_table[w_sel].addr)` — the 32-bit slot address zero-extended to 64 bits

14.4.4 Ownership of State

The block holds no state of its own. The transaction table and the `event_reported` feedback mask both live in the top reporter (`axi_monitor_reporter`), which consumes `sel_idx` to set the reported bit after the packet is accepted. Keeping this cone combinational lets several such cones (completion, error, timeout, ...) share the same table snapshot each cycle.

14.5 Timing Characteristics

This module is **purely combinational** – it contains no `always_ff` and no latch, so it holds no state and adds no clock cycles. Its outputs settle a propagation delay after its inputs, and it introduces no latency into a pipeline that instantiates it.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

14.6 Usage Examples

*// Instantiated inside axi_monitor_reporter, gated by
ENABLE_COMPL_LOGIC.*

```
axi_monitor_reporter_compl #(
    .MAX_TRANSACTIONS (MAX_TRANSACTIONS)
) u_compl (
    .trans_table      (trans_table),
    .event_reported   (event_reported),
    .cfg_compl_enable(cfg_compl_enable),

    .pkt_valid        (compl_valid),
    .pkt_type          (compl_type),
    .pkt_event_code    (compl_event_code),
    .pkt_channel       (compl_channel),
```

```
        .pkt_data      (compl_data),  
        .sel_idx       (compl_sel_idx)    // top reporter sets  
event_reported[compl_sel_idx]  
);
```

14.7 Design Notes

14.7.1 Compile-Out Path

The parent instantiates this cone under `ENABLE_COMPL_LOGIC`. Setting it to 0 removes the `MAX_TRANSACTIONS`-wide qualifier scan and priority encoder entirely, so a monitor build that never reports completions pays no LUT cost for them. `cfg_compl_enable` is the softer runtime mask when the logic is present.

14.7.2 Combinational by Design

All state lives upstream; this block is a pure combinational detect-and-select cone. Multiple packet cones can therefore examine the same table each cycle without duplicating storage, and the top reporter arbitrates which one wins the MonBus this cycle.

14.7.3 One Packet Per Cycle

The strict first-match encoder emits at most one completion per cycle. Simultaneous completions are serialized across cycles as each winner is marked reported.

14.8 Related Modules

Used by:

- **axi_monitor_reporter.sv** - Top reporter that pushes the packet into the MonBus FIFO and drives `event_reported`

Uses:

- **monitor_common_pkg** - `PktTypeCompletion`, `TRANS_COMPLETE`, `bus_transaction_t`
- **monitor_amba4_pkg** - `EVT_TRANS_COMPLETE` event code

See also:

- **axi_monitor_reporter_error.sv** - Error/orphan detection cone (sibling sub-block)

- **axi_monitor_reporter_debug.sv** - State-change debug emitter (sibling sub-block)
 - **axi_monitor_base.sv** - The monitor scaffold that houses trans_mgr + reporter
-

14.9 Testing

No dedicated testbench for this module. It has no `val/**/test_axi_monitor_reporter_compl.py`. It is exercised indirectly, through the tests of modules that instantiate it (directly or further up):

- `axi4_master_rd_mon` – `val/**/test_axi4_master_rd_mon.py`
- `axi4_master_wr_mon` – `val/**/test_axi4_master_wr_mon.py`
- `axi4_slave_rd_mon` – `val/**/test_axi4_slave_rd_mon.py`
- `axi4_slave_wr_mon` – `val/**/test_axi4_slave_wr_mon.py`
- `axi5_master_rd_mon` – `val/**/test_axi5_master_rd_mon.py`

Indirect coverage exercises this module only in the configurations those parents elaborate. A parameter or mode no parent uses is untested.

Treat any behaviour described on this page as unverified by simulation.

14.10 References

14.10.1 Source Code

- RTL: `rtl/amba/monitor/axi_monitor_reporter_compl.sv`

14.10.2 Documentation

- Packet format: `docs/markdown/rtl-amba/includes/monitor_package_spec.md`
 - Monitor base: `docs/markdown/rtl-amba/monitor/axi_monitor_base.md`
 - Known issues: `rtl/amba/KNOWN_ISSUES/axi_monitor_reporter.md`
-

Last Updated: 2026-07-15

14.11 Navigation

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15 AXI Monitor Reporter — Debug State-Change Emitter

Module: `axi_monitor_reporter_debug.sv` **Location:** `rtl/amba/monitor/` **Status:** Production Ready

15.1 Overview

The AXI Monitor Reporter Debug Emitter is the state-change (trace) sub-block of `axi_monitor_reporter` — the sixth cone alongside error, timeout, completion, threshold, and perf. It gives integrators (and the compression-analysis flow) a packet stream that mirrors the live transaction-table FSM: one `PktTypeDebug` packet per (slot, state-change) event. It holds a per-slot previous-state vector, compares it against the live state each cycle, priority-encodes the first slot that transitioned, and emits a packet carrying that slot's address plus the `(prev_state, new_state)` tuple. Unlike the combinational completion/error cones, this block is sequential (it flops the previous state).

Key features:

- One `PktTypeDebug` packet per per-slot state transition, mirroring the `trans_table` FSM
- Per-slot 3-bit `prev_state` vector, `MAX_TRANSACTIONS` deep, flopped every cycle
- First-match priority encoder selects one changed slot per cycle
- Packs `(prev_state, new_state)` in the high bits of the 64-bit data field, address in the low bits
- Direct-inject path with an `output_busy` gate (bypasses the FIFO, like threshold/perf)
- Runtime-gated by `cfg_debug_enable`; compile-time removable via the parent's `ENABLE_DEBUG_LOGIC`

Deep debugging and compression research both benefit from a packet stream that reflects the internal monitor FSM rather than just terminal events. This block produces exactly that: whenever any transaction slot changes state, it emits a debug packet naming the slot's address and the state edge it took. Slot-free transitions are handled cleanly — when a slot is freed, its `prev_state` resets to `TRANS_IDLE` so the next allocation produces a fresh `IDLE -> ADDR_PHASE`

edge rather than a spurious IDLE -> IDLE. The compression analysis can use either the state tuple or the address as its dictionary key, whichever carries more entropy.

Use cases:

- FSM-level trace of the transaction table for deep debugging
- Feeding the MonBus compression-analysis flow with a rich event stream
- Correlating state transitions with observed protocol behavior during bring-up

Key benefit: surfaces the live per-slot FSM as MonBus packets, giving debuggers and the compressor a per-transition trace that terminal-event cones cannot provide.

15.2 Parameters

Parameter	Type	Default	Description
MAX_TRANSACTION	int	16	Depth of the transaction table monitored
IDX_W	int	$\lceil \log_2(\text{MAX_TRANSACTIONS}) \rceil$	Width of the internal selected-slot index

15.3 Ports

15.3.1 Clock and Reset

Port	Direction	Width	Description
aclk	Input	1	Clock
aresetn	Input	1	Active-low asynchronous reset

15.3.2 Inputs (shared monitor state)

Port	Direction	Width	Description
trans_table	Input	bus_transaction_tx MAX_TRANSACTION	The outstanding-transaction table

Port	Direction	Width	Description
cfg_debug_enable	Input	1	Runtime enable / mask for debug packet emission

15.3.3 Direct-Inject Handshake

Port	Direction	Width	Description
output_busy	Input	1	High when the shared MonBus output is occupied; gates pkt_valid
pkt_taken	Input	1	Pulsed when this block's packet is accepted (currently unused — see Design Notes)

15.3.4 Outputs (packet fields)

Port	Direction	Width	Description
pkt_valid	Output	1	A state change was found and the output bus is free
pkt_type	Output	4	Packet type = PktTypeDebug
pkt_event_code	Output	8	Event code = AXI_DEBUG_STATE_CHANGE
pkt_channel	Output	9	Channel id {3'b0, slot.channel[5:0]}
pkt_data	Output	64	{prev_state[2:0], new_state[2:0], 26'h0, addr[31:0]}

15.4 Functional Description

15.4.1 Change Detection

When `cfg_debug_enable` is set, the block scans every slot: a slot is flagged changed when it is valid and its live state differs from the captured `r_prev_state[idx]`. All changed slots collect into `w_changed`, and a first-match priority encoder selects the lowest-index changed slot into `w_sel` (asserting `w_has_event`). When `cfg_debug_enable` is low, no slot is flagged, so no packets are produced.

15.4.2 Previous-State Flop

Each cycle `r_prev_state[idx]` snapshots the slot's live state — **except** when the slot is not valid, in which case it resets to `TRANS_IDLE`. This is the mechanism that makes freed slots behave: an emptied slot returns to IDLE, so its next allocation registers as a real `IDLE -> ADDR_PHASE` transition rather than an aliased `IDLE -> IDLE` non-event.

15.4.3 Packet Assembly

When a change is found and the output bus is free (`w_has_event && !output_busy`):

- `pkt_valid = 1`
- `pkt_type = PktTypeDebug`
- `pkt_event_code = AXI_DEBUG_STATE_CHANGE`
- `pkt_channel = {3'b0, trans_table[w_sel].channel[5:0]}`
- `pkt_data = {3'(r_prev_state[w_sel]), 3'(trans_table[w_sel].state), 26'h0, trans_table[w_sel].addr}`

The (prev, new) state tuple sits in the top six bits and the 32-bit address in the low bits, leaving a 26-bit zero gap between them. The compression analysis can key on either field.

15.4.4 Direct-Inject and output_busy

Like the threshold and perf cones, debug packets bypass the shared FIFO and inject directly onto the MonBus, so the block must observe `output_busy` to avoid overwriting an in-flight packet — `pkt_valid` is gated by `!output_busy`.

15.5 Timing Characteristics

This module is **sequential**: it contains clocked logic (via `always_ff` or the repository's `ALWAYS_FF_RST` macro) and therefore holds state. Outputs driven from those blocks are registered and appear one clock after the inputs that produced them.

Per-path cycle counts are not enumerated here; read the block that drives the signal you care about. No synthesis frequency or area figures are quoted – none have been measured against a target device.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

15.6 Usage Examples

```
// Instantiated inside axi_monitor_reporter, gated by
ENABLE_DEBUG_LOGIC.
axi_monitor_reporter_debug #(
    .MAX_TRANSACTIONS (MAX_TRANSACTIONS)
) u_debug (
    .aclk                (aclk),
    .aresetn             (aresetn),

    .trans_table         (trans_table),
    .cfg_debug_enable    (cfg_debug_enable),

    // Direct-inject arbitration with the top reporter
    .output_busy         (monbus_output_busy),
    .pkt_taken           (debug_pkt_taken), // currently unused inside
the block

    .pkt_valid           (dbg_valid),
    .pkt_type            (dbg_type),
    .pkt_event_code      (dbg_event_code),
    .pkt_channel         (dbg_channel),
    .pkt_data            (dbg_data)
);
```

Note: enabling debug packets alongside completions/perf can congest the MonBus. See the AXI Monitor Configuration Guide before turning `cfg_debug_enable` on in a real run.

15.7 Design Notes

15.7.1 pkt_taken Is Intentionally Unused

The `r_prev_state` flop advances every cycle regardless of whether the emitted packet was accepted, so a missed transition simply gets aliased into the next change. `pkt_taken` is therefore not consumed today — it is kept on the port list (with an explicit `UNUSED` lint waiver) for symmetry with the threshold/perf cones and for future hooks such as backpressure on debug bursts.

15.7.2 Sequential, Not Combinational

Unlike the completion and error cones (pure combinational), this block flops `r_prev_state`, which is inherently required to detect edges. It therefore takes `aclk/aresetn`.

15.7.3 Compile-Out Path

Instantiated under `ENABLE_DEBUG_LOGIC` in the parent; setting it to 0 drops the block entirely. `cfg_debug_enable` is the runtime mask when the logic is present — emissions are gated but the logic still synthesizes.

15.7.4 State Tuple Encoding

The 3-bit state codes follow `transaction_state_t` (`TRANS_IDLE=0`, `ADDR_PHASE=1`, `DATA_PHASE=2`, `COMPLETE=3`, `ERROR=4`, `ORPHANED=5`), packed as `{prev, new}` in `pkt_data[63:58]`.

15.8 Related Modules

Used by:

- **axi_monitor_reporter.sv** - Top reporter that arbitrates the direct-inject MonBus path

Uses:

- **monitor_common_pkg** - `PktTypeDebug`, `transaction_state_t`, `TRANS_IDLE`, `bus_transaction_t`
- **monitor_amba4_pkg** - `AXI_DEBUG_STATE_CHANGE` event code
- **reset_defs.svh** - `ALWAYS_FF_RST` / `RST_ASSERTED` reset macros

See also:

- **axi_monitor_reporter_compl.sv** - Completion detection cone (sibling sub-block)
 - **axi_monitor_reporter_error.sv** - Error/orphan detection cone (sibling sub-block)
 - **axi_monitor_base.sv** - The monitor scaffold that houses trans_mgr + reporter
-

15.9 Testing

No dedicated testbench for this module. It has no `val/**/test_axi_monitor_reporter_debug.py`. It is exercised indirectly, through the tests of modules that instantiate it (directly or further up):

- `axi4_master_rd_mon` – `val/**/test_axi4_master_rd_mon.py`
- `axi4_master_wr_mon` – `val/**/test_axi4_master_wr_mon.py`
- `axi4_slave_rd_mon` – `val/**/test_axi4_slave_rd_mon.py`
- `axi4_slave_wr_mon` – `val/**/test_axi4_slave_wr_mon.py`
- `axi5_master_rd_mon` – `val/**/test_axi5_master_rd_mon.py`

Indirect coverage exercises this module only in the configurations those parents elaborate. A parameter or mode no parent uses is untested.

Treat any behaviour described on this page as unverified by simulation.

15.10 References

15.10.1 Source Code

- RTL: `rtl/amba/monitor/axi_monitor_reporter_debug.sv`

15.10.2 Documentation

- Packet format: `docs/markdown/rtl-amba/includes/monitor_package_spec.md`
 - Monitor base: `docs/markdown/rtl-amba/monitor/axi_monitor_base.md`
 - Configuration: `docs/user-guides/AXI_Monitor_Configuration_Guide.md`
-

Last Updated: 2026-07-15

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16 AXI Monitor Reporter — Error Cone

Module: `axi_monitor_reporter_error.sv` **Location:** `rtl/amba/monitor/` **Status:** Production Ready

16.1 Overview

The AXI Monitor Reporter Error Cone is the error-packet detection sub-block of `axi_monitor_reporter`. It scans the outstanding-transaction table for unreported transactions in the `TRANS_ERROR` (genuine error, not a timeout) or `TRANS_ORPHANED` state, priority-encodes the first match, and emits the fields for a `PktTypeError` MonBus packet. It was split out of the monolithic reporter so integrators can drop it with `ENABLE_ERROR_LOGIC=0` and pay zero LUT cost for the per-slot scan and priority encoder. The block is purely combinational.

error is a **fault class** ([taxonomy](#)): in correct operation it never fires. **An error, by definition, hangs the system** — a transaction that returns `SLVERR/DECERR` or is never answered does not retire, so exercising this cone means *deliberately injecting a fault* (a slave forced to return a bad response), and the traffic that provoked it is expected to wedge. That is exactly what this cone exists to catch: it emits the `PktTypeError` packet as the hang happens. (For the address-range/allowlist flavor of error injection, which lives in `axi_monitor_addr_check` (generate-gated by `N_ADDR_RANGES > 0`, so it is built only when ranges are configured) and does *not* require this cone, see that page.)

Key features:

- Scans for unreported `TRANS_ERROR` (non-timeout) and `TRANS_ORPHANED` slots
- Uses `timeout_detected` to keep genuine errors distinct from timeouts (which the timeout cone claims)
- First-match priority encoder selects one slot per cycle
- Emits `PktTypeError` with the slot's own `event_code.raw_code`
- Runtime-gated by `cfg_error_enable`; compile-time removable via the parent's `ENABLE_ERROR_LOGIC`
- Pure combinational — table, `event_reported`, and threshold state stay in the top reporter

The transaction manager marks a slot `TRANS_ERROR` when a transaction returns `SLVERR/DECERR` or violates protocol, and `TRANS_ORPHANED` when data or a response arrives with no matching command. These must become MonBus error packets exactly once each. This cone performs the detect-and-select half: it finds the error/orphan slots that have not yet been reported, excludes any slot already flagged as a timeout so the timeout cone can own those without aliasing, picks the first match, and hands the packet fields plus the chosen slot index to the top reporter for FIFO push and `event_reported` feedback. Splitting it out lets the wide scan and encoder be compiled away when error reporting is not required.

Use cases:

- Emitting `SLVERR / DECERR / protocol-violation` error packets
- Reporting orphaned data or response beats (no matching command)
- Functional-verification configurations that must catch bus errors

Key benefit: isolates the error/orphan scan/encoder for `ENABLE_ERROR_LOGIC=0` compile-out, and cleanly separates genuine errors from timeouts via the `timeout_detected` mask.

16.2 Parameters

Parameter	Type	Default	Description
<code>MAX_TRANSACTION</code>	int	16	Depth of the shared transaction table scanned
<code>IDX_W</code>	int	$\$clog2(MAX_TRANSACTIONS)$	Width of the selected-slot index

16.3 Ports

16.3.1 Inputs (shared monitor state)

Port	Direction	Width	Description
<code>trans_table</code>	Input	<code>bus_transaction_tx</code> <code>MAX_TRANSACTION</code>	The outstanding-transaction table
<code>event_reported</code>	Input	<code>MAX_TRANSACTION</code>	Per-slot “already reported” mask

Port	Direction	Width	Description
			(from the top reporter)
timeout_detected	Input	MAX_TRANSACTION	Per-slot timeout mask; excludes timeout slots from the error scan
cfg_error_enable	Input	1	Runtime enable for error reporting

16.3.2 Outputs (packet fields to the top reporter)

Port	Direction	Width	Description
pkt_valid	Output	1	An unreported error/orphan slot was found this cycle
pkt_type	Output	4	Packet type = PktTypeError
pkt_event_code	Output	8	Event code = selected slot's event_code.raw_code
pkt_channel	Output	9	Channel id {3'b0, slot.channel[5:0]}
pkt_data	Output	64	Slot address, zero-padded to 64 bits
sel_idx	Output	IDX_W	Index of the selected slot (for event_reported feedback)

16.4 Functional Description

16.4.1 Error / Orphan Scan

For every slot, the slot qualifies as an error event when it is valid, its `event_reported` bit is clear, `cfg_error_enable` is set, and **either**:

- `state == TRANS_ERROR` **and** its `timeout_detected` bit is clear (a genuine error, not a timeout), **or**
- `state == TRANS_ORPHANED` (data/response with no matching command).

Qualifying slots collect into `w_events`. Consulting `timeout_detected` is what keeps this cone from aliasing timeouts: a transaction that errored because it timed out is left for the timeout sub-block to report, so the same slot is never emitted twice under two packet types.

16.4.2 First-Match Selection

A second loop takes the first set bit of `w_events` into `w_sel` and asserts `w_has_event` (with the local `WIDTHTRUNC` lint waiver on the index narrowing). Strict low-index-first priority means exactly one error is emitted per cycle; additional errors are serialized on later cycles as each winner is marked reported upstream.

16.4.3 Emitted Fields

- `pkt_valid = w_has_event`
- `sel_idx = w_sel`
- `pkt_type = PktTypeError`
- `pkt_event_code = trans_table[w_sel].event_code.raw_code` — the specific AXI error code captured by the trans manager (e.g. `AXI_ERR_RESP_SLVERR`, `AXI_ERR_RESP_DECERR`, `AXI_ERR_DATA_ORPHAN`, `AXI_ERR_RESP_ORPHAN`)
- `pkt_channel = {3'b0, trans_table[w_sel].channel[5:0]}`
- `pkt_data = pad_address(trans_table[w_sel].addr)`

Unlike the completion and debug cones, the error cone forwards the transaction's own captured raw error code rather than a fixed constant, so the packet carries the precise error kind.

16.4.4 Ownership of State

The block is stateless. The transaction table, the `event_reported` feedback, the `timeout_detected` mask, and the threshold flags all live in the top reporter, which consumes `sel_idx` to set the reported bit after acceptance. Keeping the cone combinational lets it share the table snapshot with the sibling cones each cycle.

16.5 Timing Characteristics

This module is **purely combinational** – it contains no `always_ff` and no latch, so it holds no state and adds no clock cycles. Its outputs settle a propagation delay after its inputs, and it introduces no latency into a pipeline that instantiates it.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

16.6 Usage Examples

```
// Instantiated inside axi_monitor_reporter, gated by  
ENABLE_ERROR_LOGIC.  
axi_monitor_reporter_error #(  
    .MAX_TRANSACTIONS (MAX_TRANSACTIONS)  
) u_error (  
    .trans_table      (trans_table),  
    .event_reported   (event_reported),  
    .timeout_detected (timeout_detected), // keeps timeouts out of  
the error scan  
    .cfg_error_enable (cfg_error_enable),  
  
    .pkt_valid        (err_valid),  
    .pkt_type         (err_type),  
    .pkt_event_code   (err_event_code),  
    .pkt_channel      (err_channel),  
    .pkt_data         (err_data),  
    .sel_idx          (err_sel_idx)      // top reporter sets  
event_reported[err_sel_idx]  
);
```

16.7 Design Notes

16.7.1 Timeout De-Aliasing

The `timeout_detected` input is the mechanism that prevents a timed-out transaction (which the trans manager may also mark `TRANS_ERROR`) from being reported by both this cone and the timeout cone. Only errors with a clear timeout bit are claimed here.

16.7.2 Compile-Out Path

Instantiated under `ENABLE_ERROR_LOGIC` in the parent; setting it to 0 removes the wide scan and encoder for zero LUT cost. `cfg_error_enable` is the runtime mask when the logic is present.

16.7.3 Raw Event Code Forwarding

Because `pkt_event_code` comes from the slot's `event_code.raw_code`, this cone reports the exact error type (SLVERR, DECERR, orphan variants, protocol errors, ...) rather than a single generic error code.

16.8 Related Modules

Used by:

- **axi_monitor_reporter.sv** - Top reporter that pushes the packet into the MonBus FIFO and drives `event_reported`

Uses:

- **monitor_common_pkg** - `PktTypeError`, `TRANS_ERROR`, `TRANS_ORPHANED`, `bus_transaction_t`
- **monitor_amba4_pkg** - AXI error event codes (via the slot's `event_code`)

See also:

- **axi_monitor_reporter_compl.sv** - Completion detection cone (sibling sub-block)
 - **axi_monitor_reporter_debug.sv** - State-change debug emitter (sibling sub-block)
 - **axi_monitor_base.sv** - The monitor scaffold that houses `trans_mgr` + `reporter`
-

16.9 Testing

No dedicated testbench for this module. It has no `val/**/test_axi_monitor_reporter_error.py`. It is exercised indirectly, through the tests of modules that instantiate it (directly or further up):

- `axi4_master_rd_mon` – `val/**/test_axi4_master_rd_mon.py`
- `axi4_master_wr_mon` – `val/**/test_axi4_master_wr_mon.py`
- `axi4_slave_rd_mon` – `val/**/test_axi4_slave_rd_mon.py`
- `axi4_slave_wr_mon` – `val/**/test_axi4_slave_wr_mon.py`
- `axi5_master_rd_mon` – `val/**/test_axi5_master_rd_mon.py`

Indirect coverage exercises this module only in the configurations those parents elaborate. A parameter or mode no parent uses is untested.

Treat any behaviour described on this page as unverified by simulation.

16.10 References

16.10.1 Source Code

- RTL: rtl/amba/monitor/axi_monitor_reporter_error.sv

16.10.2 Documentation

- Packet format: docs/markdown/rtl-amba/includes/monitor_package_spec.md
 - Monitor base: docs/markdown/rtl-amba/monitor/axi_monitor_base.md
 - Known issues: rtl/amba/KNOWN_ISSUES/axi_monitor_reporter.md
-

Last Updated: 2026-07-15

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17 AXI Monitor Reporter (Dispatcher + 6 Sub-blocks)

Modules: - axi_monitor_reporter.sv — thin top-level dispatcher -
axi_monitor_reporter_error.sv — error-packet detection (combinational) -
axi_monitor_reporter_timeout.sv — timeout-packet detection (combinational) -
axi_monitor_reporter_compl.sv — completion-packet detection (combinational) -
axi_monitor_reporter_threshold.sv — threshold-packet detection (16 latency flops + edge flags) - axi_monitor_reporter_perf.sv — legacy perf-rollup packets (completion/error lifetime counters + 5-state FSM) - axi_monitor_reporter_debug.sv — debug-packet generation

Location: rtl/amba/monitor/ **Category:** Core Infrastructure **Status:** Production Ready
(refactored to sub-blocks 2026-06-06)

17.1 Overview

The `axi_monitor_reporter` family generates Monitor-bus packets. The top-level `axi_monitor_reporter.sv` is a **thin dispatcher** that multiplexes one packet stream out of (up to) six packet-type-specific sub-blocks onto the monbus output — three classes through the shared FIFO, three straight into the 128-bit output register (see below). It also reports each emitted event back to `axi_monitor_trans_mgr` so the transaction table can release entries.

Per-packet-type detection lives in the six sub-blocks. The dispatcher gates each sub-block via an `ENABLE_*_LOGIC` parameter, so integrators can drop any combination at elaboration time and pay zero LUT/FF cost for the unused detection cones.

The bridge case (`ENABLE_ERROR_LOGIC=1`, all others 0) synthesises away the timeout, completion, threshold, perf and debug detection cones entirely.

Key features:

- 128-bit standardized `monitor_packet_t` formatting (via package helpers)
- Packet type multiplexing across six sub-blocks (error / timeout / compl / threshold / perf / debug) with per-type elaboration gates
- Protocol identification (AXI4, AXI5, APB, AXIS, CORE)
- Event code and data field population from the active sub-block
- Unit ID and Agent ID insertion (caller-configured constants)
- Packet valid/ready handshaking
- Internal monbus FIFO for packet queuing
- Event-reported feedback to `axi_monitor_trans_mgr` (closes the transaction-table loop documented in FIX-001)

What the dispatcher actually provides:

1. **Per-type detection gating** — only the enabled sub-blocks consume LUT/FF, so a single-purpose deployment (e.g. error-only on the bridge) is lean.
2. **Packet formatting** — pulls type / protocol / event code / event data / channel id from the active sub-block and packs into the 128-bit `monitor_packet_t`.
3. **Routing IDs** — inserts the static `UNIT_ID` / `AGENT_ID` so the downstream arbiter and the host can route packets back to their source.
4. **Queuing** — buffers up to `INTR_FIFO_DEPTH` **error, timeout and completion** packets when the downstream monbus is back-pressured. Threshold, perf and debug packets bypass the queue entirely (see above).

Note that this FIFO's fill level has **no** path to the monitor's block_ready flow control — that is driven purely by transaction-table occupancy.

5. **Event acknowledge** — drives event_reported_* back to axi_monitor_trans_mgr so the transaction table can release its entry once the packet is accepted into the FIFO (the queued classes are the ones that mark entries reported).
6. **Auto-retire** — releases terminal entries whose packet class cannot report (see below), so disabled classes never leak table slots.

17.2 Parameters

Parameter	Type	Default	Description
MAX_TRANSACTIONS	int	16	Transaction-table size shared with axi_monitor_trans_mgr
ADDR_WIDTH	int	32	Address width carried in event_data
UNIT_ID	logic [7:0]	8'h09	8-bit unit identifier (static)
AGENT_ID	logic [15:0]	16'h0063	16-bit agent identifier (static)
IS_READ	bit	1	1 = read-channel monitor, 0 = write-channel
INTR_FIFO_DEPTH	int	8	Reporter packet FIFO depth
ENABLE_ERROR_LOGIC	bit	1	Instantiate the error detection sub-block
ENABLE_TIMEOUT_LOGIC	bit	1	Instantiate the timeout detection sub-block
ENABLE_COMPL_LOGIC	bit	1	Instantiate the completion

Parameter	Type	Default	Description
ENABLE_THRESHOLD_LOGIC	bit	1	detection sub-block Instantiate the threshold detection sub-block
ENABLE_PERF_LOGIC	bit	(alias)	Instantiate the perf sub-block; defaults to ENABLE_PERF_PACKETS for legacy compat
ENABLE_DEBUG_LOGIC	bit	0	Instantiate the debug sub-block

17.3 Ports

All ports, from rtl/amba/monitor/axi_monitor_reporter.sv.

Clock and reset

Port	Dir	Width	Description
aclk	In	1	Clock
aresetn	In	1	Active-low asynchronous reset

Transaction-table inputs

Port	Dir	Width	Description
trans_table	In	bus_transaction_t[MAX_TRANSACTIONS]	The transaction table, read as a whole. This is the port behind the reporter's second full copy of the table

Port	Dir	Width	Description
			(r_trans_table_local) – the reason a monitored interface costs twice the table area
timeout_detected	In	MAX_TRANSACTIONS	Per-slot timeout flags from axi_monitor_timeout
filtered_mask	In	MAX_TRANSACTIONS	Per-slot mask of entries the ID/address filters have excluded

Runtime configuration

Port	Dir	Width	Description
cfg_error_enable	In	1	Emit error packets
cfg_compl_enable	In	1	Emit completion packets
cfg_threshold_enable	In	1	Emit threshold packets
cfg_timeout_enable	In	1	Emit timeout packets
cfg_perf_enable	In	1	Emit performance packets
cfg_debug_enable	In	1	Runtime mask for the debug emitter; live only when ENABLE_DEBUG_LOGIC=1
active_trans_threshold	In	16	Active-transaction count that trips a threshold packet

Port	Dir	Width	Description
latency_threshold	In	32	Latency that trips a threshold packet

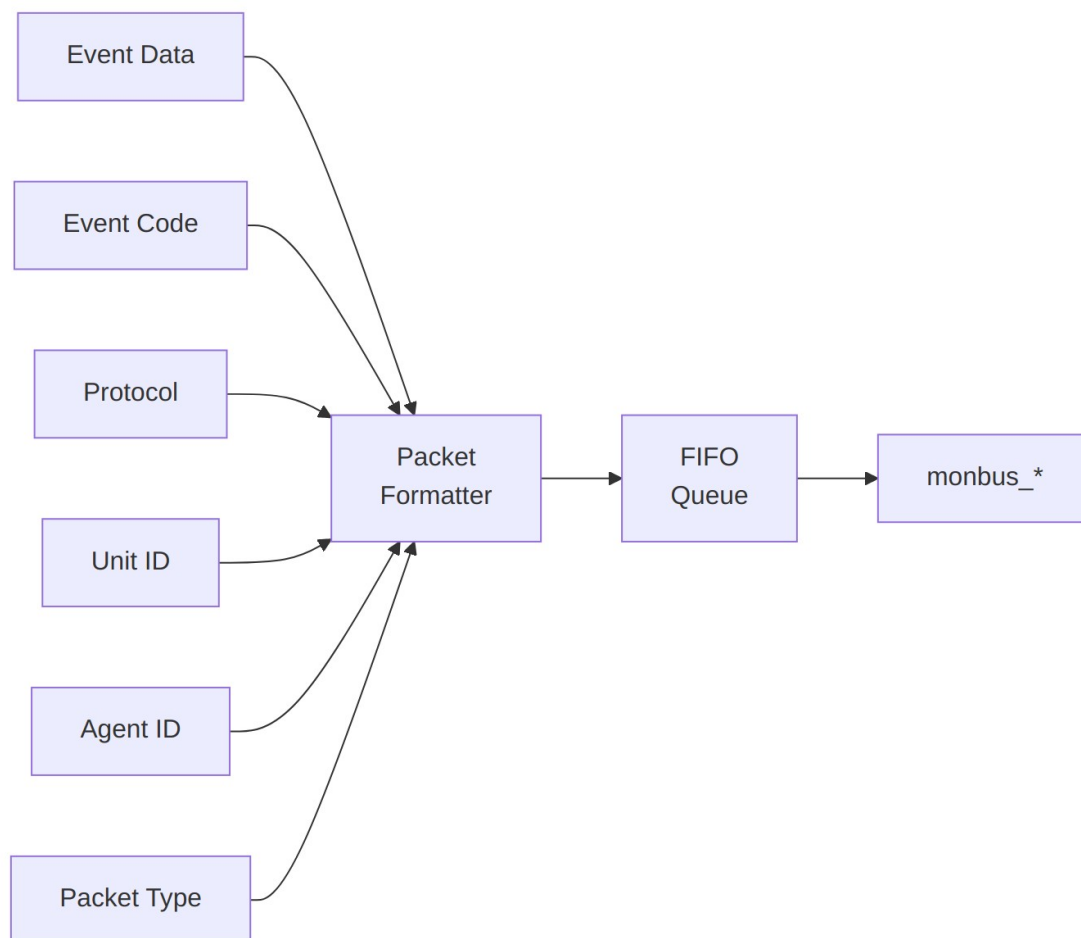
Monitor bus output

Port	Dir	Width	Description
monbus_valid	Out	1	Packet valid
monbus_ready	In	1	Downstream accepts the packet
monbus_packet	Out	monitor_packet_t	128-bit monitor packet

Status outputs

Port	Dir	Width	Description
event_count	Out	16	Total events emitted
perf_completed_count	Out	16	Completed transactions counted
perf_error_count	Out	16	Error events counted
event_reported_flags	Out	MAX_TRANSACTIONS	Per-slot “already reported” feedback to the transaction manager, so one event is not emitted twice

17.4 Functional Description



Architecture

Packet Format (128-bit `monitor_packet_t`):

Bits	Width	Field
[127:124]	4	Packet Type (error / completion / timeout / perf / etc.)
[123:109]	15	Reserved (forward-compat slack)
[108:105]	4	Protocol (AXI / AXIS / APB / ARB / CORE)
[104:97]	8	Event Code (protocol-specific)

Bits	Width	Field
[96:88]	9	Channel ID (AXI ID or channel index)
[87:72]	16	Agent ID
[71:64]	8	Unit ID
[63:0]	64	Event Data (full address, latency, counter value, etc.)

The reporter drives `monbus_packet` (128b) and `monbus_timestamp` (64b) together so the side-band timestamp travels paired with each packet through the arbiter and into the [monbus_group family](#).

17.4.1 Sub-blocks

Sub-block	Gate parameter	Generates pkt_type	Logic shape
<code>axi_monitor_reporter_error</code>	<code>ENABLE_ERROR_LOGIC</code>	<code>PktTypeError</code>	combinational
<code>axi_monitor_reporter_timeout</code>	<code>ENABLE_TIMEOUT_LOGIC</code>	<code>PktTypeTimeout</code>	combinational
<code>axi_monitor_reporter_compl</code>	<code>ENABLE_COMPL_LOGIC</code>	<code>PktTypeCompletion</code>	combinational
<code>axi_monitor_reporter_threshold</code>	<code>ENABLE_THRESHOLD_LOGIC</code>	<code>PktTypeThreshold</code>	16 latency flops + edge detect
<code>axi_monitor_reporter_perf</code>	<code>ENABLE_PERF_LOGIC</code> (alias <code>ENABLE_PERF_PACKETS</code>)	<code>PktTypePerf</code>	two 16-bit lifetime counters + 5-state FSM (the <code>perfmon_window</code> counters live in <code>axi_monitor_base</code> and are not gated by this parameter)
<code>axi_monitor_reporter_debug</code>	<code>ENABLE_DEBUG_LOGIC</code> (default 0)	<code>PktTypeDebug</code>	event-encoded debug points

Each sub-block presents the same “raise a request with packet payload” contract to the dispatcher, but the dispatcher routes them **two different ways**:

- **Error, timeout and completion** contend for the FIFO write port, in that priority order, and are queued.

- **Threshold, perf and debug** never touch the FIFO. They load the 128-bit output register directly, and only when it and the FIFO read port are both idle (!monbus_valid && !w_fifo_rd_valid) — which is what their output_busy input tells them.

That asymmetry is why an error is buffered while the other three are not. It does not mean all three are lossy — they differ, and the difference matters when you are reading a capture:

- **Perf DEFERS.** Its FSM holds while its packet is unaccepted (if (! (pkt_valid && !pkt_taken)) r_state <= w_next_state;), so a rollup that loses arbitration is emitted later carrying the then-current counts. Nothing is lost, but the timestamp is the emission, not the event.
- **Threshold** also defers while the crossing persists — the crossed flags latch on pkt_taken, not on detection. A crossing that lifts during congestion is the only case that goes unreported.
- **Debug** is genuinely lossy. r_prev_state advances every cycle regardless of acceptance (pkt_taken is unused in that block), so a state change that cannot be emitted is gone.

None of these sub-blocks are intended to be instantiated directly by integrators — they are private to the reporter family.

17.4.2 Auto-Retire: Disabled Classes Never Leak Slots

The transaction manager frees a terminal-state slot only once its event_reported flag is set, and the normal producer of that flag is an accepted FIFO write. A terminal entry whose reporting sub-block is **compiled out** (ENABLE_*_LOGIC = 0) or **runtime-disabled** (cfg_*_enable = 0) would therefore never be marked, never freed, and would permanently leak a table slot — before commit 95c9490a, the documented “performance mode” (ENABLE_COMPL_LOGIC=1 + cfg_compl_enable=0) leaked every completed entry until the table pinned and block_ready wedged the monitored bus after roughly MAX_TRANSACTION transactions.

The reporter now **auto-retires** such entries. The semantics:

- An entry marked in filtered_mask (the address-range packet filter in axi_monitor_trans_mgr) retires on the same rule the moment it is terminal. A filtered entry will never be emitted, so nothing is owed for it. Its packet is separately suppressed before the FIFO write mux, via effective valids that drop a filtered slot — suppressing the packet **without** this retire arm would leak the slot, which is the same failure the disabled-class arms exist to prevent.
- An entry in a terminal state (TRANS_COMPLETE / TRANS_ERROR / TRANS_ORPHANED) whose claiming packet class is unavailable — compiled out **or** runtime-disabled — is marked reported immediately, **without emitting a packet and without bumping event_count or the perf completion/error counters** (those count only packets actually emitted).

- The class-to-entry mapping mirrors the reporter cones' claim predicates exactly: TRANS_COMPLETE retires when the completion cone is unavailable; TRANS_ERROR retires on the **timeout** cone's availability when timeout_detected is set for that slot, else on the **error** cone's; TRANS_ORPHANED retires on the error cone's. (A naive "both cones unavailable" formula under-retires: with errors disabled but timeouts enabled, a genuine-error slot is claimable by neither cone.)
- The check is **continuous**, not edge-triggered: an entry that completed while its class was enabled but whose packet lost the FIFO-full race is still unmarked when the class is later disabled, and retires then. Accepted, documented consequence: **toggleing an enable mid-flight may drop that one entry's packet — it can never leak the slot.**

Runtime-disabling a packet class is therefore safe and makes the monitor passive for that class. If you want to keep marking and counting while suppressing emission, use the packet-type drop mask (cfg_axi_pkt_mask in [axi_monitor_filtered](#)) downstream of the reporter instead.

Directed tests: val/amba/test_axi_monitor_runtime_disable.py (fails on pre-95c9490a RTL) and val/amba/test_axi_monitor_pktgen.py.

17.5 Timing Characteristics

Metric	Value	Notes
Latency	1-2 cycles	Typical processing delay
Throughput	1 packet per 2 cycles	The registered output stage cannot reload on the same cycle its packet is accepted, so sustained output is at most one packet every other cycle even with the FIFO full

17.6 Usage Examples

This module is instantiated automatically within higher-level monitor modules — axi_monitor_base owns it, and users configure behavior through top-level monitor parameters. Configuration is typically handled at the top-level monitor instantiation; see the individual monitor documentation for configuration examples.

17.7 Design Notes

This module is where half the monitor's area goes. It keeps `r_trans_table_local`, a second full copy of `bus_transaction_t x MAX_TRANSACTION`s, alongside the transaction manager's. Budget a monitored interface as a multiple of the unmonitored one, not a percentage on top; `MAX_TRANSACTION`s is the knob and the cost is linear in it.

event_reported_flags is feedback, not status. It tells the transaction manager which slots have already emitted, so one event is not reported twice. Leaving it unconnected reintroduces a defect the family has already had.

Packet classes contend for one bus. The monbus sustains at most one packet per two cycles; enabling completion and performance together under heavy traffic drops packets. `cfg_axi_pkt_mask` suppresses a class while keeping its marking and counting.

17.8 Related Modules

- [axi_monitor_base](#)
- [arbiter_monbus_common](#)

Used by:

- **axi_monitor_base**

See also:

- **Monitor Architecture:** [docs/markdown/rtl-amba/overview.md](#)
 - **Monitor Configuration Guide:** [Monitor Base Configuration](#)
 - **Packet Format Specification:**
[docs/markdown/rtl-amba/includes/monitor_package_spec.md](#)
-

17.9 Testing

- Functional correctness of core logic
- Boundary conditions (min/max values)
- Error handling and recovery
- Interface protocol compliance

See: `val/amba/test_axi_monitor_pktgen.py` and `val/amba/test_axi_monitor_runtime_disable.py` for verification tests

17.10 Navigation

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18 AXI Monitor Reporter — Performance Packets

Module: `axi_monitor_reporter_perf.sv` **Location:** `rtl/amba/monitor/` **Status:** Production Ready

18.1 Overview

The `axi_monitor_reporter_perf` module is the performance-packet emitter for the AXI/AXIL monitor family. It is one of the per-packet-type sub-blocks that the top-level `axi_monitor_reporter` dispatches to. It owns two lifetime counters — completed transactions and error transactions — and a small cycle FSM that periodically publishes count-rollup packets of type `PktTypePerf` onto the monitor bus (`MonBus`).

This block was split out of the original monolithic reporter so integrators who do not need performance counters can drop it (compile it out with `ENABLE_PERF_LOGIC=0` at the reporter level) and reclaim the counter area.

Key features:

- Lifetime completion and error transaction counters (16-bit each)
- Counters driven from mark-reported masks supplied by the top reporter
- 5-state cycle FSM that paces packet publication
- Emits one packet at a time via a simple `pkt_valid / pkt_taken` handshake
- Backpressure aware: only advances when the output bus is free (`output_busy` low)
- Emits `AXI_PERF_COMPLETED_COUNT` and `AXI_PERF_ERROR_COUNT` event codes
- Counters exposed as ports for status / debug read-back

Performance analysis of an AXI interface needs a periodic summary of how many transactions have completed and how many ended in error over the life of the monitor. This block accumulates those two counts and, when the output path is idle, walks a fixed FSM that emits a completion-count packet followed by an error-count packet.

Use cases:

- Long-run throughput and error-rate characterization of an AXI/AXIL master or slave
- Coarse “health” telemetry streamed to a host over the MonBus
- Regression sanity: confirming the number of completions matches the stimulus
- Cross-checking the window-bucket perfmon path (Stage A/B) against a lifetime rollup

Key benefit: real area savings when disabled (the counters and FSM are removed), while providing a lightweight lifetime performance rollup that runs in parallel with the window-bucket perfmon counters in `axi_monitor_base`.

18.2 Parameters

Parameter	Type	Default	Description
MAX_TRANSACTIONS	int	16	Number of transaction slots; sets the width of the <code>error_marked_mask / compl_marked_mask</code> inputs

18.3 Ports

18.3.1 Clock and Reset

Port	Direction	Width	Description
<code>aclk</code>	Input	1	Monitor clock
<code>aresetn</code>	Input	1	Active-low asynchronous reset

18.3.2 Control / Status Inputs

Port	Direction	Width	Description
<code>cfg_perf_enable</code>	Input	1	Enable

Port	Direction	Width	Description
			performance packet generation (also gates the FSM)
output_busy	Input	1	Output path busy (FIFO has data or monbus_valid asserted); FSM stalls while high
pkt_taken	Input	1	Strobed by the top reporter when this block's packet is accepted. Load-bearing — it holds the emit FSM; see Design Notes
error_marked_mask	Input	MAX_TRANSACTION	Per-slot bit set the cycle an error event is marked-reported into the FIFO
compl_marked_mask	Input	MAX_TRANSACTION	Per-slot bit set the cycle a completion event is marked-reported into the FIFO

18.3.3 Packet Output

Port	Direction	Width	Description
pkt_valid	Output	1	A performance packet is available this cycle
pkt_type	Output	4	Packet type — constant

Port	Direction	Width	Description
			PktTypePerf (4'h4)
pkt_event_code	Output	8	Event code: AXI_PERF_COMPLETED_COUNT (8'h7) or AXI_PERF_ERROR_COUNT (8'h8)
pkt_channel	Output	9	Channel field (unused here — driven to 0)
pkt_data	Output	64	Zero-extended count value being reported

18.3.4 Lifetime Counters (Status / Debug)

Port	Direction	Width	Description
perf_completed_count	Output	16	Running total of completed transactions
perf_error_count	Output	16	Running total of error transactions

18.4 Functional Description

18.4.1 Lifetime Counters

Two 16-bit registers, `r_completed_count` and `r_error_count`, accumulate over the life of the monitor. On every clock, the block scans the `error_marked_mask` and `compl_marked_mask` inputs across all `MAX_TRANSACTIONS` slots and increments the relevant counter for each asserted bit. Because the counters advance directly from the mark masks (not from `pkt_taken`), they track every event the top reporter accepted into its FIFO regardless of whether a rollup packet is emitted. Entries **auto-retired** because their packet class is disabled do NOT feed the mark masks, so these counters count packets actually emitted, not transactions observed. Timeout packets roll up into `r_error_count` (timeout slots sit in `TRANS_ERROR`).

Both counters are exposed on the port list for status read-back, and — since commit 95c9490a — are plumbed through `axi_monitor_base` and `axi_monitor_filtered` to drive every `*_mon`

wrapper's `error_count / transaction_count` status outputs (which read 0 when `ENABLE_PERF_LOGIC=0` or `USE_MONITOR=0`).

18.4.2 Cycle FSM

A 5-state counter (`r_state`) paces packet publication and matches the original monolithic reporter's behavior one-for-one. The FSM only advances while `cfg_perf_enable` is asserted and `output_busy` is low:

State	Name	Action
3'h0	ADDR_LATENCY	No packet (placeholder state)
3'h1	DATA_LATENCY	No packet (placeholder state)
3'h2	TOTAL_LATENCY	No packet (placeholder state)
3'h3	COMPLETED_COUNT	Assert <code>w_gen_completed</code> if <code>r_completed_count > 0</code>
3'h4	ERROR_COUNT	Assert <code>w_gen_errors</code> if <code>r_error_count > 0</code> , then wrap to 3'h0

The three latency states are placeholders retained for behavioral compatibility; they emit no packet. Only the completed-count and error-count states can produce output.

18.4.3 Output Multiplexer

The output mux prioritizes the completion-count packet over the error-count packet. When `w_gen_completed` is set, `pkt_valid` asserts with event code `AXI_PERF_COMPLETED_COUNT` and `pkt_data` carrying the zero-extended completed count. Otherwise, when `w_gen_errors` is set, `pkt_valid` asserts with `AXI_PERF_ERROR_COUNT` and the error count. `pkt_type` is always `PktTypePerf` and `pkt_channel` is always 0.

18.4.4 Handshake

The block presents one packet at a time. The top reporter samples `pkt_valid`, forwards the packet fields onto the MonBus, and strobes `pkt_taken` when the packet is accepted. Publication rate is naturally throttled by the FSM (which only steps when the output is not busy), so the emitter cannot flood the bus.

18.5 Timing Characteristics

This module is **sequential**: it contains clocked logic (via `always_ff` or the repository's `ALWAYS_FF_RST` macro) and therefore holds state. Outputs driven from those blocks are registered and appear one clock after the inputs that produced them.

Per-path cycle counts are not enumerated here; read the block that drives the signal you care about. No synthesis frequency or area figures are quoted – none have been measured against a target device.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

18.6 Usage Examples

This block is not instantiated directly by users; it is instantiated inside `axi_monitor_reporter`. The pattern is:

```
axi_monitor_reporter_perf #(
    .MAX_TRANSACTIONS (MAX_TRANSACTIONS)
) u_reporter_perf (
    .aclk                (aclk),
    .aresetn             (aresetn),

    .cfg_perf_enable     (cfg_perf_enable),
    .output_busy         (w_output_busy),      // FIFO data or
monbus_valid
    .pkt_taken           (w_perf_pkt_taken),   // top reporter
accepted our packet
    .error_marked_mask   (w_error_marked_mask),
    .compl_marked_mask   (w_compl_marked_mask),

    .pkt_valid           (w_perf_pkt_valid),
    .pkt_type            (w_perf_pkt_type),
    .pkt_event_code      (w_perf_pkt_event_code),
    .pkt_channel         (w_perf_pkt_channel),
    .pkt_data            (w_perf_pkt_data),

    .perf_completed_count (perf_completed_count),
    .perf_error_count     (perf_error_count)
);
```

18.7 Design Notes

18.7.1 Completion + Performance Packets: Bandwidth, and the Two Ways to Suppress

Enabling both completion packets (`cfg_compl_enable`) and performance packets (`cfg_perf_enable`) simultaneously can congest the monitor bus under heavy traffic (the reporter sustains at most one packet per two cycles). The usual split is a functional-debug config (error + completion + timeout) and a performance config (error + perf, completions suppressed). There are two suppression mechanisms with different semantics:

- **Runtime disable** (`cfg_compl_enable = 0` with `ENABLE_COMPL_LOGIC = 1`): the monitor is passive for that class. Since commit 95c9490a this is **safe** — terminal entries of a disabled class auto-retire (are marked reported without emitting a packet and without bumping this block's counters), so the transaction table never leaks and `block_ready` never wedges. Before that commit this exact configuration leaked every completed entry and stalled the monitored bus after roughly `MAX_TRANSACTIONS` transactions. See the auto-retire section in [axi_monitor_reporter](#).
- **Packet-type drop mask** (`cfg_axi_pkt_mask` in [axi_monitor_filtered](#)): completions are still detected, marked, and **counted** (this block's `perf_completed_count` keeps advancing, and the wrapper's `transaction_count` output stays live); only the emission is dropped downstream of the reporter. Use this when you want the lifetime counters while suppressing the packet stream.

See docs/user-guides/AXI_Monitor_Configuration_Guide.md.

18.7.2 `pkt_taken` Holds the Emit FSM — Do Not Tie It Off

`pkt_taken` does not gate the *counters* (those update from the mark masks unconditionally), but it **does** gate the FSM state register:

```
// Hold the state whenever we are presenting a packet that was not
// accepted (threshold beats perf in the top reporter's output mux).
// Advancing regardless silently dropped the packet.
if (!(pkt_valid && !pkt_taken)) begin
    r_state <= w_next_state;
end
```

The hold exists because threshold outranks perf in the top reporter's output mux: without it, a perf packet that lost that arbitration was generated, never emitted, and the FSM walked past it.

So the port must be driven correctly. Tie it **low** and the FSM deadlocks the first time it presents a packet — the state holds forever. Tie it **high** and the silently-dropped-packet bug returns.

(The “retained for a future hook, tied to an unused net” pattern this section used to describe belongs to `axi_monitor_reporter_debug`, a different module, whose own page documents it correctly.)

18.7.3 Relationship to the Window-Bucket Perfmon

This block emits the legacy `PktTypePerf` count-rollup packets that summarize completion/error counts over the monitor’s lifetime. It runs in parallel with the Stage A/B window-bucket perfmon counters in `axi_monitor_base`. Those buckets are **readable as counters only** — nothing packetizes them onto the MonBus yet, so no `PktTypePerfWin` / `PktTypePerfHist` packets are emitted by any module today. The two mechanisms are complementary, not redundant.

18.7.4 Counter Wrap

Both counters are 16-bit and wrap on overflow. For very long runs the host should account for wrap or read the counters periodically via the status ports.

18.8 Related Modules

Used by:

- **`axi_monitor_reporter.sv`** — instantiates this block as its performance-packet sub-emitter
- **`axi_monitor_base.sv`** — top-level monitor scaffold that wires the reporter into the MonBus

Uses:

- **`monitor_common_pkg`** — `PktTypePerf` and shared packet definitions
- **`monitor_amba4_pkg`** — `AXI_PERF_COMPLETED_COUNT` / `AXI_PERF_ERROR_COUNT` event codes
- **`reset_defs.svh`** — reset macros (`ALWAYS_FF_RST`, `RST_ASSERTED`)

See also:

- **`axi_monitor_reporter_threshold.sv`** — threshold-crossing packet emitter (sibling)
 - **`axi_monitor_reporter_timeout.sv`** — timeout packet emitter (sibling)
-

18.9 Testing

No dedicated testbench for this module. It has no `val/**/test_axi_monitor_reporter_perf.py`. It is exercised indirectly, through the tests of modules that instantiate it (directly or further up):

- `axi4_master_rd_mon` – `val/**/test_axi4_master_rd_mon.py`
- `axi4_master_wr_mon` – `val/**/test_axi4_master_wr_mon.py`
- `axi4_slave_rd_mon` – `val/**/test_axi4_slave_rd_mon.py`
- `axi4_slave_wr_mon` – `val/**/test_axi4_slave_wr_mon.py`
- `axi5_master_rd_mon` – `val/**/test_axi5_master_rd_mon.py`

Indirect coverage exercises this module only in the configurations those parents elaborate. A parameter or mode no parent uses is untested.

Treat any behaviour described on this page as unverified by simulation.

18.10 References

18.10.1 Source Code

- RTL: `rtl/amba/monitor/axi_monitor_reporter_perf.sv`
- Parent: `rtl/amba/monitor/axi_monitor_reporter.sv`
- Packages: `rtl/amba/includes/monitor_common_pkg.sv`,
`rtl/amba/includes/monitor_amba4_pkg.sv`

18.10.2 Documentation

- Architecture: `docs/markdown/rtl-amba/shared/README.md`
 - Monitor Base: `docs/markdown/rtl-amba/monitor/axi_monitor_base.md`
 - Configuration: `docs/user-guides/AXI_Monitor_Configuration_Guide.md`
 - Packet Format:
`docs/markdown/rtl-amba/includes/monitor_package_spec.md`
-

Last Updated: 2026-07-15

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19 AXI Monitor Reporter — Threshold Packets

Module: `axi_monitor_reporter_threshold.sv` **Location:** `rtl/amba/monitor/` **Status:**
Production Ready

19.1 Overview

The `axi_monitor_reporter_threshold` module is the threshold-crossing packet emitter for the AXI/AXIL monitor family. It is one of the per-packet-type sub-blocks dispatched by the top-level `axi_monitor_reporter`. It watches two conditions — the number of currently active transactions crossing a configured limit, and any transaction's measured latency exceeding a configured limit — and emits `PktTypeThreshold` packets when either crossing occurs.

The block was split out of the original monolithic reporter so integrators can drop it (`ENABLE_THRESHOLD_LOGIC=0`) and recover real area, because it owns the per-slot latency pipeline (16 x 32-bit latency flops plus 16 threshold flags).

`threshold` is a **fault class** ([taxonomy](#)) — an early-warning fault, one step below a hard timeout. Exercise it by **injecting a slow slave**: hold responses long enough that latency crosses `LATENCY_THRESH` but stays under the timeout window (or push enough concurrent traffic to cross the active-count limit). It does not occur under healthy traffic, so like the other faults it must be provoked deliberately.

Key features:

- Active-transaction-count threshold detection (instantaneous condition)
- Per-slot latency threshold detection with a registered latency pipeline
- Edge-sticky flags to fire once per crossing rather than every cycle
- Read/write latency measurement selectable via `IS_READ`
- Internal arbitration: active-count wins over latency events
- One packet at a time via `pkt_valid` / `pkt_taken` handshake
- Emits `AXI_THRESH_ACTIVE_COUNT` and `AXI_THRESH_LATENCY` event codes

Monitoring a bus for congestion and latency outliers requires two distinct checks: how many transactions are in flight right now, and whether any individual transaction took too long. This block performs both. The active count is computed combinationally by scanning the transaction table; latency is computed per slot in a registered pipeline (splitting the wide subtract-and-compare across a cycle to close timing) and compared against the configured latency threshold.

Use cases:

- Detecting bus congestion when outstanding-transaction count exceeds a budget
- Flagging latency-outlier transactions that exceed a service-level target
- Feeding threshold-crossing telemetry to a host for QoS monitoring
- Regression checks that latency stays within expected bounds

Key benefit: real area savings when disabled (the 16-deep latency pipeline is removed), plus a timing-friendly split of the latency compare so the wide carry chain does not sit on the packet-generation critical path.

19.2 Parameters

Parameter	Type	Default	Description
MAX_TRANSACTIONS	int	16	Number of transaction slots scanned for active count and latency
IS_READ	bit	1'b1	1 = measure read latency (data - addr timestamp); 0 = measure write latency (resp - addr timestamp)
IDX_W	int	$\$clog2(MAX_TRANSACTIONS)$	Derived width of the selected-slot index

19.3 Ports

19.3.1 Clock and Reset

Port	Direction	Width	Description
aclk	Input	1	Monitor clock
aresetn	Input	1	Active-low asynchronous reset

19.3.2 Transaction Table and Configuration

Port	Direction	Width	Description
trans_table	Input	bus_transaction_t [MAX_TRANSACTION S]	Live outstanding-transaction table (valid, state, timestamps, channel)
cfg_threshold_enable	Input	1	Enable threshold detection
active_trans_threshold	Input	16	Active-transaction-count limit; crossing above it fires a packet
latency_threshold	Input	32	Per-transaction latency limit (in timestamp ticks)

19.3.3 Handshake / Backpressure

Port	Direction	Width	Description
output_busy	Input	1	Output bus busy (FIFO read or monbus_valid); threshold packets inject directly, so this prevents overwriting an in-flight one
pkt_taken	Input	1	Pulsed by the top reporter when this block's packet was accepted; sets the edge-sticky flag so the same crossing cannot re-fire. The flag

Port	Direction	Width	Description
			clears only when the underlying condition lifts

19.3.4 Packet Output

Port	Direction	Width	Description
pkt_valid	Output	1	A threshold packet is available this cycle
pkt_type	Output	4	Packet type — constant PktTypeThreshold (4'h2)
pkt_event_code	Output	8	Event code: AXI_THRESH_ACTIVE_COUNT (8'h0) or AXI_THRESH_LATENCY (8'h1)
pkt_channel	Output	9	Channel of the selected transaction (0 for active-count events)
pkt_data	Output	64	Zero-extended active count, or the zero-extended latency value

19.4 Functional Description

19.4.1 Active-Count Detection

A combinational loop scans `trans_table` and counts slots that are `valid` and in a state other than `TRANS_COMPLETE` or `TRANS_ERROR` — that is, transactions genuinely in flight. `w_active_detect` asserts when threshold detection is enabled, the count strictly exceeds

active_trans_threshold, the edge flag r_active_crossed is not already set, and the output is not busy. Because this reflects an instantaneous condition, it takes priority in the output mux.

19.4.2 Per-Slot Latency Pipeline

For each slot, a registered pipeline computes latency from the live transaction table: data_timestamp - addr_timestamp for reads (IS_READ=1) or resp_timestamp - addr_timestamp for writes. The result is stored in r_latency[idx], and a companion flag r_latency_over_thresh[idx] is set when the slot is valid, in TRANS_COMPLETE state, and its latency exceeds latency_threshold. It is deliberately **not** qualified by the latency edge flag — folding the flag in here would make the condition self-clearing, since the flag would suppress the very term that keeps it set. r_latency_crossed gates only the output mux. Registering the subtract and compare splits what would otherwise be a 16-wide carry chain plus output mux across a clock cycle.

19.4.3 Latency Event Selection

A combinational priority encoder (w_lat_sel / w_has_lat) picks the first slot with r_latency_over_thresh asserted, off the registered pipeline so it does not recreate the wide combinational path.

19.4.4 Edge-Sticky Flags

Two flags prevent re-firing on the same crossing every cycle:

- r_active_crossed — set when an active-count packet is accepted (pkt_taken with matching type/event code); cleared automatically when the active count drops back to or below the threshold.
- r_latency_crossed — set when a latency packet is accepted; gates further latency detections until re-armed.

19.4.5 Output Multiplexer

Active-count beats latency. When w_active_detect is asserted, pkt_valid fires with AXI_THRESH_ACTIVE_COUNT, pkt_data = zero-extended active count, and pkt_channel = 0. Otherwise, when a latency event is pending and the output is free (w_has_lat && ! output_busy), pkt_valid fires with AXI_THRESH_LATENCY, pkt_data = the selected slot's latency, and pkt_channel = that slot's channel (low 6 bits, zero-extended).

19.5 Timing Characteristics

This module is **sequential**: it contains clocked logic (via always_ff or the repository's ALWAYS_FF_RST macro) and therefore holds state. Outputs driven from those blocks are registered and appear one clock after the inputs that produced them.

Per-path cycle counts are not enumerated here; read the block that drives the signal you care about. No synthesis frequency or area figures are quoted – none have been measured against a target device.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

19.6 Usage Examples

This block is instantiated inside `axi_monitor_reporter`, not by users directly:

```
axi_monitor_reporter_threshold #(
    .MAX_TRANSACTIONS (MAX_TRANSACTIONS),
    .IS_READ          (IS_READ)
) u_reporter_threshold (
    .aclk              (aclk),
    .aresetn           (aresetn),

    .trans_table       (w_trans_table),
    .cfg_threshold_enable (cfg_threshold_enable),
    .active_trans_threshold (cfg_active_trans_threshold),
    .latency_threshold  (cfg_latency_threshold),

    .output_busy        (w_output_busy),
    .pkt_taken          (w_thresh_pkt_taken),

    .pkt_valid          (w_thresh_pkt_valid),
    .pkt_type           (w_thresh_pkt_type),
    .pkt_event_code     (w_thresh_pkt_event_code),
    .pkt_channel        (w_thresh_pkt_channel),
    .pkt_data           (w_thresh_pkt_data)
);
```

19.7 Design Notes

19.7.1 Threshold Packets Bypass the FIFO

Unlike error/completion packets, threshold packets inject directly into the output rather than routing through the reporter FIFO. That is why the block needs `output_busy` — to avoid overwriting a packet already in flight on the MonBus.

19.7.2 Active Count vs Latency Priority

Active count is an instantaneous, whole-table condition; latency is a per-slot event that has already been captured in the pipeline flag. The mux deliberately favors the active-count crossing so the more time-sensitive congestion signal is never starved by a queue of latency reports.

19.7.3 Latency Pipeline Is the Area Cost

The 16 x 32-bit latency registers plus 16 threshold flags are the reason disabling this block (ENABLE_THRESHOLD_LOGIC=0) yields a meaningful area reduction. When the feature is not needed, dropping the block removes the entire pipeline.

19.7.4 Read vs Write Latency

IS_READ selects which timestamp pair defines latency. Instantiate a read-monitor threshold block with IS_READ=1 and a write-monitor block with IS_READ=0 so the measured interval matches the protocol phase of interest.

19.8 Related Modules

Used by:

- **axi_monitor_reporter.sv** — instantiates this block as its threshold-packet sub-emitter
- **axi_monitor_base.sv** — top-level monitor scaffold

Uses:

- **monitor_common_pkg** — PktTypeThreshold, bus_transaction_t, transaction states
- **monitor_amba4_pkg** — AXI_THRESH_ACTIVE_COUNT / AXI_THRESH_LATENCY event codes
- **reset_defs.svh** — reset macros

See also:

- **axi_monitor_reporter_perf.sv** — performance packet emitter (sibling)
 - **axi_monitor_reporter_timeout.sv** — timeout packet emitter (sibling)
 - **axi_monitor_trans_mgr.sv** — maintains the trans_table consumed here
-

19.9 Testing

No dedicated testbench for this module. It has no `val/**/test_axi_monitor_reporter_threshold.py`. It is exercised indirectly, through the tests of modules that instantiate it (directly or further up):

- `axi4_master_rd_mon` – `val/**/test_axi4_master_rd_mon.py`
- `axi4_master_wr_mon` – `val/**/test_axi4_master_wr_mon.py`
- `axi4_slave_rd_mon` – `val/**/test_axi4_slave_rd_mon.py`
- `axi4_slave_wr_mon` – `val/**/test_axi4_slave_wr_mon.py`
- `axi5_master_rd_mon` – `val/**/test_axi5_master_rd_mon.py`

Indirect coverage exercises this module only in the configurations those parents elaborate. A parameter or mode no parent uses is untested.

Treat any behaviour described on this page as unverified by simulation.

19.10 References

19.10.1 Source Code

- RTL: `rtl/amba/monitor/axi_monitor_reporter_threshold.sv`
- Parent: `rtl/amba/monitor/axi_monitor_reporter.sv`
- Packages: `rtl/amba/includes/monitor_common_pkg.sv`,
`rtl/amba/includes/monitor_amba4_pkg.sv`

19.10.2 Documentation

- Architecture: `docs/markdown/rtl-amba/shared/README.md`
 - Monitor Base: `docs/markdown/rtl-amba/monitor/axi_monitor_base.md`
 - Packet Format:
`docs/markdown/rtl-amba/includes/monitor_package_spec.md`
-

Last Updated: 2026-07-15

19.11 Navigation

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20 AXI Monitor Reporter — Timeout Packets

Module: `axi_monitor_reporter_timeout.sv` **Location:** `rtl/amba/monitor/` **Status:**
Production Ready

20.1 Overview

The `axi_monitor_reporter_timeout` module is the timeout-packet emitter for the AXI/AXIL monitor family. It is one of the per-packet-type sub-blocks dispatched by the top-level `axi_monitor_reporter`. It is a pure combinational cone that scans the transaction table for unreported error slots the timeout detector has flagged, priority-encodes the first match, and drives the corresponding `PktTypeTimeout` packet fields.

The block was split out of the original monolithic reporter so integrators can drop it (`ENABLE_TIMEOUT_LOGIC=0`).

timeout is a **fault class** ([taxonomy](#)): it never fires in correct operation. Exercise it by **injecting a non-responsive slave** — hold the R/B response beats past the configured timeout window so the transaction stays outstanding long enough for the timeout detector to flag it. Like all faults, this stalls the traffic; the cone's value is emitting `PktTypeTimeout` so the capture records *which* transaction hung.

Key features:

- Pure combinational detection (no internal state)
- Scans for valid, unreported, error-state slots flagged as timed out
- Priority-encodes the first matching slot
- Emits the packet's event code, channel, and address directly from the table
- Exposes the selected slot index (`sel_idx`) for the top reporter's mark-reported feedback
- Shares the `timeout_detected` vector with the error sub-block to avoid double-reporting

When a transaction stalls, the `axi_monitor_timeout` block asserts a per-slot `timeout_detected` bit and the transaction manager moves that slot into the `TRANS_ERROR` state. This reporter block turns that condition into a MonBus packet: it finds the first slot that is valid, in error, timed out, and not yet reported, and emits a timeout packet describing it (event code, channel, and address).

Use cases:

- Reporting stuck AXI transactions (missing R/B response, hung handshake)
- Surfacing protocol hangs to a host IRQ handler for recovery

- Regression detection of deadlock or backpressure faults
- Correlating a timeout with the offending address and channel

Key benefit: zero-state, low-cost detection that shares its `timeout_detected` and `event_reported` vectors with the error emitter, guaranteeing each stuck transaction is reported exactly once.

20.2 Parameters

Parameter	Type	Default	Description
MAX_TRANSACTIONS	int	16	Number of transaction slots scanned
IDX_W	int	$\$clog2(MAX_TRANSACTIONS)$	Derived width of the selected-slot index

20.3 Ports

20.3.1 Inputs

Port	Direction	Width	Description
trans_table	Input	bus_transaction_t [MAX_TRANSACTIONS]	Live outstanding-transaction table (valid, state, event_code, channel, addr)
event_reported	Input	MAX_TRANSACTIONS	Per-slot bit indicating the slot's event was already emitted (suppresses re-reporting)
timeout_detected	Input	MAX_TRANSACTIONS	Per-slot timeout flags from <code>axi_monitor_timeout</code>

Port	Direction	Width	Description
cfg_timeout_enable	Input	1	Enable timeout packet generation

20.3.2 Packet Output

Port	Direction	Width	Description
pkt_valid	Output	1	A timeout packet is available (a matching slot was found)
pkt_type	Output	4	Packet type — constant PktTypeTimeout (4'h3)
pkt_event_code	Output	8	Event code taken from the selected slot's event_code.raw_code
pkt_channel	Output	9	Channel of the selected slot (low 6 bits, zero-extended)
pkt_data	Output	64	Zero-extended address of the selected slot
sel_idx	Output	IDX_W	Index of the selected slot, returned to the top reporter for mark-reported feedback

20.4 Functional Description

20.4.1 Candidate Detection

A combinational loop builds a one-hot-eligible mask `w_events`: a slot qualifies when it is `valid`, its `event_reported` bit is clear, its state is `TRANS_ERROR`, timeout reporting is enabled (`cfg_timeout_enable`), and the slot's `timeout_detected` bit is set. This intersection ensures only genuinely-timed-out, not-yet-reported error slots are candidates.

20.4.2 Priority Encoding

A second loop scans `w_events` from index 0 upward and latches the first asserted slot into `w_sel`, asserting `w_has_event`. This gives a deterministic, lowest-index-first selection.

20.4.3 Output Assignment

The outputs are pure continuous assignments off the selected slot:

- `pkt_valid = w_has_event`
- `sel_idx = w_sel`
- `pkt_type = PktTypeTimeout`
- `pkt_event_code = trans_table[w_sel].event_code.raw_code`
- `pkt_channel = {3'b0, trans_table[w_sel].channel[5:0]}`
- `pkt_data = pad_address(trans_table[w_sel].addr) (zero-extended to 64 bits)`

20.4.4 No Double-Reporting

The error sub-block masks the same `timeout_detected` slots, so a stuck transaction that is reported as a timeout here is not also reported as a plain error. The top reporter uses `sel_idx` to set the slot's `event_reported` bit, retiring the candidate.

20.5 Timing Characteristics

This module is **purely combinational** – it contains no `always_ff` and no latch, so it holds no state and adds no clock cycles. Its outputs settle a propagation delay after its inputs, and it introduces no latency into a pipeline that instantiates it.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

20.6 Usage Examples

This block is instantiated inside `axi_monitor_reporter`, not by users directly:

```
axi_monitor_reporter_timeout #(
    .MAX_TRANSACTIONS (MAX_TRANSACTIONS)
) u_reporter_timeout (
    .trans_table      (w_trans_table),
    .event_reported   (w_event_reported),
    .timeout_detected (w_timeout_detected),
    .cfg_timeout_enable (cfg_timeout_enable),

    .pkt_valid        (w_timeout_pkt_valid),
    .pkt_type         (w_timeout_pkt_type),
    .pkt_event_code   (w_timeout_pkt_event_code),
    .pkt_channel      (w_timeout_pkt_channel),
    .pkt_data         (w_timeout_pkt_data),
    .sel_idx          (w_timeout_sel_idx)
);
```

20.7 Design Notes

20.7.1 Pure Combinational

The block holds no state; all edge-tracking and mark-reported bookkeeping lives in the top reporter. This keeps the timeout cone cheap and lets the top reporter arbitrate timeout packets against the other sub-emitters within one cycle.

20.7.2 Event Code Comes From the Slot

`pkt_event_code` is not a fixed constant — it is the slot's captured `event_code.raw_code`, so the packet carries the specific timeout reason recorded when the transaction manager moved the slot to `TRANS_ERROR`.

20.7.3 Shared Vectors Are Load-Bearing

Correct single-reporting depends on the top reporter driving `event_reported` and the timeout detector driving `timeout_detected` consistently. Historically, a missing `event_reported` feedback path caused transaction-table exhaustion; that feedback is now in place (see `rtl/amba/KNOWN_ISSUES/axi_monitor_reporter.md`).

20.8 Related Modules

Used by:

- **axi_monitor_reporter.sv** — instantiates this block as its timeout-packet sub-emitter
- **axi_monitor_base.sv** — top-level monitor scaffold

Uses:

- **monitor_common_pkg** — PktTypeTimeout, bus_transaction_t, TRANS_ERROR
- **monitor_amba4_pkg** — AMBA event-code definitions
- (No sequential logic — no reset macros required)

See also:

- **axi_monitor_timeout.sv** — produces the timeout_detected vector consumed here
 - **axi_monitor_reporter_perf.sv** — performance packet emitter (sibling)
 - **axi_monitor_reporter_threshold.sv** — threshold packet emitter (sibling)
-

20.9 Testing

No dedicated testbench for this module. It has no val/**/test_axi_monitor_reporter_timeout.py. It is exercised indirectly, through the tests of modules that instantiate it (directly or further up):

- axi4_master_rd_mon – val/**/test_axi4_master_rd_mon.py
- axi4_master_wr_mon – val/**/test_axi4_master_wr_mon.py
- axi4_slave_rd_mon – val/**/test_axi4_slave_rd_mon.py
- axi4_slave_wr_mon – val/**/test_axi4_slave_wr_mon.py
- axi5_master_rd_mon – val/**/test_axi5_master_rd_mon.py

Indirect coverage exercises this module only in the configurations those parents elaborate. A parameter or mode no parent uses is untested.

Treat any behaviour described on this page as unverified by simulation.

20.10 References

20.10.1 Source Code

- RTL: rtl/amba/monitor/axi_monitor_reporter_timeout.sv
- Parent: rtl/amba/monitor/axi_monitor_reporter.sv
- Packages: rtl/amba/includes/monitor_common_pkg.sv, rtl/amba/includes/monitor_amba4_pkg.sv

20.10.2 Documentation

- Architecture: docs/markdown/rtl-amba/shared/README.md
 - Monitor Base: docs/markdown/rtl-amba/monitor/axi_monitor_base.md
 - Known Issues: rtl/amba/KNOWN_ISSUES/axi_monitor_reporter.md
 - Packet Format: docs/markdown/rtl-amba/includes/monitor_package_spec.md
-

Last Updated: 2026-07-15

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21 AXI Monitor Timeout

Module: axi_monitor_timeout.sv **Location:** rtl/amba/monitor/ **Category:** Core Infrastructure **Status:** Production Ready

21.1 Overview

The axi_monitor_timeout module provides configurable timeout detection for stuck transactions — the block that notices when a protocol phase stops making progress.

This is a **shared infrastructure module** used internally by the AXI/AXIL monitors. You won't instantiate it directly, but it's central to how the monitor catches hung traffic.

Key features:

- Per-phase timeout detection (AR/AW, R/W, B)
- Configurable timeout thresholds
- Frequency scaling for timeout counts
- Timeout event reporting with transaction ID
- Active transaction tracking
- Timeout clear on transaction completion

What it's for:

1. **Per-Phase Monitoring:** separate timeouts for AR/AW, R/W, B channels
2. **Configurable Thresholds:** adjustable timeout values per phase
3. **Event Reporting:** generates timeout packets with transaction details
4. **Active Tracking:** monitors all outstanding transactions
5. **Frequency Scaling:** adapts timeout counts to system clock frequency

21.2 Parameters

Parameter	Type	Default	Description
MAX_TRANSACTION	int	16	Number of transactions to monitor
ADDR_WIDTH	int	32	Address width for reporting
IS_READ	bit	1	1 = read channel, 0 = write channel

21.3 Ports

21.3.1 Clock and Reset

Port	Width	Direction	Description
aclk	1	Input	Monitor clock
aresetn	1	Input	Active-low asynchronous reset

21.3.2 Transaction Table

Port	Width	Direction	Description
trans_table	bus_transaction_tx MAX_TRANSACTIONS	Input	The transaction manager's table, read directly as an unpacked array. This module only reads it; axi_monitor_trans_mgr owns the entries.

21.3.3 Timer

Port	Width	Direction	Description
timer_tick	1	Input	One-cycle pulse from the frequency-invariant timer. Every threshold below counts these, not aclk cycles, so a timeout means the same wall-clock interval at any bus frequency.

21.3.4 Timeout Configuration

Each threshold is a count of timer_tick pulses for one phase of a transaction.

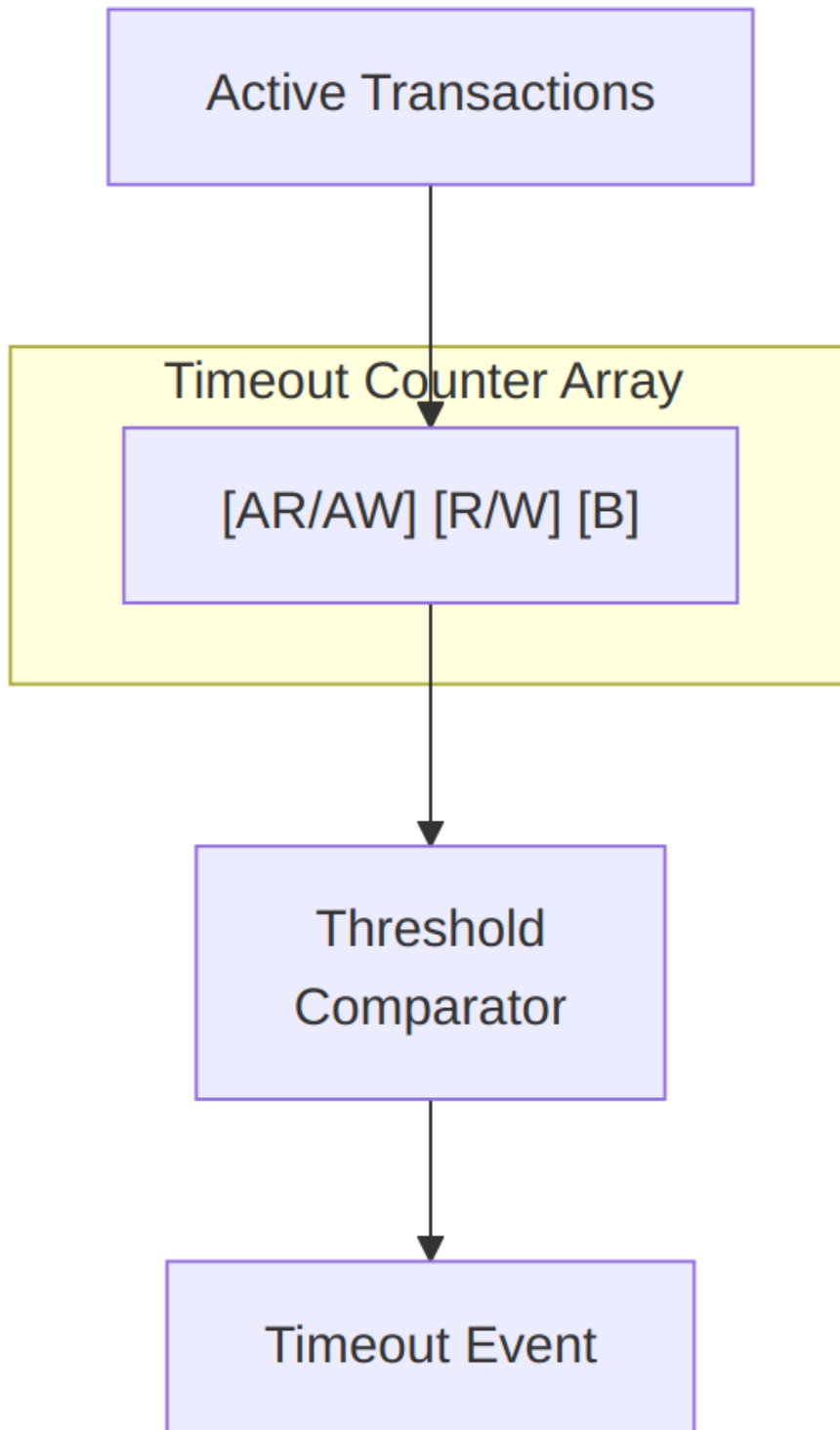
Port	Width	Direction	Description
cfg_addr_cnt	16	Input	Address-phase timeout threshold
cfg_data_cnt	16	Input	Data-phase timeout threshold
cfg_resp_cnt	16	Input	Response-phase timeout threshold
cfg_timeout_enable	1	Input	Master enable for timeout detection. Low does not merely mask the output: the per-phase timers and all recorded detections are flushed, so the block is inert rather than holding

Port	Width	Direction	Description
			stale state. A detection registered while disabled would resurface the instant the enable returned – a detection with no timer_tick behind it, which the formal property ap_no_set_without_tick rejects.

21.3.5 Outputs

Port	Width	Direction	Description
timeout_detected	MAX_TRANSACTIONS	Output	One bit per table slot, high for a transaction that has timed out. Consumed by the transaction manager to retire the entry.

21.4 Functional Description



Architecture

Each transaction has 3 independent timeout counters for different phases. The timers are dedicated per-slot state owned by this module (`r_addr_timer` / `r_data_timer` / `r_resp_timer`, **16-bit** each — `TIMER_W = 16`, sized to hold the full microsecond threshold); the rest of the transaction record is read live off `trans_table`. Timers count `timer_tick` events (from the frequency-invariant timer, scaled by `cfg_freq_sel`) while their phase is pending, and fire at `timer >= cfg_addr_cnt / cfg_data_cnt / cfg_resp_cnt`.

21.4.1 Detection Semantics

The per-phase “still waiting” conditions, read straight off the live table:

Phase	Condition
Address	<code>valid && state == TRANS_ADDR_PHASE && !cmd_received</code> (command issued, not yet accepted)
Data	<code>valid && state in {TRANS_ADDR_PHASE, TRANS_DATA_PHASE} && cmd_received && !data_completed</code>
Response	<code>write monitors only: valid && state == TRANS_DATA_PHASE && data_completed && !resp_received</code>

Notable, deliberate properties:

- **“Command accepted, first beat never arrives” is detectable.** The data condition intentionally does NOT require `data_started`. With that term (the pre-cb29e226 behavior) this stall class could never fire: the entry sat in `TRANS_ADDR_PHASE` forever, pinned its table slot, and enough of them held `block_ready` low permanently. A transaction whose command just handshake gets the full `cfg_data_cnt` window for its first beat, same as every later beat.
- **Runtime disable flushes state.** When `cfg_timeout_enable` is 0, all timers and detection flags are cleared, not merely masked. A stale detection computed against old timer state can therefore never resurface the instant the enable comes back (enforced by the `ap_no_set_without_tick` formal property). Disable means inert.
- **Detection is sticky until the slot retires** — the flag is held through `TRANS_ERROR` on purpose (that is the state a detected timeout puts the transaction into, and the reporter uses this vector to split timeout packets

from genuine-error packets); it clears when the slot empties or reaches TRANS_COMPLETE.

- **This module cannot modify the table.** It exports timeout_detected per slot; axi_monitor_trans_mgr consumes that vector (its i_timeout_detected input) and moves the entry to TRANS_ERROR with EVT_CMD_TIMEOUT / EVT_DATA_TIMEOUT / EVT_RESP_TIMEOUT so it becomes cleanup-eligible.
-

21.5 Timing Characteristics

Metric	Value	Notes
Latency	1-2 cycles	Typical processing delay
Throughput	1 operation/cycle	Maximum rate

21.6 Usage Examples

This module is instantiated automatically within higher-level monitor modules — axi_monitor_base owns it, and users configure behavior through top-level monitor parameters. See the individual monitor documentation for configuration examples.

21.7 Design Notes

These are phase DURATION limits, not stall detectors. Each timer is zeroed only while its phase is not pending (if (!w_data_pending[idx]) r_data_timer[idx] <= '0;); a beat handshake does not reset it. A long burst that is making steady progress still trips EVT_DATA_TIMEOUT once the phase as a whole exceeds the threshold. Size cfg_timeout_cycles for your longest legitimate burst, not for your worst tolerable inter-beat gap.

The thresholds are microseconds, not clocks. timer_tick comes from counter_freq_invariant, so a timeout means the same wall-clock time at any aclk – provided ACLK_MHZ matches the real frequency. Leave it at 100 on a 90 MHz part and every timeout is wrong, silently.

TIMER_W is 16 deliberately. These counters were 4 bits, and every wrapper squashed the host's 16-bit request into them with a saturating truncation, so any value >= 16 became 15 and the entire configurable range collapsed onto 1..15 us.

21.8 Related Modules

- [axi_monitor_base](#)

Used by:

- **axi_monitor_base**

See also:

- **Monitor Architecture:** docs/markdown/rtl-amba/overview.md
 - **Monitor Configuration Guide:** [Monitor Base Configuration](#)
 - **Packet Format Specification:**
docs/markdown/rtl-amba/includes/monitor_package_spec.md
-

21.9 Testing

- Functional correctness of core logic
- Boundary conditions (min/max values)
- Error handling and recovery
- Interface protocol compliance

See: val/amba/test_axi_monitor_trans_mgr.py (timeout-to-terminal transitions) and the *_mon wrapper suites for verification tests

21.10 Navigation

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22 AXI Monitor Timer

Module: axi_monitor_timer.sv **Location:** rtl/amba/monitor/ **Status:** Production Ready

22.1 Overview

The AXI Monitor Timer provides frequency-invariant timing for the AXI monitor infrastructure by generating consistent timer ticks regardless of the system clock frequency. It combines a cycle-

accurate global timestamp counter with a frequency-invariant timer tick generator to support both precise latency measurement and consistent timeout detection across different clock frequencies.

Key features:

- Frequency-invariant timer tick generation
- Fixed 1 MHz (1 us) tick; the 4-bit `cfg_freq_sel` selects the **clock frequency** the divider is matched to, not the tick period
- Global timestamp counter (32-bit cycle counter)
- Uses `counter_freq_invariant` for portable timing
- Zero-latency timestamp (combinational output)
- Consistent timeout behavior across frequencies
- Single clock domain operation
- Minimal resource utilization

The Timer module solves a portability problem that bites every monitor architecture eventually: timeout thresholds specified in cycles become frequency-dependent, so they need reconfiguration every time the clock frequency changes. By using frequency-invariant timer ticks, timeout thresholds can be specified in abstract “ticks” that represent consistent time periods regardless of the underlying clock frequency.

The global timestamp counter provides cycle-accurate timing for latency measurement and performance analysis, while the timer tick output enables timeout detection with configurable resolution.

22.2 Parameters

Runtime frequency selection is via `cfg_freq_sel`. The module also exposes elaboration parameters that size the clock-frequency-invariant (CFI) counter table:

Parameter	Type	Default	Description
<code>CFI_MIN_FREQ_MHZ</code>	<code>int</code>	5	Minimum frequency (MHz) in the CFI table
<code>CFI_MAX_FREQ_MHZ</code>	<code>int</code>	220	Maximum frequency (MHz) in the CFI table
<code>CFI_NUM_FREQ_ENTRIES</code>	<code>int</code>	16	Number of frequency entries

Parameter	Type	Default	Description
CFI_FREQ_STRATEGY	int	0	Frequency-spacing strategy selector

SEL_WIDTH is derived from CFI_NUM_FREQ_ENTRIES and sizes cfg_freq_sel.

22.3 Ports

22.3.1 Global Clock and Reset

Port	Direction	Width	Description
acclk	Input	1	Clock signal
aresetn	Input	1	Active-low asynchronous reset

22.3.2 Timer Configuration

Port	Direction	Width	Description
cfg_freq_sel	Input	4	Frequency selection for timer tick (0-15)

22.3.3 Timer Outputs

Port	Direction	Width	Description
timer_tick	Output	1	Timer tick signal (frequency-invariant)
timestamp	Output	32	Global timestamp counter (cycle count)

22.4 Functional Description

The Timer module implements two independent timing functions that work together to support the monitor infrastructure.

22.4.1 Global Timestamp Counter

The timestamp counter provides cycle-accurate time reference:

Implementation:

- 32-bit up-counter
- Increments every clock cycle
- Resets to zero on aresetn assertion
- Registered output (r_timestamp)

Purpose:

- Transaction timestamp marking (addr_timestamp, data_timestamp, resp_timestamp)
- Latency calculation (timestamp differences)
- Performance metric computation
- Event correlation across transactions

Range and Wraparound:

- Maximum count: $2^{32} - 1 = 4,294,967,295$ cycles
- At 1 GHz: ~4.3 seconds before wraparound
- At 100 MHz: ~43 seconds before wraparound
- Wraparound handled naturally in latency calculations (32-bit arithmetic)

Usage Example:

```
// Transaction manager records timestamp
addr_timestamp <= timestamp; // When address phase completes

// Later, calculate latency
latency = data_timestamp - addr_timestamp; // Handles wraparound
```

22.4.2 Frequency-Invariant Timer Tick

The timer tick provides consistent periodic signal across frequencies:

Implementation:

- Uses counter_freq_invariant module from rtl/common/

- Configured via `cfg_freq_sel` (4-bit selection: 0-15)
- Generates single-cycle pulse (`timer_tick`) at configured interval
- Combinational output (`w_timer_tick`)

Frequency Selection:

`cfg_freq_sel` does **not** select a tick period. The tick rate is **fixed at 1 MHz — one tick per microsecond** — and `cfg_freq_sel` indexes a lookup table of *clock frequencies in MHz* so the divider can produce that 1 us tick from whatever clock this instance actually runs on. That is the whole point of the “frequency-invariant” name: a timeout of N is N microseconds on every clock.

The table is built at elaboration from `CFI_MIN_FREQ_MHZ`, `CFI_MAX_FREQ_MHZ`, `CFI_NUM_FREQ_ENTRIES` and `CFI_FREQ_STRATEGY`. With this module’s defaults (5-220 MHz, 16 entries, LINEAR) the entries are $\text{freq}[i] = 5 + (220 - 5) * i / 15$:

<code>cfg_freq_sel</code>	Clock (MHz)	Divisor (cycles/tick)	<code>cfg_freq_sel</code>	Clock (MHz)	Divisor
0	5	5	8	119	119
1	19	19	9	134	134
2	33	33	10	148	148
3	48	48	11	162	162
4	62	62	12	177	177
5	76	76	13	191	191
6	91	91	14	205	205
7	105	105	15	220	220

The divisor **is** the frequency in MHz, because dividing an F-MHz clock by F gives 1 MHz. So the rule for setting it is simply: **pick the entry closest to your actual `aclk` frequency**.

Instantiated through `axi_monitor_base`, the table is flat. The base module defaults `CFI_MIN_FREQ_MHZ` = `CFI_MAX_FREQ_MHZ` = 100, so every entry is 100 and the tick is exactly 1 us at 100 MHz *regardless of `cfg_freq_sel`*. Override the CFI parameters only if the design changes `aclk` at runtime. A monitor running at a clock the table cannot express ticks at the wrong rate, and every timeout scales with it.

Purpose:

- Timeout detection timing base
- Consistent timeout duration across frequencies
- Periodic performance metric reporting

- Threshold comparison timing

Tick Characteristics:

- Single cycle pulse (asserts high for 1 cycle)
- Precise period (no jitter or drift)
- Synchronous to aclk
- Immediately reflects cfg_freq_sel changes

22.4.3 Counter Freq Invariant Integration

The module instantiates counter_freq_invariant with specific configuration:

Instance Configuration:

```
counter_freq_invariant #(
    .COUNTER_WIDTH      (1),                // Only need tick output
    .MIN_FREQ_MHZ       (CFI_MIN_FREQ_MHZ),
    .MAX_FREQ_MHZ       (CFI_MAX_FREQ_MHZ),
    .NUM_FREQ_ENTRIES   (CFI_NUM_FREQ_ENTRIES),
    .FREQ_STRATEGY       (CFI_FREQ_STRATEGY)
) timer_counter (
    .clk                 (aclk),
    .rst_n               (aresetn),
    .sync_reset_n        (1'b1),
    .freq_sel            (cfg_freq_sel),
    .tick                (w_timer_tick),
    .o_counter           ()                 // counter output unused
);
```

Design Rationale:

- COUNTER_WIDTH=1: only the tick pulse is needed, not the counter value
- The CFI_* parameters are forwarded so the frequency table is sized by this module's parameters rather than hard-coded
- sync_reset_n tied high: no need for a synchronous reset
- o_counter left open: only the tick matters (note the o_ prefix — there is no port named counter)

counter_freq_invariant Module:

- Located in rtl/common/counter_freq_invariant.sv
- Divides an arbitrary clock down to a **1 MHz tick**
- freq_sel indexes a table of clock frequencies in MHz; the entry is used directly as the divisor

- Supports LINEAR (uniform) and POW2 (doubling) table spacing
- See rtl/common/ documentation for detailed behavior

22.4.4 Integration with Monitor Architecture

The Timer module provides fundamental timing services to the entire monitor infrastructure.

22.4.4.1 *Output Integration*

To Transaction Manager (axi_monitor_trans_mgr):

- timestamp output provides time reference
- Used to mark transaction phase completion times
- Enables latency calculation in reporter

To Timeout Module (axi_monitor_timeout):

- timer_tick output drives timeout detection
- Timeout module increments counters on each tick
- Enables frequency-invariant timeout thresholds

To Reporter Module (axi_monitor_reporter):

- Indirect use via transaction table timestamps
- Reporter calculates latencies from timestamp differences
- Performance metrics derived from timestamp data

22.4.4.2 *Configuration Strategy*

There is one rule: **set `cfg_freq_sel` to the table entry closest to your actual `aclk` frequency.** The tick is 1 us at that setting, and every timeout threshold is then read directly in microseconds.

aclk	Default table (5-220 MHz, LINEAR)	Resulting tick
100 MHz	<code>cfg_freq_sel = 4'd7</code> (105 MHz — nearest entry)	1.05 us (5% long)
200 MHz	<code>cfg_freq_sel = 4'd14</code> (205 MHz)	1.025 us (2.5% long)
50 MHz	<code>cfg_freq_sel = 4'd3</code> (48 MHz)	0.96 us (4% short)

The tick period is **divisor / f_clk** — the selected entry is a cycle count, so a divisor above your real clock rate makes the tick (and every timeout built on it) run *long*, and one below makes it run *short*. The residual error is the gap between your clock and the nearest table entry; a table built

for one exact frequency (CFI_MIN = CFI_MAX = aclk_mhz, which is what axi_monitor_base does at 100 MHz) has none.

Sizing timeouts. Because the tick is 1 us, the timeout counts are microseconds directly — no cycle arithmetic:

```
cfg_addr_cnt = 100    // 100 us on the command phase
cfg_data_cnt = 500    // 500 us on the data phase
```

The counts are 16-bit, so the range is 1..65535 us (~65 ms), and 16'hFFFF is the conventional “effectively never” setting.

22.5 Timing Characteristics

This module is **sequential**: it contains clocked logic (via always_ff or the repository’s ALWAYS_FF_RST macro) and therefore holds state. Outputs driven from those blocks are registered and appear one clock after the inputs that produced them.

Per-path cycle counts are not enumerated here; read the block that drives the signal you care about. No synthesis frequency or area figures are quoted – none have been measured against a target device.

Timing closure is therefore a question of the surrounding logic’s slack, not of this module’s cycle count. No synthesis figures are quoted; none have been measured.

22.6 Usage Examples

Every parameter and port below is read from the module declaration.

```
axi_monitor_timer #(
    .CFI_MIN_FREQ_MHZ      (5),
    .CFI_MAX_FREQ_MHZ      (220),
    .CFI_NUM_FREQ_ENTRIES  (16),
    .CFI_FREQ_STRATEGY     (0)
) u_axi_monitor_timer (
    .aclk                   (aclk),
    .aresetn                (aresetn),
    .cfg_freq_sel           (cfg_freq_sel),
    .timer_tick             (timer_tick),
    .timestamp              (timestamp)
);
```

22.7 Design Notes

22.7.1 Timestamp Wraparound Handling

The 32-bit timestamp counter wraps to zero after ~4.3 seconds at 1 GHz:

Latency Calculation: Latency calculations naturally handle wraparound through unsigned arithmetic:

```
latency = end_timestamp - start_timestamp; // Works across wraparound
```

Example:

```
start_timestamp = 0xFFFF_FFFF (max value)
end_timestamp   = 0x0000_0005 (after wraparound)
latency = 0x0000_0005 - 0xFFFF_FFFF = 0x0000_0006 (6 cycles)
```

Limitations:

- Maximum measurable latency: 2^{31} cycles (half of counter range)
- Exceeding this causes incorrect results
- In practice, transaction latencies $\ll 2^{31}$ cycles

Mitigation: If longer-term timestamps needed, extend timestamp to 64 bits (requires parameter addition).

22.7.2 Frequency Selection Is Not a Granularity Trade-off

`cfg_freq_sel` is a **clock-matching** control, not a resolution control. Every valid setting produces the same 1 us tick; a setting that does not match `aclk` does not buy finer or coarser timeouts, it simply makes every timeout wrong by the ratio between the selected entry and the real clock. There is no power/precision trade-off to tune here.

Timeout *resolution* is fixed at 1 us. If you need finer resolution than that, the lever is the CFI table (build it for a sub-multiple of the real clock so the tick is shorter), not `cfg_freq_sel`.

22.7.3 Dynamic Frequency Scaling

When the system clock changes at runtime (DFS):

Timestamp counter: keeps counting *cycles*, so latency measurements are in cycles and their absolute-time meaning shifts with the clock.

Timer tick: the divisor is whatever `cfg_freq_sel` selected, so after a clock change the tick is no longer 1 us — it is off by the ratio of the new clock to the selected table entry, and every timeout scales with it.

Recommendation: re-point `cfg_freq_sel` at the table entry matching the new frequency whenever the clock changes; this is exactly the case the CFI table exists for, and it is why the

default table spans 5-220 MHz rather than being built for a single frequency. (`axi_monitor_base` instantiates a flat 100 MHz table by default, which is correct for a fixed-clock design and wrong for a DFS one — override the `CFI_*` parameters there.)

22.7.4 Zero Timestamp on Reset

The timestamp counter resets to zero on `aresetn` assertion:

Implications:

- First transaction after reset has timestamp near zero
- Timestamps from before reset invalid after reset
- No timestamp preservation across resets

Best Practice: Reset all monitors simultaneously to ensure consistent timestamp domain across the system.

22.7.5 Timer Tick Duty Cycle

The `timer_tick` output is a single-cycle pulse:

Characteristics:

- High for exactly 1 cycle
- Low for (`tick_period` - 1) cycles
- Duty cycle = $1 / \text{tick_period}$

Usage: Modules should use `timer_tick` as an enable/trigger signal, not as a clock. It's synchronous to `ack` but much slower, making it unsuitable as a clock source.

22.7.6 Resource Utilization

The timer module has minimal resource cost:

Registers:

- 32-bit timestamp counter
- 1-bit counter in `counter_freq_invariant`
- Prescaler state in `counter_freq_invariant`

Combinational Logic:

- Increment logic for timestamp
- Frequency divider in `counter_freq_invariant`

22.7.7 Multiple Timer Instances

Each monitor instance can have its own timer if needed:

Advantages of Separate Timers:

- Independent frequency selection per monitor
- No timing constraints between monitors
- Simpler physical design (local timing)

Advantages of Shared Timer:

- Single timestamp domain across monitors
- Easier event correlation
- Reduced resource utilization

Recommendation: Use shared timer unless monitors operate at different frequencies or in different clock domains.

22.8 Related Modules

Used by:

- axi_monitor_base.sv (primary user)
- axi_monitor_filtered.sv (via axi_monitor_base)
- All AXI/AXIL monitor variants

Uses:

- counter_freq_invariant.sv (from rtl/common/)
- Provides frequency-invariant counting capability

Critical interfaces:

- **To axi_monitor_trans_mgr:**
 - timestamp output (32-bit time reference)
- **To axi_monitor_timeout:**
 - timer_tick output (timeout trigger)

Related infrastructure:

- axi_monitor_trans_mgr.sv (timestamp consumer)
- axi_monitor_timeout.sv (timer tick consumer)

- [axi_monitor_reporter.sv](#) (indirect via timestamps)
-

22.9 Testing

No dedicated testbench for this module. It has no `val/**/test_axi_monitor_timer.py`. It is exercised indirectly, through the tests of modules that instantiate it (directly or further up):

- `axi4_master_rd_mon` – `val/**/test_axi4_master_rd_mon.py`
- `axi4_master_wr_mon` – `val/**/test_axi4_master_wr_mon.py`
- `axi4_slave_rd_mon` – `val/**/test_axi4_slave_rd_mon.py`
- `axi4_slave_wr_mon` – `val/**/test_axi4_slave_wr_mon.py`
- `axi5_master_rd_mon` – `val/**/test_axi5_master_rd_mon.py`

Indirect coverage exercises this module only in the configurations those parents elaborate. A parameter or mode no parent uses is untested.

Treat any behaviour described on this page as unverified by simulation.

22.10 References

22.10.1 Specifications

- Base Module: [axi_monitor_base.md](#)
- Timeout Module: [axi_monitor_timeout.md](#)
- Transaction Manager: [axi_monitor_trans_mgr.md](#)

22.10.2 Source Code

- RTL: `rtl/amba/monitor/axi_monitor_timer.sv`
- Counter: `rtl/common/counter_freq_invariant.sv`
- Documentation: `docs/markdown/rtl-common/counter_freq_invariant.md`
- Tests: `val/amba/test_axi4_master_rd_mon.py` (integration)

22.10.3 Configuration Guides

- [Monitor Base Configuration](#) - Timeout configuration
 - `docs/markdown/rtl-amba/index.md` - Subsystem requirements
-

Last Updated: 2025-10-24

22.11 Navigation

- [Back to axi_monitor_base](#)
- [Previous: axi_monitor_timeout](#)
- [Back to Shared Infrastructure Index](#)
- [Back to rtl-amba Index](#)

23 AXI Monitor Transaction Manager

Module: `axi_monitor_trans_mgr.sv` **Location:** `rtl/amba/monitor/` **Category:** Core Infrastructure **Status:** Production Ready (CAM-backed revision, 2026-06-08)

23.1 Overview

`axi_monitor_trans_mgr` tracks outstanding AXI transactions through their `addr` → `data` → `resp` lifecycle. It exposes a registered table of in-flight transactions (`trans_table[N]`) for downstream consumers (reporter, debug, timeout) and feeds the monitor bus with state-change events.

This is shared infrastructure — every AXI4 / AXI5 / AXI-Lite monitor uses it internally. You don't instantiate it directly; you configure its behaviour through the top-level monitor wrapper.

The current revision delegates per-transaction keying + storage to the shared `monitor_trans_cam` module. The previous in-place revision, once parked in `mon_temp/`, was deleted in d246a72d along with its equivalence test and the `TRANS_MGR_VARIANT` rollback knob — there is no legacy variant anymore. Don't go looking for the escape hatch; it isn't there.

What it does for you:

- Tracks up to `MAX_TRANSACTIONS` outstanding (default 16)
- 3 independent ID lookups per cycle (`addr` / `data` / `resp`) via CAM
- Out-of-order transaction support
- **Multiple outstanding same-ID transactions**, each in its own slot, with oldest-first (rank-based) attribution of data/response beats
- Same-cycle `AW+W` first-beat capture for write monitors (combinational bypass)
- Command-entry cap implementing the saturation-recovery contract (see [axi_monitor_base](#))
- Burst beat counting

- Per-phase timestamps for latency reporting
 - Orphan-data / orphan-resp detection (per AXI4 / AXI-Lite rules)
 - Timeout-to-terminal-state transition via `i_timeout_detected` feedback
 - State-change events for downstream packet generation
 - Active-transaction counter (exact CAM occupancy, registered pop-count)
 - Cleanup-when-event-reported handshake with the reporter
 - AXI4 / AXI4-Lite / read / write variants via parameters
-

23.2 Parameters

Parameter	Type	Default	Description
MAX_TRANSACTIONS	int	16	Transaction table depth
ADDR_WIDTH	int	32	Width of the address bus tracked. Values above 32 truncate the REPORTED address: <code>bus_transaction_t.addr</code> is a 32-bit field, so allocation stores <code>32' (cmd_addr)</code> and packets carry only the low 32 bits. Tracking itself is unaffected (the CAM keys on ID, not address), and <code>ADDR_WIDTH=64</code> is a supported, tested configuration —

Parameter	Type	Default	Description
ID_WIDTH	int	8	but a 64-bit integrator should not expect full addresses in monitor packets Width of the AXI ID. Hard maximum 8 — bus_transaction_t.id is 8 bits, so a wider key would disagree with the payload and mis-attribute transactions. ID_WIDTH > 8 is refused at elaboration with an \$error, deliberately: there is no safe degraded behaviour
IS_READ	bit	1	1 for read monitors, 0 for write
IS_AXI	bit	1	1 for AXI4, 0 for AXI-Lite
USE_WDATA_ORDER_Q	bit	0	Write monitors: attribute W beats via an AWID FIFO instead of the table-wide state predicate. Required when NUM_BANKS > 1

Parameter	Type	Default	Description
NUM_BANKS	int	1	Generate the CAM this many times, MAX_TRANSACTIONS / NUM_BANKS deep each. Power of 2, must divide the table
ENABLE_PERF_PACKETS	bit	0	Reserved — perf packet generation hook
ADDR_FILTER_ENABLE	bit	0	Address-range packet filter. 0 leaves filtered_mask permanently zero and the build bit-identical. See Address filtering for why this suppresses packets rather than refusing admission

The AW and IW short-alias parameters are retained for API stability with prior revisions; they default to ADDR_WIDTH and ID_WIDTH.

23.3 Ports

The module exports a registered trans_table of bus_transaction_t entries (see monitor_amba4_pkg.sv) plus aggregate status:

```

module axi_monitor_trans_mgr
    import monitor_common_pkg::*;
    import monitor_amba4_pkg::*;
#(
    parameter int MAX_TRANSACTIONS = 16,
    parameter int ADDR_WIDTH       = 32,

```

```

parameter int ID_WIDTH          = 8,
parameter bit IS_READ           = 1'b1,
parameter bit IS_AXI           = 1'b1,
...
) (
    input logic                aclk,
    input logic                aresetn,

    // Synchronous clear: empty the transaction CAM and zero the
    // active-count pipeline on the next edge (no full reset needed).
    // Pulse one cycle while idle.
    input logic                clear,

    // Address channel
    input logic                cmd_valid,
    input logic                cmd_ready,
    input logic [IW-1:0]      cmd_id,
    input logic [AW-1:0]      cmd_addr,
    input logic [7:0]         cmd_len,
    input logic [2:0]         cmd_size,
    input logic [1:0]         cmd_burst,

    // Data channel
    input logic                data_valid,
    input logic                data_ready,
    input logic [IW-1:0]      data_id,
    input logic                data_last,
    input logic [1:0]         data_resp,

    // Response channel (write only)
    input logic                resp_valid,
    input logic                resp_ready,
    input logic [IW-1:0]      resp_id,
    input logic [1:0]         resp_code,

    input logic [31:0]         timestamp,
    input logic [MAX_TRANSACTIONS-1:0] i_event_reported_flags,

    // Timeout feedback from axi_monitor_timeout. The timeout block
    // detects
    // the stall but cannot modify the table; trans_mgr consumes this
    // vector
    // and moves the flagged entry to TRANS_ERROR with the appropriate
    // EVT_*_TIMEOUT code so it becomes cleanup-eligible instead of
    // leaking.
    input logic [MAX_TRANSACTIONS-1:0] i_timeout_detected,

```



```

// Address-range packet filter (active when ADDR_FILTER_ENABLE=1).
// Inclusive [low, high]; a command whose address falls OUTSIDE
the
// window has its packets suppressed. See "Address filtering".
input logic                                cfg_addr_filter_enable,
input logic [AW-1:0]                      cfg_addr_filter_low,
input logic [AW-1:0]                      cfg_addr_filter_high,

// One bit per slot: 1 = suppress this entry's packets. Consumed
by
// axi_monitor_reporter, which also retires filtered entries so
they do
// not leak their slot.
output logic [MAX_TRANSACTIONS-1:0]      filtered_mask,

output bus_transaction_t
trans_table[MAX_TRANSACTIONS],
output logic [7:0]                        active_count
);

```

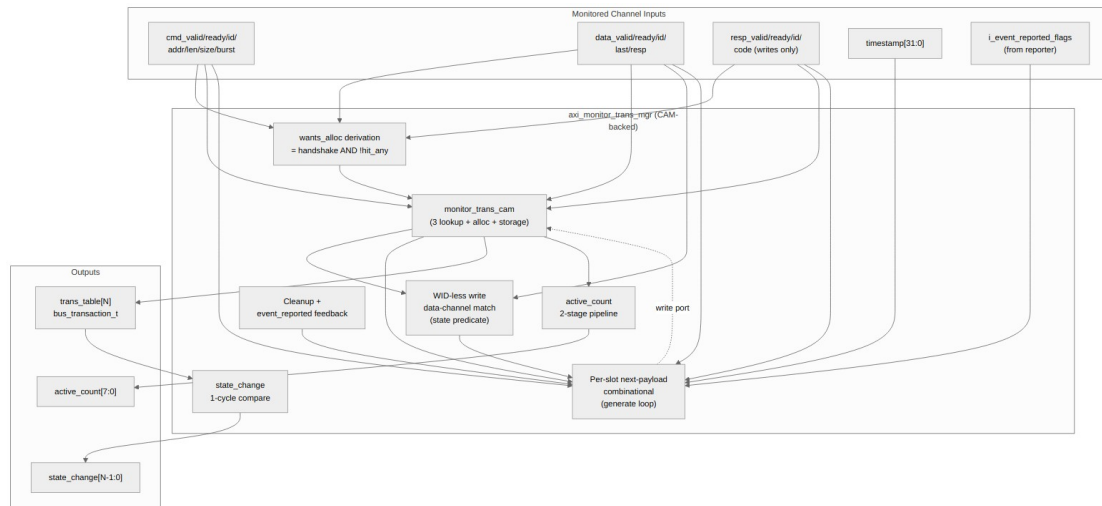
There is no per-slot state-change output, and none is planned: the reporters scan `trans_table` themselves. To act on a slot moving, compare `trans_table[i].state` — which is what `axi_monitor_reporter_debug` does to raise its `PktTypeDebug` packets.

The `bus_transaction_t` struct (defined in `monitor_amba4_pkg.sv`) contains: - State flags: `valid`, `cmd_received`, `data_started`, `data_completed`, `resp_received`, `event_reported`, `eos_seen` - FSM state: `state` (`TRANS_IDLE` / `ADDR_PHASE` / `DATA_PHASE` / `COMPLETE` / `ERROR` / `ORPHANED`) - Captured fields: `addr`, `id`, `len`, `size`, `burst`, `channel` - Timers and timestamps: `addr_timer`, `data_timer`, `resp_timer`, `addr_timestamp`, `data_timestamp`, `resp_timestamp` - Beat tracking: `expected_beats`, `data_beat_count` - event_code union (`axi_error` / `axi_timeout` / etc.)

Total: 285 bits per entry. With `MAX_TRANSACTIONS=16`, the `trans_table` is ~4.5 Kb of registered state.

23.4 Functional Description

23.4.1 Architecture



axi_monitor_trans_mgr block diagram

Source: axi_monitor_trans_mgr.mmd

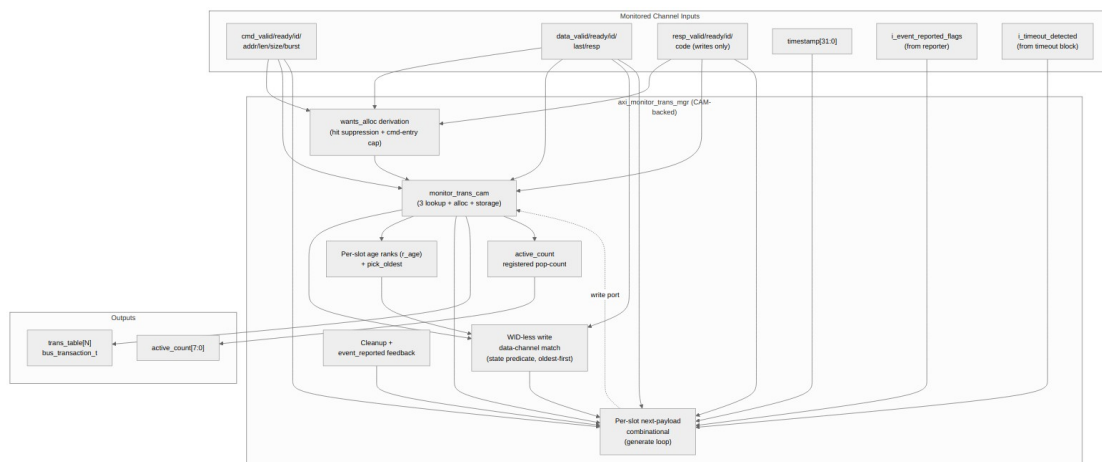


Diagram 6

The split of responsibilities is the whole story here. The trans_mgr owns:

- The **WID-less write data-channel match** (state predicate over the payload, not an id match — the CAM only sees ids), resolved oldest-first.
- The **per-slot age ranks** (r_age) and the **pick_oldest** selector that attribute each data/response beat to the oldest matching entry.

- The per-slot **next-payload computation** (the per-phase if/else chain that says “what should slot i’s bus_transaction_t look like next cycle”).
- The **wants_alloc** derivation (hit suppression + the command-entry cap; see [Allocation and Same-ID Tracking](#)).
- Cleanup eligibility, event_reported feedback, the registered active_count pop-count, and the filtered_mask verdict.

The CAM owns:

- Per-slot (valid, id, payload) storage.
- The 3 parallel ID lookups (addr_match_oh, data_match_oh, resp_match_oh).
- The free-slot vector and 3-way priority-encoded alloc one-hots.

This split makes the parallel-match shape that closes 100 MHz on the xc7a100t-1 ((* keep = "true" *) per-bit match vectors, per-slot generate-loop storage) explicit and reusable.

23.4.2 Transaction Lifecycle

A typical AXI4 read transaction, walked phase by phase:

```

                                addr_alloc fires (free CAM slot picked)
                                valid      = 1
                                state      = TRANS_ADDR_PHASE
cmd_valid handshake            id      = cmd_id
(cmd_id, cmd_addr, ...)       cmd_received= cmd_ready
                                expected_beats = cmd_len + 1
                                addr_timestamp = timestamp

                                addr_update fires (the SAME entry, still
cmd held valid                awaiting its handshake: cmd_received=0)
across stalled cycles         cmd_received <= 1 on the handshake cycle
                                addr_timer   <= 0
                                addr_timestamp <= timestamp

                                data_update fires (oldest matching open
entry)
data_valid handshake           data_started <= 1
(data_id == cmd_id,            data_beat_count++
 data_last, data_resp)         state      <= TRANS_DATA_PHASE
                                if data_last:
                                    data_completed <= 1
                                    state      <= TRANS_COMPLETE
                                if data_resp[1]: # RESP error
                                    state      <= TRANS_ERROR
                                    event_code  <= EVT_RESP_*
```

```

                                (later, reporter handles the event)
                                cleanup fires
event_reported flag    valid <= 0      # slot returned to free
pool
from reporter          # CAM sees the entry as free on next
cycle

```

A second AR/AW with the **same ID** allocates its **own slot** — same-ID outstanding transactions are legal AXI4 and are never merged. Data and response beats are attributed oldest-first (see the next section).

For AXI4 writes the data phase uses a state predicate match (not id), since AXI4 W has no WID. For AXI-Lite writes, the data channel CAN allocate an orphan slot if data arrives before the AW handshake.

A transaction flagged by `i_timeout_detected` (and not already terminal) is moved to `TRANS_ERROR` with an event code derived from its progress: `EVT_CMD_TIMEOUT` if the command never handshook, `EVT_RESP_TIMEOUT` if data completed but the B response is outstanding, else `EVT_DATA_TIMEOUT`. This makes timed-out entries cleanup-eligible instead of leaking their slot.

23.4.3 Allocation and Same-ID Tracking

23.4.3.1 *Separate slot per same-ID transaction*

The pre-CAM design merged a new command into any existing entry with the same ID. That was a **defect** (issue #41 defect 1, fixed): AXI4 permits several outstanding transactions with the same ID, so a second AR/AW must get its own slot. Allocation is now suppressed only for the one legitimate case — a command held valid across several stalled cycles must not allocate a fresh slot every cycle. The actual derivations:

```

// A "pending" entry is the same-ID entry still awaiting its handshake
// (cmd_received=0) and not being freed this cycle.
addr_hit_any      = |w_addr_pend_oh;
addr_wants_alloc  = cmd_valid && !addr_hit_any && w_cmd_headroom;

// Reads: allocate an orphan when a data beat matches no entry.
data_wants_alloc  = data_valid && data_ready && !data_hit_any;
// IS_READ
data_wants_alloc  = data_valid && data_ready && !IS_AXI && !
data_hit_any; // write
resp_wants_alloc  = !IS_READ && resp_valid && resp_ready && !
resp_hit_any;

```

`w_cmd_headroom` is the command-entry cap: command-originated entries may occupy at most `MAX_TRANSACTIONS - cmd_entry_reserve(MAX_TRANSACTIONS)` slots (reserve = 4 for tables of 16+, 0 below). A command seen while the cap is reached is simply not tracked until a command entry retires. This cap is one half of the saturation-recovery contract; the other half is the

block_ready reopen threshold in axi_monitor_base — see the canonical description in [axi_monitor_base](#).

23.4.3.2 *Oldest-first (rank-based) attribution*

Because same-ID entries coexist, an incoming beat can match several slots. AXI4 orders same-ID data/responses by issue order, so each beat belongs to the **oldest** matching entry that is still open. Slot index carries no ordering information (allocation takes the lowest free index, so a recycled low slot can be younger than a live high slot); instead each slot carries a dense rank `r_age[i]` (0 = oldest live entry), and a `pick_oldest()` function selects the lowest-rank candidate. Two tiers: prefer the oldest match whose data/response phase is still open; if every match has already closed, a stray **last** beat falls back to the oldest match (re-running completion — harmless, and it reconstructs untracked bursts under ID oversubscription), while a stray **non-last** beat is absorbed with no update at all (the pre-fix behavior re-opened a terminal entry into an unclosable state — the production saturation-wedge mechanism, guarded by the in-RTL formal property `ap_no_reopened_complete`).

23.4.3.3 *Write-data attribution by AWID FIFO (USE_WDATA_ORDER_Q)*

AXI4 W beats carry no WID, so the entry a beat belongs to cannot come from the beat. With `USE_WDATA_ORDER_Q=1` the manager records the **AWID** on each AW handshake and pops it on W-LAST; the head AWID keys the write-data candidate set, and `pick_oldest()` then resolves among that ID's entries.

This is required whenever the table is banked. `pick_oldest()` compares **same-bank only** — the cross-bank comparators are constant-folded away at elaboration — and that is sound precisely because candidates are ID-matched and all of an ID's entries live in one bank. The legacy write path built its candidates from a state predicate over the *whole* table, which is not ID-matched, so at `NUM_BANKS=B` the select returned one winner **per bank** and a single W beat advanced up to B transactions. `NUM_BANKS > 1` on a write monitor therefore refuses to elaborate without this parameter (`$error`), because the combination has no correct fallback.

Bus requirement — W must not lead AW. Attribution is by AW order, so the AW naming a beat must already have been seen. Same-cycle AW+W is supported (below); W strictly *before* its AW has no AWID to attribute it to and is treated as a stray. This is the restriction commercial VIPs commonly impose.

Regression: `val/amba/test_axi_monitor_trans_mgr_wr_bank.py` (attribution at `NUM_BANKS` 1 and 4, plus the refusal of the illegal combination).

23.4.3.4 *Same-cycle AW+W bypass (write monitors)*

The WID-less write predicate runs over registered entries, so a W beat arriving in the **same cycle** as its AW used to match nothing and be silently lost (the entry then waited for data forever and the B response fabricated an `EVT_PROT0COL` error on legal traffic — routine for single-beat writes from skid-buffered masters). Fixed in 95c9490a: when no registered entry matches, the beat binds combinationally to the slot the command path touches this cycle — either the slot being allocated for this AW (via a local mirror of the CAM's allocation pick, cross-checked by the

ap_bypass_alloc_mirror formal property) or the pending same-ID entry, state-qualified to TRANS_ADDR_PHASE so orphan adoption keeps its legacy behavior. An entry whose AW has not yet handshake is held in TRANS_ADDR_PHASE even after taking the beat, preserving address-phase timeout coverage (formal property ap_wr_data_phase_has_cmd).

23.4.4 Address filtering

ADDR_FILTER_ENABLE adds an address window, cfg_addr_filter_low..cfg_addr_filter_high inclusive, gated by cfg_addr_filter_enable. A command whose address falls **outside** the window sets that slot's bit in filtered_mask, and the reporter suppresses every packet for it.

The verdict is taken **at allocation** and held for the life of the slot. It has to be: the address appears only on the command channel, so once the entry exists there is nothing left to re-test it against.

The filter suppresses packets; it does **not** refuse admission. That looks like the weaker choice and is the correct one. Address exists only on the command channel, so refusing to allocate would leave that transaction's data and response beats arriving with no entry to match. Unmatched data is deliberately ungated — a monitor must never stall returning data — so those beats would be reported as orphan errors, and a filter meant to cut packet traffic would increase it.

Cost of deciding here rather than at admission is MAX_TRANSACTIONS flops (16 at the default). Admission filtering would need a counter per *possible* ID — $2 * ID_WIDTH$ counters, 1280 flops at $ID_WIDTH=8/MAX_TRANSACTIONS=16$ — because one ID can hold several outstanding transactions whose addresses straddle the window, so a flag per ID is not enough.

Because the verdict is latched, widening the window at runtime does not un-filter entries already in the table; only transactions admitted afterwards see the new window. That is what makes the window safe to reprogram live — an entry's fate is fixed when it is admitted, so the retire accounting cannot be corrupted underneath it.

The consumer must retire filtered entries as well as silence them. A slot is freed only once event_reported is set, and the only producer of that flag is an accepted monbus write — so suppressing the packet without retiring the entry leaks the slot. axi_monitor_reporter handles this by folding filtered terminal entries into its auto-retire path.

23.5 Timing Characteristics

Everything downstream of the trans_mgr is registered, so the externally visible latencies are short and fixed:

Metric	Value	Notes
Throughput	1 transaction-event/cycle	Per-phase handshake; up

Metric	Value	Notes
trans_table latency	1 cycle	to 3 phases (addr / data / resp) can act per cycle
active_count latency	1 cycle	Output is registered
Resource	~5 Kb storage	Registered pop-count of CAM occupancy
		16 × 285-bit struct, in the CAM

23.6 Usage Examples

Every parameter and port below is taken from the module declaration.

```
axi_monitor_trans_mgr #(
    .MAX_TRANSACTIONS    (16),
    .ADDR_WIDTH          (32),
    .ID_WIDTH            (8),
    .IS_READ              (1'b1),
    .IS_AXI               (1'b1),
    .ENABLE_PERF_PACKETS (1'b0),
    .USE_WDATA_ORDER_Q   (1'b0),
    .NUM_BANKS           (1),
    .AW                   (ADDR_WIDTH),
    .IW                   (ID_WIDTH)
) u_axi_monitor_trans_mgr (
    .aclk                (aclk),
    .aresetn              (aresetn),
    .clear                (clear),
    .cmd_valid             (cmd_valid),
    .cmd_ready             (cmd_ready),
    .cmd_id                (cmd_id),
    .cmd_addr              (cmd_addr),
    .cmd_len               (cmd_len),
    .cmd_size              (cmd_size),
    .cmd_burst             (cmd_burst),
    .data_valid            (data_valid),
    .data_ready            (data_ready),
    .data_id               (data_id),
    .data_last             (data_last),
    .data_resp             (data_resp),
    .resp_valid            (resp_valid),
    .resp_ready            (resp_ready),
    .resp_id               (resp_id),
```

```

        .resp_code            (resp_code),
        .timestamp            (timestamp),
        .i_event_reported_flags(i_event_reported_flags),
        .i_timeout_detected    (i_timeout_detected),
        .cfg_addr_filter_enable(cfg_addr_filter_enable),
        .cfg_addr_filter_low    (cfg_addr_filter_low),
        .cfg_addr_filter_high    (cfg_addr_filter_high),
        .filtered_mask          (filtered_mask),
        .trans_table            (trans_table),
        .active_count           (active_count)
    );

```

23.7 Design Notes

23.7.1 Synthesis Notes

The CAM-backed revision preserves the 2026-04-23 WNS fix:

Construct	Rationale
Per-slot always_comb for next-payload (in a generate loop)	N independent small cones; synth cannot fuse them across slots
Per-slot CAM storage via monitor_trans_cam (generate-loop always_ff)	Same property at the registered storage layer
(* keep = "true" *) on CAM match vectors	Prevents Vivado from fusing match-result usage into the update cones, which would re-introduce the 12-LUT-level WNS issue
active_count = registered pop-count of cam_entry_valid	Derived directly from live CAM occupancy, not an alloc-minus-cleanup accumulator. The former accumulator could desync from true occupancy and underflow to 0xFF under legal AXI (found by the SymbiYosys proof — see rtl/amba/KNOWN_ISSUES/axi_monitor_active_count_underflow.md); a registered pop-count is structurally bounded to [0, N]. The valid bits are

Construct	Rationale
Synchronous clear zeroes active_count alongside the CAM	already registers, so the adder tree sits cleanly between flops; active_count lags occupancy by 1 cycle, which the block_ready margin absorbs. clear invalidates every CAM slot and zeroes the registered count and age ranks on the same edge, so an empty table is never published with a stale nonzero active_count.

The combined effect is that no signal in the trans_mgr has more than ~6 LUT levels between flops at typical configurations — closes 100 MHz on xc7a100t-1 with margin.

23.7.2 Formal Properties

The saturation-recovery and same-cycle-bypass contracts are encoded as in-RTL formal properties (under `ifdef FORMAL`, flattened into the SymbiYosys proofs and mutation-checked):

Property	Guarantee
ap_no_reopened_complete	No read entry is ever in TRANS_DATA_PHASE with data_completed set — the unclosable “poison” state behind the production saturation wedge is unreachable
ap_wr_data_phase_has_cmd	A write entry in TRANS_DATA_PHASE always has its command handshake recorded (the same-cycle bypass cannot create timeout-coverage holes)
ap_bypass_alloc_mirror	The bypass’s local allocation mirror always agrees with the CAM’s own pick
ap_cmd_entry_cap	Command-originated entries never exceed $N - \text{cmd_entry_reserve}(N)$ (vacuous by design when the reserve is 0, i.e. $N < 16$)

23.8 Related Modules

Module	Role
monitor_trans_cam	Per-slot keying + storage with 3 lookup ports and alloc priority encoder
axi_monitor_base	Top-level monitor wrapper that instantiates trans_mgr + timer + reporter
axi_monitor_reporter	Scans trans_table, gated by filtered_mask, to generate monbus packets
axi_monitor_timeout	Watches the per-phase timers in trans_table for timeout events

Further reading:

- **Monitor Architecture:** docs/markdown/rtl-amba/overview.md
- **Monitor Configuration Guide:** axi_monitor_base.md
- **Packet Format Specification:** monitor_package_spec.md
- **Saturation-recovery contract:** axi_monitor_base.md and monitor_common_pkg::cmd_entry_reserve()

23.9 Testing

Test	Coverage
val/amba/ test_axi_monitor_trans_mgr.py	Directed trans_mgr suite: same-ID slot separation, oldest-first attribution, stray-beat absorption, phase_saturation_recovers (block_ready reopen), timeout-to-terminal transitions
val/amba/ test_monitor_trans_cam.py	The CAM sub-module in isolation
val/amba/ test_axi4_master_rd_mon.py and all *_mon* tests	End-to-end through the full monitor stack

Run the directed suite to validate trans_mgr changes:

```
pytest val/amba/test_axi_monitor_trans_mgr.py -v
```

23.10 Navigation

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24 Monitor Bus Round-Robin Arbiter

Module: `monbus_arbiter.sv` **Location:** `rtl/amba/monitor/` **Status:** Production Ready

24.1 Overview

`monbus_arbiter` merges monitor bus packet streams from multiple clients — AXI monitors, APB monitors, arbiter monitors, whatever you’ve got — into a single output stream, using fair round-robin arbitration with an ACK protocol. Optional skid buffers on the inputs and the output improve timing closure and give you elasticity against backpressure.

The ACK protocol is the point: a granted client gets to finish its packet before the arbiter moves on to the next requester, so multi-cycle transfers never fragment. Combined with the integrated buffering and proper backpressure handling, packets don’t get lost on the way through.

- Round-robin arbitration for monitor bus packet streams
 - ACK mode operation (grants held until acknowledged)
 - Optional input skid buffers per client (2..8 entries)
 - Optional output skid buffer (2..8 entries)
 - 128-bit packet + 64-bit side-band timestamp, carried atomically through a 192-bit skid
 - Parameterizable client count (2-64; 1 does not elaborate)
 - Skid buffers optional (disabling them removes the skid registers; the arbitration cycle itself remains, since `grant_valid` is registered)
-

24.2 Parameters

Parameter	Type	Default	Description
CLIENTS	int	4	Number of monitor bus

Parameter	Type	Default	Description
			clients (2-64). CLIENTS=1 does not elaborate: $N = \lceil \log_2(1) \rceil = 0$ makes grant_id a logic [-1:0] and the priority encoder's generic branch does an illegal i[-1:0] part-select
INPUT_SKID_ENABLE	int	1	Enable skid buffers on input interfaces
OUTPUT_SKID_ENABLE	int	1	Enable skid buffer on output interface
INPUT_SKID_DEPTH	int	2	Depth of input skid buffers (2..8 inclusive)
OUTPUT_SKID_DEPTH	int	2	Depth of output skid buffer (2..8 inclusive)
SKID_DATA_WIDTH	int	MONBUS_PKT_WIDTH + MONBUS_TS_WIDTH	Width of the packed skid-buffer payload.

24.3 Ports

24.3.1 Clock and Reset

Port	Direction	Width	Description
axi_aclk	input	1	Clock signal
axi_aresetn	input	1	Active-low asynchronous

Port	Direction	Width	Description
			reset

24.3.2 Block Arbitration

Port	Direction	Width	Description
block_arb	input	1	Block arbitration when asserted

24.3.3 Monitor Bus Inputs (Array of CLIENTS)

Port	Direction	Width	Description
monbus_valid_in[CLIENTS]	input	CLIENTS	Per-client packet valid signals
monbus_ready_in[CLIENTS]	output	CLIENTS	Per-client ready signals
monbus_packet_in[CLIENTS]	input	CLIENTS × 128	Per-client monitor_packet_t packets
monbus_timestamp_in[CLIENTS]	input	CLIENTS × 64	Per-client monbus_timestamp_t, sampled at each client's emission time

24.3.4 Monitor Bus Output (Aggregated)

Port	Direction	Width	Description
monbus_valid	output	1	Aggregated packet valid
monbus_ready	input	1	Aggregated ready from downstream
monbus_packet	output	128	Aggregated monitor_packet_t
monbus_timestamp	output	64	Aggregated monbus_timestamp_t, paired atomically with monbus_packet

24.3.5 Debug/Status Outputs

Port	Direction	Width	Description
grant_valid	output	1	Grant is active this cycle
grant	output	CLIENTS	One-hot grant vector
grant_id	output	$\$clog2(CLIENTS)$	Binary encoded grant ID
last_grant	output	CLIENTS	Previous grant (for round-robin rotation)

24.4 Functional Description

24.4.1 Arbiter Request and Grant ACK Logic

The module maps monitor bus protocol onto arbiter protocol.

Request Mapping: each client's valid becomes an arbiter request — `request[i] = int_monbus_valid_in[i]`, i.e. the signal *after* the optional input skid, not the port itself.

Grant ACK Logic: ACK occurs only when the beat is actually **consumed** — grant, valid, and downstream ready together:

```
grant_ack[i] = grant[i] && int_monbus_valid_in[i] && int_monbus_ready;
```

The `int_monbus_ready` term is load-bearing and must not be dropped. Without it (`grant[i] && int_monbus_valid_in[i]` alone) the ack fires on every cycle the sink is back-pressuring, so the grant rotates continuously while **zero** transfers take place — breaking the grant-hold contract. That was a real defect; `val/amba/test_monbus_arbiter_grant_hold.py` is the regression that pins it.

24.4.2 Round-Robin Arbiter Instance

Uses `arbiter_round_robin` with `WAIT_GNT_ACK=1`:

- Fair rotation through active requests
- Grant held until acknowledged
- `Block_arb` input for external control

24.4.3 Client Ready Signal Generation

Each client's ready signal asserts when:

1. That client is currently granted AND
 2. Downstream (internal output) is ready to accept data
- ```
int_monbus_ready_in[i] = grant[i] && int_monbus_ready;
```

This ensures only the granted client can transfer data, preventing collisions.

**With the default `INPUT_SKID_ENABLE=1`, this is not what the port does.** The rule above governs the *internal* ready behind the skid buffer. The actual port `monbus_ready_in[i]` is the skid's `wr_ready`, which asserts whenever the skid has room — **independently of any grant**. A client can therefore be accepted while ungranted; the skid holds the beat until its grant comes. Only with `INPUT_SKID_ENABLE=0` do the port and the internal signal coincide.

### 24.4.4 Output Multiplexer

Selects data from the granted client:

```
if (grant_valid)
 int_monbus_packet = int_monbus_packet_in[grant_id]; // post-skid
```

### 24.4.5 Optional Skid Buffers

All skid buffers in this arbiter carry the **packet and timestamp atomically** in a 192-bit payload (`MONBUS_PKT_WIDTH + MONBUS_TS_WIDTH = 128 + 64`). The arbiter never separates a packet from its sampled timestamp, so consumers downstream of the [monbus\\_group family](#) see a coherent (pkt, ts) tuple even after multiple levels of skid.

**Input Skid Buffers** (per client):

- Uses `gaxi_skid_buffer` instances configured for 192-bit data
- Provides elasticity for clients with bursty traffic
- Improves timing closure by breaking long paths
- Depth configurable: 2..8 entries inclusive (odd depths are legal)

**Output Skid Buffer:**

- Buffers aggregated stream before output (also 192-bit)
- Prevents backpressure propagation to arbiter
- Same depth options as input buffers

### When to Enable Skid Buffers:

- Enable INPUT\_SKID if clients have high-latency paths or bursty traffic
  - Enable OUTPUT\_SKID if downstream consumer has variable latency
  - Disable both for minimum latency. That removes the skid registers only; it is NOT a combinational pass-through. `monbus_valid` comes from `arbiter_round_robin's grant_valid`, which is registered, so a request still costs one arbitration cycle. See “Skid-Free Configuration”.
- 

## 24.5 Timing Characteristics

---

This module is **sequential**: it contains clocked logic (via `always_ff` or the repository's `ALWAYS_FF_RST` macro) and therefore holds state. Outputs driven from those blocks are registered and appear one clock after the inputs that produced them.

Per-path cycle counts are not enumerated here; read the block that drives the signal you care about. No synthesis frequency or area figures are quoted – none have been measured against a target device.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

---

## 24.6 Usage Examples

---

```
// Aggregate monitor bus streams from 4 clients
monbus_arbiter #(
 .CLIENTS (4),
 .INPUT_SKID_ENABLE (1), // Enable input buffers
 .OUTPUT_SKID_ENABLE (1), // Enable output buffer
 .INPUT_SKID_DEPTH (4), // 4-entry input buffers
 .OUTPUT_SKID_DEPTH (4) // 4-entry output buffer
) u_monbus_arb (
 .axi_aclk (clk),
 .axi_aresetn (rst_n),
 .block_arb (1'b0), // Not blocked

 // Connect 4 client monitor bus streams (packet + timestamp per
 // client).
 // NOTE the asymmetry: the three INPUT arrays take assignment
 // patterns,
 // which are legal rvalues for an unpacked-array input.
 monbus_ready_in is
 // an OUTPUT, so its connection must be an lvalue -- an assignment
```



```

pattern
 // there does not compile. Declare the array and connect it whole:
 // logic mon_ready [4];
 // assign {mon0_ready, mon1_ready, mon2_ready, mon3_ready} =
 // {mon_ready[0], mon_ready[1], mon_ready[2],
mon_ready[3]};
 .monbus_valid_in ('{mon0_valid, mon1_valid, mon2_valid,
mon3_valid}),
 .monbus_ready_in (mon_ready),
 .monbus_packet_in ('{mon0_packet, mon1_packet, mon2_packet,
mon3_packet}),
 .monbus_timestamp_in ('{mon0_ts, mon1_ts, mon2_ts,
mon3_ts}),

 // Aggregated output stream (packet + timestamp paired atomically)
 .monbus_valid (agg_valid),
 .monbus_ready (agg_ready),
 .monbus_packet (agg_packet),
 .monbus_timestamp (agg_ts),

 // Debug outputs
 .grant_valid (arb_grant_valid),
 .grant (arb_grant_onehot),
 .grant_id (arb_grant_id),
 .last_grant (arb_last_grant)
);

// Downstream consumer is typically the monbus_group family, which
// expects the
// 128-bit packet on monbus_packet and the 64-bit timestamp on
// monbus_timestamp. No additional FIFO is needed at this boundary –
// raw-mode groups have no ingress skid of their own; only the
// compressed build adds one (u_comp_in_skid in monbus_group_core).

```

---

## 24.7 Design Notes

---

### 24.7.1 ACK Mode Operation

The ACK mode is what guarantees clean packet transfer:

**Grant Assertion:** When arbiter selects a client, grant[i] asserts **Client Response:** Client's monbus\_valid\_in[i] must be high **Grant ACK:** grant\_ack[i] = grant[i] && int\_monbus\_valid\_in[i] && int\_monbus\_ready — the ready term is required (see above) **Hold Grant:** Arbiter holds grant[i] until grant\_ack[i] asserts **Next Client:** After ACK, arbiter moves to next requesting client

This prevents:

- Packet fragmentation (grant switching mid-transfer)
- Lost packets (grant removed before client ready)
- Unfair arbitration (clients getting multiple back-to-back grants)

### 24.7.2 Skid Buffer Depth Selection

Choose depth based on system characteristics:

#### 2 Entries (Minimum):

- Minimal buffering for timing closure only
- Use when clients have low, consistent latency

#### 4 Entries (Recommended):

- Good balance of buffering and area
- Handles typical backpressure scenarios
- Recommended for most systems

#### 6-8 Entries:

- High buffering for very bursty traffic
- Use when downstream has high variable latency
- Higher area cost

### 24.7.3 Skid-Free Configuration

For minimum latency:

```
.INPUT_SKID_ENABLE (0),
.OUTPUT_SKID_ENABLE (0)
```

This removes the skid registers, but it does **not** make the valid path combinational. `monbus_valid` is driven from the arbiter's `grant_valid`, which is a REGISTERED output of `arbiter_round_robin`, so a request still takes an arbitration cycle to appear as a grant:

- `monbus_valid_in[i] -> request[i] -> (registered grant) -> monbus_valid`
- `monbus_packet_in[grant_id] -> monbus_packet` (combinational mux on `grant_id`)
- `monbus_ready -> monbus_ready_in[grant_id]` (combinational, gated by `grant`)

Use when:

- Timing is not critical
- Clients and downstream have matched latencies
- Minimum latency required for event ordering

#### 24.7.4 Assertions

The module includes comprehensive assertions:

**Grant One-Hot:** Verifies grant vector is one-hot when valid **Grant ID Consistency:** Verifies grant\_id corresponds to asserted grant bit **Ready Exclusivity:** Verifies only granted client receives ready signal

---

## 24.8 Related Modules

---

Used by:

- System-level monitor bus aggregation hierarchies
- Multi-subsystem monitoring infrastructures

Uses:

- arbiter\_round\_robin.sv (ACK-mode arbiter)
- gaxi\_skid\_buffer.sv (optional input/output buffering)

Related:

- arbiter\_monbus\_common.sv (generates monitor bus packets)
  - arbiter\_rr\_pwm\_monbus.sv (arbiter with integrated monitoring)
- 

## 24.9 Testing

---

The grant-hold defect described above is pinned by `val/amba/test_monbus_arbiter_grant_hold.py` — run it after any change to the ACK or ready logic.

---

## 24.10 References

---

### 24.10.1 Specifications

- Internal: docs/markdown/rtl-amba/index.md (AMBA subsystem requirements)
- Internal: docs/markdown/rtl-amba/includes/monitor\_package\_spec.md (monitor bus protocol)

### 24.10.2 Source Code

- RTL: rtl/amba/monitor/monbus\_arbiter.sv
  - Tests: val/amba/test\_monbus\_arbiter\_grant\_hold.py (if exists)
- 

Last Updated: 2025-10-24

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## 24.11 Navigation

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## 25 MonBus Group — AXI4 / AXI4

**Module:** monbus\_axi4\_axi4\_group.sv **Location:** rtl/amba/monitor/ **Status:** Production Ready

---

### 25.1 Overview

---

monbus\_axi4\_axi4\_group is the AXI4-slave-read + AXI4-master-write wrapper around monbus\_group\_core. It pairs a full AXI4 slave-read leaf (the error/interrupt drain port, supporting burst reads) with a full AXI4 master-write leaf (the bulk-capture port, emitting large write bursts). Both leaves are true AXI4, so their FUB interfaces pass straight through to the core's AXI4-shaped FUB ports; the wrapper only supplies safe constant defaults for the AXI4 sideband fields the core does not produce.

An SoC monitoring subsystem needs to expose captured monitor packets to two AXI4 consumers: a CPU that reads error records over a slave port, and a DMA-style write path that dumps a bulk trace to memory. This wrapper wires the two AXI4 leaf protocol adapters (axi4\_slave\_rd, axi4\_master\_wr) to the shared capture core, giving you a drop-in AXI4/AXI4 capture block.

Use it for:

- AXI4-connected monitoring block where both the error-read and trace-write masters are AXI4
- High-throughput trace capture that benefits from long master-write bursts
- CPU IRQ handler reading error records via AXI4 burst reads

The key benefit: true AXI4 on both sides with zero protocol translation loss — the leaves pass through to the core, so burst geometry, IDs, and full-size bursts are available end to end.

Feature summary:

- Full AXI4 on both the error-drain (slave read) and bulk-capture (master write) sides
  - Burst AR support on the slave-read drain (walk multiple error records per burst)
  - Large write bursts on the master-write side (MAX\_BURST\_BEATS up to 256)
  - Thin structural adapter — all logic lives in monbus\_group\_core
  - Configurable skid depths on all five AXI channels
  - Optional runtime compression (cfg\_compress\_en) with USE\_COMPRESSION
  - Full per-protocol filter mask set (AXI / AXIS / CORE) passed to the core
- 

## 25.2 Parameters

| Parameter        | Type | Default | Description                         |
|------------------|------|---------|-------------------------------------|
| FIFO_DEPTH_ERR   | int  | 64      | Error FIFO depth in 192-bit records |
| FIFO_DEPTH_WRITE | int  | 96      | Write FIFO depth in 64-bit beats    |
| ADDR_WIDTH       | int  | 32      | Address width for both AXI4 ports   |
| AXI_ID_WIDTH_S   | int  | 8       | Slave-read ID width                 |
| AXI_ID_WIDTH_M   | int  | 8       | Master-write ID width               |
| AXI_USER_WIDTH   | int  | 1       | AXI user-signal width               |

| Parameter            | Type | Default | Description                                          |
|----------------------|------|---------|------------------------------------------------------|
| MAX_BURST_BEATS      | int  | 64      | Max beats per master-write sub-burst                 |
| FLUSH_TIMEOUT_CYCLES | int  | 1024    | Cycles before a timeout-triggered flush              |
| NUM_PROTOCOLS        | int  | 3       | Informational only                                   |
| USE_COMPRESSION      | int  | 0       | 0 = raw 3-beat records; 1 = elaborate the compressor |
| SKID_DEPTH_AR        | int  | 2       | Slave-read AR channel skid depth                     |
| SKID_DEPTH_R         | int  | 4       | Slave-read R channel skid depth                      |
| SKID_DEPTH_AW        | int  | 2       | Master-write AW channel skid depth                   |
| SKID_DEPTH_W         | int  | 4       | Master-write W channel skid depth                    |
| SKID_DEPTH_B         | int  | 2       | Master-write B channel skid depth                    |

## 25.3 Ports

### 25.3.1 Clock, Reset, Clear

| Port     | Direction | Width | Description |
|----------|-----------|-------|-------------|
| axi_aclk | input     | 1     | Clock       |

| Port        | Direction | Width | Description                              |
|-------------|-----------|-------|------------------------------------------|
| axi_aresetn | input     | 1     | Active-low asynchronous reset            |
| cam_clear   | input     | 1     | Synchronous compressor CAM + stats clear |

### 25.3.2 MonBus Ingress and Timestamp

| Port             | Direction | Width                   | Description                                   |
|------------------|-----------|-------------------------|-----------------------------------------------|
| monbus_valid     | input     | 1                       | Incoming packet valid                         |
| monbus_ready     | output    | 1                       | Ready to accept                               |
| monbus_packet    | input     | monitor_packet_t (128)  | Monitor packet                                |
| monbus_timestamp | input     | monbus_timestamp_t (64) | Side-band timestamp                           |
| mon_time_out     | output    | monbus_timestamp_t (64) | Free-running counter for wrappers' i_mon_time |

### 25.3.3 S-Side — AXI4 Slave Read (error record drain)

| Port          | Direction | Width          | Description              |
|---------------|-----------|----------------|--------------------------|
| s_axi_arid    | input     | AXI_ID_WIDTH_S | Read address ID          |
| s_axi_araddr  | input     | ADDR_WIDTH     | Read address             |
| s_axi_arlen   | input     | 8              | Burst length (beats – 1) |
| s_axi_arsize  | input     | 3              | Burst size               |
| s_axi_arburst | input     | 2              | Burst type               |
| s_axi_arlock  | input     | 1              | Lock                     |
| s_axi_arcache | input     | 4              | Cache attributes         |

| Port           | Direction | Width          | Description                     |
|----------------|-----------|----------------|---------------------------------|
| s_axi_arprot   | input     | 3              | Protection                      |
| s_axi_arqos    | input     | 4              | QoS                             |
| s_axi_arregion | input     | 4              | Region                          |
| s_axi_aruser   | input     | AXI_USER_WIDTH | User                            |
| s_axi_arvalid  | input     | 1              | AR valid                        |
| s_axi_arready  | output    | 1              | AR ready                        |
| s_axi_rid      | output    | AXI_ID_WIDTH   | Read data ID                    |
| s_axi_rdata    | output    | 64             | Read data (sliced error record) |
| s_axi_rresp    | output    | 2              | Read response                   |
| s_axi_rlast    | output    | 1              | Read last                       |
| s_axi_ruser    | output    | AXI_USER_WIDTH | User (tied to 0)                |
| s_axi_rvalid   | output    | 1              | R valid                         |
| s_axi_rready   | input     | 1              | R ready                         |

#### 25.3.4 M-Side — AXI4 Master Write (bulk trace capture)

| Port          | Direction | Width        | Description              |
|---------------|-----------|--------------|--------------------------|
| m_axi_awid    | output    | AXI_ID_WIDTH | Write address ID         |
| m_axi_awaddr  | output    | ADDR_WIDTH   | Write address            |
| m_axi_awlen   | output    | 8            | Burst length (beats – 1) |
| m_axi_awsz    | output    | 3            | Burst size               |
| m_axi_awburst | output    | 2            | Burst type               |
| m_axi_awlock  | output    | 1            | Lock                     |
| m_axi_awcache | output    | 4            | Cache attributes         |
| m_axi_awprot  | output    | 3            | Protection               |



| Port           | Direction | Width          | Description       |
|----------------|-----------|----------------|-------------------|
| m_axi_awqos    | output    | 4              | QoS               |
| m_axi_awregion | output    | 4              | Region            |
| m_axi_awuser   | output    | AXI_USER_WIDTH | User              |
| m_axi_awvalid  | output    | 1              | AW valid          |
| m_axi_awready  | input     | 1              | AW ready          |
| m_axi_wdata    | output    | 64             | Write data        |
| m_axi_wstrb    | output    | 8              | Write strobes     |
| m_axi_wlast    | output    | 1              | Write last        |
| m_axi_wuser    | output    | AXI_USER_WIDTH | User              |
| m_axi_wvalid   | output    | 1              | W valid           |
| m_axi_wready   | input     | 1              | W ready           |
| m_axi_bid      | input     | AXI_ID_WIDTH_M | Write response ID |
| m_axi_bresp    | input     | 2              | Write response    |
| m_axi_buser    | input     | AXI_USER_WIDTH | User (ignored)    |
| m_axi_bvalid   | input     | 1              | B valid           |
| m_axi_bready   | output    | 1              | B ready           |

### 25.3.5 Status and Egress Configuration

| Port                | Direction | Width      | Description              |
|---------------------|-----------|------------|--------------------------|
| irq_out             | output    | 1          | Error FIFO non-empty     |
| cfg_base_addr       | input     | ADDR_WIDTH | Capture window base      |
| cfg_limit_addr      | input     | ADDR_WIDTH | Capture window limit     |
| cfg_flush_watermark | input     | 16         | Flush-trigger beat count |
| cfg_compress_en     | input     | 1          | Runtime                  |

| Port                                      | Direction | Width   | Description                                                                           |
|-------------------------------------------|-----------|---------|---------------------------------------------------------------------------------------|
|                                           |           |         | compression enable                                                                    |
| cfg_axi_* /<br>cfg_axis_* /<br>cfg_core_* | input     | 16 each | Per-protocol filter masks (drop / err-select / per-event-code); see monbus_group_core |
| err_fifo_full                             | output    | 1       | Error FIFO full                                                                       |
| write_fifo_full                           | output    | 1       | Write FIFO full                                                                       |
| err_fifo_count                            | output    | 16      | Error FIFO occupancy (records)                                                        |
| write_fifo_count                          | output    | 16      | Write FIFO occupancy (beats)                                                          |
| mon_compressor_statistics_*               | output    | 32 each | Compressor statistics (0 in raw mode)                                                 |

## 25.4 Functional Description

### 25.4.1 S-Side Error Drain (AXI4 slave read)

The `axi4_slave_rd` leaf terminates the incoming AXI4 read channel and presents a clean FUB read interface (`fub_axi_ar*` / `fub_axi_r*`) to the core. The core's slave-read slicer serves each 192-bit error record as three 64-bit beats (timestamp, packet-hi, packet-lo). Because the leaf is full AXI4, a single AR burst can request several records at once. The AXI4-only AR sideband fields (`arlock`/`arcache`/`arqos`/`arregion`/`aruser`) are consumed by the leaf and not forwarded to the core; `s_axi_ruser` is tied to 0.

### 25.4.2 M-Side Bulk Capture (AXI4 master write)

The core's master-write FUB drives the `axi4_master_wr` leaf, which emits the full AXI4 write channel. The wrapper hard-wires the AXI4 sideband fields the core does not generate to safe defaults at the leaf's FUB inputs: `awlock=0`, `awcache=4'b0010` (normal non-cacheable bufferable), `awprot=0`, `awqos=0`, `awregion=0`, `awuser=0`, `wuser=0`. `MAX_BURST_BEATS` defaults to 64, so a flush typically drains as one large sub-burst.

### 25.4.3 Shared Egress Configuration

All filtering, FIFO sizing, address-window, watermark, timeout, and compression behavior is set by the config ports and handed straight to `monbus_group_core`. The three per-protocol mask groups (`cfg_axi_*`, `cfg_axis_*`, `cfg_core_*`) select which packet types drop, which route to the error FIFO, and which event codes are masked — identical semantics across all four group wrappers.

---

## 25.5 Timing Characteristics

---

| Skid parameter | Default depth |
|----------------|---------------|
| SKID_DEPTH_AR  | 2 entries     |
| SKID_DEPTH_R   | 4 entries     |
| SKID_DEPTH_AW  | 2 entries     |
| SKID_DEPTH_W   | 4 entries     |
| SKID_DEPTH_B   | 2 entries     |

Each channel traverses one `gaxi_skid_buffer`, which registers both `rd_valid` and its storage. The **1-cycle input-to-output latency therefore applies on every transfer, including the unstalled case** – there is no combinational bypass. Depth buys backpressure absorption, not throughput; full rate is sustained once the pipeline is primed. Legal range is 2..8 inclusive, odd values included.

Clocking: `aclk`, reset `aresetn` (active-low asynchronous).

No synthesis numbers are quoted here. Frequency and area depend on the target device and the parameters you elaborate with; run your own build.

---

## 25.6 Usage Examples

---

```
monbus_axi4_axi4_group #(
 .ADDR_WIDTH (32),
 .AXI_ID_WIDTH_S (8),
 .AXI_ID_WIDTH_M (8),
 .MAX_BURST_BEATS (64),
 .USE_COMPRESSION (0)
) u_mon_group (
 .axi_aclk (aclk),
 .axi_aresetn (aresetn),
 .cam_clear (1'b0),
```

```

.monbus_valid (agg_valid), // from monbus_arbiter
.monbus_ready (agg_ready),
.monbus_packet (agg_packet),
.monbus_timestamp (agg_timestamp),
.mon_time_out (mon_time),

// AXI4 slave-read drain -> CPU
.s_axi_arid
(cpu_arid), .s_axi_araddr(cpu_araddr), .s_axi_arlen(cpu_arlen),
.s_axi_arvalid(cpu_arvalid), .s_axi_arready(cpu_arready),
.s_axi_rid
(cpu_rid), .s_axi_rdata(cpu_rdata), .s_axi_rlast(cpu_rlast),
.s_axi_rvalid(cpu_rvalid), .s_axi_rready(cpu_rready),
// ... AR sideband ...

// AXI4 master-write bulk capture -> memory
.m_axi_awaddr(cap_awaddr), .m_axi_awlen(cap_awlen), .m_axi_awvalid
(cap_awvalid),
.m_axi_awready(cap_awready), .m_axi_wdata(cap_wdata), .m_axi_wlast
(cap_wlast),
.m_axi_wvalid(cap_wvalid), .m_axi_wready(cap_wready),
.m_axi_bvalid(cap_bvalid), .m_axi_bready(cap_bready),
// ... AW sideband ...

.irq_out (mon_irq),
.cfg_base_addr (cap_base),
.cfg_limit_addr (cap_limit),
.cfg_flush_watermark (16'd48),
.cfg_compress_en (1'b0)
// ... per-protocol masks ...
);

```

---

## 25.7 Design Notes

### 25.7.1 Both Sides Are Full AXI4

Unlike the AXIL variants, neither leaf forces MAX\_BURST\_BEATS=1 and neither injects single-beat defaults into the core — the AXI4 IDs and burst lengths flow through. This is the highest-throughput member of the group family.

### 25.7.2 AW Sideband Defaults Live in the Wrapper

The core deliberately does not model awlock/awcache/awqos/awregion/awuser. The wrapper supplies conservative constants at the master-write leaf so downstream interconnect sees a well-formed AXI4 burst.

### 25.7.3 Skid Depths Are Tunable

The five `SKID_DEPTH_*` parameters size the per-channel skid buffers in the leaves; increase them to absorb more backpressure on congested interconnect.

---

## 25.8 Related Modules

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Used by:

- SoC monitoring subsystems requiring an AXI4/AXI4 MonBus capture block

Uses:

- **`monbus_group_core.sv`** — protocol-agnostic capture core (all logic)
- **`axi4_slave_rd.sv`** — AXI4 slave-read leaf (error drain)
- **`axi4_master_wr.sv`** — AXI4 master-write leaf (bulk capture)
- **`monitor_common_pkg`** — packet and timestamp types

See also:

- **`monbus_axi4_axil4_group.sv`** — AXI4 read + AXIL write variant
  - **`monbus_axil4_axi4_group.sv`** — AXIL read + AXI4 write variant
  - **`monbus_axil4_axil4_group.sv`** — AXIL read + AXIL write variant
  - **`monbus_arbiter.sv`** — upstream aggregation
- 

## 25.9 Testing

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**No test coverage.** There is no `val/**/test_monbus_axi4_axi4_group.py`, and no module that instantiates this one has a testbench either, so nothing in the repository exercises it.

Treat any behaviour described on this page as unverified by simulation.

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## 25.10 References

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### 25.10.1 Source Code

- RTL: `rtl/amba/monitor/monbus_axi4_axi4_group.sv`
- Core: `rtl/amba/monitor/monbus_group_core.sv`

### 25.10.2 Documentation

- Architecture: docs/markdown/rtl-amba/shared/README.md
  - Group Core: docs/markdown/rtl-amba/monitor/monbus\_group\_core.md
  - Packet Format:  
docs/markdown/rtl-amba/includes/monitor\_package\_spec.md
- 

**Last Updated:** 2026-07-15

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## 25.11 Navigation

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- [Back to rtl-amba Index](#)

## 26 MonBus Group — AXI4 / AXIL

**Module:** monbus\_axi4\_axil4\_group.sv **Location:** rtl/amba/monitor/ **Status:** Production Ready

---

### 26.1 Overview

---

monbus\_axi4\_axil4\_group is the AXI4-slave-read + AXIL-master-write wrapper around monbus\_group\_core. It pairs a full AXI4 slave-read leaf (the burst-capable error/interrupt drain port) with a single-beat AXI4-Lite master-write leaf (the bulk-capture port). The slave-read FUB passes through to the core's AXI4-shaped read FUB, while the master-write side forces MAX\_BURST\_BEATS=1 so the core emits one beat per AW — matching AXIL's single-beat semantics.

Some systems read error records over a high-throughput AXI4 port but write the bulk trace to a simple AXI4-Lite peripheral bus — a register-file-style capture sink, a low-cost bridge, or a control-plane fabric. This wrapper mixes the two: an axi4\_slave\_rd leaf for burst error reads and an axil4\_master\_wr leaf for single-beat trace writes, both driven by the shared capture core.

Use it for:

- Trace capture to an AXIL-only memory-mapped sink while reading errors over AXI4
- Control-plane monitoring where the write path is a lightweight AXIL bus
- Mixed-fabric SoCs pairing an AXI4 CPU read port with an AXIL capture port

The key benefit: this reuses the exact same capture core as the all-AXI4 variant — only MAX\_BURST\_BEATS and the master-write leaf differ — so behavior and configuration stay identical across the family, with the write side degrading gracefully to single beats.

Feature summary:

- AXI4 burst reads on the error-drain side (walk multiple records per burst)
- AXIL single-beat writes on the bulk-capture side (MAX\_BURST\_BEATS=1)
- Thin structural adapter — all logic lives in monbus\_group\_core
- AXI4-only AR sideband fields dropped at the wrapper boundary
- Master-write ID forced to width 1 (AXIL has no ID)
- Optional runtime compression (cfg\_compress\_en) with USE\_COMPRESSION
- Full per-protocol filter mask set (AXI / AXIS / CORE) passed to the core

## 26.2 Parameters

| Parameter            | Type | Default | Description                               |
|----------------------|------|---------|-------------------------------------------|
| FIFO_DEPTH_ERR       | int  | 64      | Error FIFO depth in 192-bit records       |
| FIFO_DEPTH_WRITE     | int  | 96      | Write FIFO depth in 64-bit beats          |
| ADDR_WIDTH           | int  | 32      | Address width for both ports              |
| AXI_ID_WIDTH         | int  | 8       | Slave-read ID width                       |
| AXI_USER_WIDTH       | int  | 1       | AXI user-signal width (slave-read side)   |
| FLUSH_TIMEOUT_CYCLES | int  | 1024    | Cycles before a timeout-triggered flush   |
| NUM_PROTOCOLS        | int  | 3       | Informational only                        |
| USE_COMPRESSION      | int  | 0       | 0 = raw 3-beat records; 1 = elaborate the |

| Parameter     | Type | Default | Description                                          |
|---------------|------|---------|------------------------------------------------------|
| SKID_DEPTH_AR | int  | 2       | compressor<br>Slave-read AR<br>channel skid<br>depth |
| SKID_DEPTH_R  | int  | 4       | Slave-read R<br>channel skid<br>depth                |
| SKID_DEPTH_AW | int  | 2       | AXIL master-write<br>AW channel skid<br>depth        |
| SKID_DEPTH_W  | int  | 2       | AXIL master-write<br>W channel skid<br>depth         |
| SKID_DEPTH_B  | int  | 2       | AXIL master-write<br>B channel skid<br>depth         |

Internally, the core is instantiated with AXI\_ID\_WIDTH\_M=1, AXI\_ID\_WIDTH\_S=AXI\_ID\_WIDTH, and MAX\_BURST\_BEATS=1.

## 26.3 Ports

### 26.3.1 Clock, Reset, Clear

| Port        | Direction | Width | Description                                       |
|-------------|-----------|-------|---------------------------------------------------|
| axi_aclk    | input     | 1     | Clock                                             |
| axi_aresetn | input     | 1     | Active-low<br>asynchronous<br>reset               |
| cam_clear   | input     | 1     | Synchronous<br>compressor<br>CAM + stats<br>clear |



### 26.3.2 MonBus Ingress and Timestamp

| Port             | Direction | Width                   | Description                                   |
|------------------|-----------|-------------------------|-----------------------------------------------|
| monbus_valid     | input     | 1                       | Incoming packet valid                         |
| monbus_ready     | output    | 1                       | Ready to accept                               |
| monbus_packet    | input     | monitor_packet_t (128)  | Monitor packet                                |
| monbus_timestamp | input     | monbus_timestamp_t (64) | Side-band timestamp                           |
| mon_time_out     | output    | monbus_timestamp_t (64) | Free-running counter for wrappers' i_mon_time |

### 26.3.3 S-Side — AXI4 Slave Read (error record drain)

| Port           | Direction | Width          | Description              |
|----------------|-----------|----------------|--------------------------|
| s_axi_arid     | input     | AXI_ID_WIDTH   | Read address ID          |
| s_axi_araddr   | input     | ADDR_WIDTH     | Read address             |
| s_axi_arlen    | input     | 8              | Burst length (beats – 1) |
| s_axi_arsize   | input     | 3              | Burst size               |
| s_axi_arburst  | input     | 2              | Burst type               |
| s_axi_arlock   | input     | 1              | Lock                     |
| s_axi_arcache  | input     | 4              | Cache attributes         |
| s_axi_arprot   | input     | 3              | Protection               |
| s_axi_arqos    | input     | 4              | QoS                      |
| s_axi_arregion | input     | 4              | Region                   |
| s_axi_aruser   | input     | AXI_USER_WIDTH | User                     |
| s_axi_arvalid  | input     | 1              | AR valid                 |
| s_axi_arready  | output    | 1              | AR ready                 |

| Port         | Direction | Width          | Description                        |
|--------------|-----------|----------------|------------------------------------|
| s_axi_rid    | output    | AXI_ID_WIDTH   | Read data ID                       |
| s_axi_rdata  | output    | 64             | Read data<br>(sliced error record) |
| s_axi_rresp  | output    | 2              | Read response                      |
| s_axi_rlast  | output    | 1              | Read last                          |
| s_axi_ruser  | output    | AXI_USER_WIDTH | User (tied to 0)                   |
| s_axi_rvalid | output    | 1              | R valid                            |
| s_axi_rready | input     | 1              | R ready                            |

#### 26.3.4 M-Side — AXIL Master Write (bulk trace capture)

| Port           | Direction | Width      | Description    |
|----------------|-----------|------------|----------------|
| m_axil_awvalid | output    | 1          | AW valid       |
| m_axil_awready | input     | 1          | AW ready       |
| m_axil_awaddr  | output    | ADDR_WIDTH | Write address  |
| m_axil_awprot  | output    | 3          | Protection     |
| m_axil_wvalid  | output    | 1          | W valid        |
| m_axil_wready  | input     | 1          | W ready        |
| m_axil_wdata   | output    | 64         | Write data     |
| m_axil_wstrb   | output    | 8          | Write strobes  |
| m_axil_bvalid  | input     | 1          | B valid        |
| m_axil_bready  | output    | 1          | B ready        |
| m_axil_bresp   | input     | 2          | Write response |

#### 26.3.5 Status and Egress Configuration

| Port          | Direction | Width      | Description          |
|---------------|-----------|------------|----------------------|
| irq_out       | output    | 1          | Error FIFO non-empty |
| cfg_base_addr | input     | ADDR_WIDTH | Capture window base  |

| Port                                      | Direction | Width      | Description                                                                           |
|-------------------------------------------|-----------|------------|---------------------------------------------------------------------------------------|
| cfg_limit_addr                            | input     | ADDR_WIDTH | Capture window limit                                                                  |
| cfg_flush_watermark                       | input     | 16         | Flush-trigger beat count                                                              |
| cfg_compress_en                           | input     | 1          | Runtime compression enable                                                            |
| cfg_axi_* /<br>cfg_axis_* /<br>cfg_core_* | input     | 16 each    | Per-protocol filter masks (drop / err-select / per-event-code); see monbus_group_core |
| err_fifo_full                             | output    | 1          | Error FIFO full                                                                       |
| write_fifo_full                           | output    | 1          | Write FIFO full                                                                       |
| err_fifo_count                            | output    | 16         | Error FIFO occupancy (records)                                                        |
| write_fifo_count                          | output    | 16         | Write FIFO occupancy (beats)                                                          |
| mon_compressor_statistics_*               | output    | 32 each    | Compressor statistics (0 in raw mode)                                                 |

## 26.4 Functional Description

### 26.4.1 S-Side Error Drain (AXI4 slave read)

Identical to the AXI4/AXI4 variant on the read side: the `axi4_slave_rd` leaf terminates the AXI4 read channel and hands a clean read FUB to the core, whose slicer serves each 192-bit error record as three 64-bit beats. Full AXI4 bursts can request multiple records at once. The AXI4-only AR sideband fields (`arlock/arcache/arqos/arregion/aruser`) are dropped at the wrapper boundary; `s_axi_ruser` is tied to 0.

## 26.4.2 M-Side Bulk Capture (AXIL master write)

The core is parameterized with `MAX_BURST_BEATS=1`, so its master-write burst writer emits one AW per beat — the single-beat pattern AXIL requires. The core's AXI4-shaped master-write FUB is bridged to the `axil4_master_wr` leaf: `awaddr`, `awvalid/awready`, `wdata`, `wstrb`, `wvalid/wready`, and `bresp/bvalid/bready` connect through, while the AXI4-only FUB outputs the core still drives (`awid`, `awlen`, `awsz`, `awburst`, `wlast`) are captured in unused nets and discarded. The AXIL leaf drives `awprot=0`, and the core's `bid` input is tied to 0.

## 26.4.3 Shared Egress Configuration

Filtering, FIFO sizing, address-window, watermark, timeout, and compression are all handled by `monbus_group_core` via the same config ports as every other group wrapper. The three per-protocol mask groups (`cfg_axi_*`, `cfg_axis_*`, `cfg_core_*`) behave identically here.

---

## 26.5 Timing Characteristics

---

| Skid parameter             | Default depth |
|----------------------------|---------------|
| <code>SKID_DEPTH_AR</code> | 2 entries     |
| <code>SKID_DEPTH_R</code>  | 4 entries     |
| <code>SKID_DEPTH_AW</code> | 2 entries     |
| <code>SKID_DEPTH_W</code>  | 2 entries     |
| <code>SKID_DEPTH_B</code>  | 2 entries     |

Each channel traverses one `gaxi_skid_buffer`, which registers both `rd_valid` and its storage. The **1-cycle input-to-output latency therefore applies on every transfer, including the un stalled case** – there is no combinational bypass. Depth buys backpressure absorption, not throughput; full rate is sustained once the pipeline is primed. Legal range is 2..8 inclusive, odd values included.

Clocking: `aclk`, reset `aresetn` (active-low asynchronous).

No synthesis numbers are quoted here. Frequency and area depend on the target device and the parameters you elaborate with; run your own build.

---

## 26.6 Usage Examples

---

```
monbus_axi4_axil4_group #(
 .ADDR_WIDTH (32),
 .AXI_ID_WIDTH (8),
 .USE_COMPRESSION (0)
```

```

) u_mon_group (
 .axi_aclk (aclk),
 .axi_aresetn (aresetn),
 .cam_clear (1'b0),

 .monbus_valid (agg_valid),
 .monbus_ready (agg_ready),
 .monbus_packet (agg_packet),
 .monbus_timestamp (agg_timestamp),
 .mon_time_out (mon_time),

 // AXI4 slave-read drain -> CPU
 .s_axi_arid(cpu_arid), .s_axi_araddr(cpu_araddr), .s_axi_arlen(cpu
_arlen),
 .s_axi_arvalid(cpu_arvalid), .s_axi_arready(cpu_arready),
 .s_axi_rdata(cpu_rdata), .s_axi_rlast(cpu_rlast),
 .s_axi_rvalid(cpu_rvalid), .s_axi_rready(cpu_rready),
 // ... AR sideband ...

 // AXIL master-write bulk capture -> peripheral sink
 .m_axil_awaddr(cap_awaddr), .m_axil_awvalid(cap_awvalid), .m_axil_
awready(cap_awready),
 .m_axil_wdata(cap_wdata), .m_axil_wstrb(cap_wstrb),
 .m_axil_wvalid(cap_wvalid), .m_axil_wready(cap_wready),
 .m_axil_bvalid(cap_bvalid), .m_axil_bready(cap_bready), .m_axil_br
esp(cap_bresp),

 .irq_out (mon_irq),
 .cfg_base_addr (cap_base),
 .cfg_limit_addr (cap_limit),
 .cfg_flush_watermark (16'd48),
 .cfg_compress_en (1'b0)
 // ... per-protocol masks ...
);

```

---

## 26.7 Design Notes

### 26.7.1 AXIL Forces Single-Beat Writes

The only functional difference from the AXI4/AXI4 variant is MAX\_BURST\_BEATS=1 and the AXIL master-write leaf. Every flush emits one beat per AW handshake, so throughput on the capture side is lower than the AXI4/AXI4 group — appropriate for a lightweight AXIL sink.

### 26.7.2 Core Still Drives AXI4 FUB Fields

Because the core is AXI4-shaped internally, it always drives `awlen/awsize/awburst/awid/wlast`. On the AXIL side these are simply not routed to the leaf (captured in `_unused` nets). This is intentional and keeps the core single-source.

### 26.7.3 Read Side Is Unchanged

The error-drain path is byte-for-byte the same as the AXI4/AXI4 group — the record slicer, burst AR handling, and IRQ behavior are all in the shared core.

---

## 26.8 Related Modules

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Used by:

- SoC monitoring subsystems reading errors over AXI4 but capturing the trace over AXIL

Uses:

- **monbus\_group\_core.sv** — protocol-agnostic capture core (all logic), `MAX_BURST_BEATS=1`
- **axi4\_slave\_rd.sv** — AXI4 slave-read leaf (error drain)
- **axil4\_master\_wr.sv** — AXIL master-write leaf (bulk capture)
- **monitor\_common\_pkg** — packet and timestamp types

See also:

- **monbus\_axi4\_axi4\_group.sv** — AXI4 read + AXI4 write variant
  - **monbus\_axil4\_axi4\_group.sv** — AXIL read + AXI4 write variant
  - **monbus\_axil4\_axil4\_group.sv** — AXIL read + AXIL write variant
  - **monbus\_arbiter.sv** — upstream aggregation
- 

## 26.9 Testing

---

`val/amba/test_monbus_axi4_axil4_group.py` exercises this module. It collects 1 parameter cases at the default `REG_LEVEL`.

```
source env_python
pytest val/amba/test_monbus_axi4_axil4_group.py -v
```

---

## 26.10 References

---

### 26.10.1 Source Code

- RTL: rtl/amba/monitor/monbus\_axi4\_axil4\_group.sv
- Core: rtl/amba/monitor/monbus\_group\_core.sv

### 26.10.2 Documentation

- Architecture: docs/markdown/rtl-amba/shared/README.md
  - Group Core: docs/markdown/rtl-amba/monitor/monbus\_group\_core.md
  - Packet Format:  
docs/markdown/rtl-amba/includes/monitor\_package\_spec.md
- 

**Last Updated:** 2026-07-15

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## 26.11 Navigation

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## 27 Monitor Bus Group — AXIL Slave-Read / AXI4 Master-Write

**Module:** monbus\_axil4\_axi4\_group.sv **Location:** rtl/amba/monitor/ **Status:** Production Ready

---

### 27.1 Overview

---

monbus\_axil4\_axi4\_group is the mixed-fabric wrapper of the monitor-bus delivery family. It wraps the protocol-agnostic monbus\_group\_core with an **AXIL slave-read err-drain port** (the CPU IRQ handler pops error records) and a **full AXI4 burst master-write port** (streams captured records into a memory ring, bunching multiple records into multi-beat bursts to amortize address-channel overhead). It is the delivery block stream\_char / rapids\_char instantiate for MonBus egress when the drain fabric is AXIL but the memory ring lives behind an AXI4 fabric.

Like the AXIL/AXIL variant, this wrapper splits the arbitrated monitor-bus stream into a CPU-facing error drain and a memory-ring bulk dump. The difference is the dump side: a full AXI4

burst master lets the burst writer emit many records in a single  $AW + N \times W + B$ , which is throughput-optimal when the ring sits behind an AXI4 fabric.

The core's FUBs are AXI4-shaped internally. On the read side the wrapper bridges the AXIL leaf into the core's AXI4-shaped read FUB (`arid=0` / `arlen=0` / `arsize=3` / `arburst=INCR`). On the write side the core drives most AXI4 fields directly; the fields the core does not produce (`awlock` / `awcache` / `awqos` / `awregion` / `awuser` / `wuser`) are tied to safe defaults at the AXI4 leaf boundary.

Use it for:

- MonBus egress for `stream_char` / `rapids_char` with an AXIL drain and an AXI4 ring fabric
- CPU IRQ error drain plus high-throughput burst dump into memory
- Trace capture where amortizing AXI4 address-channel overhead matters

The key benefit: a lightweight AXIL CPU drain paired with a full AXI4 burst dump, so trace records land in memory with minimal address-channel overhead.

Feature summary:

- AXIL slave-read err-FIFO drain (CPU-facing IRQ path)
- AXI4 burst master-write bulk-capture (`MAX_BURST_BEATS` up to 256, default 64)
- Selectable err-drain data width: 64-bit (one beat per record slice) or 32-bit (2:1 read serializer for a 32-bit host crossbar)
- Shared per-protocol egress packet filter (AXI / AXIS / CORE)
- `irq_out` asserted whenever the err FIFO is non-empty
- Optional runtime compression (`USE_COMPRESSION` + `cfg_compress_en`)
- Compressor statistics and FIFO status/count outputs

## 27.2 Parameters

| Parameter                                      | Type | Default | Description                                            |
|------------------------------------------------|------|---------|--------------------------------------------------------|
| <code>FIFO_DEPTH_ERR</code>                    | int  | 64      | Error FIFO depth, in records                           |
| <code>FIFO_DEPTH_WRI</code><br><code>TE</code> | int  | 96      | Write FIFO depth, in beats (96 beats = 32 raw records) |
| <code>ADDR_WIDTH</code>                        | int  | 32      | Address width on                                       |



| Parameter            | Type | Default | Description                                                                                                                                                                          |
|----------------------|------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                      |      |         | the slave-read and master-write ports                                                                                                                                                |
| S_AXIL_DATA_WIDTH    | int  | 64      | Err-drain read data width. 64 = one beat per 64-bit record slice; 32 = a 2:1 read serializer presents each slice as a low then high beat (6 beats/record) for a 32-bit host crossbar |
| AXI_ID_WIDTH         | int  | 8       | Master-write AXI4 ID width                                                                                                                                                           |
| AXI_USER_WIDTH       | int  | 1       | Master-write AXI4 USER width                                                                                                                                                         |
| MAX_BURST_BEATS      | int  | 64      | Maximum beats per master-write AW (AXI4 protocol max is 256)                                                                                                                         |
| FLUSH_TIMEOUT_CYCLES | int  | 1024    | Cycles since last accepted W handshake before a timeout-driven flush                                                                                                                 |
| NUM_PROTOCOLS        | int  | 3       | Informational — AXI / AXIS / CORE filter configs are unconditional                                                                                                                   |
| USE_COMPRESSION      | int  | 0       | Elaboration: 0 omits the compressor (raw-                                                                                                                                            |

| Parameter     | Type | Default | Description                                                                   |
|---------------|------|---------|-------------------------------------------------------------------------------|
|               |      |         | only); 1 elaborates it for runtime selection via <code>cfg_compress_en</code> |
| SKID_DEPTH_AR | int  | 2       | Slave-read AR skid depth                                                      |
| SKID_DEPTH_R  | int  | 4       | Slave-read R skid depth                                                       |
| SKID_DEPTH_AW | int  | 2       | Master-write AW skid depth                                                    |
| SKID_DEPTH_W  | int  | 4       | Master-write W skid depth (deeper to absorb burst W)                          |
| SKID_DEPTH_B  | int  | 2       | Master-write B skid depth                                                     |

## 27.3 Ports

### 27.3.1 Clock, Reset, and CAM Clear

| Port        | Direction | Width | Description                                                                           |
|-------------|-----------|-------|---------------------------------------------------------------------------------------|
| axi_aclk    | input     | 1     | Clock                                                                                 |
| axi_aresetn | input     | 1     | Active-low asynchronous reset                                                         |
| cam_clear   | input     | 1     | Synchronous clear of the compressor CAM + stat counters (tied off in raw-only builds) |

### 27.3.2 Monitor-Bus Input + Timestamp

| Port             | Direction | Width              | Description                                     |
|------------------|-----------|--------------------|-------------------------------------------------|
| monbus_valid     | input     | 1                  | Monitor-bus packet valid                        |
| monbus_ready     | output    | 1                  | Monitor-bus ready (backpressure to the arbiter) |
| monbus_packet    | input     | monitor_packet_t   | 128-bit monitor packet                          |
| monbus_timestamp | input     | monbus_timestamp_t | 64-bit source timestamp                         |
| mon_time_out     | output    | monbus_timestamp_t | Free-running 64-bit timestamp counter           |

### 27.3.3 AXIL Slave Read — Err-FIFO Drain

| Port           | Direction | Width             | Description                                            |
|----------------|-----------|-------------------|--------------------------------------------------------|
| s_axil_arvalid | input     | 1                 | Read-address valid                                     |
| s_axil_arready | output    | 1                 | Read-address ready                                     |
| s_axil_araddr  | input     | ADDR_WIDTH        | Read address                                           |
| s_axil_arprot  | input     | 3                 | Read protection attributes                             |
| s_axil_rvalid  | output    | 1                 | Read-data valid                                        |
| s_axil_rready  | input     | 1                 | Read-data ready                                        |
| s_axil_rdata   | output    | S_AXIL_DATA_WIDTH | Read data (record slice, or half-slice in 32-bit mode) |
| s_axil_rresp   | output    | 2                 | Read response                                          |

### 27.3.4 AXI4 Master Write — Burst Bulk Capture

| Port       | Direction | Width        | Description      |
|------------|-----------|--------------|------------------|
| m_axi_awid | output    | AXI_ID_WIDTH | Write-address ID |

| Port           | Direction | Width          | Description                                          |
|----------------|-----------|----------------|------------------------------------------------------|
| m_axi_awaddr   | output    | ADDR_WIDTH     | Write burst base address (within the ring window)    |
| m_axi_awlen    | output    | 8              | Burst length minus 1<br>(sub_burst_beats - 1)        |
| m_axi_awsz     | output    | 3              | Beat size (fixed at 3 = 8 bytes)                     |
| m_axi_awburst  | output    | 2              | Burst type (fixed INCR)                              |
| m_axi_awlock   | output    | 1              | Lock (tied 0 at the leaf)                            |
| m_axi_awcache  | output    | 4              | Cache (tied 4'b0010 Normal Non-cacheable Bufferable) |
| m_axi_awprot   | output    | 3              | Protection (tied 0)                                  |
| m_axi_awqos    | output    | 4              | QoS (tied 0)                                         |
| m_axi_awregion | output    | 4              | Region (tied 0)                                      |
| m_axi_awuser   | output    | AXI_USER_WIDTH | User (tied 0)                                        |
| m_axi_awvalid  | output    | 1              | Write-address valid                                  |
| m_axi_awready  | input     | 1              | Write-address ready                                  |
| m_axi_wdata    | output    | 64             | Write data (one 64-bit beat)                         |
| m_axi_wstrb    | output    | 8              | Write strobe                                         |
| m_axi_wlast    | output    | 1              | Last beat of the sub-burst                           |
| m_axi_wuser    | output    | AXI_USER_WIDTH | Write-data user (tied 0)                             |

| Port         | Direction | Width          | Description                  |
|--------------|-----------|----------------|------------------------------|
| m_axi_wvalid | output    | 1              | Write-data valid             |
| m_axi_wready | input     | 1              | Write-data ready             |
| m_axi_bid    | input     | AXI_ID_WIDTH   | Write-response ID            |
| m_axi_bresp  | input     | 2              | Write response               |
| m_axi_buser  | input     | AXI_USER_WIDTH | Write-response user (unused) |
| m_axi_bvalid | input     | 1              | Write-response valid         |
| m_axi_bready | output    | 1              | Write-response ready         |

### 27.3.5 Interrupt

| Port    | Direction | Width | Description                                 |
|---------|-----------|-------|---------------------------------------------|
| irq_out | output    | 1     | Asserted whenever the err FIFO is non-empty |

### 27.3.6 Configuration

| Port                | Direction | Width      | Description                                                       |
|---------------------|-----------|------------|-------------------------------------------------------------------|
| cfg_base_addr       | input     | ADDR_WIDTH | Master-write ring base address                                    |
| cfg_limit_addr      | input     | ADDR_WIDTH | Master-write ring limit address (writer wraps, does not saturate) |
| cfg_flush_watermark | input     | 16         | Write-FIFO depth (beats) at which a flush fires                   |
| cfg_compress_en     | input     | 1          | Runtime compression enable (effective only when USE_COMPRESSION=  |

| Port | Direction | Width | Description |
|------|-----------|-------|-------------|
| 1)   |           |       |             |

### 27.3.7 Per-Protocol Filter Masks

Three protocols (AXI, AXIS, CORE), each with a packet-type mask, an err-select mask, and per-event-category masks. All are 16-bit inputs — same shape as the AXIL/AXIL variant.

| Port                                                                                                                                                                | Protocol | Role                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------|
| cfg_axi_pkt_mask                                                                                                                                                    | AXI      | Drop by packet type              |
| cfg_axi_err_select                                                                                                                                                  | AXI      | Route to err FIFO by packet type |
| cfg_axi_error_mask /<br>cfg_axi_timeout_mask /<br>cfg_axi_compl_mask /<br>cfg_axi_thresh_mask /<br>cfg_axi_perf_mask /<br>cfg_axi_addr_mask /<br>cfg_axi_debug_mask | AXI      | Per-event-code drop masks        |
| cfg_axis_pkt_mask                                                                                                                                                   | AXIS     | Drop by packet type              |
| cfg_axis_err_select                                                                                                                                                 | AXIS     | Route to err FIFO by packet type |
| cfg_axis_error_mask /<br>cfg_axis_timeout_mask /<br>cfg_axis_compl_mask /<br>cfg_axis_credit_mask /<br>cfg_axis_channel_mask /<br>cfg_axis_stream_mask              | AXIS     | Per-event-code drop masks        |
| cfg_core_pkt_mask                                                                                                                                                   | CORE     | Drop by packet type              |
| cfg_core_err_select                                                                                                                                                 | CORE     | Route to err FIFO by packet type |
| cfg_core_error_mask /<br>cfg_core_timeout_mask /<br>cfg_core_compl_mask /<br>cfg_core_thresh_mask /<br>cfg_core_perf_mask /<br>cfg_core_debug_mask                  | CORE     | Per-event-code drop masks        |

### 27.3.8 Status / Debug

| Port                                                                                                                                    | Direction | Width   | Description                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------|-----------|---------|----------------------------------------------------------|
| err_fifo_full                                                                                                                           | output    | 1       | Err FIFO write port not ready                            |
| write_fifo_full                                                                                                                         | output    | 1       | Write FIFO write port not ready                          |
| err_fifo_count                                                                                                                          | output    | 16      | Err FIFO entry count (records)                           |
| write_fifo_count                                                                                                                        | output    | 16      | Write FIFO entry count (beats)                           |
| mon_compressor_stat_tier1_a /<br>_tier1_b / _tier1_c /<br>_tier0 / _cam_miss<br>/ _delta_ts_ovf /<br>_event_data_ovf /<br>_ed_delta_ovf | output    | 32 each | Compressor statistics (live only when USE_COMPRESSION=1) |

## 27.4 Functional Description

### 27.4.1 S-Side Err-Drain Read Protocol

Each monbus err record is three 64-bit slices — {tag, source\_ts}, packet[127:64], packet[63:0]. A 64-bit axil4\_slave\_rd leaf bridges the external AXIL read onto the core's 64-bit read FUB, and the core's slice counter returns the three slices in sequence; the err-FIFO record is popped only after the packet-low slice is read.

- **64-bit drain (S\_AXIL\_DATA\_WIDTH == 64):** the leaf is wired straight through — 3 beats/record.
- **32-bit drain (S\_AXIL\_DATA\_WIDTH == 32):** a phase bit splits each 64-bit leaf beat into a low then high 32-bit external read — 6 beats/record. The AR is forwarded to the leaf only on the low phase (one core read per pair, no prefetch); the low read consumes the beat and latches its high half; the high read replays the latch, gating R on its own accepted AR (r\_hi\_ar) to honor AXIL AR-before-R ordering. This is the identical mechanism used in monbus\_axil4\_axil4\_group.

## 27.4.2 M-Side Bulk-Capture Protocol (AXI4 Burst)

Surviving (non-err) packets go to the write FIFO and are streamed out the AXI4 master-write port by the core's burst writer. Because this is a full AXI4 master, one drain cycle typically becomes one large sub-burst — e.g. 24 beats = 8 raw records in a single AW +  $24 \times W + B$  — capped per AW by MAX\_BURST\_BEATS. awsize is fixed at 3 (8 bytes/beat), awburst at INCR, awlen = sub\_burst\_beats - 1, and wlast asserts on the final beat of each sub-burst. The address is **circular** within [cfg\_base\_addr, cfg\_limit\_addr] (wraps, does not saturate). The writer fires on watermark or timeout. See monbus\_group for the full burst-writer geometry (3-stage pipeline, mod-3 rounding, 4KB-boundary respect, rewind-snap).

The core drives awid / awaddr / awlen / awsize / awburst and wdata / wstrb / wlast / wvalid. The AXI4 fields the core does not produce are tied to safe defaults at the leaf: awlock=0, awcache=4'b0010 (Normal Non-cacheable Bufferable), awprot=0, awqos=0, awregion=0, awuser=0, wuser=0.

## 27.4.3 Interrupt

irq\_out is asserted by the core whenever the err FIFO is non-empty.

## 27.4.4 Shared Egress Packet Filter

The per-protocol filter (AXI, AXIS, CORE) decides for each packet: drop, route to the err FIFO (err\_select), or route to the write FIFO (default). Protocols not in the supported set (APB, ARB) are always dropped. The filter config is identical across all four family wrappers — see monbus\_group for the per-event category tables and masking rules.

## 27.4.5 Compression

When USE\_COMPRESSION=1, monbus\_compressor is selectable at runtime via cfg\_compress\_en, emitting a 64-bit self-tagged slot per COMPRESSED record instead of three raw beats (an incompressible record escapes to tier-0 and still costs three); cam\_clear synchronously empties the template CAM and zeroes stats between runs. In raw-only builds these fold away. (Unlike the AXIL/AXIL variant, this wrapper does not expose HALF\_BEAT\_EN.)

---

## 27.5 Timing Characteristics

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| Skid parameter | Default depth |
|----------------|---------------|
| SKID_DEPTH_AR  | 2 entries     |
| SKID_DEPTH_R   | 4 entries     |
| SKID_DEPTH_AW  | 2 entries     |
| SKID_DEPTH_W   | 4 entries     |
| SKID_DEPTH_B   | 2 entries     |



Each channel traverses one `gaxi_skid_buffer`, which registers both `rd_valid` and its storage. The **1-cycle input-to-output latency therefore applies on every transfer, including the unstalled case** – there is no combinational bypass. Depth buys backpressure absorption, not throughput; full rate is sustained once the pipeline is primed. Legal range is 2..8 inclusive, odd values included.

Clocking: `aclk`, reset `aresetn` (active-low asynchronous).

No synthesis numbers are quoted here. Frequency and area depend on the target device and the parameters you elaborate with; run your own build.

---

## 27.6 Usage Examples

---

```
monbus_axil4_axi4_group #(
 .FIFO_DEPTH_ERR (64),
 .FIFO_DEPTH_WRITE (96), // beats (3x for raw mode)
 .ADDR_WIDTH (32),
 .S_AXIL_DATA_WIDTH (64), // or 32 for a narrow host crossbar
 .AXI_ID_WIDTH (8),
 .MAX_BURST_BEATS (64),
 .FLUSH_TIMEOUT_CYCLES (1024),
 .USE_COMPRESSION (0)
) u_monbus_egress (
 .axi_aclk (aclk),
 .axi_aresetn (aresetn),
 .cam_clear (1'b0),

 // Arbitrated monitor-bus input
 .monbus_valid (mon_valid),
 .monbus_ready (mon_ready),
 .monbus_packet (mon_packet),
 .monbus_timestamp (mon_timestamp),
 .mon_time_out (mon_time),

 // CPU-facing AXIL err drain
 .s_axil_arvalid (cpu_arvalid),
 .s_axil_arready (cpu_arready),
 .s_axil_araddr (cpu_araddr),
 .s_axil_arprot (cpu_arprot),
 .s_axil_rvalid (cpu_rvalid),
 .s_axil_rready (cpu_rready),
 .s_axil_rdata (cpu_rdata),
 .s_axil_rresp (cpu_rresp),

 // AXI4 burst memory-ring dump
 .m_axi_awid (ring_awid),
```

```

.m_axi_awaddr (ring_awaddr),
.m_axi_awlen (ring_awlen),
/* ... remaining m_axi_aw*, m_axi_w*, m_axi_b* ... */

.irq_out (monbus_irq),
.cfg_base_addr (RING_BASE),
.cfg_limit_addr (RING_LIMIT),
.cfg_flush_watermark (16'd24),
.cfg_compress_en (1'b0),

// Per-protocol filter masks (AXI / AXIS / CORE) ...
.cfg_axi_pkt_mask (axi_pkt_mask),
.cfg_axi_err_select (axi_err_select),
/* ... remaining masks ... */

.err_fifo_count (err_count),
.write_fifo_count (wr_count)
/* ... compressor stats ... */
);

```

---

## 27.7 Design Notes

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### 27.7.1 When to Pick This Over AXIL/AXIL

Choose this wrapper when the memory ring lives behind an AXI4 fabric and you want to bunch records into multi-beat bursts to amortize address-channel overhead. The memory image is identical to the AXIL-master case; only the address-channel handshake count differs. Pick `MAX_BURST_BEATS` to taste (up to 256).

### 27.7.2 AXI4 Leaf Default Tie-Offs

The core only produces the AXI4 fields it needs. The remaining AXI4 qualifier fields (`awlock` / `awcache` / `awqos` / `awregion` / `awuser` / `wuser`) are tied at the `axi4_master_wr` leaf boundary, with `awcache` set to Normal Non-cacheable Bufferable — appropriate for a trace-ring write that must reach memory.

### 27.7.3 Why One Core + Four Wrappers

The filter, FIFOs, burst writer, and drain live once in `monbus_group_core`; the AXIL read leaf, the AXI4 write leaf, and the two FUB bridges live here. Each family wrapper carries only the exact port shape its fabric demands.

---

## 27.8 Related Modules

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Used by:

- `stream_char / rapids_char` MonBus egress (AXIL drain + AXI4 ring)
- Trace-capture topologies needing high-throughput burst dump into AXI4 memory

Uses:

- **`monbus_group_core.sv`** — Protocol-agnostic filter + FIFO + burst-writer + drain
- **`axil4_slave_rd.sv`** — Slave-read (err-drain) skid leaf
- **`axi4_master_wr.sv`** — Master-write (burst bulk-capture) skid leaf
- **`monbus_compressor.sv`** — Optional runtime compressor (`USE_COMPRESSION=1`)
- **`monitor_common_pkg`** — `monitor_packet_t` / `monbus_timestamp_t` types
- **`reset_defs.svh`** — Reset macros

See also:

- **`monbus_axil4_axil4_group.sv`** — Same drain, AXIL single-beat master-write
  - **`monbus_axi4_axil4_group.sv` / `monbus_axi4_axi4_group.sv`** — AXI4-burst slave-read variants
  - **`monbus_arbiter.sv`** — Upstream multi-source merge (instantiate before this wrapper for  $N > 1$  sources)
  - **`axi4_intf_master_observer.sv`** — Instantiates this wrapper as its central filter + err FIFO + dump writer (`projects/components/misc/rtl/`)
- 

## 27.9 Testing

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- `val/amba/test_monbus_axil4_axi4_group.py`
- 

## 27.10 References

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### 27.10.1 Source Code

- RTL: `rtl/amba/monitor/monbus_axil4_axi4_group.sv`

- Core: rtl/amba/monitor/monbus\_group\_core.sv

### 27.10.2 Documentation

- Family spec: docs/markdown/rtl-amba/monitor/monbus\_group.md
  - Architecture: docs/markdown/rtl-amba/shared/README.md
  - Packet format:  
docs/markdown/rtl-amba/includes/monitor\_package\_spec.md
- 

**Last Updated:** 2026-07-15

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## 27.11 Navigation

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- [Back to rtl-amba Index](#)

## 28 Monitor Bus Group — AXIL Slave-Read / AXIL Master-Write

**Module:** monbus\_axil4\_axil4\_group.sv **Location:** rtl/amba/monitor/ **Status:** Production Ready

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### 28.1 Overview

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monbus\_axil4\_axil4\_group is the all-AXI4-Lite wrapper of the monitor-bus delivery family. It wraps the protocol-agnostic monbus\_group\_core with an **AXIL slave-read err-drain port** (for the CPU IRQ handler to pop error records) and an **AXIL single-beat master-write port** (to stream captured records into a memory ring). It is the delivery block that stream\_char / rapids\_char instantiate for MonBus egress when both the drain and dump fabrics are AXIL.

This wrapper replaces the legacy monbus\_axil\_group.sv, which fused the core and the AXIL skids into one module.

Here's the problem it solves. The monitor bus produces a single arbitrated stream of 128-bit packets plus a 64-bit timestamp. That stream has to be split into two destinations: error/interrupt records the CPU polls, and bulk trace records written into a memory ring for later offline analysis. monbus\_group\_core does the filtering, FIFOing, watermark/timeout burst-writing, and slave-read drain. This wrapper gives the core an exact AXIL port shape on both the CPU-facing (slave-read) and memory-facing (master-write) sides, so the caller ties off no spurious AXI4-only fields.

The core's FUBs are AXI4-shaped internally; the wrapper bridges the AXIL leaves to those FUBs (supplying `arid=0` / `arlen=0` / `arsize=3` / `arburst=INCR` on the read side and forcing `MAX_BURST_BEATS=1` on the write side).

Use it for:

- MonBus egress for `stream_char` / `rapids_char` on an AXIL fabric
- CPU IRQ-driven error drain with a memory-mapped ring dump
- Minimal-footprint monitor delivery where both fabrics are AXIL
- 32-bit host crossbar drain via the built-in 2:1 read serializer

The key benefit: exact AXIL port shape on both sides with no fake AXI4 fields, plus a built-in 32-bit err-drain serializer for narrow host buses.

Feature summary:

- AXIL slave-read err-FIFO drain (CPU-facing IRQ path)
- AXIL single-beat master-write bulk-capture (memory-ring egress)
- Selectable err-drain data width: 64-bit (one beat per record slice) or 32-bit (2:1 read serializer for a 32-bit host crossbar)
- Shared per-protocol egress packet filter (AXI / AXIS / CORE)
- `irq_out` asserted whenever the err FIFO is non-empty
- Optional runtime compression (`USE_COMPRESSION` + `cfg_compress_en`) with half-beat packing (`HALF_BEAT_EN`)
- Compressor statistics and FIFO status/count outputs

## 28.2 Parameters

| Parameter                      | Type | Default | Description                                            |
|--------------------------------|------|---------|--------------------------------------------------------|
| <code>FIFO_DEPTH_ERR</code>    | int  | 64      | Error FIFO depth, in records                           |
| <code>FIFO_DEPTH_WRITE</code>  | int  | 96      | Write FIFO depth, in beats (96 beats = 32 raw records) |
| <code>ADDR_WIDTH</code>        | int  | 32      | Address width on the slave-read and master-write ports |
| <code>S_AXIL_DATA_WIDTH</code> | int  | 64      | Err-drain read                                         |

| Parameter            | Type | Default | Description                                                                                                                                                                            |
|----------------------|------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DTH                  |      |         | data width. 64 = one beat per 64-bit record slice (3 beats/record); 32 = a 2:1 read serializer presents each slice as a low then high beat (6 beats/record) for a 32-bit host crossbar |
| FLUSH_TIMEOUT_CYCLES | int  | 1024    | Cycles since last accepted W handshake before a timeout-driven flush                                                                                                                   |
| NUM_PROTOCOLS        | int  | 3       | Informational — AXI / AXIS / CORE filter configs are unconditional                                                                                                                     |
| USE_COMPRESSION      | int  | 0       | Elaboration: 0 omits the compressor (raw-only); 1 elaborates it for runtime selection via <code>cfg_compress_en</code>                                                                 |
| HALF_BEAT_EN         | int  | 0       | Elaboration: pack two 30-bit half-slots per 64-bit beat downstream of the compressor (requires <code>USE_COMPRESSION=1</code> )                                                        |
| SKID_DEPTH_AR        | int  | 2       | Slave-read AR                                                                                                                                                                          |

| Parameter     | Type | Default | Description                           |
|---------------|------|---------|---------------------------------------|
| SKID_DEPTH_R  | int  | 4       | skid depth<br>Slave-read R skid depth |
| SKID_DEPTH_AW | int  | 2       | Master-write AW skid depth            |
| SKID_DEPTH_W  | int  | 2       | Master-write W skid depth             |
| SKID_DEPTH_B  | int  | 2       | Master-write B skid depth             |

The AXIL master forces MAX\_BURST\_BEATS=1 in the core, so this wrapper has no MAX\_BURST\_BEATS parameter (unlike the AXI4-master variant).

## 28.3 Ports

### 28.3.1 Clock, Reset, and CAM Clear

| Port        | Direction | Width | Description                                                                           |
|-------------|-----------|-------|---------------------------------------------------------------------------------------|
| axi_aclk    | input     | 1     | Clock                                                                                 |
| axi_aresetn | input     | 1     | Active-low asynchronous reset                                                         |
| cam_clear   | input     | 1     | Synchronous clear of the compressor CAM + stat counters (tied off in raw-only builds) |

### 28.3.2 Monitor-Bus Input + Timestamp

| Port         | Direction | Width | Description              |
|--------------|-----------|-------|--------------------------|
| monbus_valid | input     | 1     | Monitor-bus packet valid |
| monbus_ready | output    | 1     | Monitor-bus ready        |

| Port             | Direction | Width              | Description                           |
|------------------|-----------|--------------------|---------------------------------------|
|                  |           |                    | (backpressure to the arbiter)         |
| monbus_packet    | input     | monitor_packet_t   | 128-bit monitor packet                |
| monbus_timestamp | input     | monbus_timestamp_t | 64-bit source timestamp               |
| mon_time_out     | output    | monbus_timestamp_t | Free-running 64-bit timestamp counter |

### 28.3.3 AXIL Slave Read — Err-FIFO Drain

| Port           | Direction | Width             | Description                                                                    |
|----------------|-----------|-------------------|--------------------------------------------------------------------------------|
| s_axil_arvalid | input     | 1                 | Read-address valid                                                             |
| s_axil_arready | output    | 1                 | Read-address ready                                                             |
| s_axil_araddr  | input     | ADDR_WIDTH        | Read address (record-slice index; address value not memory-mapped by the core) |
| s_axil_arprot  | input     | 3                 | Read protection attributes                                                     |
| s_axil_rvalid  | output    | 1                 | Read-data valid                                                                |
| s_axil_rready  | input     | 1                 | Read-data ready                                                                |
| s_axil_rdata   | output    | S_AXIL_DATA_WIDTH | Read data (record slice, or half-slice in 32-bit mode)                         |
| s_axil_rresp   | output    | 2                 | Read response                                                                  |

### 28.3.4 AXIL Master Write — Bulk Capture

| Port           | Direction | Width | Description   |
|----------------|-----------|-------|---------------|
| m_axil_awvalid | output    | 1     | Write-address |



| Port           | Direction | Width      | Description                                            |
|----------------|-----------|------------|--------------------------------------------------------|
|                |           |            | valid                                                  |
| m_axil_awready | input     | 1          | Write-address ready                                    |
| m_axil_awaddr  | output    | ADDR_WIDTH | Write address (within [cfg_base_addr, cfg_limit_addr]) |
| m_axil_awprot  | output    | 3          | Write protection attributes                            |
| m_axil_wvalid  | output    | 1          | Write-data valid                                       |
| m_axil_wready  | input     | 1          | Write-data ready                                       |
| m_axil_wdata   | output    | 64         | Write data (one 64-bit beat)                           |
| m_axil_wstrb   | output    | 8          | Write strobe                                           |
| m_axil_bvalid  | input     | 1          | Write-response valid                                   |
| m_axil_bready  | output    | 1          | Write-response ready                                   |
| m_axil_bresp   | input     | 2          | Write response                                         |

### 28.3.5 Interrupt

| Port    | Direction | Width | Description                                 |
|---------|-----------|-------|---------------------------------------------|
| irq_out | output    | 1     | Asserted whenever the err FIFO is non-empty |

### 28.3.6 Configuration

| Port           | Direction | Width      | Description                                    |
|----------------|-----------|------------|------------------------------------------------|
| cfg_base_addr  | input     | ADDR_WIDTH | Master-write ring base address                 |
| cfg_limit_addr | input     | ADDR_WIDTH | Master-write ring limit address (writer wraps, |

| Port                | Direction | Width | Description                                                           |
|---------------------|-----------|-------|-----------------------------------------------------------------------|
| cfg_flush_watermark | input     | 16    | does not saturate)<br>Write-FIFO depth (beats) at which a flush fires |
| cfg_compress_en     | input     | 1     | Runtime compression enable (effective only when USE_COMPRESSION=1)    |

### 28.3.7 Per-Protocol Filter Masks

Three protocols (AXI, AXIS, CORE), each with a packet-type mask, an err-select mask, and per-event-category masks. All are 16-bit inputs.

| Port                                                                                                                                                                | Protocol | Role                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------|
| cfg_axi_pkt_mask                                                                                                                                                    | AXI      | Drop by packet type              |
| cfg_axi_err_select                                                                                                                                                  | AXI      | Route to err FIFO by packet type |
| cfg_axi_error_mask /<br>cfg_axi_timeout_mask /<br>cfg_axi_compl_mask /<br>cfg_axi_thresh_mask /<br>cfg_axi_perf_mask /<br>cfg_axi_addr_mask /<br>cfg_axi_debug_mask | AXI      | Per-event-code drop masks        |
| cfg_axis_pkt_mask                                                                                                                                                   | AXIS     | Drop by packet type              |
| cfg_axis_err_select                                                                                                                                                 | AXIS     | Route to err FIFO by packet type |
| cfg_axis_error_mask /<br>cfg_axis_timeout_mask /<br>cfg_axis_compl_mask /<br>cfg_axis_credit_mask /<br>cfg_axis_channel_mask /<br>cfg_axis_stream_mask              | AXIS     | Per-event-code drop masks        |

| Port                                                                                                                                               | Protocol | Role                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------|
| cfg_core_pkt_mask                                                                                                                                  | CORE     | Drop by packet type              |
| cfg_core_err_select                                                                                                                                | CORE     | Route to err FIFO by packet type |
| cfg_core_error_mask /<br>cfg_core_timeout_mask /<br>cfg_core_compl_mask /<br>cfg_core_thresh_mask /<br>cfg_core_perf_mask /<br>cfg_core_debug_mask | CORE     | Per-event-code drop masks        |

### 28.3.8 Status / Debug

| Port                                                                                                                                    | Direction | Width   | Description                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------|-----------|---------|----------------------------------------------------------|
| err_fifo_full                                                                                                                           | output    | 1       | Err FIFO write port not ready                            |
| write_fifo_full                                                                                                                         | output    | 1       | Write FIFO write port not ready                          |
| err_fifo_count                                                                                                                          | output    | 16      | Err FIFO entry count (records)                           |
| write_fifo_count                                                                                                                        | output    | 16      | Write FIFO entry count (beats)                           |
| mon_compressor_stat_tier1_a /<br>_tier1_b / _tier1_c /<br>_tier0 / _cam_miss /<br>_delta_ts_ovf /<br>_event_data_ovf /<br>_ed_delta_ovf | output    | 32 each | Compressor statistics (live only when USE_COMPRESSION=1) |

## 28.4 Functional Description

### 28.4.1 S-Side Err-Drain Read Protocol

Each monbus err record is three 64-bit slices — {tag, source\_ts}, packet[127:64], packet[63:0]. A 64-bit axil4\_slave\_rd leaf bridges the external AXIL read onto the core's 64-

bit read FUB, and the core's slice counter returns the three slices in sequence; the err-FIFO record is popped only after slice 2 (packet low half) is read.

- **64-bit drain (`S_AXIL_DATA_WIDTH == 64, g_drain_direct`):** the leaf is wired straight through, one beat per slice — 3 beats/record.
- **32-bit drain (`S_AXIL_DATA_WIDTH == 32, g_drain_2to1`):** a phase bit splits each 64-bit leaf beat into a low then high 32-bit external read — 6 beats/record. The external AR is forwarded to the leaf only on the low phase, so the leaf issues exactly one core read per pair and never prefetches. The low read streams and consumes the beat while latching its high half; the high read replays the latch and gates R on its own accepted AR (`r_hi_ar`) to honor AXIL AR-before-R ordering (a master still holding `r_ready` from the low beat cannot consume the high beat early, which would desync the pair and double the core reads).

### 28.4.2 M-Side Bulk-Capture Protocol

Surviving (non-err) packets go to the write FIFO and are streamed out the AXIL master-write port by the core's burst writer. Because this is an AXIL master, `MAX_BURST_BEATS=1` — each record is emitted as single-beat AW/W/B handshakes at consecutive addresses (three per raw record). The writer fires on watermark (`write_fifo_count >= cfg_flush_watermark`) or timeout (`FLUSH_TIMEOUT_CYCLES`), and the address is **circular** within [`cfg_base_addr`, `cfg_limit_addr`] — it wraps rather than saturating. Host-side ring pointers must match this wrap behavior. See `monbus_group` for the full burst-writer geometry.

### 28.4.3 Interrupt

`irq_out` is asserted by the core whenever the err FIFO is non-empty, so a CPU can take an interrupt and drain records over the slave-read port.

### 28.4.4 Shared Egress Packet Filter

The per-protocol filter (AXI, AXIS, CORE) decides for each packet: drop (via `pkt_mask` or a per-event mask), route to the err FIFO (via `err_select`), or route to the write FIFO (default). Protocols not in the supported set (APB, ARB) are always dropped. The filter config is identical across all four family wrappers — see `monbus_group` for the per-event category tables and masking rules.

### 28.4.5 Compression

When `USE_COMPRESSION=1`, `monbus_compressor` can be selected at runtime via `cfg_compress_en`, emitting a 64-bit self-tagged slot per COMPRESSED record into the write FIFO instead of three raw beats (an incompressible record escapes to tier-0 and still costs three); `HALF_BEAT_EN=1` further packs two 30-bit half-slots per beat. `cam_clear` synchronously empties the compressor template CAM and zeroes its stat counters between runs. In raw-only builds these fold away.

---

## 28.5 Timing Characteristics

---

| Skid parameter | Default depth |
|----------------|---------------|
| SKID_DEPTH_AR  | 2 entries     |
| SKID_DEPTH_R   | 4 entries     |
| SKID_DEPTH_AW  | 2 entries     |
| SKID_DEPTH_W   | 2 entries     |
| SKID_DEPTH_B   | 2 entries     |

Each channel traverses one `gaxi_skid_buffer`, which registers both `rd_valid` and its storage. The **1-cycle input-to-output latency therefore applies on every transfer, including the uninstalled case** – there is no combinational bypass. Depth buys backpressure absorption, not throughput; full rate is sustained once the pipeline is primed. Legal range is 2..8 inclusive, odd values included.

Clocking: `aclk`, reset `aresetn` (active-low asynchronous).

No synthesis numbers are quoted here. Frequency and area depend on the target device and the parameters you elaborate with; run your own build.

---

## 28.6 Usage Examples

---

```
monbus_axil4_axil4_group #(
 .FIFO_DEPTH_ERR (64),
 .FIFO_DEPTH_WRITE (96), // beats (3x for raw mode)
 .ADDR_WIDTH (32),
 .S_AXIL_DATA_WIDTH (64), // or 32 for a narrow host crossbar
 .FLUSH_TIMEOUT_CYCLES (1024),
 .USE_COMPRESSION (0)
) u_monbus_egress (
 .axi_aclk (aclk),
 .axi_aresetn (aresetn),
 .cam_clear (1'b0),

 // Arbitrated monitor-bus input
 .monbus_valid (mon_valid),
 .monbus_ready (mon_ready),
 .monbus_packet (mon_packet),
 .monbus_timestamp (mon_timestamp),
 .mon_time_out (mon_time),
```

```

// CPU-facing err drain
.s_axil_arvalid (cpu_arvalid),
.s_axil_arready (cpu_arready),
.s_axil_araddr (cpu_araddr),
.s_axil_arprot (cpu_arprot),
.s_axil_rvalid (cpu_rvalid),
.s_axil_rready (cpu_rready),
.s_axil_rdata (cpu_rdata),
.s_axil_rresp (cpu_rresp),

// Memory-ring dump
.m_axil_awvalid (ring_awvalid),
.m_axil_awready (ring_awready),
.m_axil_awaddr (ring_awaddr),
/* ... m_axil_w*, m_axil_b* ... */

.irq_out (monbus_irq),
.cfg_base_addr (RING_BASE),
.cfg_limit_addr (RING_LIMIT),
.cfg_flush_watermark (16'd24),
.cfg_compress_en (1'b0),

// Per-protocol filter masks (AXI / AXIS / CORE) ...
.cfg_axi_pkt_mask (axi_pkt_mask),
.cfg_axi_err_select (axi_err_select),
/* ... remaining masks ... */

.err_fifo_count (err_count),
.write_fifo_count (wr_count)
/* ... compressor stats ... */
);

```

---

## 28.7 Design Notes

---

### 28.7.1 Legacy Replacement

This wrapper (plus its three siblings) replaces the fused `monbus_axil_group.sv`. Key surface changes from the legacy module: `cfg_flush_watermark` is a new required input, `err_fifo_count` / `write_fifo_count` are 16 bits, and `FIFO_DEPTH_WRITE` is in beats (not records). See the migration section in `monbus_group`.

## 28.7.2 Why One Core + Four Wrappers

SystemVerilog cannot conditionally include ports in a single module's port list, so each protocol combination gets its own thin wrapper with the exact port shape. The filter, FIFOs, burst writer, and drain live once in `monbus_group_core`; the AXIL leaves and the FUB bridge live here.

## 28.7.3 32-Bit Drain Serializer

The `g_drain_2to1` path exists for 32-bit host crossbars. It is a careful 2:1 serializer: consuming the leaf beat on the low read (rather than holding it across both) makes `s_axil_rvalid` pulse once per external beat, so a master holding `rready` cannot mis-toggle the phase. See the shared mechanism note in the AXIL/AXI4 wrapper — the two wrappers use identical code.

---

## 28.8 Related Modules

---

Used by:

- `stream_char / rapids_char` MonBus egress (AXIL fabric)
- Any monitor-delivery topology with AXIL drain and AXIL dump fabrics

Uses:

- **`monbus_group_core.sv`** — Protocol-agnostic filter + FIFO + burst-writer + drain
- **`axil4_slave_rd.sv`** — Slave-read (err-drain) skid leaf
- **`axil4_master_wr.sv`** — Master-write (bulk-capture) skid leaf
- **`monbus_compressor.sv`** — Optional runtime compressor (`USE_COMPRESSION=1`)
- **`monitor_common_pkg`** — `monitor_packet_t / monbus_timestamp_t` types
- **`reset_defs.svh`** — Reset macros

See also:

- **`monbus_axil4_axi4_group.sv`** — Same drain, AXI4-burst master-write
  - **`monbus_axi4_axil4_group.sv / monbus_axi4_axi4_group.sv`** — AXI4-burst slave-read variants
  - **`monbus_arbiter.sv`** — Upstream multi-source merge (instantiate before this wrapper for  $N > 1$  sources)
  - **`sdpram_slave_axil_axil.sv`** — Canonical memory-ring backend for the master-write port
-

## 28.9 Testing

---

- `val/amba/test_monbus_axil4_axil4_group.py`
  - `val/amba/test_monbus_axil4_axil4_group_compressed.py`
  - `val/amba/test_monbus_axil4_axil4_group_master_write.py`
- 

## 28.10 References

---

### 28.10.1 Source Code

- RTL: `rtl/amba/monitor/monbus_axil4_axil4_group.sv`
- Core: `rtl/amba/monitor/monbus_group_core.sv`

### 28.10.2 Documentation

- Family spec: `docs/markdown/rtl-amba/monitor/monbus_group.md`
  - Architecture: `docs/markdown/rtl-amba/shared/README.md`
  - Packet format:  
`docs/markdown/rtl-amba/includes/monitor_package_spec.md`
- 

Last Updated: 2026-07-15

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## 28.11 Navigation

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- [Back to Shared Infrastructure Index](#)
- [Back to rtl-amba Index](#)

# 29 Monitor Bus LRU CAM

**Module:** `monbus_cam.sv` **Location:** `rtl/amba/monitor/` **Category:** Bulk-Trace Compression Infrastructure **Status:** Reference design — superseded in-production by `monbus_cam_pipe`

---

## 29.1 Overview

---

**Deprecation note.** `monbus_cam.sv` is the single-cycle reference design. The in-production CAM inside `monbus_compressor` is now `monbus_cam_pipe.sv`, a 2-cycle pipelined variant that splits the 49-bit



compare → priority-encode → move-to-front chain into two stages so the path closes at 100 MHz on Nexys A7 (f909f01f / 0d1c0a1a). The two share LRU semantics and per-entry storage, but they are **not** interchangeable in interface: `monbus_cam_pipe` adds a cycle of latency and has **no ACTION\_NONE** — every `access_en` cycle is a commit, with TOUCH/INSTALL derived from hit/miss, because the commit lands before the tier decision exists. It also has **no skid and no credit logic**; the credit-gated result skid (`u_res_skid / r_credit`) lives in `monbus_compressor`, and it is what sets sustained tier-1 throughput. At `SKID_DEPTH = 3` (the current value) that is 1 record/cycle measured; at the earlier depth of 2 it measured 0.67. See `monbus_compressor`. Both files are kept in tree: this single-cycle module serves as the executable spec for the LRU semantics and the compressor's CAM behavior, and is what the algorithmic tests (`test_monbus_cam.py`) target. New code should instantiate `monbus_cam_pipe`.

`monbus_cam` is a **true-LRU caching content-addressable memory** that defines the template-index semantics the `monbus_compressor` depends on (the compressor instantiates the pipelined `monbus_cam_pipe`; this module is the executable spec). It assigns 5-bit template indices (`tmpl_idx`) to the 49-bit template keys it extracts from monitor packets. “True LRU” means the eviction victim is the least recently *accessed* entry (matched, touched, or installed) — not the least recently *inserted*. This matters because the bulk-trace compression format relies on both encoder and decoder maintaining identical CAM state from the slot stream alone; if the two sides diverge on eviction order, the decoder produces garbage.

The module is a **bit-exact mirror** of the Python Cam class in `bin/TBClasses/monbus/monbus_compressor.py`. Any divergence between the two implementations is a regression.

At a glance:

- 32-entry capacity (locked by the bulk-trace format spec)
  - 49-bit key, 64-bit payload (both parameterizable)
  - **Per-entry timestamp storage** (`TS_WIDTH=24`) for per-template `delta_ts` — see the dedicated section below
  - Single combinational access port: lookup + commit in one cycle
  - True LRU eviction via **position-indexed storage** — the slot index IS the recency rank (slot 0 = MRU, slot `DEPTH-1` = LRU)
  - 3 caller-driven actions: NONE, TOUCH, INSTALL
  - Eviction pulse output for stats / instrumentation
  - Simulation-only protocol assertions on the caller
-

## 29.2 Parameters

The default values (KEY\_WIDTH=49, DATA\_WIDTH=64, DEPTH=32) are what the compressor instantiates and what the locked format spec mandates. The parameters exist so the module can be reused in other contexts (e.g. a smaller on-chip event cache) but monbus\_compressor itself doesn't override them.

| Parameter  | Type | Default                                                | Description                                      |
|------------|------|--------------------------------------------------------|--------------------------------------------------|
| KEY_WIDTH  | int  | 49                                                     | Template key width (locked by the format spec)   |
| DATA_WIDTH | int  | 64                                                     | Payload width                                    |
| TS_WIDTH   | int  | 24                                                     | Per-entry last_ts width (per-template delta_ts)  |
| DEPTH      | int  | 32                                                     | Entry capacity (locked at 32 by the format spec) |
| IDX_WIDTH  | int  | (DEPTH > 1) ? $\lceil \log_2(\text{DEPTH}) \rceil$ : 1 | Derived index width                              |
| CNT_WIDTH  | int  | $\lceil \log_2(\text{DEPTH} + 1) \rceil$               | Derived count width                              |

If you change DEPTH, the corresponding change in the Python golden's DEFAULT\_CAM\_SIZE constant must happen in lockstep — otherwise the encoded slot stream will diverge.

## 29.3 Ports

```
module monbus_cam #(
 parameter int KEY_WIDTH = 49,
 parameter int DATA_WIDTH = 64,
 parameter int TS_WIDTH = 24, // per-entry last_ts width (per-
template delta_ts)
 parameter int DEPTH = 32,
 parameter int IDX_WIDTH = (DEPTH > 1) ? $clog2(DEPTH) : 1,
 parameter int CNT_WIDTH = $clog2(DEPTH + 1)
) (
 input logic clk,
 input logic rst_n,
```

```

 // Access port (one combinational lookup + one commit per cycle)
 input logic [KEY_WIDTH-1:0] access_key,
 output logic access_hit,
 output logic [IDX_WIDTH-1:0] access_idx, // position rank
 (only valid on hit)
 output logic [DATA_WIDTH-1:0] access_old_data, // pre-commit
 payload at access_idx
 output logic [TS_WIDTH-1:0] access_old_ts, // pre-commit
 timestamp at access_idx

 input logic [1:0] access_action,
 input logic [DATA_WIDTH-1:0] access_new_data,
 input logic [TS_WIDTH-1:0] access_new_ts, // timestamp to
 write on TOUCH / INSTALL

 // Status
 output logic cam_full,
 output logic [CNT_WIDTH-1:0] cam_count,
 output logic evicted, // pulses on full-
 CAM INSTALL

 // Counting-consumer ports (additive; the compressor ties/ignores
 these)
 output logic [KEY_WIDTH-1:0] evict_key, // victim key,
 valid when evicted
 output logic [DATA_WIDTH-1:0] evict_data, // victim data,
 valid when evicted
 input logic [IDX_WIDTH-1:0] dump_idx, // position to
 observe
 output logic dump_valid, // entry at
 dump_idx is occupied
 output logic [KEY_WIDTH-1:0] dump_key,
 output logic [DATA_WIDTH-1:0] dump_data,
 input logic soft_clear // synchronous
 invalidate-all
);

```

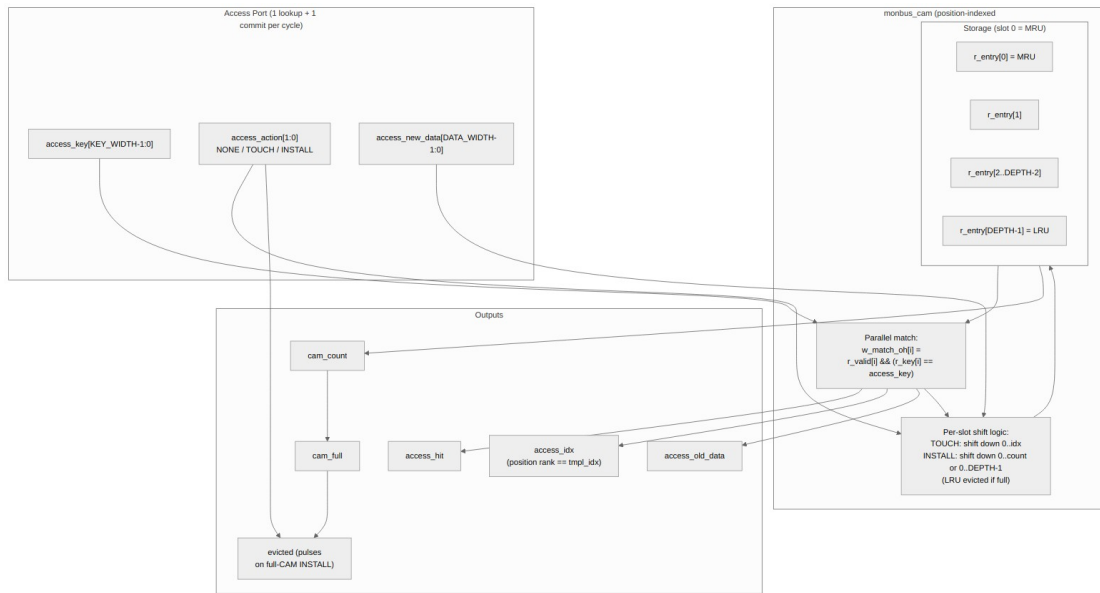
### 29.3.1 Counting-Consumer Ports

The compressor uses only the access port + evicted. A second class of consumer — a **counting** cache such as `monbus_pkt_tally`, where the payload is a partial count that must survive eviction — needs to see the victim and to walk live entries. These ports were added **additively** for that use; they do not change the LRU behaviour the compressor golden depends on, and any instantiation that does not need them may tie `dump_idx/soft_clear` low and leave the new outputs open. (The compressor no longer instantiates *this* module – it uses `monbus_cam_pipe`, which has neither port.)

| Port                                    | Direction | Width                         | Description                                                                                                                                                                   |
|-----------------------------------------|-----------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| evict_key /<br>evict_data               | output    | KEY_WIDTH /<br>DATA_WIDTH     | The LRU victim (position DEPTH-1), combinational, valid the cycle evicted is high. A counting consumer folds evict_data back into its backing store before the entry is lost. |
| dump_idx                                | input     | IDX_WIDTH                     | Position to observe (for a freeze/flush walk over all entries).                                                                                                               |
| dump_valid /<br>dump_key /<br>dump_data | output    | 1 / KEY_WIDTH /<br>DATA_WIDTH | Occupancy + contents of dump_idx, purely observational (no state change).                                                                                                     |
| soft_clear                              | input     | 1                             | Synchronous invalidate-all (cam_count → 0) without an async reset pulse; re-arms the cache between capture windows. Takes priority over any concurrent access_action.         |

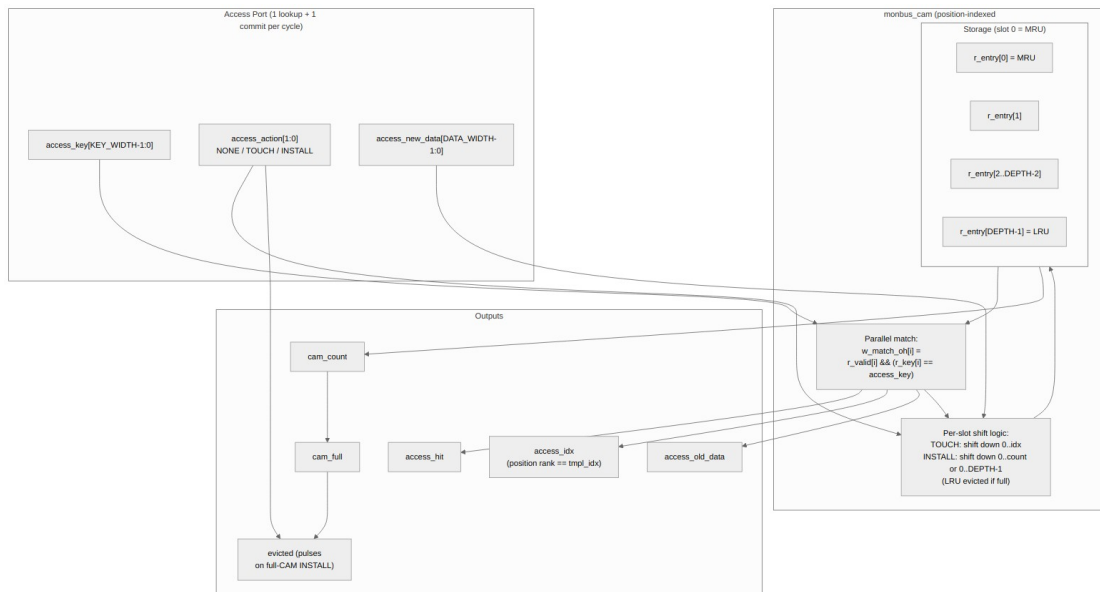
## 29.4 Functional Description

### 29.4.1 Architecture



*monbus\_cam block diagram*

Source: `monbus_cam.mmd`



*Diagram 7*

## 29.4.2 Storage Model: Position-Indexed LRU

The fundamental design choice is that **the slot index IS the position rank**:

```
r_entry[0] = most-recently-used (MRU)
r_entry[1] .. r_entry[count-1] = ordered, newer first
r_entry[count..DEPTH-1] = invalid (empty slots)
```

This means:

- The `access_idx` output on a hit is the entry's current position rank, which is exactly what `tmpl_idx` needs to be in the compressed slot stream.
- On TOUCH or INSTALL, the matched/new entry moves to slot 0 and older entries shift down by one position. This is one structural operation that updates *both* the storage AND the rank simultaneously.
- The LRU victim is always whoever sits at slot DEPTH-1. No tag pointers, no doubly-linked list, no per-entry counter — pure structural.

The trade-off: every TOUCH/INSTALL performs a shift of up to DEPTH-1 entries on a single clock edge. For DEPTH=32 this is well within timing budget (parallel per-slot updates in a generate loop) but it's the reason the format spec locks the size at 32 and not 64 or 128.

## 29.4.3 Actions

The caller drives exactly one action per cycle on the `access_action` port:

| Action         | Encoding | Caller protocol           | State change on commit                                                                                     |
|----------------|----------|---------------------------|------------------------------------------------------------------------------------------------------------|
| ACTION_NONE    | 2'b00    | Pure lookup — anytime     | None. Just samples hit/idx/old_data.                                                                       |
| ACTION_TOUCH   | 2'b01    | Must coincide with a hit  | Matched entry moves to slot 0, payload updated with access_new_data.                                       |
| ACTION_INSTALL | 2'b10    | Must coincide with a miss | New entry installed at slot 0. If <code>cam_full</code> , the entry at slot DEPTH-1 is evicted and evicted |

| Action | Encoding | Caller protocol | State change on commit                                        |
|--------|----------|-----------------|---------------------------------------------------------------|
| 2'b11  | reserved | —               | pulses high that cycle.<br>Treated as NONE (no state change). |

**Caller protocol enforcement** (simulation-only, via \$error):

- TOUCH without access\_hit is illegal — the caller saw a miss but is claiming to be touching an existing entry.
- INSTALL while access\_hit is illegal — the key is already present and reinstalling would create a duplicate.
- The internal match vector must be at most one-hot.

The natural compressor pattern is TOUCH-on-hit / INSTALL-on-miss, which satisfies all three constraints by construction.

#### 29.4.4 Per-Slot Update Mechanics

On every clock edge:

| Action                 | What happens to slot i                                                                                                                                                                                                                                              |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NONE or reserved       | Unchanged.                                                                                                                                                                                                                                                          |
| TOUCH matching slot P  | Slots 1..P each take their predecessor's contents (slot i ≤ slot i - 1), so the entries formerly at 0..P-1 shift down by one and the matched entry's old position P is overwritten. Slot 0 becomes the (matched key, new_data). Slots <b>above</b> P are untouched. |
| INSTALL when !cam_full | Insertion position is cam_count. Slots 1..cam_count shift down from 0..cam_count-1. Slot 0 becomes the new entry. cam_count++.                                                                                                                                      |
| INSTALL when cam_full  | Insertion position is DEPTH-1 (overwriting the LRU). Slots 1..DEPTH-1 shift down from 0..DEPTH-2. Slot 0 becomes the new entry. evicted                                                                                                                             |

| Action | What happens to slot <i>i</i>                       |
|--------|-----------------------------------------------------|
|        | pulses high. <code>cam_count</code> stays at DEPTH. |

A single `ALWAYS_FF_RST` block contains a `for` loop over slots 1..DEPTH-1, each iteration gated by `do_shift && (CNT_WIDTH'(i) <= shift_to)` (slot 0 is handled separately). Synthesis infers the same per-slot enables. This compiles to ~one LUT level per slot on the per-bit datapath — Vivado synthesises the whole shift as DEPTH parallel small update cones.

### 29.4.5 Per-Entry Timestamp Storage

Earlier revisions of the CAM stored only (key, data) per entry; the compressor measured `delta_ts` against a single global `r_last_ts`. That worked for single-source streams but collapsed compression to raw whenever multiple sources interleaved templates with non-monotonic absolute timestamps (the 4-channel STREAM characterization case).

The current CAM adds a **per-entry `r_ts[TS_WIDTH=24]` array** that shifts in lockstep with the key and data on every TOUCH / INSTALL:

`r_key[i], r_data[i], r_ts[i]` shift together when slot *i* moves

`access_old_ts` outputs the *pre-commit* timestamp at the matched slot (valid only when `access_hit`) — i.e. the timestamp of the previous record that used this template. The compressor uses it to compute `delta_ts = src_ts_lo - cam_access_old_ts`, then writes the current record's `source_ts[23:0]` back into the slot via `access_new_ts` in the same cycle.

**TS\_WIDTH = 24 bits.** Format-B (the 23-bit-delta Tier-1 format) needs 24 bits to *detect* its delta overflow. 16 bits silently aliases large gaps to wrong encodes.

The CAM is therefore no longer pure opaque-payload — its caller needs to drive `access_new_ts` along with `access_new_data` on every TOUCH / INSTALL. The compressor wires this directly from the incoming record's low 24 timestamp bits.

### 29.4.6 Match Logic

The match vector is a parallel one-hot:

```
always_comb begin
 for (int i = 0; i < DEPTH; i++) begin
 w_match_oh[i] = r_valid[i] && (r_key[i] == access_key);
 end
end
```

That's 1 LUT per bit × 32 bits × 49-bit equality. Vivado fuses each equality into ~3 LUT levels; the whole match is independent of the rest of the cycle's logic (no chained dependencies on shift/count signals). The priority encoder feeding `access_idx` is then DEPTH-1-input.



## 29.5 Timing Characteristics

---

This module is **sequential**: it contains clocked logic (via `always_ff` or the repository's `ALWAYS_FF_RST` macro) and therefore holds state. Outputs driven from those blocks are registered and appear one clock after the inputs that produced them.

Per-path cycle counts are not enumerated here; read the block that drives the signal you care about. No synthesis frequency or area figures are quoted – none have been measured against a target device.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

---

## 29.6 Usage Examples

---

Every parameter and port below is taken from the module declaration.

```
monbus_cam #(
 .KEY_WIDTH (49),
 .DATA_WIDTH (64),
 .TS_WIDTH (24),
 .DEPTH (32),
 .IDX_WIDTH ((DEPTH > 1)),
 .CNT_WIDTH ($clog2(DEPTH + 1))
) u_monbus_cam (
 .clk (clk),
 .rst_n (rst_n),
 .access_key (access_key),
 .access_hit (access_hit),
 .access_idx (access_idx),
 .access_old_data (access_old_data),
 .access_old_ts (access_old_ts),
 .access_action (access_action),
 .access_new_data (access_new_data),
 .access_new_ts (access_new_ts),
 .cam_full (cam_full),
 .cam_count (cam_count),
 .evicted (evicted),
 .evict_key (evict_key),
 .evict_data (evict_data),
 .dump_idx (dump_idx),
 .dump_valid (dump_valid),
 .dump_key (dump_key),
 .dump_data (dump_data),
 .soft_clear (soft_clear)
);
```

---

## 29.7 Design Notes

---

**cam\_clear** is level-sensitive and must be driven. Sample it for one cycle while the CAM is idle; it empties the template store and zeroes the compressor statistics so the next run starts from a clean hit/miss profile, without asserting `axi_aresetn`. Leaving it unconnected means the profile never resets between runs.

**Tied off in raw-only builds.** With `USE_COMPRESSION=0` there is no CAM and the port folds away.

---

## 29.8 Testing

---

`val/amba/test_monbus_cam.py` runs 10 sub-tests covering:

1. Reset state (all empty, `cam_full=0`, no matches)
2. Install + lookup basic round-trip
3. Fill to `DEPTH-1` (no overflow path exercised)
4. `TOUCH` updates payload, `idx` becomes 0
5. `cam_full` asserts on the `Nth` install
6. LRU eviction on `INSTALL` when full
7. evicted pulses **only** on full-CAM install (not on lookup, not on touch)
8. Miss on absent key (no state change)
9. `TOUCH` moves the entry to MRU (the LRU-specific invariant)
10. Random stress (~500 ops at `FUNC`, 5000 at `FULL`) cross-checked against a Python LRU model

The Python model in the test is a bit-exact mirror of the RTL — same storage semantics, same shift rules. Random stress is constrained to the caller protocol (no `INSTALL` on hit, no `TOUCH` on miss) so the protocol assertions never fire spuriously.

```
pytest val/amba/test_monbus_cam.py -v
```

`REG_LEVEL` parameter sweep (gate / func / full):

- **GATE:** 1 config (default 49/64/32)
  - **FUNC:** 2 configs (+ small `DEPTH=8`)
  - **FULL:** 6 configs (`DEPTH` 4/8/16/32, key 16/32/49/64, data 16/32/64)
-

## 29.9 Related Modules

| Module                                                    | Role                                                                                                              |
|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| monbus_compressor                                         | Consumer of the CAM <i>semantics</i> this module specifies — but it instantiates monbus_cam_pipe, not this module |
| bin/TBClasses/monbus/<br>monbus_compressor.py (Cam class) | Python golden mirror                                                                                              |
| monitor_trans_cam                                         | Sister CAM, different use case (AXI ID matching, multi-port, no LRU)                                              |

## 29.10 Navigation

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## 30 Monitor Bus LRU CAM (Pipelined)

**Module:** monbus\_cam\_pipe.sv **Location:** rtl/amba/monitor/ **Category:** Bulk-Trace  
Compression Infrastructure **Status:** Production — replaces monbus\_cam inside the compressor

### 30.1 Overview

monbus\_cam\_pipe is a **2-cycle pipelined variant** of monbus\_cam. It implements the same true-LRU CAM semantics — same key/data/timestamp storage, same MRU-at-slot-0 move-to-front, same hit/idx/old\_\* result — but splits the combinational match → priority-encode → commit chain into two registered stages so the path closes at 100 MHz on Nexys A7.

It is the in-production CAM inside monbus\_compressor. The single-cycle monbus\_cam.sv is retained as a reference design and as the algorithmic oracle for unit tests.

```
single-cycle monbus_cam : one cycle: compare → encode → commit +
result
pipelined monbus_cam_pipe : cycle T : 32-way compare
 cycle T+1 : priority-encode + commit +
result reg
```

The latency cost is **+1 cycle**. Throughput is unchanged: a forwarding network corrects each new lookup for the in-flight commit, so back-to-back accesses see bit-exact (hit, idx, old\_data,

old\_ts) sequences identical to monbus\_cam. Validated by val/amba/test\_monbus\_cam\_pipe.py.

---

## 30.2 Parameters

---

Same defaults as the single-cycle reference — and the same lock: these are what the compressor instantiates and what the format spec mandates.

| Parameter  | Type | Default | Description               |
|------------|------|---------|---------------------------|
| KEY_WIDTH  | int  | 49      | Template key width        |
| DATA_WIDTH | int  | 64      | Payload width             |
| TS_WIDTH   | int  | 24      | Per-entry timestamp width |
| DEPTH      | int  | 32      | Entry capacity            |

## 30.3 Ports

---

```
module monbus_cam_pipe #(
 parameter int KEY_WIDTH = 49,
 parameter int DATA_WIDTH = 64,
 parameter int TS_WIDTH = 24,
 parameter int DEPTH = 32
) (
 input logic clk,
 input logic rst_n,

 // Synchronous clear: invalidate all entries and flush the
 // lookup pipeline on the next edge. Lets the host rebase the
 // template table without asserting rst_n. Takes priority over
 // access_en.
 input logic clear,

 // Access presented this cycle (one per cycle when access_en=1).
 input logic access_en,
 input logic [KEY_WIDTH-1:0] access_key,
 input logic [DATA_WIDTH-1:0] access_new_data,
 input logic [TS_WIDTH-1:0] access_new_ts,
```

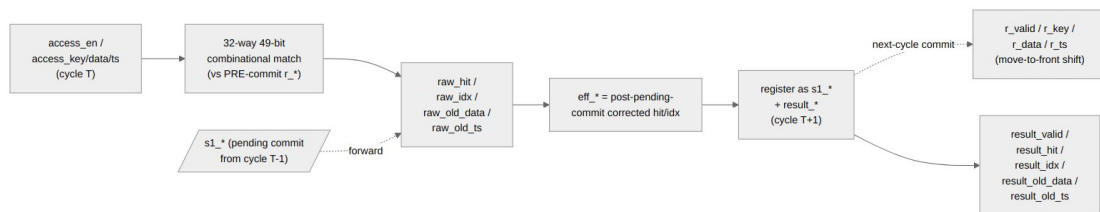
```

// Result of the access presented LAST cycle.
output logic result_valid,
output logic result_hit,
output logic [IDX_WIDTH-1:0] result_idx,
output logic [DATA_WIDTH-1:0] result_old_data,
output logic [TS_WIDTH-1:0] result_old_ts,

output logic cam_full,
output logic [CNT_WIDTH-1:0] cam_count
);

```

## 30.4 Functional Description



### Pipeline Diagram

The two registered stages are:

| Stage       | Cycle | Work                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 (compare) | T     | <p>32-way 49-bit key compare against the <b>pre-commit</b> register array. Combinational priority encode picks the <b>lowest-numbered</b> match: the descending loop's last write wins, so slot 0 (the MRU end) takes priority – identical to the single-cycle module, whose comment states the same rule. Functionally moot, since the match vector is at most one-hot by construction.</p> |

| Stage      | Cycle | Work                                                                                                                                                                                                     |
|------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 (commit) | T+1   | Apply the <i>pending</i> commit from the previous access (move-to-front shift). Register the <b>forward-corrected</b> result of cycle T's access. Capture cycle T+1's access as the next pending commit. |

The cycle T+1 commit and the cycle T+1 result registration both reference the same  $r_*$  snapshot, so the result reflects the array state *after* the in-flight move-to-front.

### 30.4.1 Forwarding (depth-1)

When access **A** is compared at cycle T+1, the previous access **P** (captured at cycle T) is committing this same cycle. A's combinational match sees the *pre-P* array, so the forward-correct adjustment is needed:

| Case                                                      | Effect on A                                                                                                           |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| A.key == P.key                                            | P just wrote slot 0 with its new data/ts<br>→ A hits with $idx=0$ , $old\_data = P.new\_data$ , $old\_ts = P.new\_ts$ |
| A matched the LRU entry that P evicted (full-CAM install) | A is now a miss (the evicted key is gone)                                                                             |
| A.raw_idx < P.shift_to                                    | A's entry shifted one slot down →<br>$eff\_idx = raw\_idx + 1$                                                        |
| Otherwise                                                 | A's match position is unchanged                                                                                       |

P.shift\_to and the actual commit shift both read the **same** pre-commit  $r_*$  this cycle, so the forwarding and the commit always agree. Bubbles ( $access\_en=0$ ) still let P commit and clear the pending slot, so a later access sees a fully-committed array and no forwarding is needed.

The end result: the ( $hit, idx, old\_data, old\_ts$ ) output sequence for any access trace is **byte-exact** with the single-cycle monbus\_cam for the same input trace (modulo the +1-cycle latency). This is the property the equivalence test guards.

### 30.4.2 Self-Derived Action

The two-bit access\_action port from monbus\_cam is gone. Instead:

```
s1_action = eff_hit ? ACTION_TOUCH : ACTION_INSTALL (registered)
```

On a hit, the matched entry moves to slot 0 with the new data/ts written through (TOUCH). On a miss, a new entry is installed at slot 0 (INSTALL), evicting the LRU if the CAM is full. The caller cannot suppress the commit — every `access_en=1` cycle is a commit.

For the compressor this is exactly the desired behavior: every record TOUCHes on hit and INSTALLs on a CAM miss; the previous CAM's ACTION\_NONE lookups were not used in production. If a future caller needs lookups without commits, it would need a separate input port gating the commit (not present today).

### 30.4.3 Synchronous Clear

`clear` is a single-cycle synchronous reset:

- All `r_valid[i]` cleared (so the CAM appears empty next cycle).
- `r_count` zeroed.
- `sl_valid` cleared (any pending commit is dropped).
- `result_valid` cleared (any in-flight result is dropped).
- Storage (`r_key`, `r_data`, `r_ts`) is **not** cleared — it's ignored while `r_valid[i]=0` so leaving it stale is harmless.

`clear` takes priority over `access_en`: an access pulse arriving in the same cycle as a clear is dropped. The host should hold `clear` high for one cycle while the upstream compression source is idle, which is the contract for the Stream CTRL [4] bit driving this through the monbus group wrapper.

---

## 30.5 Timing Characteristics

---

This module is **sequential**: it contains clocked logic (via `always_ff` or the repository's ALWAYS\_FF\_RST macro) and therefore holds state. Outputs driven from those blocks are registered and appear one clock after the inputs that produced them.

Per-path cycle counts are not enumerated here; read the block that drives the signal you care about. No synthesis frequency or area figures are quoted – none have been measured against a target device.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

---

## 30.6 Usage Examples

---

Every parameter and port below is taken from the module declaration.

```

monbus_cam_pipe #(
 .KEY_WIDTH (49),
 .DATA_WIDTH (64),
 .TS_WIDTH (24),
 .DEPTH (32),
 .IDX_WIDTH ((DEPTH > 1)),
 .CNT_WIDTH ($clog2(DEPTH + 1))
) u_monbus_cam_pipe (
 .clk (clk),
 .rst_n (rst_n),
 .clear (clear),
 .access_en (access_en),
 .access_key (access_key),
 .access_new_data (access_new_data),
 .access_new_ts (access_new_ts),
 .result_valid (result_valid),
 .result_hit (result_hit),
 .result_idx (result_idx),
 .result_old_data (result_old_data),
 .result_old_ts (result_old_ts),
 .cam_full (cam_full),
 .cam_count (cam_count)
);

```

---

## 30.7 Design Notes

### 30.7.1 How It Differs from monbus\_cam

| Aspect           | monbus_cam                        | monbus_cam_pipe                                                        |
|------------------|-----------------------------------|------------------------------------------------------------------------|
| Latency          | 0 (combinational lookup + commit) | 1 cycle                                                                |
| Throughput       | 1 access/cycle                    | 1 access/cycle                                                         |
| Action selection | Caller drives action[1:0]         | <b>Self-derived</b> from forwarded hit (TOUCH on hit, INSTALL on miss) |
| clear input      | (added in same commit)            | yes — sync invalidate + pipe flush                                     |

The self-derived action drop matters: a pipelined caller cannot know the hit at present-cycle time (exposing a combinational hit would defeat the pipeline). The compressor always TOUCHes on hit and INSTALLs on miss, so the CAM derives this internally and removes the two action bits from its own port surface.



---

## 30.8 Testing

---

```
pytest val/amba/test_monbus_cam_pipe.py -v
```

The test drives random access traces through both `monbus_cam` and `monbus_cam_pipe` in parallel and asserts the `(hit, idx, old_data, old_ts)` sequence matches cycle-for-cycle (modulo the +1-cycle latency on the pipelined side). Both modules are fed from the same random key/data/ts stream so any divergence shows up immediately.

Additional coverage:

- `test_monbus_compressor.py` — uses `monbus_cam_pipe` end-to-end inside the compressor and cross-checks the slot stream against the Python golden encoder.
  - `test_monbus_axil4_axil4_group_compressed.py` — same coverage at the group-wrapper level, including window wrap.
- 

## 30.9 Related Modules

---

| Module                           | Role                                                                             |
|----------------------------------|----------------------------------------------------------------------------------|
| <code>monbus_cam</code>          | Single-cycle reference design — same LRU semantics, used by the unit-test oracle |
| <code>monbus_compressor</code>   | Production caller of this CAM                                                    |
| <code>monbus_group family</code> | Host of the compressor + master writer                                           |

---

## 30.10 Navigation

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# 31 Monitor Bus Compressor

**Module:** `monbus_compressor.sv` **Location:** `rtl/amba/monitor/` **Category:** Bulk-Trace Compression **Status:** Production Ready

---

## 31.1 Overview

---

`monbus_compressor` is a **hardware bulk-trace encoder** for the monitor bus. It consumes streamed (`monitor_packet_t`, `monbus_timestamp_t`) records, exploits the heavy templating that real workloads exhibit, and emits a 64-bit slot stream that compresses by roughly **2.6×** with **byte-identical** equivalence to a Python golden encoder.

The compressor sits in front of the master writer inside the `monbus_group` family when a wrapper opts in via the `USE_COMPRESSION=1` parameter. The slot stream is what eventually lands in the SRAM dump ring on the FPGA, so smaller slots → longer recording windows in the same memory budget.

The format was **locked** at commit 5bbb83d1 after a sweep across CAM sizes and a real-silicon validation run on a Nexys A7. Once locked, the slot stream is *the* wire format — any RTL change here that diverges from the Python golden constitutes a regression.

### 31.1.1 Why Compress Monitor Traces?

A 128-bit monitor packet plus a 64-bit timestamp is 24 bytes per event. At even modest event rates (a few million events/sec), an AXI4 monitor running all packet types enabled can saturate a 32-bit AXIL writer’s bandwidth and fill a typical SRAM dump region in milliseconds.

**The key observation behind this compressor:** real workloads are *extremely* repetitive at the monitor-packet level. In the locked validation dataset (`desc_axi_16desc_8ch_1MB`, 682 records of descriptor-fetch traffic):

| Metric                                                                                                                                                               | Value                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Distinct ( <code>pkt_type</code> , <code>protocol</code> , <code>event_code</code> , <code>channel_id</code> , <code>agent_id</code> , <code>unit_id</code> ) tuples | <b>32</b> (4.7 % unique) |
| Records emitted                                                                                                                                                      | 682                      |
| Compression ratio                                                                                                                                                    | <b>2.66×</b>             |
| Tier-1 hit rate                                                                                                                                                      | <b>93.5 %</b>            |
| Tier-0 escapes                                                                                                                                                       | 6.5 % (44 records)       |

The 32 distinct “templates” fit *exactly* in a 32-entry CAM. Going wider buys zero incremental hits on this workload. That’s the design point.

---

## 31.2 Parameters

The 64-bit slot format and CAM size (32 entries × 49-bit key) are locked to match the Python golden. Only HALF\_BEAT\_EN is selectable: it gates an *additional* output port pair that feeds an optional downstream packer; the main out\_slot stream is unchanged.

| Parameter    | Type | Default | Description                                                                                                                                                                                     |
|--------------|------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HALF_BEAT_EN | int  | 0       | 0 = default; 64-bit-slot codec only (the committed, timing-closed path). 1 = also expose a 30-bit half-slot sideband (out_half_*) for the downstream monbus_halfbeat_packer. Folds away when 0. |

## 31.3 Ports

```
module monbus_compressor
 import monitor_common_pkg::*;
#(
 // 0 = default; 64-bit-slot codec only (the committed, timing-
 // closed path).
 // 1 = also expose a 30-bit half-slot sideband (out_half_*) for
 the
 // downstream `monbus_halfbeat_packer`. Folds away when 0.
 parameter int HALF_BEAT_EN = 0
) (
 input logic clk,
 input logic rst_n,
 // Synchronous clear: empty the template CAM (and zero the stat
 counters)
 // without asserting rst_n. Pulse one cycle while idle between
 runs.
 input logic clear,

 // Records in (one valid/ready handshake)
 input logic in_valid,
 output logic in_ready,
```

```

input monitor_packet_t in_packet, // 128 bits
input monbus_timestamp_t in_source_ts, // 64 bits

// Slots out (one valid/ready handshake)
output logic out_valid,
input logic out_ready,
output logic [63:0] out_slot,

// Half-slot sideband (valid alongside out_valid on a tier-1 beat
that
// also fits a 30-bit half-slot). Used by monbus_halfbeat_packer
when
// HALF_BEAT_EN=1; tied 0 when HALF_BEAT_EN=0.
output logic out_half_valid,
output logic [29:0] out_half_slot,

// Statistics counters (combinational from registers)
output logic [31:0] stat_tier1_a,
output logic [31:0] stat_tier1_b,
output logic [31:0] stat_tier1_c,
output logic [31:0] stat_tier0,
output logic [31:0] stat_cam_miss,
output logic [31:0] stat_delta_ts_ovf,
output logic [31:0] stat_event_data_ovf,
output logic [31:0] stat_ed_delta_ovf
);

```

### 31.3.1 Synchronous CAM Clear

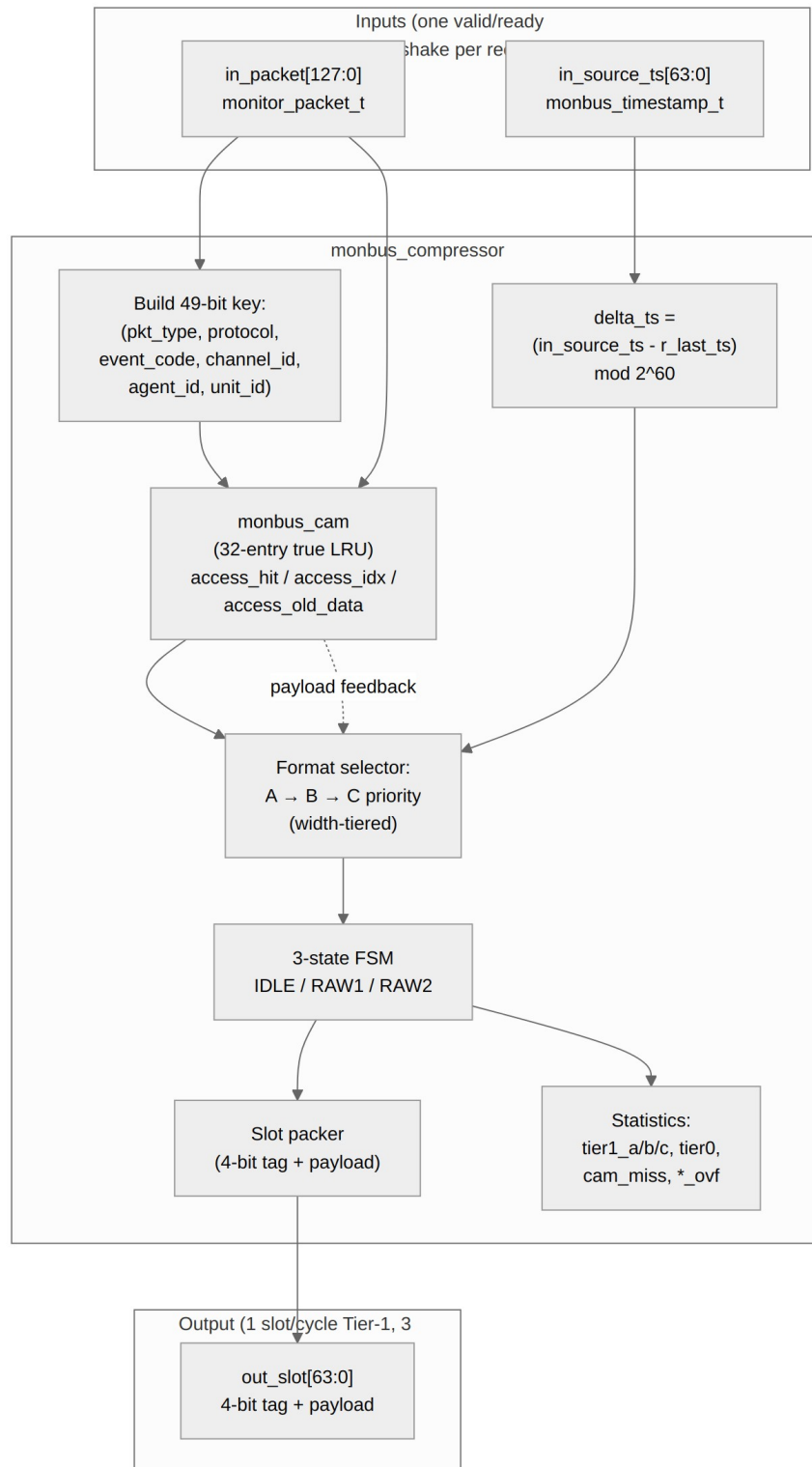
clear is a synchronous, level-sensitive input. Holding it for one cycle (while idle) invalidates every CAM entry — `r_valid`, `last_event_data`, and `last_ts` — *and* zeroes the stat counters. This is the path the Stream characterization block uses (CTRL[4]): the host can rebase compression statistics between runs without toggling `rst_n` (which would also reset the surrounding monbus pipeline, FIFOs, and AXI skids). Drive it low at all other times.

---

## 31.4 Functional Description

---

### 31.4.1 Architecture



*monbus\_compressor block diagram*

Source: monbus\_compressor.mmd

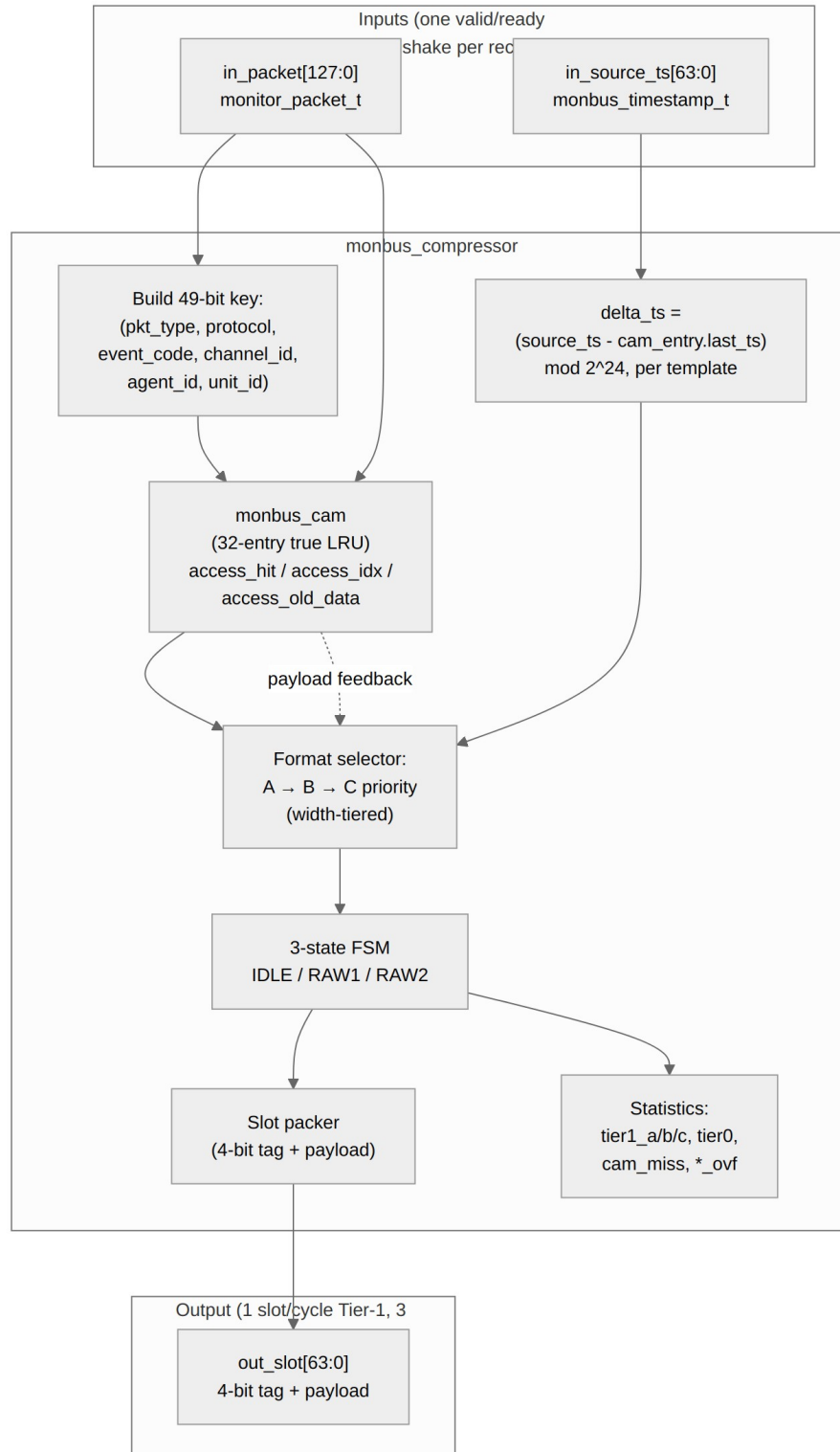


Diagram 9



## 31.4.2 Compression Techniques

The compressor combines four ideas, applied in priority order per record:

1. **Template extraction with CAM-backed indexing**
2. **Delta-encoding of the timestamp**
3. **Width-tiered payload encoding** (Tier-1 formats A/B/C)
4. **Differential encoding of the payload** (Tier-1 format C for monotonic counters / sequential addresses)
5. **Tier-0 RAW escape** when none of the above apply

Each successful Tier-1 encoding squeezes 24 bytes (raw record) into 8 bytes (one slot) — that's the 3× upper bound. The 2.66× achieved ratio reflects the 6.5 % escape rate, plus the framing overhead in Tier-0.

### 31.4.2.1 1. Template extraction

The 128-bit monitor packet has six “low-entropy” fields that repeat heavily across events generated by the same logical agent:

| Field                     | Bits      | Description                                             |
|---------------------------|-----------|---------------------------------------------------------|
| packet_type               | 4         | error / timeout / completion / threshold / perf / debug |
| protocol                  | 4         | AXI / AXIS / APB / CORE                                 |
| event_code                | 8         | sub-category within packet type                         |
| channel_id                | 9         | per-engine channel identifier                           |
| agent_id                  | 16        | per-IP agent identifier                                 |
| unit_id                   | 8         | sub-unit within an IP                                   |
| <b>template key total</b> | <b>49</b> |                                                         |

Two events from the same agent reporting the same condition share all 49 bits. The only differences cycle-to-cycle are the 64-bit `event_data` field (addresses, counters, byte counts) and the timestamp. The compressor identifies each template once, assigns it a **5-bit `tmpl_idx`** (0..31), and from then on sends only (`tmpl_idx`, `event_data`, `delta_timestamp`) instead of the full 49-bit key.

The mapping (key) → `tmpl_idx` lives in a 32-entry **caching CAM** (`monbus_cam`). Both the encoder and the decoder maintain the same CAM state, driven by identical (action, key, data)

sequences derived from the slot stream. No out-of-band table sync is required — the slot stream is self-describing.

### 31.4.2.2 2. Delta-encoded timestamp

Absolute timestamps are 60 bits, which covers any practical recording window ( $2^{60}$  cycles at 100 MHz is roughly 365 years). Deltas between consecutive events on the same *template* rarely exceed a few thousand cycles in the steady state, so the delta needs only 15 to 23 bits.

The delta is measured **per template**, against the matched CAM entry's stored 24-bit timestamp — not against a single global last-timestamp:

```
delta_ts = source_ts[23:0] - cam_entry.last_ts // mod 2^{24}
```

Each CAM entry carries its own `last_ts`, updated when that entry is touched, so interleaved templates from different sources cannot corrupt each other's deltas. A decoder must therefore chain deltas **per template**, with 24-bit wrap. See [Per-template delta\\_ts](#) for the full rationale and the history of the global-delta scheme this replaced.

### 31.4.2.3 3. Width-tiered Tier-1 formats

Three Tier-1 slot formats each cover a different (`delta_ts_width`, `event_data_width`) trade-off. The encoder picks the **first one that fits** in this priority order:

| Format | Tag | tmpl_idx        | delta_ts         | event_data                                 | Notes              |
|--------|-----|-----------------|------------------|--------------------------------------------|--------------------|
| A      | 0x1 | 5b @<br>[59:55] | 15b @<br>[54:40] | 40b @<br>[39:0]                            | Common case        |
| B      | 0x2 | 5b @<br>[59:55] | 23b @<br>[54:32] | 32b @<br>[31:0]                            | Big delta_ts       |
| C      | 0x3 | 5b @<br>[59:55] | 15b @<br>[54:40] | 40b @<br>[39:0]<br><b>signed<br/>delta</b> | Monotonic counters |

The choice is governed by:

```
delta_ts ≤ 215−1 AND event_data ≤ 240−1 → Format A (most common)
delta_ts ≤ 223−1 AND event_data ≤ 232−1 → Format B
delta_ts ≤ 215−1 AND ed_delta in [−239, 239−1] → Format C
otherwise → Tier-0 escape
```

**Format A** assumes the agent has been active recently ( $\text{delta\_ts} < \sim 328 \mu\text{s}$  @ 100 MHz) and reports an `event_data` value that fits in 40 bits. This covers most steady-state traffic where the same template keeps firing.

**Format B** handles longer idle periods — up to ~84 ms between events on the same template. The event\_data budget shrinks from 40 to 32 bits, which still covers most addresses on 32-bit address spaces.

**Format C** is the differential payload encoder, described next.

#### 31.4.2.4 4. Differential payload encoding (Format C)

For monotonically increasing counters (transaction counts, accumulated byte totals, sequential addresses), absolute event\_data may exceed 40 bits but the *delta* between consecutive events on the same template fits in 40 signed bits. Format C exploits this:

```
ed_delta = event_data - CAM[tmpl_idx].last_event_data # signed
if $(-2^{39}) \leq \text{ed_delta} < 2^{39}$:
 emit Format C slot containing (tmpl_idx, delta_ts, ed_delta)
 update CAM[tmpl_idx].last_event_data = event_data
```

The encoder records the previous event\_data per template (stored as the CAM's **payload** field — 64 bits per slot). The decoder maintains the same CAM, so when it sees a Format C slot it reconstructs:

```
event_data = CAM[tmpl_idx].last_event_data + ed_delta
```

The locked validation dataset has 0 Format C hits — descriptor-fetch traffic isn't monotonic enough. But for stream-data or descriptor-write traces, this format kicks in heavily.

#### 31.4.2.5 5. Tier-0 RAW escape

If none of the Tier-1 formats fit (CAM miss, or delta\_ts > 8M cycles, or event\_data >= 2<sup>40</sup>), the compressor falls back to a 3-beat RAW record:

```
beat 0 (tag = 0x0): [63:60] = 0x0 | [59:0] = source_ts[59:0]
beat 1 : packet[127:64]
beat 2 : packet[63:0]
```

On a Tier-0 escape **caused by a CAM miss**, the encoder also **installs** the new template at the MRU position of the CAM (evicting the LRU if the CAM is full). The decoder does the identical install on its side, so subsequent Tier-1 references to the new tmpl\_idx resolve correctly.

On a Tier-0 escape **caused by an overflow** (delta\_ts or event\_data too big), the encoder does NOT re-install — the key was already in the CAM with the right tmpl\_idx, and the escape is just a 3-beat fallback for the one unusually-large record. Statistics counters record which overflow reason caused each escape, so the host can tell the difference.

### 31.4.3 Slot Bit Layouts

The 64-bit slot is always self-describing through the 4-bit tag in [63:60]:

```
Tag 0x0 — RAW (Tier-0 escape, 3 beats total)
beat 0: [63:60] tag=0x0
```

```

[59:0] source_ts[59:0]
beat 1: packet[127:64]
beat 2: packet[63:0]

```

Tag 0x1 – Tier-1 Format A (1 beat)

```

beat 0: [63:60] tag=0x1
 [59:55] tmpl_idx[4:0]
 [54:40] delta_ts[14:0]
 [39:0] event_data[39:0]

```

Tag 0x2 – Tier-1 Format B (1 beat)

```

beat 0: [63:60] tag=0x2
 [59:55] tmpl_idx[4:0]
 [54:32] delta_ts[22:0]
 [31:0] event_data[31:0]

```

Tag 0x3 – Tier-1 Format C (1 beat)

```

beat 0: [63:60] tag=0x3
 [59:55] tmpl_idx[4:0]
 [54:40] delta_ts[14:0]
 [39:0] ed_delta[39:0] (signed)

```

Tags 0x4-0xF – Reserved (decoder must error on these)

The decoder reads one 64-bit slot, inspects bits [63:60], and knows immediately whether to read 0 more beats (Tier-1) or 2 more beats (Tier-0 RAW). No lookahead is required.

### 31.4.4 CAM Design

The 32-entry caching CAM lives in `rtl/amba/monitor/monbus_cam_pipe.sv`. It is a **true LRU** CAM with position-indexed storage — the position rank IS the `tmpl_idx`. This matters because both encoder and decoder must agree on the rank assignment when they install or touch a template, and position-indexed storage makes the rank update trivially atomic with the storage update.

**Production CAM is pipelined.** Earlier builds used the single-cycle `monbus_cam` directly in the encoder’s combinational critical path. To close 100 MHz on Nexys A7, the in-production CAM is now `monbus_cam_pipe` — a 2-cycle variant that registers the 32-way compare → priority-encode → move-to-front sequence into two stages, and feeds the encoder through a credit-gated result skid (SKID\_DEPTH=2). A `r_credit` counter tracks records in flight + skid occupancy so `cam_en` only fires when there is a slot to land the result, even under back-to-back compression bursts. Sustained throughput is **1 record/cycle** (measured): the pipelined CAM’s +1 cycle is absorbed, and SKID\_DEPTH=3 gives three credits against the 3-cycle credit round trip (present → CAM result → registered skid pop → credit returned), which is exactly what the input handshake can consume. At the previous depth of 2 the measured rate was 0.67/cycle. The single-cycle `monbus_cam.sv` is kept in-tree as the reference design (see its doc for the deprecation note).

Per-entry size:

| Field                                           | Bits       |
|-------------------------------------------------|------------|
| valid                                           | 1          |
| key (concatenated 6-tuple)                      | 49         |
| last_event_data                                 | 64         |
| last_ts (r_ts, 24-bit, per-template delta base) | 24         |
| <b>total per entry</b>                          | <b>138</b> |

No rank is stored: the slot index **is** the recency rank (entries shift on move-to-front), so recency costs no bits.

**Total CAM storage:** 32 entries × 138 bits = **4416 bits ≈ 4.3 Kb (552 bytes)**.

The production CAM (monbus\_cam\_pipe) has **no action input**. It derives the action itself, on every access, from whether the lookup hit:

| Derived action | When     | Effect                                                          |
|----------------|----------|-----------------------------------------------------------------|
| ACTION_TOUCH   | any hit  | Matched entry moves to MRU; last_event_data and last_ts updated |
| ACTION_INSTALL | any miss | New entry installed at MRU (evicts LRU if full)                 |

```
s1_action <= eff_hit ? ACTION_TOUCH : ACTION_INSTALL; // on every access_en
```

This is a consequence of pipelining, not an oversight: the CAM commits at **presentation** time, one cycle before the tier/format decision exists, so the commit *cannot* depend on it. Every presented record commits — including a Tier-0 overflow escape, which still touches its matched entry. There is no ACTION\_NONE path and no FSM-issued action; RAW beat expansion stalls the input through credit/skid backpressure instead.

### 31.4.5 Encoder Decision Tree

Per input record (one cycle):

1. Build the 49-bit template key from in\_packet.  
Compute event\_data = in\_packet[63:0] (the lower 64 bits).  
(delta\_ts is computed in step 2, against the matched entry's last\_ts.)
2. CAM lookup (pipelined, `monbus\_cam\_pipe`) – the CAM commits here, deriving

its own action from hit/miss, one cycle BEFORE the tier decision below:

- MISS → CAM self-installs (key, event\_data). Tier-0 escape.  
FSM enters S\_RAW1 → S\_RAW2; emits 3 slots over 3 cycles.  
Bump stat\_cam\_miss + stat\_tier0.
- HIT(idx) → CAM self-touches (moves to MRU, updates last\_event\_data and last\_ts). delta\_ts = source\_ts[23:0] - entry.last\_ts (mod 2<sup>24</sup>).  
Continue to step 3.

3. Try Tier-1 formats A → B → C in order:

- A. if delta\_ts < 2<sup>15</sup> AND event\_data < 2<sup>40</sup>:  
emit Format A slot. Bump stat\_tier1\_a. Done.  
(The CAM already touched the entry in step 2.)
- B. else if delta\_ts < 2<sup>23</sup> AND event\_data < 2<sup>32</sup>:  
emit Format B slot. Bump stat\_tier1\_b. Done.
- C. else if delta\_ts < 2<sup>15</sup>:  
ed\_delta = event\_data - CAM[idx].last\_event\_data (signed)  
if -2<sup>39</sup> ≤ ed\_delta < 2<sup>39</sup>:  
emit Format C slot. Bump stat\_tier1\_c. Done.
- else:  
Tier-0 escape (CAM hit but record didn't fit any Tier-1).  
Do NOT install (key already in CAM).  
Bump stat\_delta\_ts\_ovf / stat\_event\_data\_ovf / stat\_ed\_delta\_ovf  
depending on which overflow was the trigger, + stat\_tier0.

These three things happen on **three different clock edges**, not one: the CAM commits at presentation (inside monbus\_cam\_pipe), the format select is registered into q\_\* at enc\_commit the next cycle, and out\_valid = q\_valid drives the slot the cycle after that.

### 31.4.6 Statistics Counters

Each output stat is a 32-bit registered counter with no saturation guard, so they **wrap** at 2<sup>32</sup>. A capture long enough to overflow one (2<sup>32</sup> records of a single format) silently rolls that counter back through 0 – read them as free-running counters, not as protected totals.

| Counter      | Increments when       |
|--------------|-----------------------|
| stat_tier1_a | Format A slot emitted |
| stat_tier1_b | Format B slot emitted |

| Counter             | Increments when                                                                                                                  |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------|
| stat_tier1_c        | Format C slot emitted                                                                                                            |
| stat_tier0          | Tier-0 RAW escape emitted (3 slots)                                                                                              |
| stat_cam_miss       | Tier-0 escape caused by CAM miss                                                                                                 |
| stat_delta_ts_ovf   | Tier-0 escape caused by $\text{delta\_ts} \geq 2^{23}$<br>(the fit test is $< 2^{23}$ , so $2^{23}$ exactly does <i>not</i> fit) |
| stat_event_data_ovf | Tier-0 escape caused by $\text{event\_data} \geq 2^{40}$ (delta_ts fit)                                                          |
| stat_ed_delta_ovf   | Tier-0 escape caused by ed_delta out of $\pm 2^{39}$                                                                             |

The host firmware reads these via a configurable register block at the end of a capture run to characterize compression effectiveness per workload. The ratio  $(\text{stat\_tier1\_a} + \text{stat\_tier1\_b} + \text{stat\_tier1\_c}) / \text{total\_records}$  is the hit rate; on the validation dataset it's 93.5 %.

## 31.5 Timing Characteristics

The encoder is split into **three stages** (1, 2a, 2b in the table below), with a tier-1 record ~3 cycles in flight. Per-record slot counts are unchanged from the original single-cycle design, and so is the sustained input rate, at 1 record/cycle measured (SKID\_DEPTH=3).

| Stage                      | Logic                                                                                         |
|----------------------------|-----------------------------------------------------------------------------------------------|
| 1 — lookup / commit        | key build → CAM lookup → CAM commit (including per-template last_ts update via access_new_ts) |
| 2a — encode register (q_*) | delta_ts (per-template) / fits / format-select / slot pack — REGISTERED                       |
| 2b — output                | drive the slot(s); RAW (tier-0) 3-beat expansion                                              |

| Path                                 | Latency                                | Throughput                         |
|--------------------------------------|----------------------------------------|------------------------------------|
| Tier-1 record (CAM hit, Tier-1 fits) | 1 record → 1 slot, ~3 cycles in flight | <b>1 record / cycle</b> (measured) |

| Path                                             | Latency                                    | Throughput          |
|--------------------------------------------------|--------------------------------------------|---------------------|
| Tier-0 record (CAM miss, or all Tier-1 overflow) | 1 record → 3 slots, 2 cycles + 2 RAW beats | 1 record / 3 cycles |

There is **no CAM read-after-write hazard**: the commit action depends only on hit/miss (the stage-1 lookup), not on the stage-2 format. The next record's lookup one cycle later sees the committed state — exactly as in the single-cycle design. Bit-exact slot stream against the Python golden either way.

### 31.5.1 Per-template `delta_ts`

Earlier revisions measured `delta_ts` against a single global `r_last_ts`. With interleaved templates from multiple sources (the 4-channel STREAM characterization case), that global delta could go apparently-negative between two interleaved templates and escape to raw — collapsing compression.

The current encoder measures `delta_ts` against **the matched CAM entry's stored `last_ts`** (`cam_access_old_ts`), and writes back the current record's `source_ts[23:0]` to that slot via `access_new_ts` in the same cycle. The CAM's per-entry `r_ts[TS_WIDTH=24]` shifts in lockstep with the LRU move-to-front. The Python encoder/decoder were updated in lockstep — 24 / 24 golden + 1 / 1 RTL compressor + 3 / 3 group cosim tests pass, bit-exact.

`TS_STORE_BITS = 24`. Format-B (the 23-bit-delta format) needs 24 to *detect* its overflow — 16 silently aliases large gaps to wrong encodes.

### 31.5.2 Input skid

A 2-deep `gaxi_skid_buffer` registers the (`source_ts`, `packet`) feed into the compressor. The monbus aggregator's output skid sits far from this compressor's CAM in the Nexys A7 floorplan, and aggregator → `in_key` → 32-way CAM match/commit was the route-dominated 100 MHz worst path (WNS swung ±0.25 ns on placement alone). The input skid ends that long hop at a local flop.

### 31.5.3 pblock (Nexys A7 build only)

The expanded per-template CAM (32 entries × `TS_STORE_BITS=24`) crowded the descriptor monitor's column on Nexys A7 and pushed its internal `trans_mgr/u_cam → r_alloc_cnt` path route-bound. The characterization build adds a `pblock_monbus` constraint pinning the monbus group to `CLOCKREGION_X0Y2:X0Y3` (out of column X1), so the monitor can recompact. The monitor → group crossing is registered through the input skid above, so the longer inter-region hop is fine. 100 MHz closes (WNS +0.012 ns, 0 failing endpoints). See [projects/NexysA7/stream\\_characterization/](#) for the constraint file and floorplan details.

## 31.6 Usage Examples

Every parameter and port below is taken from the module declaration.



```

monbus_compressor #(
 .HALF_BEAT_EN (0)
) u_monbus_compressor (
 .clk (clk),
 .rst_n (rst_n),
 .clear (clear),
 .in_valid (in_valid),
 .in_ready (in_ready),
 .in_packet (in_packet),
 .in_source_ts (in_source_ts),
 .out_valid (out_valid),
 .out_ready (out_ready),
 .out_slot (out_slot),
 .out_half_valid (out_half_valid),
 .out_half_slot (out_half_slot),
 .stat_tier1_a (stat_tier1_a),
 .stat_tier1_b (stat_tier1_b),
 .stat_tier1_c (stat_tier1_c),
 .stat_tier0 (stat_tier0),
 .stat_cam_miss (stat_cam_miss),
 .stat_delta_ts_ovf (stat_delta_ts_ovf),
 .stat_event_data_ovf (stat_event_data_ovf),
 .stat_ed_delta_ovf (stat_ed_delta_ovf)
);

```

---

## 31.7 Design Notes

---

**Compression is a runtime choice, not a build one.** With `USE_COMPRESSION=1` the compressor is selectable via `cfg_compress_en`, emitting one 64-bit self-tagged slot per compressed record instead of three raw beats. An incompressible record escapes to tier 0 and still costs three, so the worst case is no worse than raw.

**The template CAM is the timing-critical path in this family.** On the STREAM characterisation build this route is the chronic critical path; the fix there was floorplanning, not pipelining.

**cam\_clear zeroes the statistics as well as the templates,** which is what makes per-run hit-rate figures comparable.

---

## 31.8 Testing

---

The acceptance criterion is **byte-identical equivalence** between the RTL slot stream and the Python golden encoder (`bin/TBClasses/monbus/monbus_compressor.py`) for the same input record sequence. There is no other golden — the Python class IS the spec.

Two layers of validation:

### 31.8.1 Unit-level (the compressor in isolation)

```
pytest val/amba/test_monbus_compressor.py -v
```

Two phases:

- **Phase 1:** small synthesized stream that exercises all 4 slot tags and CAM eviction. ~9 slots, easy to debug.
- **Phase 2:** the real-silicon dataset desc\_axi\_16desc\_8ch\_1MB.json (682 records → 770 golden slots). Cross-checks each slot bit-exact, plus asserts the per-tier statistics counters match the Python encoder's per-tier counts exactly.

```
$ pytest val/amba/test_monbus_compressor.py -v
```

```
...
INFO records=682, golden_slots=770
INFO rtl_a=628, rtl_b=10, rtl_c=0, tier0=44
INFO === Phase 2: PASS ===
INFO === ALL PHASES PASSED ===
```

### 31.8.2 Integration-level (compressor + AXIL writer + base/limit wrap)

```
pytest val/amba/test_monbus_axil4_axil4_group_compressed.py -v
```

Three phases:

- Generous window (no wrap): 9 slots from the synthesized stream
- Tight 8-slot window: forces a mid-stream wrap back to cfg\_base\_addr
- Real-silicon dataset: 770 slots through the full monbus\_axil4\_axil4\_group with USE\_COMPRESSION=1

Both layers run as part of the standard amba regression and must pass before any compressor change merges.

### 31.8.3 Validation Numbers (locked dataset)

Workload: desc\_axi\_16desc\_8ch\_1MB — 8 channels × 16 descriptors × 1 MB DMA, descriptor-fetch monitor on a Nexys A7.

| Metric            | Value        |
|-------------------|--------------|
| Input records     | 682          |
| Output slots      | 770          |
| Compression ratio | 2.66×        |
| Tier-1 A hits     | 628 (92.1 %) |

| Metric                        | Value                      |
|-------------------------------|----------------------------|
| Tier-1 B hits                 | 10 (1.5 %)                 |
| Tier-1 C hits                 | 0 (0 %)                    |
| Tier-0 escapes                | 44 (6.5 %)                 |
| caused by CAM miss            | 32                         |
| caused by delta_ts overflow   | 12                         |
| caused by event_data overflow | 0                          |
| caused by ed_delta overflow   | 0                          |
| Distinct templates seen       | 32 (perfectly fits CAM=32) |

The dataset and acceptance recipe live at  
[projects/NexysA7/stream\\_characterization/reports/compression\\_dataset/](#).

### 31.8.4 Test Suite Summary

- **Acceptance (byte-identical vs golden):**  
[val/amba/test\\_monbus\\_compressor.py](#)
- **End-to-end (with AXIL writer + wrap):**  
[val/amba/test\\_monbus\\_axil4\\_axil4\\_group\\_compressed.py](#)
- **CAM sub-module:** [val/amba/test\\_monbus\\_cam.py](#)

Run all three:

```
pytest val/amba/test_monbus_cam.py \
 val/amba/test_monbus_compressor.py \
 val/amba/test_monbus_axil4_axil4_group_compressed.py -v
```

## 31.9 Related Modules

| Module                    | Role                                                                                                      |
|---------------------------|-----------------------------------------------------------------------------------------------------------|
| monbus_cam_pipe           | Pipelined 32-entry LRU CAM —<br>production CAM in the compressor                                          |
| monbus_cam                | Single-cycle reference CAM<br>(superseded; kept for comparison)                                           |
| monbus_halfbeat_packer.sv | Pairs two 30-bit half-slots into one 64-bit beat downstream of the compressor (enabled by HALF_BEAT_EN=1) |

| Module                                                 | Role                                                                       |
|--------------------------------------------------------|----------------------------------------------------------------------------|
| <a href="#">monbus_group family</a>                    | Host of the compressor, plus the master writer that drains slots to memory |
| <code>sdpram_slave_axil_axil</code>                    | Typical SRAM-ring backend for the compressed slot stream                   |
| <code>bin/TBClasses/monbus/monbus_compressor.py</code> | Python golden encoder/decoder — the format spec                            |

## 31.10 Navigation

- [Back to Shared Infrastructure Index](#)
- [Back to rtl-amba Index](#)

## 32 monbus\_group\_core

**Module:** `monbus_group_core.sv` **Location:** `rtl/amba/monitor/` **Status:** Production Ready

### 32.1 Overview

The `monbus_group_core` module is the protocol-agnostic heart of the `monbus_*_*_group` family. It receives a single monitor-bus (MonBus) stream plus a side-band timestamp, applies per-protocol filter masks, and routes each accepted packet into one of two destinations: a record-granular error/interrupt FIFO drained over an AXI4-shaped slave-read port, or a beat-granular write FIFO drained over an AXI4-shaped master-write port with watermark and timeout flushing.

This core is the single source of truth for filtering, FIFO management, optional compression, and the burst writer/slicer state machines. The `monbus_<p1>_<p2>_group.sv` wrappers are pure structural adapters — they bridge its two AXI4-shaped FUB ports into protocol-specific leaf skids, and nothing more.

#### 32.1.1 Key Features

- Single MonBus ingress with per-protocol filter masks (AXI, AXIS, CORE)
- Free-running timestamp generator distributed to every monitor (`mon_time_out`)
- Error/interrupt FIFO: record-granular (192-bit records = 128-bit packet + 64-bit timestamp)

- Write FIFO: beat-granular (one entry = one 64-bit beat)
- AXI4-shaped slave-read drain with burst AR support and a 3-slice record slicer
- AXI4-shaped master-write burst writer with address-window, 4KB, and watermark controls
- Runtime-selectable compression (cfg\_compress\_en) with optional half-beat packing
- Raw mode emits complete 24-byte (3-beat) records; compressed mode emits self-tagged 8-byte slots
- Timing-closed 3-stage geometry pipeline for burst planning at 100 MHz

Aggregated monitor packets have to serve two consumers with very different needs: a CPU interrupt handler that walks recent error records, and a bulk-capture memory buffer that stores a long trace. This core splits the filtered stream between an error FIFO (read on demand by an IRQ handler) and a write FIFO (flushed to memory as AXI bursts), and handles all the address arithmetic, record framing, and optional compression in one place.

**Use Cases:** - Central MonBus capture block behind monbus\_arbiter in an SoC monitoring subsystem - Streaming error records to a CPU IRQ handler while bulk-capturing a full trace to DRAM - Compressed trace capture to fit more monitor history in a fixed buffer - Building protocol-specific group wrappers (AXI4/AXI4, AXI4/AXIL, AXIL/AXI4, AXIL/AXIL) on a shared core

**Key Benefit:** All the hard logic — filtering, dual-FIFO routing, burst geometry, compression — lives once in this core. The four protocol-pair wrappers reduce to thin structural adapters, so a bug fix or feature lands in one file for every group.

## 32.2 Parameters

| Parameter        | Type | Default | Description                                      |
|------------------|------|---------|--------------------------------------------------|
| FIFO_DEPTH_ERR   | int  | 64      | Error FIFO depth in 192-bit records              |
| FIFO_DEPTH_WRITE | int  | 96      | Write FIFO depth in 64-bit beats (beat-granular) |
| ADDR_WIDTH       | int  | 32      | Address width for both FUB ports                 |
| AXI_ID_WIDTH_M   | int  | 1       | Master-write ID width (1 in AXIL builds)         |
| AXI_ID_WIDTH_S   | int  | 1       | Slave-read ID width (1 in                        |

| Parameter            | Type | Default | Description                                                                           |
|----------------------|------|---------|---------------------------------------------------------------------------------------|
| MAX_BURST_BEATS      | int  | 1       | AXIL builds)<br>Max beats per master-write sub-burst (1 for AXIL, up to 256 for AXI4) |
| FLUSH_TIMEOUT_CYCLES | int  | 1024    | Cycles since last accepted W beat before forcing a flush                              |
| NUM_PROTOCOLS        | int  | 3       | Informational only                                                                    |
| USE_COMPRESSION      | int  | 0       | 0 = raw 3-beat records only; 1 = elaborate the compressor hardware                    |
| HALF_BEAT_EN         | int  | 0       | 1 = pack two 30-bit half-slots per beat (requires USE_COMPRESSION==1)                 |

## 32.3 Ports

### 32.3.1 Clock, Reset, and Clear

| Port        | Direction | Width | Description                                                                                                   |
|-------------|-----------|-------|---------------------------------------------------------------------------------------------------------------|
| axi_aclk    | input     | 1     | Core clock                                                                                                    |
| axi_aresetn | input     | 1     | Active-low asynchronous reset                                                                                 |
| cam_clear   | input     | 1     | Synchronous clear of the compressor template CAM + stats (no effect when USE_COMPRESSION==0); pulse when idle |

### 32.3.2 MonBus Ingress and Timestamp

| Port         | Direction | Width | Description                     |
|--------------|-----------|-------|---------------------------------|
| monbus_valid | input     | 1     | Incoming packet valid           |
| monbus_ready | output    | 1     | Core ready to accept the packet |

| Port             | Direction | Width                   | Description                                               |
|------------------|-----------|-------------------------|-----------------------------------------------------------|
| monbus_packet    | input     | monitor_packet_t (128)  | Monitor packet                                            |
| monbus_timestamp | input     | monbus_timestamp_t (64) | Side-band timestamp for the packet                        |
| mon_time_out     | output    | monbus_timestamp_t (64) | Free-running counter; drive to every wrapper's i_mon_time |

### 32.3.3 Status / IRQ / Debug

| Port             | Direction | Width | Description                                |
|------------------|-----------|-------|--------------------------------------------|
| irq_out          | output    | 1     | Asserted while the error FIFO is non-empty |
| err_fifo_full    | output    | 1     | Error FIFO cannot accept a write           |
| write_fifo_full  | output    | 1     | Write FIFO cannot accept a beat            |
| err_fifo_count   | output    | 16    | Error FIFO occupancy (records)             |
| write_fifo_count | output    | 16    | Write FIFO occupancy (beats)               |

### 32.3.4 Address Window and Flush Configuration

| Port                | Direction | Width       | Description                                        |
|---------------------|-----------|-------------|----------------------------------------------------|
| cfg_base_addr       | input     | ADDR_WIDT H | Base address of the master-write capture window    |
| cfg_limit_addr      | input     | ADDR_WIDT H | Inclusive limit address of the capture window      |
| cfg_flush_watermark | input     | 16          | Beat count in the write FIFO that triggers a flush |

| Port               | Direction | Width | Description                                                                                                                     |
|--------------------|-----------|-------|---------------------------------------------------------------------------------------------------------------------------------|
| cfg_compression_en | input     | 1     | burst<br>Runtime compression select (meaningful only when<br>USE_COMPRESSION==1);<br>hold stable while the write path is active |

### 32.3.5 Per-Protocol Filter Masks

Three protocol groups (AXI = protocol 0, AXIS = protocol 1, CORE = protocol 4), each with a packet-drop mask, an error-select mask, and per-packet-type event masks. All are 16 bits.

| Port                    | Direction | Width | Description                                        |
|-------------------------|-----------|-------|----------------------------------------------------|
| cfg_axi_pkt_mask        | input     | 16    | AXI: per-packet-type drop mask (bit = packet type) |
| cfg_axi_err_select      | input     | 16    | AXI: per-packet-type route-to-error-FIFO select    |
| cfg_axi_err_mask        | input     | 16    | AXI: per-event-code mask for Error packets         |
| cfg_axi_timeout_mask    | input     | 16    | AXI: per-event-code mask for Timeout packets       |
| cfg_axi_completion_mask | input     | 16    | AXI: per-event-code mask for Completion packets    |
| cfg_axi_threshold_mask  | input     | 16    | AXI: per-event-code mask for Threshold packets     |
| cfg_axi_perf_mask       | input     | 16    | AXI: per-event-code mask for Perf packets          |
| cfg_axi_addr_match_mask | input     | 16    | AXI: per-event-code mask for AddrMatch packets     |
| cfg_axi_debug_mask      | input     | 16    | AXI: per-event-code mask for Debug packets         |
| cfg_axis_pkt_mask       | input     | 16    | AXIS: per-packet-type drop mask                    |
| cfg_axis_               | input     | 16    | AXIS: per-packet-type                              |



| Port                  | Direction | Width | Description                                      |
|-----------------------|-----------|-------|--------------------------------------------------|
| err_select            |           |       | route-to-error-FIFO select                       |
| cfg_axis_error_mask   | input     | 16    | AXIS: per-event-code mask for Error packets      |
| cfg_axis_timeout_mask | input     | 16    | AXIS: per-event-code mask for Timeout packets    |
| cfg_axis_compl_mask   | input     | 16    | AXIS: per-event-code mask for Completion packets |
| cfg_axis_credit_mask  | input     | 16    | AXIS: per-event-code mask for Credit packets     |
| cfg_axis_channel_mask | input     | 16    | AXIS: per-event-code mask for Channel packets    |
| cfg_axis_stream_mask  | input     | 16    | AXIS: per-event-code mask for Stream packets     |
| cfg_core_pkt_mask     | input     | 16    | CORE: per-packet-type drop mask                  |
| cfg_core_err_select   | input     | 16    | CORE: per-packet-type route-to-error-FIFO select |
| cfg_core_error_mask   | input     | 16    | CORE: per-event-code mask for Error packets      |
| cfg_core_timeout_mask | input     | 16    | CORE: per-event-code mask for Timeout packets    |
| cfg_core_compl_mask   | input     | 16    | CORE: per-event-code mask for Completion packets |
| cfg_core_thresh_mask  | input     | 16    | CORE: per-event-code mask for Threshold packets  |
| cfg_core_perf_mask    | input     | 16    | CORE: per-event-code mask for Perf packets       |
| cfg_core_             | input     | 16    | CORE: per-event-code                             |

| Port       | Direction | Width | Description            |
|------------|-----------|-------|------------------------|
| debug_mask |           |       | mask for Debug packets |

### 32.3.6 Compressor Statistics

| Port                               | Direction | Width | Description                                  |
|------------------------------------|-----------|-------|----------------------------------------------|
| mon_compressor_stat_tier1_a        | output    | 32    | Tier-1 compression counter A (0 in raw mode) |
| mon_compressor_stat_tier1_b        | output    | 32    | Tier-1 compression counter B                 |
| mon_compressor_stat_tier1_c        | output    | 32    | Tier-1 compression counter C                 |
| mon_compressor_stat_tier0          | output    | 32    | Tier-0 (raw escape) count                    |
| mon_compressor_stat_cam_miss       | output    | 32    | Template CAM miss count                      |
| mon_compressor_stat_delta_ts_ovf   | output    | 32    | Delta-timestamp overflow count               |
| mon_compressor_stat_event_data_ovf | output    | 32    | Event-data overflow count                    |
| mon_compressor_stat_ed_delta_ovf   | output    | 32    | Event-data-delta overflow count              |

### 32.3.7 AXI4-Shaped Master-Write FUB (bulk capture)

| Port         | Direction | Width          | Description                    |
|--------------|-----------|----------------|--------------------------------|
| fub_m_awid   | output    | AXI_ID_WIDTH_M | Write address ID (driven to 0) |
| fub_m_awaddr | output    | ADDR_WIDTH_H   | Write burst start address      |
| fub_m_awl    | output    | 8              | Sub-burst length (beats –      |

| Port              | Direction | Width              | Description                          |
|-------------------|-----------|--------------------|--------------------------------------|
| en                |           |                    | 1)                                   |
| fub_m_aws<br>ize  | output    | 3                  | Fixed 3 (8 bytes/beat)               |
| fub_m_awb<br>urst | output    | 2                  | Fixed INCR (2'b01)                   |
| fub_m_awv<br>alid | output    | 1                  | AW valid                             |
| fub_m_awr<br>eady | input     | 1                  | AW ready                             |
| fub_m_wda<br>ta   | output    | 64                 | Write beat data (from<br>write FIFO) |
| fub_m_wst<br>rb   | output    | 8                  | Write strobes (fixed 8'hFF)          |
| fub_m_wla<br>st   | output    | 1                  | Last beat of the sub-burst           |
| fub_m_wva<br>lid  | output    | 1                  | W valid                              |
| fub_m_wre<br>ady  | input     | 1                  | W ready                              |
| fub_m_bid         | input     | AXI_ID_WID<br>TH_M | Write response ID<br>(ignored)       |
| fub_m_bre<br>sp   | input     | 2                  | Write response (ignored)             |
| fub_m_bva<br>lid  | input     | 1                  | B valid                              |
| fub_m_bre<br>ady  | output    | 1                  | B ready                              |

### 32.3.8 AXI4-Shaped Slave-Read FUB (error drain)

| Port             | Direction | Width              | Description                                 |
|------------------|-----------|--------------------|---------------------------------------------|
| fub_s_ari<br>d   | input     | AXI_ID_WID<br>TH_S | Read address ID                             |
| fub_s_ara<br>ddr | input     | ADDR_WIDT<br>H     | Read address (unused;<br>drain is a stream) |
| fub_s_arl<br>en  | input     | 8                  | Burst length (beats – 1)                    |
| fub_s_ars<br>ize | input     | 3                  | Burst size (unused; always<br>64-bit)       |

| Port            | Direction | Width          | Description                       |
|-----------------|-----------|----------------|-----------------------------------|
| fub_s_arburst   | input     | 2              | Burst type (unused; always INCR)  |
| fub_s_arvalid   | input     | 1              | AR valid                          |
| fub_s_arready   | output    | 1              | AR ready (accepts only when idle) |
| fub_s_rid       | output    | AXI_ID_WIDTH_S | Read data ID (echoes burst ID)    |
| fub_s_rdata     | output    | 64             | Sliced record beat                |
| fub_s_rresponse | output    | 2              | Read response (fixed OKAY)        |
| fub_s_rlast     | output    | 1              | Last beat of the burst            |
| fub_s_rvalid    | output    | 1              | R valid                           |
| fub_s_rready    | input     | 1              | R ready                           |

## 32.4 Functional Description

### 32.4.1 Free-Running Timestamp

A counter (`r_ts_counter`) increments every clock and is exposed as `mon_time_out`. Every monitor wrapper drives its `i_mon_time` from this output, so all packets across the group share one time base.

### 32.4.2 Packet Filtering

Each incoming packet is decomposed into `pkt_type`, `pkt_protocol` (bits [108:105]), and `pkt_event_code`. Filtering runs in three tiers per protocol:

1. **Drop mask** (`cfg_*_pkt_mask[pkt_type]`) — if set, the packet is dropped outright.
2. **Error-select** (`cfg_*_err_select[pkt_type]`) — if set (and not dropped), the packet routes to the error FIFO; otherwise it routes to the write path.
3. **Event mask** — when the event code's high nibble is 0 (`ec_in_mask_range`), the low nibble indexes a per-packet-type event mask; a set bit forces a drop and clears the error-FIFO routing.

An unrecognized protocol drops the packet. The final routing decision is `pkt_to_write_path = !pkt_drop && !pkt_to_err_fifo`.

### 32.4.3 Error FIFO and Slave-Read Drain

Accepted error packets are stored as 192-bit records {timestamp, packet} in `u_err_fifo`, and `irq_out` asserts whenever the FIFO is non-empty. The slave-read drain presents each record as three 64-bit slices over a burst read:

- slice 0 = {tag=4'h0, source\_ts[59:0]}
- slice 1 = packet[127:64]
- slice 2 = packet[63:0] (record is popped here)

AR is accepted whenever no burst is in flight (`fub_s_arready = !r_rd_in_burst`) — there is no slice-position or FIFO-occupancy condition. That has two consequences worth knowing before you write the driver: an AR on an empty FIFO is accepted and `rvalid` simply stalls until a record arrives, and `rvalid` drops mid-burst if the FIFO underruns, resuming when a new record shows up. `rlast` asserts on the  $(arlen+1)$ -th beat. Size `arlen` as a multiple of 3 minus one so the burst lands cleanly on record boundaries.

### 32.4.4 Write Path — Raw Expander vs Compressor

Two producers feed the beat-granular write FIFO, selected at runtime by `w_use_comp = (USE_COMPRESSION != 0) && cfg_compress_en`:

- **Raw expander** (always elaborated): a 3-state FSM (EXP\_TS/EXP\_HI/EXP\_LO) pushes {ts, pkt\_hi, pkt\_lo} beats atomically — a record is never split across backpressure. It uses per-beat `wr_ready` rather than a “3 slots free” precheck, because the precheck would build a count → valid → count combinational loop.
- **Compressor** (elaborated only when `USE_COMPRESSION==1`): `monbus_compressor` sits between the input and the FIFO, emitting self-tagged 64-bit slots. A compressed (tier-1) record is ONE slot; an incompressible record escapes to tier-0 and emits THREE, the same as raw mode. Size the write FIFO and `cfg_flush_watermark` for the traffic you expect: 1 beat/record only holds while records compress, and escape-heavy traffic reaches 3 beats/record. A 2-deep input skid registers (`source_ts, packet`) right at the compressor boundary so the route-dominated aggregator → CAM path ends at a local flop (+1 cycle latency, throughput preserved, slot stream bit-exact). When `HALF_BEAT_EN==1`, `monbus_halfbeat_packer` packs two 30-bit half-slots per beat downstream of the compressor.

Only one path is active at a time; the two FIFO-write outputs are muxed by `w_use_comp`. `monbus_ready` is driven by whichever path accepted the packet (drop, error-FIFO write ready, or write-path term).

### 32.4.5 Master-Write Burst Writer

The write FIFO is flushed to memory as AXI4 bursts. A flush fires when either the FIFO occupancy reaches `cfg_flush_watermark` or `FLUSH_TIMEOUT_CYCLES` elapse since the last accepted W beat — and at least one whole record (`BEATS_PER_UNIT`, 3 in raw mode, 1 compressed) is available.

Burst length is bounded by the minimum of FIFO occupancy, `MAX_BURST_BEATS`, distance to `cfg_limit_addr`, and distance to the next 4KB boundary, then rounded down to whole records. The address arithmetic is pipelined over 3 registered stages — the plan trails `r_wr_addr`, which moves in `WR_IDLE` (at each drain commit and on the rewind-snap and base-step-over branches) — keeping the wide window/4KB/round-to-record chain off the 100 MHz critical path. The fresh FIFO-occupancy cap is applied combinationally at commit, so a stale count cannot short the burst. The mod-3 whole-record rounding uses `math_mod_3_compress` carry-save instances rather than a wide divider.

The writer FSM (`WR_IDLE` → `WR_AW` → `WR_W` → `WR_B`) emits as many  $AW + N \times W + B$  sub-bursts as needed to drain the planned beat count, advancing the address one 8-byte beat per accepted W. When the current window/4KB region cannot fit a whole record, the writer rewinds `r_wr_addr` to `cfg_base_addr` and re-settles the geometry pipeline.

---

## 32.5 Timing Characteristics

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This module is **sequential**: it contains clocked logic (via `always_ff` or the repository's `ALWAYS_FF_RST` macro) and therefore holds state. Outputs driven from those blocks are registered and appear one clock after the inputs that produced them.

Per-path cycle counts are not enumerated here; read the block that drives the signal you care about. No synthesis frequency or area figures are quoted – none have been measured against a target device.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

---

## 32.6 Usage Examples

---

You don't instantiate `monbus_group_core` directly — the group wrappers do. A wrapper connects the two AXI4-shaped FUBs to protocol leaves:

```

monbus_group_core #(
 .FIFO_DEPTH_ERR (64),
 .FIFO_DEPTH_WRITE (96),
 .ADDR_WIDTH (32),
 .AXI_ID_WIDTH_M (8),
 .AXI_ID_WIDTH_S (8),
 .MAX_BURST_BEATS (64), // AXI4 build; use 1 for AXIL
master-write
 .FLUSH_TIMEOUT_CYCLES (1024),
 .USE_COMPRESSION (0)
) u_core (
 .axi_aclk (axi_aclk),
 .axi_aresetn (axi_aresetn),
 .cam_clear (cam_clear),

 .monbus_valid (monbus_valid),
 .monbus_ready (monbus_ready),
 .monbus_packet (monbus_packet),
 .monbus_timestamp (monbus_timestamp),
 .mon_time_out (mon_time_out),

 .cfg_base_addr (cfg_base_addr),
 .cfg_limit_addr (cfg_limit_addr),
 .cfg_flush_watermark (cfg_flush_watermark),
 .cfg_compress_en (cfg_compress_en),
 // ... per-protocol masks ...

 // AXI4-shaped master-write FUB -> axi4_master_wr leaf
 .fub_m_awaddr (wr_fub_awaddr), .fub_m_awlen (wr_fub_awlen),
 .fub_m_awvalid (wr_fub_awvalid), .fub_m_awready (wr_fub_awready),
 .fub_m_wdata (wr_fub_wdata), .fub_m_wlast (wr_fub_wlast),
 // ...

 // AXI4-shaped slave-read FUB <- axi4_slave_rd leaf
 .fub_s_arlen (rd_fub_arlen), .fub_s_arvalid (rd_fub_arvalid),
 .fub_s_rdata (rd_fub_rdata), .fub_s_rlast (rd_fub_rlast)
 // ...
);

```

---

## 32.7 Design Notes

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### 32.7.1 Hold `cfg_compress_en` Stable

`cfg_compress_en` changes both the record framing (raw 3-beat vs compressed 1-beat) and `BEATS_PER_UNIT`. Flip it mid-stream and you mix formats in the write FIFO, which corrupts burst sizing. Program it once before monitoring starts.

### 32.7.2 AXIL Builds Use the Same Core

AXIL group wrappers instantiate this same AXI4-shaped core with `MAX_BURST_BEATS=1` and ID widths of 1, supplying single-beat defaults (`len=0`, `size=$clog2(8)`, `INCR`, `id=0`) at the leaf. There is no AXIL-specific variant of the core, and there doesn't need to be.

### 32.7.3 FIFO Granularity Difference

The error FIFO is record-granular (192-bit); the write FIFO is beat-granular (64-bit). `err_fifo_count` reports records while `write_fifo_count` reports beats — don't compare them directly.

### 32.7.4 Timing-Motivated Structure

Several structural choices exist purely to close 100 MHz timing: the 3-stage geometry pipeline, local registering of `cfg_base_addr/cfg_limit_addr` with a `max_fanout` cap, the registered raw FIFO count shared by both the flush trigger and the burst cap, and the compressor input skid. A prior split (fresh trigger against a lagged cap) shorted bursts to 21/24 beats — both now derive from the same registered count.

### 32.7.5 4KB and Window Compliance

The burst is always sized so its last byte stays at or below `cfg_limit_addr` and never crosses a 4KB boundary, satisfying AXI4 addressing rules without mid-burst wrapping.

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## 32.8 Related Modules

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### 32.8.1 Used By

- **`monbus_axi4_axi4_group.sv`** — AXI4 slave-read + AXI4 master-write wrapper
- **`monbus_axi4_axil4_group.sv`** — AXI4 slave-read + AXIL master-write wrapper
- **`monbus_axil4_axi4_group.sv`** — AXIL slave-read + AXI4 master-write wrapper



- **monbus\_axil4\_axil4\_group.sv** — AXIL slave-read + AXIL master-write wrapper

### 32.8.2 Uses

- **gaxi\_fifo\_sync.sv** — error FIFO and write FIFO
- **gaxi\_skid\_buffer.sv** — compressor input skid
- **monbus\_compressor.sv** — optional packet compressor (present when `USE_COMPRESSION==1`)
- **monbus\_halfbeat\_packer.sv** — optional half-beat packer (`HALF_BEAT_EN==1`)
- **math\_mod\_3\_compress.sv** (`rtl/math/`) — carry-save mod-3 for whole-record rounding
- **monitor\_common\_pkg** — packet types, protocols, `monitor_packet_t`, `monbus_timestamp_t`

### 32.8.3 See Also

- **monbus\_arbiter.sv** — aggregates multiple monitor streams upstream of this core
  - **monbus\_group.md** — group-level overview
- 

## 32.9 Testing

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**No dedicated testbench for this module.** It has no `val/**/test_monbus_group_core.py`; it is exercised indirectly, through the tests of the modules that instantiate it:

- `monbus_axi4_axil4_group` – `val/amba/test_monbus_axi4_axil4_group.py`
- `monbus_axil4_axi4_group` – `val/amba/test_monbus_axil4_axi4_group.py`
- `monbus_axil4_axil4_group` – `val/amba/test_monbus_axil4_axil4_group.py`

Indirect coverage exercises this module only in the configurations its parents elaborate. A parameter or mode no parent uses is untested.

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## 32.10 References

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### 32.10.1 Source Code

- RTL: `rtl/amba/monitor/monbus_group_core.sv`

- Package: rtl/amba/includes/monitor\_common\_pkg.sv

### 32.10.2 Documentation

- Architecture: docs/markdown/rtl-amba/shared/README.md
  - Compressor: docs/markdown/rtl-amba/monitor/monbus\_compressor.md
  - Packet Format:  
docs/markdown/rtl-amba/includes/monitor\_package\_spec.md
- 

**Last Updated:** 2026-07-15

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## 32.11 Navigation

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- [Back to Shared Infrastructure Index](#)
- [Back to rtl-amba Index](#)

## 33 monbus\_group

**Modules:** - monbus\_group\_core.sv — protocol-agnostic backend (filter, FIFOs, watermark/timeout burst writer, slice-counter slave-read drain) -  
monbus\_axil4\_axil4\_group.sv — wrapper: AXIL slave-read + AXIL master-write -  
monbus\_axil4\_axi4\_group.sv — wrapper: AXIL slave-read + AXI4-burst master-write -  
monbus\_axi4\_axil4\_group.sv — wrapper: AXI4-burst slave-read + AXIL master-write -  
monbus\_axi4\_axi4\_group.sv — wrapper: AXI4-burst slave-read + AXI4-burst master-write

**Location:** rtl/amba/monitor/ **Category:** Monitor Bus → AXI / AXIL Aggregation **Status:**  
Production Ready

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### 33.1 Overview

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The monbus\_group family is the **delivery layer** for the monitor bus. It takes a single monbus stream — already arbitrated upstream if multiple sources need to be merged — and:

1. **Filters** packets per protocol (drop / route-to-err-FIFO / route-to-write-FIFO)
2. **Drains** the error FIFO over a slave-read port for CPU IRQ handlers
3. **Streams** the write FIFO over a master-write port into a memory ring
4. **Optionally compresses** the write-path traffic before it lands in memory (gated by USE\_COMPRESSION=1)

The family replaces the previous single `monbus_axil_group.sv` module. The new design exposes the same behavior under four protocol-permutation wrappers, so you pick the exact slave/master shape your fabric demands without tying off “fake AXI4” fields on AXIL sides.

## 33.2 Parameters

All wrappers expose the same set of behavioral knobs (per-wrapper parameters like `AXI_ID_WIDTH` / `AXI_USER_WIDTH` are only present where at least one side is AXI4):

| Parameter                         | Default                            | Notes                                                                                                                                                                                                         |
|-----------------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>FIFO_DEPTH_ERR</code>       | 64                                 | Error FIFO depth, in <b>records</b> (192-bit entries).                                                                                                                                                        |
| <code>FIFO_DEPTH_WRITE</code>     | 96                                 | Write FIFO depth, in <b>beats</b> (64-bit entries). 96 beats = 32 raw records.                                                                                                                                |
| <code>ADDR_WIDTH</code>           | 32                                 | Address width on both slave and master ports.                                                                                                                                                                 |
| <code>FLUSH_TIMEOUT_CYCLES</code> | 1024                               | Cycles since the last accepted W handshake before the writer fires a timeout-driven flush.                                                                                                                    |
| <code>MAX_BURST_BEATS</code>      | 1 (AXIL master) / 64 (AXI4 master) | Maximum beats per master-write burst. AXI4 protocol max is 256.                                                                                                                                               |
| <code>NUM_PROTOCOLS</code>        | 3                                  | Informational — AXI, AXIS, CORE filter configs are unconditional.                                                                                                                                             |
| <code>USE_COMPRESSION</code>      | 0                                  | <b>Elaboration:</b> 0 omits the compressor entirely (raw-only build, minimum area); 1 elaborates the compressor so it can be selected at runtime via <code>cfg_compress_en</code> . The raw 3-beat-per-record |

| Parameter              | Default                                              | Notes                                                                                                                                                                                                                                                                  |
|------------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HALF_BEAT_EN           | 0                                                    | path is always elaborated.<br><b>Elaboration:</b> pack two 30-bit half-slots per 64-bit beat downstream of the compressor (monbus_halfbeat_packer), pushing tier-1 bandwidth from 1 beat/record toward 0.5 beat/record. Requires USE_COMPRESSION=1; folds away when 0. |
| SKID_DEPTH_AR/R/AW/W/B | 2/4/2/2/2 (AXIL master) /<br>2/4/2/4/2 (AXI4 master) | Skid buffer depths per channel.<br>monbus_axi4_axi4_group defaults<br>SKID_DEPTH_W=4 to absorb burst W traffic; the AXIL-master wrappers keep 2, since MAX_BURST_BEATS=1 leaves no burst to absorb.                                                                    |

### 33.2.1 Runtime Config

In addition to the elaboration parameters, the group exposes runtime inputs that pick mode and shape on the fly:

| Input           | Width | Notes                                                                                                                    |
|-----------------|-------|--------------------------------------------------------------------------------------------------------------------------|
| cfg_compress_en | 1     | Runtime compression enable. Only effective when USE_COMPRESSION=1 (otherwise the compressor isn't elaborated and the bit |

| Input                                         | Width      | Notes                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| cam_clear                                     | 1          | <p>folds away). Must be stable while the write path is active.</p> <p><b>Synchronous CAM clear</b> (level-sensitive, sample one cycle while idle). Empties the compressor template CAM and zeroes the compressor stat counters so the next run starts with a fresh hit/miss profile — without asserting axi_aresetn. Tied off in raw-only builds (USE_COMPRESSION=0) where there is no CAM. Stream uses this on CTRL[4].</p> |
| cfg_base_addr /<br>cfg_limit_addr             | ADDR_WIDTH | Master-write address window.                                                                                                                                                                                                                                                                                                                                                                                                 |
| cfg_flush_watermark                           | 16         | Master-write FIFO depth (in beats) at which a flush is triggered.                                                                                                                                                                                                                                                                                                                                                            |
| cfg_<proto>*_mask /<br>cfg_<proto>_err_select | 16 each    | Per-protocol filter and err-FIFO routing masks; see the dedicated section below.                                                                                                                                                                                                                                                                                                                                             |

### 33.3 Ports

The status and debug outputs below are common to all wrappers:

| Port    | Direction | Width | Description           |
|---------|-----------|-------|-----------------------|
| irq_out | output    | 1     | High whenever the err |

| Port                  | Direction | Width   | Description                                                                       |
|-----------------------|-----------|---------|-----------------------------------------------------------------------------------|
| err_fifo_full         | output    | 1       | FIFO is non-empty<br>Err FIFO write port not ready                                |
| write_fifo_full       | output    | 1       | Write FIFO write port not ready                                                   |
| err_fifo_count        | output    | 16      | Err FIFO entry count (records)                                                    |
| write_fifo_count      | output    | 16      | Write FIFO entry count (beats)                                                    |
| mon_time_out          | output    | 64      | Free-running timestamp counter                                                    |
| mon_compressor_stat_* | output    | 32 each | Compressor statistics, only live when<br>USE_COMPRESSION=1; tied to 0 in raw mode |

The compressor stat ports stay in the port list regardless of elaboration mode, so wrapper layers see a consistent port surface; only their semantics differ.

## 33.4 Functional Description

### 33.4.1 Architecture

The diagram below is for the AXIL/AXIL variant; the other three wrappers are structurally identical except for the leaf skids and the FUB-to-leaf bridge (e.g., axi4\_slave\_rd instead of axil4\_slave\_rd).

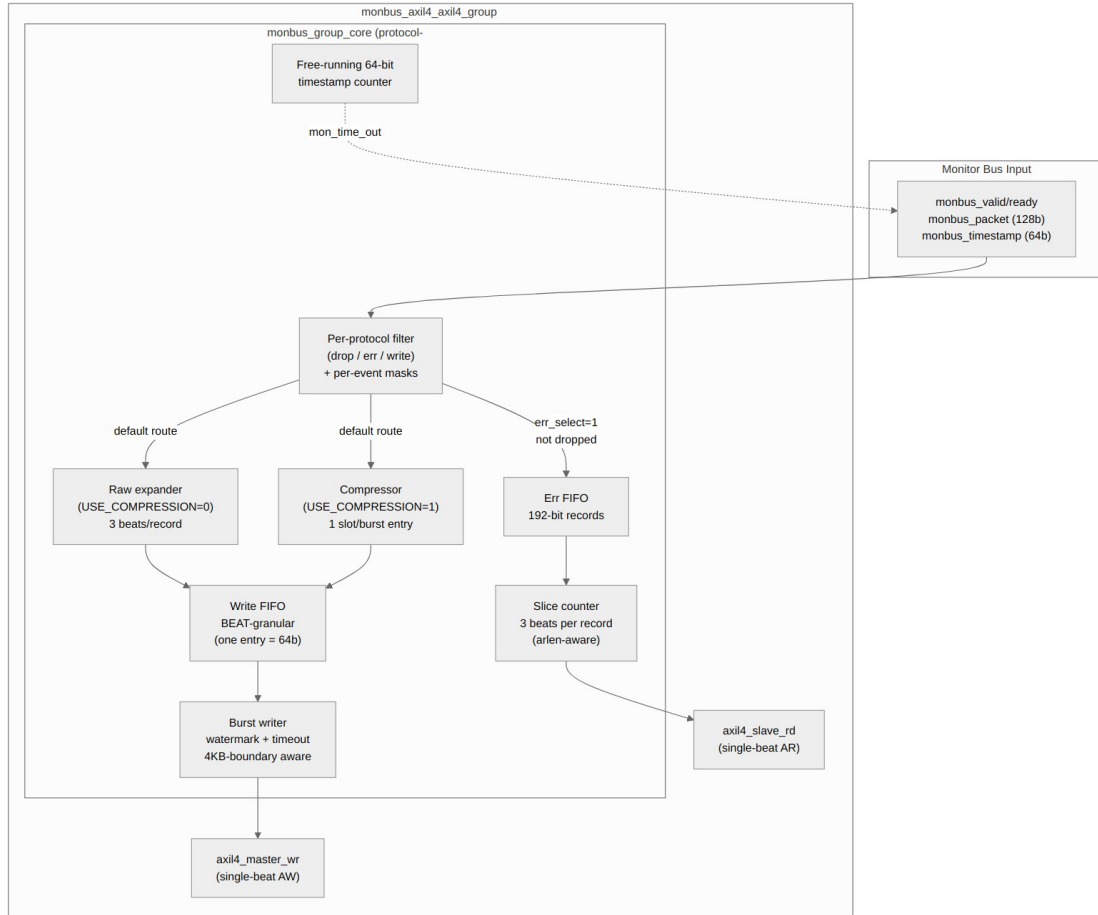


Diagram 10

The module has **no built-in arbitration** — it is a single-input block. Callers with N upstream sources (e.g., RAPIDS source+sink) instantiate a separate `monbus_arbiter` upstream to merge their streams before feeding the group. Keeping arbitration orthogonal to capture means the family never has to duplicate the arbiter for each consumer's input count.

### 33.4.2 Beat Layout

The write-FIFO and slave-read drain are **beat-granular** — one queue entry is one 64-bit beat. There are two emit modes.

#### 33.4.2.1 Raw mode (`cfg_compress_en = 0`, or `USE_COMPRESSION = 0 build`)

Each accepted monbus record becomes three sequential 64-bit beats in the write FIFO:

```
beat 0 = {tag[3:0]=4'h0, source_ts[59:0]}
beat 1 = packet[127:64]
beat 2 = packet[63:0]
```

The expander is **atomic per record**: once it commits to driving the ts beat, it drives beats 1 and 2 before accepting a new record. Partial records never appear in the FIFO, so the master-write burst writer can safely round burst lengths down to a multiple of 3.

The tag nibble in beat 0 is reserved for future on-the-wire format variants. The raw writer hardwires it to 4'h0.

#### **33.4.2.2      *Compressed mode (cfg\_compress\_en = 1, requires USE\_COMPRESSION = 1 build)***

monbus\_compressor consumes (packet, source\_ts) records via its in-handshake and emits 64-bit self-tagged slots via its out-handshake. Each slot becomes one beat directly in the write FIFO. Slot tags (bits [63:60]):

```
4'h0 = raw expansion beat (Tier-0 fallback)
4'h1 = Tier-1 format A
4'h2 = Tier-1 format B
4'h3 = Tier-1 format C
4'h4 = half-beat pair (TAG_HALF_PAIR, emitted when HALF_BEAT_EN=1)
4'h5..4'hF = reserved
```

See monbus\_compressor.md for the encoder's internal state and the byte-exact Python golden.

The raw expander is **always elaborated**; the compressor is elaborated only when USE\_COMPRESSION=1. The two paths' FIFO-write outputs and monbus\_ready are muxed on w\_use\_comp = (USE\_COMPRESSION != 0) && cfg\_compress\_en. Only one path is ever active; the inactive one is gated off at its input. BEATS\_PER\_UNIT (3 in raw, 1 in compressed) is therefore a runtime quantity (w\_beats\_per\_unit), feeding the burst-writer geometry described below.

**Note on switching modes:** cfg\_compress\_en must be stable while there is unflushed data in the write FIFO. Toggling it mid-stream would mix raw 3-beat records with compressed 1-beat slots at the same memory addresses — the host decoder couldn't reassemble them. Software should drain the FIFO (poll write\_fifo\_count to zero) before flipping the bit.

#### **33.4.2.3      *Slave-read drain (both modes)***

The slave-read port returns the same 3-beat layout per err-FIFO record, sliced by an internal counter. In AXI4 mode, arlen+1 beats are emitted across the burst with rlast asserting on the final beat; the slicer advances through records, and the FIFO entry is popped only when slice 2 (SLICE\_PKT\_LO) fires. In AXIL mode each AR returns one beat.

### **33.4.3 Master-Write Behavior**

The burst writer fires when **either**:

- flush\_trigger\_watermark: r\_fifo\_beats >= cfg\_flush\_watermark
- flush\_trigger\_timeout: r\_timeout\_cnt >= FLUSH\_TIMEOUT\_CYCLES



...and at least BEATS\_PER\_UNIT beats are available (3 in raw mode, 1 in compressed mode).

### 33.4.3.1 *Burst geometry: 3-stage pipeline + fresh-FIFO cap at commit*

The drain-plan math was originally a single combinational chain off `r_wr_addr` feeding straight back into `r_wr_addr` — the 100 MHz critical path on Nexys A7 (-1) (WNS –7.06 ns post-route). Since `r_wr_addr` is stable while the writer sits in `WR_IDLE` (only `WR_W` advances it) and the write FIFO only grows there, the **address geometry** is now a **3-stage registered pipeline**:

```
stage 1 (caps): s1_beats_to_limit / s1_beats_to_4kb from
r_wr_addr,
 plus the rewind decision (geom_addr).
stage 2 (planned): min-cap tree ONLY (min of window / 4KB).
 u_mod3_geo is fed combinationaly from this
 register and consumed by stage 3, so
 s2_beats_planned is NOT yet record-rounded.
stage 3 (rounded + addr): r_plan_geo_units (address-feasible whole-
 record count) + r_plan_addr (eff_addr).
```

A `geom_valid` settle counter holds the plan invalid for the first few cycles after the writer **leaves** `WR_IDLE`, so the pipeline reflects the settled address before the FSM commits. The counter resets on that state exit only – **not** when `r_wr_addr` moves *inside* `WR_IDLE`, which the rewind-snap and base-step-over branches below both do. On those paths `geom_valid` stays asserted and the FSM can act on a stale plan for a few cycles. That is tolerable rather than hazardous: a commit always consumes `r_plan_addr`, which travels through the pipeline with its own plan, so a stale plan can never be paired with a fresh address.

#### 33.4.3.1.1 Rewind-snap: in-window with no record-room left

If a flush fires while `r_wr_addr` is in-window but the remaining bytes to the next 4KB boundary cannot fit a whole record, the pipeline produces `r_plan_ok=false` (units rounded down to zero) and the writer would otherwise wedge in `WR_IDLE`. The FSM detects this case (`do_flush && geom_valid && !r_plan_ok && r_wr_addr != cfg_base_addr`) and **snaps `r_wr_addr` to `cfg_base_addr`**, staying in `WR_IDLE` so the pipeline re-settles with fresh geometry.

A **second branch** covers the case where `cfg_base_addr` *itself* cannot host a whole record: `do_flush && geom_valid && !r_plan_ok && (r_wr_addr == cfg_base_addr)` steps `r_wr_addr` over the short stub to the next 4KB boundary and drains there. So the host does **not** have to place `cfg_base_addr` with a record's worth of room before the next 4KB line — the hardware handles a base that lands in a stub. (The `!=`-only guard, without this second branch, wedged the writer; that is why it exists.)

This case is hit by the AXIL/AXIL master\_write stress in Phase 5, where `cfg_base_addr` is intentionally placed only a few beats below a 4KB boundary so the very first drain has to wrap-snap before it can emit anything.

The **FRESH FIFO-occupancy cap** is applied combinationaly at the `WR_IDLE` commit (not inside the pipeline):

```
// Pipeline produces r_plan_geo_units = address-feasible whole-record
beats.
// At commit time, cap by the CURRENT FIFO contents:
units_commit = min(r_plan_geo_units, w_fifo_units)
```

where  $w\_fifo\_units = r\_fifo\_beats - (r\_fifo\_beats \bmod 3)$  (or  $r\_fifo\_beats$  itself in compressed mode where the unit is 1 beat). This matters because the FIFO keeps filling while the writer sits in `WR_IDLE`; a stale (3-cycle-old) FIFO count would short the burst — for example, capping a watermark-triggered 24-beat plan to 21. The fresh path starts from a single counter (no address subtract/shift), so it doesn't recreate the critical path.

Per-AW sub-burst sizing happens **inside the FSM**, capping each AW by `MAX_BURST_BEATS`:

```
sub_burst_beats = min(remaining_in_cycle, MAX_BURST_BEATS)
awlen = sub_burst_beats - 1
```

The split between drain-cycle plan (whole records) and per-AW sub-burst (`MAX_BURST_BEATS`) is what makes the family work for both AXIL (`MAX_BURST_BEATS = 1`) and AXI4 masters in raw mode (`BEATS_PER_UNIT = 3`):

- **AXI4 master, raw mode:** one drain cycle  $\approx$  one large sub-burst. e.g. 24 beats = 8 records in a single AW +  $24 \times W + B$ . Throughput-optimal.
- **AXIL master, raw mode:** one drain cycle =  $N$  single-beat sub-bursts. With `MAX_BURST_BEATS=1` the FSM runs one single-beat sub-burst per beat, so a 3-beat raw record costs three AW + three W + three B handshakes – the W and B counts scale too, not just the address channel – at consecutive addresses. The memory image is identical to the AXI4 case; the handshake counts on all three channels differ.

### 33.4.3.2 *Address-window behavior: wrap, not saturate*

The master-write address is **circular** within `[cfg_base_addr, cfg_limit_addr]`. When the writer rolls past the window's end (or starts out of window, e.g. `r_wr_addr = 0` immediately after reset), the geometry pipeline finds `r_wr_addr` outside `[cfg_base_addr, cfg_limit_addr]` and computes against a pre-rewound `geom_addr = cfg_base_addr`. The final `r_plan_addr` reflects that rewind — so the next burst starts at `cfg_base_addr` and the dump ring keeps moving forever.

This is **not** a saturation behavior. The writer never stalls at `cfg_limit_addr`; it always wraps. Any host-side bookkeeping (SRAM-dump write pointer, ring-buffer position cursor, etc.) **must match this wrap behavior** — saturating a host-visible pointer at the SRAM cap will desynchronize it from the actual ring state. The `stream_char_harness.sv` debug pointer was fixed for exactly this reason on 2026-06-14 (d82b1e04); see that commit message for the hardware-confirmed symptom.

### 33.4.3.3 *mod-3 rounding (no /3 operator)*

Rounding a beat count down to a whole record ( $X - (X \bmod 3)$ ) is done by two `math_mod_3_compress` instances (`u_mod3_geo`, `u_mod3_fifo`) — a carry-save-compressor implementation of  $X \bmod 3$  via base-4 digit sum (each 2-bit group has weight  $4^k \equiv 1 \bmod 3$ ). No  $*$  or  $/$  operators, so Vivado neither infers a DSP48 (a combinational path through a far-placed DSP was catastrophic) nor a CARRY4 iterative divider — both used to blow timing. Exhaustively verified for all 65 536 inputs (`val/math/test_math_mod_3_compress.py`).

### 33.4.3.4 *Final port behavior*

`awsize` is fixed at 3 (8 bytes per beat) and `awburst` at INCR (2'b01). `awlen` is `sub_burst_beats - 1`. `wlast` asserts on the final beat of each sub-burst.

## 33.4.4 Per-Protocol Filter Configuration

The filter is configured per protocol (AXI, AXIS, CORE). For each protocol the caller provides:

- `cfg_<proto>_pkt_mask[15:0]` — drop the packet if the bit at `pkt_type` is set.
- `cfg_<proto>_err_select[15:0]` — route to the err FIFO if the bit at `pkt_type` is set (and not dropped). All other surviving packets go to the write FIFO.
- `cfg_<proto>_<event_type>_mask[15:0]` — per-event-code mask. If the bit at `event_code[3:0]` is set, the packet is dropped regardless of the above. Per-protocol event categories differ:

- **AXI:** error, timeout, compl, thresh, perf, addr, debug
- **AXIS:** error, timeout, compl, credit, channel, stream
- **CORE:** error, timeout, compl, thresh, perf, debug

Per-event masks are 16 bits but only `event_code[3:0]` indexes; codes  $\geq 16$  are treated as “not in mask range” (no masking applied). Protocols not in the supported set (APB=2, ARB=3) are always dropped.

---

## 33.5 Usage Examples

---

### 33.5.1 Migration from `monbus_axil_group`

The legacy `monbus_axil_group.sv` module is gone. Migrate to the wrapper matching your fabric. The port-surface changes that matter:

1. **M\_AXIL\_DATA\_WIDTH dropped; S\_AXIL\_DATA\_WIDTH retained** — the master-write data path is locked at 64 bits, so M\_AXIL\_DATA\_WIDTH is gone. S\_AXIL\_DATA\_WIDTH survives on the two AXIL-slave-read wrappers (monbus\_axil4\_axil4\_group, monbus\_axil4\_axi4\_group) with a default of 64 (one beat per record slice); set it to 32 to enable the built-in 2:1 read serializer for a 32-bit host crossbar (6 beats per record). The AXI4-slave-read wrappers have no such parameter.
2. **cfg\_flush\_watermark[15:0] is a new required input**. Set to 1 for the legacy “fire immediately” behavior; set higher to batch.
3. **err\_fifo\_count and write\_fifo\_count are now 16 bits** (were 8).
4. **FIFO\_DEPTH\_WRITE is in beats**, not records. 32 records of the legacy module = 96 beats in raw mode.

Example (AXIL/AXIL — the closest analog to the legacy module):

```
// Old
monbus_axil_group #(
 .FIFO_DEPTH_ERR (64),
 .FIFO_DEPTH_WRITE (32), // records
 .S_AXIL_DATA_WIDTH (64),
 .M_AXIL_DATA_WIDTH (64),
 .USE_COMPRESSION (0)
) u_group (...);

// New
monbus_axil4_axil4_group #(
 .FIFO_DEPTH_ERR (64),
 .FIFO_DEPTH_WRITE (96), // beats (3x for raw mode)
 .FLUSH_TIMEOUT_CYCLES (1024),
 .USE_COMPRESSION (0)
) u_group (
 .cfg_flush_watermark (16'd24), // 8 records worth, eager flush
 ...
);
```

For AXI4 burst capture — say the memory ring lives behind an AXI4 fabric and you want to bunch records into multi-beat bursts to amortize address-channel overhead — switch the wrapper to monbus\_axil4\_axi4\_group and pick MAX\_BURST\_BEATS to taste.

## 33.6 Design Notes

### 33.6.1 Why One Core Plus Four Wrappers

SystemVerilog cannot conditionally include or exclude ports from a single module's port list. The port list is a static syntactic construct, fixed at module declaration. So to give each protocol combination an exact port shape — no spurious AXI4-only fields for the caller to tie off — the family is split:

- **One common backend** (`monbus_group_core.sv`) carries the filter, FIFOs, watermark/timeout burst-writer FSM, slice-counter drain, and optional compressor. Its FUBs are AXI4-shaped on both sides (`id` / `awlen` / `awsize` / `awburst` / `wlast` / `arlen` / `rlast` / etc.). AXIL wrappers feed single-beat defaults (`len=0`, `size=3`, `burst=INCR`, `id=0`) and force `MAX_BURST_BEATS=1`.
- **Four thin wrappers**, each with the right protocol-shaped port list, that directly instantiate the matching leaf skids (`axi4_slave_rd` / `axil4_slave_rd` and `axi4_master_wr` / `axil4_master_wr`) and bridge their FUBs to the core's FUB.

| Wrapper                               | Slave-read shape                            | Master-write shape                             | MAX_BURST_BEATS (default) |
|---------------------------------------|---------------------------------------------|------------------------------------------------|---------------------------|
| <code>monbus_axil4_axil4_group</code> | AXIL<br><code>s_axil_ar*/r*</code>          | AXIL<br><code>m_axil_aw*/w*/b*</code>          | 1                         |
| <code>monbus_axil4_axi4_group</code>  | AXIL<br><code>s_axil_ar*/r*</code>          | AXI4<br><code>m_axi_aw*/w*/b*</code><br>(full) | 64                        |
| <code>monbus_axi4_axil4_group</code>  | AXI4<br><code>s_axi_ar*/r*</code><br>(full) | AXIL<br><code>m_axil_aw*/w*/b*</code>          | 1                         |
| <code>monbus_axi4_axi4_group</code>   | AXI4<br><code>s_axi_ar*/r*</code><br>(full) | AXI4<br><code>m_axi_aw*/w*/b*</code><br>(full) | 64                        |

Callers pick the wrapper module name matching their fabric. The bare `monbus_group_core` is not intended to be instantiated directly — its AXI4-shaped FUBs are an internal contract.

## 33.7 Related Modules

### 33.7.1 Sub-modules Instantiated

| Sub-module                                              | Role                                                                                                                                                                                                  |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| axi4_slave_rd / axil4_slave_rd                          | Slave-read skid (one per wrapper)                                                                                                                                                                     |
| axi4_master_wr / axil4_master_wr                        | Master-write skid (one per wrapper)                                                                                                                                                                   |
| gaxi_fifo_sync × 2                                      | Err FIFO + write FIFO inside the core                                                                                                                                                                 |
| monbus_compressor (conditional)                         | Compressed-mode encoder; only when USE_COMPRESSION=1                                                                                                                                                  |
| monbus_halfbeat_packer (conditional)                    | Pairs two 30-bit half-slots into one 64-bit beat downstream of the compressor; only when HALF_BEAT_EN=1 (which itself requires USE_COMPRESSION=1)                                                     |
| monbus_cam_pipe (transitive)                            | <b>Pipelined</b> 49-bit-key LRU CAM with per-template last_ts/last_data; replaces the unpipelined monbus_cam as the in-production CAM inside the compressor. See monbus_cam for the deprecation note. |
| math_mod_3_compress × 2 (rtl/math/)                     | Geometry / FIFO whole-record rounding ( $X - (X \bmod 3)$ )                                                                                                                                           |
| math_adder_carry_save_nbit (transitive)                 | 3:2 compressor primitive used by math_mod_3_compress                                                                                                                                                  |
| gaxi_skid_buffer (u_comp_in_skid, in monbus_group_core) | Input skid on the (source_ts, packet) feed into the compressor; breaks the aggregator → CAM long route. The compressor has a separate internal skid (u_res_skid) on the CAM-result path               |

A single canonical filelist enumerates the group core's dependency tree, so a new core dep lands in **one place**:

```
rtl/amba/filelists/monbus_group.f
```

It lists math\_adder\_carry\_save\_nbit + math\_mod\_3\_compress + monbus\_cam + monbus\_cam\_pipe + monbus\_compressor + monbus\_halfbeat\_packer + monbus\_group\_core.

All consumers (val/amba tests, RAPIDS / STREAM macro filelists) - f-include it rather than listing those sources inline.

### 33.7.2 See Also

| Module                 | Role                                                                                                          |
|------------------------|---------------------------------------------------------------------------------------------------------------|
| monbus_compressor      | Bulk-trace encoder used when USE_COMPRESSION=1                                                                |
| monbus_cam_pipe        | Pipelined LRU CAM backing the compressor (production)                                                         |
| monbus_cam             | Unpipelined reference CAM (superseded by monbus_cam_pipe)                                                     |
| monbus_arbiter         | Upstream multi-source merge (instantiate before this family if you have N>1 sources)                          |
| sdpram_slave_axil_axil | Canonical memory-ring backend for the master-write port (AXIL/AXIL variant pairs with sdpram_slave_axil_axil) |
| axi_monitor_base       | Source of the monitor packets this family captures                                                            |
| axi_monitor_reporter   | Per-protocol packet-emission frontend                                                                         |

## 33.8 Testing

Verification lives in val/amba/:

- test\_monbus\_axil4\_axil4\_group.py — basic flow + err-FIFO drain on the AXIL/AXIL wrapper (slave-read coverage; master-write side is driven into a synthetic sink in this test).
- test\_monbus\_axil4\_axil4\_group\_compressed.py — byte-exact compressed slot stream comparison against the Python Encoder golden across three phases (small synth stream, real-silicon 682-record dataset, wrap-window).
- test\_monbus\_axil4\_axil4\_group\_master\_write.py — raw-mode master-write coverage on the AXIL/AXIL wrapper. Three phases: watermark-driven flush (asserts 3\*N beats at 8-byte stride starting at cfg\_base\_addr),

timeout-driven flush, window wrap-back. The test that would have caught the AXIL-master raw-mode flush deadlock fixed on 2026-06-11.

- `test_monbus_axil4_axi4_group.py` — AXI4 burst master-write coverage on the AXIL/AXI4 wrapper. Three phases: watermark-driven multi-beat burst, timeout-triggered flush, 4KB-boundary respect.
- `test_monbus_axi4_axil4_group.py` — dedicated AXI4-slave + AXIL-master coverage. Phase 1 covers the master-write raw-mode flush; Phase 2 covers the AXI4 burst slave-read drain (asserts `rlast` timing across a 6-beat AR with custom `arid`).

```
pytest val/amba/test_monbus_axil4_axil4_group.py \
 val/amba/test_monbus_axil4_axil4_group_compressed.py \
 val/amba/test_monbus_axil4_axil4_group_master_write.py \
 val/amba/test_monbus_axil4_axi4_group.py \
 val/amba/test_monbus_axi4_axil4_group.py -v
```

`monbus_axi4_axi4_group` does not yet have a dedicated test; its master-write path is covered by `test_monbus_axil4_axi4_group.py` (same master leaf) and its AXI4-burst slave-read drain is covered by `test_monbus_axi4_axil4_group.py` Phase 2 (same slave leaf). Adding a direct test for the pure-AXI4 wrapper is a future-work item.

## 34 monbus\_halfbeat\_packer

**Module:** `monbus_halfbeat_packer.sv` **Location:** `rtl/amba/monitor/` **Status:** Production Ready

---

### 34.1 Overview

The `monbus_halfbeat_packer` module packs two 30-bit half-slots into a single 64-bit beat, sitting downstream of `monbus_compressor`. The compressor emits one 64-bit beat per tier-1 record — a 66.7% reduction ceiling, 1 beat per 3-beat raw record. Pair two compatible records into one beat and this packer pushes the reduction to as much as 83.3% (0.5 beat per record). It is bit-exact to the Python golden model `Encoder(half_beat=True)`.

#### 34.1.1 Key Features

- Pairs two 30-bit half-slots into one `TAG_HALF_PAIR (0x4)` beat
- Buffers one half-slot until a partner arrives
- Preserves record order: flushes a buffered half before forwarding a non-half beat



- Flushes a lone trailing half when the input goes idle (last record never stranded)
- Forwards full 64-bit slots and raw-escape beats verbatim
- Bit-exact to the compressor Python golden model for gap-free record streams
- Simple valid/ready handshake on both ports

Trace-capture bandwidth is precious. The compressor already reduces most records to a single 64-bit beat, but records that also fit in 30 bits can be paired two-to-a-beat. This packer performs that pairing with a one-slot holding buffer, emitting a combined beat when a second eligible half arrives while preserving strict record order for everything that cannot be paired.

**Use Cases:** - Maximizing trace history stored in a fixed capture buffer - Reducing MonBus write bandwidth to memory in compressed-capture builds - Matching the software decoder's half-beat encoding exactly (cosim parity)

**Key Benefit:** Breaks the compressor's 1-beat-per-record floor without changing record semantics — pairing is opportunistic and always order-preserving, so it is never wrong, only occasionally less-packed under input bubbles.

## 34.2 Parameters

This module has no parameters.

## 34.3 Ports

### 34.3.1 Clock and Reset

| Port  | Direction | Width | Description                   |
|-------|-----------|-------|-------------------------------|
| clk   | input     | 1     | Clock                         |
| rst_n | input     | 1     | Active-low asynchronous reset |

### 34.3.2 Input — Beats from the Compressor

| Port     | Direction | Width | Description                |
|----------|-----------|-------|----------------------------|
| in_valid | input     | 1     | Input beat valid           |
| in_ready | output    | 1     | Packer ready to accept the |

| Port          | Direction | Width | Description                                                                   |
|---------------|-----------|-------|-------------------------------------------------------------------------------|
| in_slot       | input     | 64    | input beat<br>Full 64-bit slot (tier-1 slot or one of a raw record's 3 beats) |
| in_half_valid | input     | 1     | Set when in_slot's record also fits a 30-bit half-slot                        |
| in_half_slot  | input     | 30    | The 30-bit half-slot for the current record                                   |

### 34.3.3 Output — Packed Beats to the Write FIFO

| Port      | Direction | Width | Description                                                               |
|-----------|-----------|-------|---------------------------------------------------------------------------|
| out_valid | output    | 1     | Output beat valid                                                         |
| out_ready | input     | 1     | Downstream ready                                                          |
| out_slot  | output    | 64    | Packed beat (paired half-slots, forwarded full slot, or half + NOP flush) |

## 34.4 Functional Description

### 34.4.1 Beat Layout

The paired beat matches `monbus_compressor.py`:

```

TAG_HALF_PAIR beat = {tag[63:60]=0x4, slotA[59:30], slotB[29:0]}
half-slot = {sub[29:28], idx[27:23], delta_ts[22:13],
data[12:0]}
NOP slot = 30'd0 (sub == HSUB_NOP)

```

### 34.4.2 Case Decode

Five mutually-considered cases drive the packer combinatorially, based on `in_valid`, `in_half_valid`, and whether a half is buffered (`r_pend_valid`):

| Signal   | Condition                          | Action                                                      |
|----------|------------------------------------|-------------------------------------------------------------|
| pair_now | half arrives, one already buffered | Emit the pair {TAG, pend, in_half}, consume input on accept |

| Signal     | Condition                   | Action                                                                     |
|------------|-----------------------------|----------------------------------------------------------------------------|
| buffer_now | half arrives, none buffered | Latch it, emit nothing, always consume input                               |
| fwd_now    | non-half, none buffered     | Forward in_slot verbatim, consume input on accept                          |
| flush_fwd  | non-half, one buffered      | Emit the lone buffered half (paired with NOP) first, <b>hold</b> the input |
| idle_flush | input idle, one buffered    | Emit the lone trailing half (paired with NOP)                              |

out\_valid asserts for any of pair\_now, fwd\_now, flush\_fwd, or idle\_flush.

### 34.4.3 Handshake and Ordering

in\_ready is asserted unconditionally for buffer\_now (no output is produced that cycle), and follows out\_ready for pair\_now/fwd\_now. For flush\_fwd, in\_ready is held low: the buffered half must be flushed first, so the non-half input is consumed the next cycle as fwd\_now. This is what preserves record order — a full slot can never jump ahead of a still-buffered earlier half.

### 34.4.4 Pending-Slot Register

r\_pend\_valid / r\_pend\_slot hold the first-of-pair half. The register sets on buffer\_now and clears when the buffered half is consumed — either paired (pair\_now) or flushed (flush\_fwd / idle\_flush) — gated by out\_ready.

### 34.4.5 Bit-Exactness

The golden model flushes a lone trailing half exactly once at end-of-stream. This packer flushes whenever its input is idle with a half buffered, which is identical for a gap-free record stream (the cosim drives records back-to-back, then idles once). A mid-stream input bubble would flush early — harmless (slightly less packing), never wrong.

## 34.5 Timing Characteristics

---

This module is **sequential**: it contains clocked logic (via `always_ff` or the repository's `ALWAYS_FF_RST` macro) and therefore holds state. Outputs driven from those blocks are registered and appear one clock after the inputs that produced them.

Per-path cycle counts are not enumerated here; read the block that drives the signal you care about. No synthesis frequency or area figures are quoted – none have been measured against a target device.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

---

## 34.6 Usage Examples

---

The packer is instantiated inside `monbus_group_core` under the `HALF_BEAT_EN==1` generate branch:

```
monbus_halfbeat_packer u_packer (
 .clk (axi_aclk),
 .rst_n (axi_aresetn),
 .in_valid (comp_out_valid),
 .in_ready (comp_out_ready),
 .in_slot (comp_out_slot),
 .in_half_valid (comp_out_half_valid),
 .in_half_slot (comp_out_half_slot),
 .out_valid (comp_wr_valid),
 .out_ready (write_fifo_wr_ready),
 .out_slot (comp_wr_data)
);
```

When `HALF_BEAT_EN==0`, the compressor drives the write FIFO directly (the committed, timing-closed path) and this packer is not elaborated.

---

## 34.7 Design Notes

---

### 34.7.1 Requires the Compressor

Half-beat packing only makes sense downstream of the compressor, so `HALF_BEAT_EN==1` requires `USE_COMPRESSION==1`. In raw-only builds neither block exists.

### 34.7.2 Order Preservation Is Non-Negotiable

The `flush_fwd_hold` is the crux of correctness: without it, a full slot could be emitted while an earlier half still sits buffered, reordering the record stream. The single-cycle hold guarantees the buffered half is written first.

### 34.7.3 One-Slot Buffer Only

The packer holds at most one pending half. It never accumulates more than two records' worth of state, keeping it cheap and its latency bounded to a single beat.

---

## 34.8 Related Modules

---

### 34.8.1 Used By

- **`monbus_group_core.sv`** — instantiates the packer in the `HALF_BEAT_EN==1` branch

### 34.8.2 Uses

- **`reset_defs.svh`** — reset macros (`ALWAYS_FF_RST`, `RST_ASSERTED`)

### 34.8.3 See Also

- **`monbus_compressor.sv`** — upstream compressor supplying the half sideband
  - **`monbus_group_core.sv`** — capture core that owns the compressed write path
- 

## 34.9 Testing

---

**No dedicated testbench for this module.** It has no `val/**/test_monbus_halfbeat_packer.py`. It is exercised indirectly, through the tests of modules that instantiate it (directly or further up):

- `monbus_axi4_axil4_group` – `val/**/test_monbus_axi4_axil4_group.py`
- `monbus_axil4_axi4_group` – `val/**/test_monbus_axil4_axi4_group.py`
- `monbus_axil4_axil4_group` – `val/**/test_monbus_axil4_axil4_group.py`

Indirect coverage exercises this module only in the configurations those parents elaborate. A parameter or mode no parent uses is untested.

Treat any behaviour described on this page as unverified by simulation.

---

## 34.10 References

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### 34.10.1 Source Code

- RTL: rtl/amba/monitor/monbus\_halfbeat\_packer.sv
- Golden model: monbus\_compressor.py (Encoder(half\_beat=True))

### 34.10.2 Documentation

- Architecture: docs/markdown/rtl-amba/shared/README.md
  - Compressor: docs/markdown/rtl-amba/monitor/monbus\_compressor.md
  - Group Core: docs/markdown/rtl-amba/monitor/monbus\_group\_core.md
- 

**Last Updated:** 2026-07-15

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## 34.11 Navigation

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# 35 Monitor Bus Legal-Set CAM

**Module:** monbus\_legal\_cam.sv **Location:** rtl/amba/monitor/ **Category:** Coverage / Observability **Status:** Production Ready

---

## 35.1 Overview

---

monbus\_legal\_cam is the legal-set match CAM used by monbus\_pkt\_tally in profile mode. A CSR-loaded set of legal message identities (agent / protocol / packet-type / event-code keys) is matched combinationally against an incoming key. A **hit** returns the entry's **dense index** — used directly as the tally bin, so per-agent coverage is a first-class count. A **miss** is reported so the caller can route the packet to a single UNEXPECTED bin.

The point is density: the legal message space is ~1.2% of the full protocol × type × event grid, and crossing it with agent\_id would be ~99% empty. Rather than a sparse direct-mapped SRAM, the legal set is enumerated into N\_ENTRIES dense bins, and everything else collapses to one UNEXPECTED bin — which doubles as a spec-violation detector.

---

## 35.2 Parameters

---

| Parameter | Type | Default | Description                            |
|-----------|------|---------|----------------------------------------|
| N_ENTRIES | int  | 64      | Legal-set capacity (dense bins 0..N-1) |
| KEY_WIDTH | int  | 32      | Message-identity key width             |
| IDX_WIDTH | int  | derived | $\text{clog}_2(\text{N\_ENTRIES})$     |

The tally builds the key as {agent\_id[15:0], protocol[3:0], pkt\_type[3:0], event\_code[7:0]} (32 bits).

---

## 35.3 Ports

---

| Port       | Direction | Width     | Description                          |
|------------|-----------|-----------|--------------------------------------|
| clk        | input     | 1         | Clock                                |
| rst_n      | input     | 1         | Active-low asynchronous reset        |
| load_clear | input     | 1         | Pulse: invalidate all entries        |
| load_we    | input     | 1         | Pulse: write one entry               |
| load_addr  | input     | IDX_WIDTH | Entry index to write                 |
| load_valid | input     | 1         | Entry valid bit written with the key |
| load_key   | input     | KEY_WIDTH | Legal message-identity key           |
| lookup_key | input     | KEY_WIDTH | Incoming identity                    |

| Port       | Direction | Width     | Description                                                  |
|------------|-----------|-----------|--------------------------------------------------------------|
| lookup_hit | output    | 1         | to match<br>1 when<br>lookup_key<br>matches a valid<br>entry |
| lookup_idx | output    | IDX_WIDTH | Dense index of the<br>match (valid on<br>hit)                |

## 35.4 Functional Description

- **Valid** is a packed N\_ENTRIES-bit vector, so its reset is a single-shot assign — no per-element delayed-array loop (avoids Verilator BLKLOOPINIT). Keys are gated by valid, so they need no reset.
- **Lookup** is a combinational parallel exact-match against every valid entry, priority-encoded to the low matching index. The host loads unique tuples, so at most one entry matches; the low index wins if a duplicate is ever loaded.
- A hit drives lookup\_idx = the entry index; a miss (lookup\_hit = 0) is the caller's cue to use its UNEXPECTED bin.

## 35.5 Timing Characteristics

This module is **sequential**: it contains clocked logic (via always\_ff or the repository's ALWAYS\_FF\_RST macro) and therefore holds state. Outputs driven from those blocks are registered and appear one clock after the inputs that produced them.

Per-path cycle counts are not enumerated here; read the block that drives the signal you care about. No synthesis frequency or area figures are quoted – none have been measured against a target device.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.



## 35.6 Usage Examples

---

Every parameter and port below is taken from the module declaration.

```
monbus_legal_cam #(
 .N_ENTRIES (64),
 .KEY_WIDTH (32),
 .IDX_WIDTH ((N_ENTRIES > 1))
) u_monbus_legal_cam (
 .clk (clk),
 .rst_n (rst_n),
 .load_clear (load_clear),
 .load_we (load_we),
 .load_addr (load_addr),
 .load_valid (load_valid),
 .load_key (load_key),
 .lookup_key (lookup_key),
 .lookup_hit (lookup_hit),
 .lookup_idx (lookup_idx)
);
```

---

## 35.7 Design Notes

---

Area scales with  $N\_ENTRIES \times KEY\_WIDTH$  comparators. When the full legal set exceeds  $N\_ENTRIES$ , the host loads a slice per run and reprograms for the next — the CAM is CSR-loadable at runtime.

---

## 35.8 Related Modules

---

| Module           | Relationship                                                     |
|------------------|------------------------------------------------------------------|
| monbus_pkt_tally | The sole consumer; instantiates this in PROFILE_MODE.            |
| monbus_cam       | The tally's separate LRU write-combining cache (unrelated role). |

Coverage design and the on-board scenario flow: [vault/handbook/fpga/Genesys2/stream-mon/monitor-board-coverage.md](#).

---

## 35.9 Testing

---

Exercised through the tally's fub test (`val/amba/test_monbus_pkt_tally.py`, profile-mode config): a legal set is loaded, matching and deliberately-illegal packets are driven, and the dense bins + UNEXPECTED are cross-checked against a Python golden.

---

## 35.10 Navigation

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## 36 monbus\_pkt\_tally

**Module:** `monbus_pkt_tally.sv` **Location:** `rtl/amba/monitor/` **Category:** Coverage / Observability **Status:** Production Ready

---

### 36.1 Overview

---

`monbus_pkt_tally` is an **on-chip packet-coverage histogram**. Every accepted monitor-bus packet's identity key {`agent[15:0]`, `protocol[3:0]`, `pkt_type[3:0]`, `event_code[7:0]`} is looked up in a host-loaded **legal set** (a `monbus_legal_cam`); the CAM returns a **dense bin index** — the key's position in the legal set — and the count SRAM at that index is incremented. A key that is *not* in the legal set increments a single **UNEXPECTED** bin (index `N_PROFILE`), so anything unforeseen is caught rather than silently mis-binned. The CAM is **always** in the path — there is no direct-mapped bypass and no write-combining cache.

It's the silicon twin of the sim-side packet-type coverage matrix (`bin/monbus_coverage_report.py` + `TBClasses.monbus.parse`): a bin count > 0 means “this message happened on hardware”, and one readback sweep dumps the whole coverage matrix. And because a counter absorbs any arrival rate, a coverage run can span **millions of cycles** with no capture-bandwidth limit — unlike the [compressor](#)+log path, which bounds capture to the log SRAM depth.

The intended home is the Genesys 2 monitor board-validation build; see `vault/handbook/fpga/Genesys2/stream-mon/monitor-board-coverage.md`.

---

## 36.2 Parameters

| Parameter   | Default            | Description                    | Constraints                                                            |
|-------------|--------------------|--------------------------------|------------------------------------------------------------------------|
| PKT_WIDTH   | 128                | Monitor packet width           | Locked by monitor_common_pkg                                           |
| TS_WIDTH    | 64                 | Side-band timestamp width      | Locked                                                                 |
| COUNT_WIDTH | 32                 | Saturating bin-count width     | $\geq 1$ ; sizes the SRAM word                                         |
| NUM_LATCH   | 4                  | First-event capture slots      | $\geq 1$                                                               |
| ADDR_BITS   | 7                  | Bin address width              | $\geq \lceil \log_2(N\_PROFILE+1) \rceil$ — sizes the dense count SRAM |
| N_PROFILE   | 64                 | Legal-set entries (dense bins) | Dense bins 0 . . N-1; N = UNEXPECTED                                   |
| SRAM_DEPTH  | $1 \ll ADDR\_BITS$ | Derived count-SRAM depth       | Do not override                                                        |
| PROF_KEY_W  | int                | 32                             | {agent[15:0],proto[3:0],type[3:0],event[7:0]}                          |

### 36.2.1 Derived Parameters (do not override)

These are declared as parameter so the elaborator can compute them, not so callers can set them. Each defaults to an expression over the parameters above; overriding one desynchronises it from its source and the design fails to elaborate or silently mis-sizes a bus. Set the parameters they are derived FROM and leave these alone.

| Derived parameter | Default expression                                        |
|-------------------|-----------------------------------------------------------|
| PROF_IDX_W        | $(N\_PROFILE > 1) ? \lceil \log_2(N\_PROFILE) \rceil : 1$ |
| LSEL_WIDTH        | $(NUM\_LATCH > 1) ? \lceil \log_2(NUM\_LATCH) \rceil : 1$ |
| LFILL_WIDTH       | $\lceil \log_2(NUM\_LATCH + 1) \rceil$                    |

## 36.3 Ports

---

The full interface, straight from the RTL:

```
module monbus_pkt_tally #(
 parameter int PKT_WIDTH = 128, // monitor_packet_t width
 (locked)
 parameter int TS_WIDTH = 64, // side-band timestamp width
 (locked)
 parameter int COUNT_WIDTH = 32, // saturating bin count width
 parameter int NUM_LATCH = 4, // first-event capture slots
 parameter int ADDR_BITS = 7, // bin address width (sizes the
 dense SRAM)
 parameter int N_PROFILE = 64, // legal-set entries; dense
 bins 0..N-1, UNEXPECTED = N
 parameter int SRAM_DEPTH = (1 << ADDR_BITS)
 // ... plus the legal-set load port
 (profile_clear/we/waddr/wvalid/wkey)
) (
 input logic clk,
 input logic rst_n,

 // Accepted-packet input (valid/ready; one packet per handshake)
 input logic in_valid,
 output logic in_ready,
 input logic [PKT_WIDTH-1:0] in_packet,
 input logic [TS_WIDTH-1:0] in_ts,

 // Window / snapshot control
 input logic i_freeze, // level: hold
 counting
 input logic i_flush, // no-op (no cache
 to drain)
 output logic o_flush_busy, // high while the
 clear walk runs
 input logic i_clear, // pulse: zero
 everything

 // Count readback (registered; valid one cycle after rd_addr, idle
 only)
 input logic [ADDR_BITS-1:0] rd_addr,
 output logic [COUNT_WIDTH-1:0] rd_count,

 // First-event latch
 input logic i_watch_arm, // level:
 capture enable
 input logic [15:0] i_watch_pkttype_mask, // bit p =
```

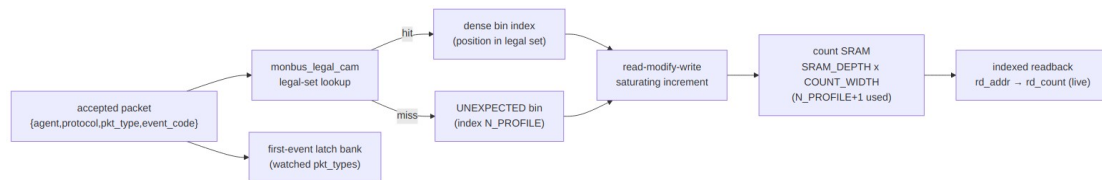
```

watch pkt_type p
input logic [$clog2(NUM_LATCH)-1:0] latch_sel,
output logic latch_valid,
output logic [PKT_WIDTH-1:0] latch_packet,
output logic [TS_WIDTH-1:0] latch_ts,
output logic [$clog2(NUM_LATCH+1)-1:0] latch_fill
);

```

---

## 36.4 Functional Description



### Architecture

A tiny FSM runs the accept: ST\_RUN reads the CAM-resolved bin, ST\_WR writes the saturating-incremented count back; a walk-counter (ST\_CLEAR) zeroes the SRAM on clear. `i_flush` is accepted for interface compatibility but is a **no-op** — there is no cache to drain.

#### 36.4.1 Data Model

```

total(bin) = SRAM[bin] # always live; no cache term
bin = legal_cam.lookup(key) ? dense_index : UNEXPECTED (index
N_PROFILE)

```

- **Legal-set hit** — increment `SRAM[dense_index]` (the key's slot in the loaded legal set).
- **Legal-set miss** — increment `SRAM[UNEXPECTED]`. A non-zero `UNEXPECTED` count means a packet arrived with a tuple the host did not load, and should be flagged loudly.

All counts **saturate**: a pegged bin never wraps.

The **legal set** is loaded by the host over the config port (clear, then one {key, index} pair per entry). The 32-bit key is {agent[15:0], protocol[3:0], pkt\_type[3:0], event\_code[7:0]} — note it includes **agent**, so per-instance identity is resolved, not just the message class. Dense bins run `0..N_PROFILE-1` with `UNEXPECTED = N_PROFILE`, so only `N_PROFILE+1` words are ever *used*, regardless of how sparse the underlying tuple space is. The SRAM itself is `SRAM_DEPTH = 1 << ADDR_BITS` words (128 at the default `ADDR_BITS = 7`), and the clear walk covers all of them.

### 36.4.2 Legal-Set (Dense) Binning

A direct-mapped {protocol, pkt\_type, event\_code} matrix cannot answer “*did every agent fire?*” (agent\_id is not in the key) and is ~99% empty (only ~245 of the 20,480 protocol×type×event cells are ever legal). So the tally does **not** use one; the CAM-resolved dense binning is the **only** mode (the earlier PROFILE\_MODE = 0 direct-mapped path was removed).

A monbus\_legal\_cam holds up to N\_PROFILE legal {agent, protocol, pkt\_type, event\_code} keys; an incoming packet is matched to the entry’s **dense bin index** (a hit) or routed to the single **UNEXPECTED bin** at index N\_PROFILE (a miss, also captured by the first-event latch). This makes per-agent coverage a first-class count, and turns any out-of-profile message — a wrong event code, an untracked agent, a protocol a unit should not speak — into a swept specification violation signal.

The legal set is loaded/cleared over the config port. Two host encodings exist: the raw profile-load pins (profile\_clear/profile\_we/profile\_waddr/profile\_wvalid/profile\_wkey), and — in the STREAM-monitor AXIL wrapper — a **register model** (CAM\_CLEAR, then per entry CAM\_KEY followed by CAM\_LOAD = (valid<<31)|index) that carries the index in the write *data* so it is immune to bus-width/stride hazards. Either way the host can reprogram slices per run when the full legal set exceeds N\_PROFILE.

Coverage gate: every expected dense bin > 0 (all agents/types fired) **and** UNEXPECTED == 0 (nothing rogue slipped through). Note that UNEXPECTED == 0 is also the *fault-free* signal — the [fault classes](#) (error/timeout/threshold) only appear here when a fault was deliberately injected.

### 36.4.3 Snapshot Protocol

The host reads a coherent coverage snapshot over the CSR/AXIL window as follows:

1. i\_freeze = 1 — stop counting at a coherent boundary.
2. Sweep rd\_addr over every bin and read rd\_count (one-cycle registered read, valid only while idle). No flush is needed — counts are live in SRAM.
3. Pulse i\_clear — zero the SRAM + the first-event latches for the next window. o\_flush\_busy is high during the SRAM walk.

rd\_count is **always** the coherent count (= SRAM[rd\_addr]); there is no cache to fold in. i\_flush is retained on the interface but is a no-op.

### 36.4.4 First-Event Latch

NUM\_LATCH slots capture the full 128-bit packet + timestamp of the first accepted packets whose pkt\_type bit is set in i\_watch\_pkttype\_mask (with i\_watch\_arm = 1). This yields the offending packet behind a nonzero error bin, not just a count. latch\_fill reports how many slots are populated; read a slot by driving latch\_sel and sampling latch\_valid / latch\_packet / latch\_ts. The bank is cleared by i\_clear.

---

## 36.5 Timing Characteristics

---

- **Accept path:** a combinational legal-set CAM lookup resolves the bin, then a 2-cycle read-then-saturating-write (ST\_RUN → ST\_WR) commits it. `in_ready` is `(r_st == ST_RUN) && !i_freeze && !i_clear && !r_clear_pend` – so a clear pulse also drops it, and it stays low until the SRAM clear walk finishes. The documented protocol freezes before clearing, which masks this, but a checker written from the “running and unfrozen” summary alone will false-flag a clear that arrives unfrozen. The clear terms are required for valid/ready correctness, not tidiness.
  - **Read:** `rd_count` is a registered read of `SRAM[rd_addr]`, always live.
  - **Clear:** walks `SRAM_DEPTH` entries writing zero (ST\_CLEAR).
  - **Flush:** no-op (no cache).
- 

## 36.6 Usage Examples

---

Every parameter and port below is taken from the module declaration.

```
monbus_pkt_tally #(
 .PKT_WIDTH (128),
 .TS_WIDTH (64),
 .COUNT_WIDTH (32),
 .NUM_LATCH (4),
 .ADDR_BITS (7),
 .N_PROFILE (64),
 .SRAM_DEPTH ((1 << ADDR_BITS)),
 .PROF_IDX_W ((N_PROFILE > 1)),
 .PROF_KEY_W (32),
 .LSEL_WIDTH ((NUM_LATCH > 1))
) u_monbus_pkt_tally (
 .clk (clk),
 .rst_n (rst_n),
 .in_valid (in_valid),
 .in_ready (in_ready),
 .in_packet (in_packet),
 .in_ts (in_ts),
 .i_freeze (i_freeze),
 .i_flush (i_flush),
 .o_flush_busy (o_flush_busy),
 .i_clear (i_clear),
 .rd_addr (rd_addr),
```

```

 .rd_count (rd_count),
 .i_watch_arm (i_watch_arm),
 .i_watch_pkttype_mask (i_watch_pkttype_mask),
 .latch_sel (latch_sel),
 .latch_valid (latch_valid),
 .latch_packet (latch_packet),
 .latch_ts (latch_ts),
 .latch_fill (latch_fill),
 .profile_clear (profile_clear),
 .profile_we (profile_we),
 .profile_waddr (profile_waddr),
 .profile_wvalid (profile_wvalid),
 .profile_wkey (profile_wkey)
);

```

---

## 36.7 Design Notes

---

### 36.7.1 Why Count Instead of Log?

A 128-bit monitor packet plus a 64-bit timestamp is 24 bytes per event. Log every one and a run tops out at  $\text{SRAM\_depth} / 24$  events — a millisecond or two at realistic rates, then you're full. But the board-validation goal is *coverage* ("did every packet type happen, and how often?"), not a trace. For that, a **saturating counter per message bin** is the right structure: it absorbs any rate, drops the compressor (and its worst-case CAM timing path) from the board build, and *is* the coverage matrix in hardware.

Each accepted packet is a read-modify-write on the count SRAM (read the bin, saturating-increment, write it back). A short two-cycle accept sequence services that RMW directly; there is no cache in front. An earlier design added an LRU write-combining cache to collapse the RMWs, but it stranded low-volume counts in the cache (a readback returned SRAM only), so it was removed in favour of the simple, always-coherent direct RMW. `rd_count` is therefore always live — it is `SRAM[rd_addr]` with nothing to drain first.

---

## 36.8 Related Modules

---

### 36.8.1 Building Blocks Reused

| Block                         | Role                                                                                                                      |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <code>monbus_legal_cam</code> | The CSR-loaded legal-set match CAM that maps {agent, protocol, pkt_type, event_code} to a dense bin index (hit) or a miss |



| Block                        | Role                                                                                                                                                  |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| monitor_common_pkg           | (UNEXPECTED). Always in the path.<br>The locked 128-bit packet field map (pkt_type[127:124], protocol[108:105], event_code[104:97], agent_id[87:72]). |
| synchronous single-port SRAM | The backing dense count matrix (N_PROFILE+1 deep).                                                                                                    |

### 36.8.2 See Also

- monbus\_legal\_cam.md — the legal-set CAM that resolves each packet to its dense bin.
- monbus\_compressor.md — the alternative capture path (bounded compressed log) this histogram replaces on the board build.
- axi\_perf\_latency\_hist.md — a companion histogram that bins latency magnitude (not message type) with the same freeze/clear window semantics.
- monbus\_group\_core.md — the filter/route front whose accepted-packet stream feeds this block.

## 36.9 Testing

**Location:** val/amba/test\_monbus\_pkt\_tally.py **Run:** pytest val/amba/test\_monbus\_pkt\_tally.py -v

The acceptance criterion is an **exact** cross-check: after a freeze/flush, the hardware bin counts must equal a pure-Python golden count of the same accepted (protocol, pkt\_type, event\_code) stream. A lost increment shows up as a per-bin mismatch. Phases: random count + readback, back-to-back same-bin accepts (the RMW hazard the two-cycle accept sequence exists to cover), saturation (a bin pegs and never wraps), first-event latch, and clear. (Earlier revisions of this section described eviction-stress phases against an LRU write-combining cache; that cache was removed – every accept is now a direct SRAM read-modify-write.)

## 36.10 Navigation

- [← Back to monitor index](#)

- [← Back to rtl-amba index](#)

## 37 monbus\_wb4\_axil4\_group / monbus\_wb4\_axi4\_group

### 37.1 Overview

The monitor-bus groups with their read-only CSR port presented as a **Wishbone B4 slave** instead of an AXI4-Lite one, so a Wishbone host can own a monitor group without an AXI bridge in front of it.

| Module                 | Host read port | Record-write master |
|------------------------|----------------|---------------------|
| monbus_wb4_axil4_group | Wishbone B4    | AXI4-Lite           |
| monbus_wb4_axi4_group  | Wishbone B4    | AXI4 (burst)        |

Both are **thin wrappers**. The drain path, compressor, FIFOs, interrupt and statistics are the existing monbus\_axil4\_axil4\_group and monbus\_axil4\_axi4\_group bodies, untouched; only the host read port changes, through monbus\_wb4\_rd\_shim. Duplicating the group body would have created a second copy to keep in step, and the copy nobody edits is the one that rots.

### 37.2 monbus\_wb4\_rd\_shim

Wishbone B4 slave in, AXI4-Lite **read** master out.

| Parameter             | Default | Description                                 |
|-----------------------|---------|---------------------------------------------|
| ADDR_WIDTH            | 32      | Address width                               |
| DATA_WIDTH            | 64      | The group's read beat; 32 for a 32-bit host |
| CMD_DEPTH / RSP_DEPTH | 2       | The internal wb4_slave's queues             |
| CLASSIC               | 0       | 0 = B4 pipelined, 1 = B4 standard           |
| AXIL_PROT             | 3'b000  | ARPROT driven on every read                 |

**Writes are refused, not ignored.** The CSR port is read-only. A Wishbone write terminates ERR in the shim and never reaches the group, so a host that writes here learns it was wrong instead of believing it worked.

**One transfer at a time.** This port is a host reading a drain register; pipelining it would add a reorder buffer to save nothing. The internal slave is built `MAX_OUTSTANDING = 1`.

This is deliberately not `wb4_to_axil4`, which carries write channels and an outstanding queue that a read-only CSR port has no use for.

### 37.3 Reading a record

---

Identical to the AXI4-Lite groups: each read pops one beat of the error FIFO, and a record is three 64-bit slices. With `S_AXIL_DATA_WIDTH = 32` the group's 2:1 serializer presents each slice as a low then a high read, so a record is six beats. The Wishbone side sees ordinary single reads either way.

The drain register is **address-insensitive**: any read in the region pops the next beat. That is the group's existing behaviour, not something the shim adds, and it means a test cannot detect a corrupted read address on this port. The data path is checked instead.

### 37.4 Notes

---

- Under `ifdef FORMAL` the shim asserts that a write never reaches the group, that only one read is outstanding, that R is consumed only while a read is open, and that a write is always refused rather than acknowledged.
- The wrappers forward every other group port straight through, so the configuration, statistics and interrupt interfaces are unchanged.

### 37.5 Related

---

- [monbus\\_axil4\\_axil4\\_group](#), [monbus\\_axil4\\_axi4\\_group](#) - the bodies these wrap
- [wb4\\_slave](#) - the Wishbone end of the shim

### 37.6 Test

---

`val/amba/test_monbus_wb4_axil4_group.py` is the AXI4-Lite group's test with its drain switched to Wishbone through the shared `MonbusGroupHarness`, which gained a `drain_proto="wb4"` path for it. Records decode correctly at both the 64-bit and 32-bit drain widths. A mutation rotating the shim's read data fails the test.

## 38 The Monitor System

**What it is:** on-chip observability that emits a fixed 128-bit packet whenever something worth knowing about happens – an error, a timeout, a completion, a performance sample, a debug trace point – and a set of transports that get those packets off the chip.

**What it is not:** protocol-specific. The packet format, the transports, the filtering and the capture back ends know nothing about AXI. A protocol wrapper supplies detection; everything downstream is shared. That is why an arbiter – which is not a bus protocol at all – rides the same infrastructure as AXI4.

**Scope of this document.** The architecture and what it can do. Per-module detail lives in the module pages; the packet field map and event-code assignments live in [monitor\\_package\\_spec](#).

---

## 38.1 The one thing to understand first

---

Everything is a `monitor_packet_t`: 128 bits, laid out identically no matter who produced it.

| Bits      | Field       | Width | Meaning                                      |
|-----------|-------------|-------|----------------------------------------------|
| [127:124] | packet_type | 4     | error, timeout, completion, perf, debug, ... |
| [123:109] | reserved    | 15    | forward-compat slack                         |
| [108:105] | protocol    | 4     | AXI / AXIS / APB / ARB / CORE                |
| [104:97]  | event_code  | 8     | which error, which event, which metric       |
| [96:88]   | channel_id  | 9     | channel, or AXI transaction ID               |
| [87:72]   | agent_id    | 16    | which instance                               |
| [71:64]   | unit_id     | 8     | which subsystem                              |
| [63:0]    | event_data  | 64    | payload – a full 64-bit address fits         |

A 64-bit timestamp travels beside the packet as side-band, not inside it. The pair (packet, timestamp) is carried atomically from the point of emission all the way to the capture back end – `monbus_arbiter` moves both through a 192-bit skid precisely so a consumer can never observe a packet against the wrong timestamp.

The consequence worth internalising: **{protocol, packet\_type, event\_code} is a message identity**. Filtering keys on it. The coverage histogram bins on it. The compressor templates on it.

Adding a new event to a new kind of block means choosing a protocol and allocating event\_code values – and the rest of the system already handles it.

---

## 38.2 Two coordinate systems and a topology

---

This is the property that makes one 128-bit word serve a whole SoC, and none of it is obvious from the field list.

**Read the first two as coordinate systems, not as a tree with a fixed root.** The packet carries a classification (*what happened*) and an identity (*who it happened to*), and the two are orthogonal. Neither is the base. It is entirely reasonable to organise by protocol and treat identity as an attribute of the event – and equally reasonable to organise by unit\_id and treat the classification as an attribute of the block. The packet is indifferent.

That choice is not cosmetic. Putting unit\_id first scopes protocol inside it and turns a hard 16-protocol ceiling into an effectively unbounded space – see [Precedence decides how many protocols exist](#) below, which is the most consequential decision on this page.

Today's *hardware* consumers happen to be classification-major: the filter masks are per-protocol then per-type (cfg\_axi\_pkt\_mask, cfg\_axis\_error\_mask, ...), and monbus\_pkt\_tally addresses its histogram with {protocol, pkt\_type, event\_code} directly. That is an implementation choice of those two blocks, not a property of the format – **nothing in the monitor hardware filters or indexes on unit\_id or agent\_id at all**. A unit-major consumer (“show me everything subsystem 9 did, whatever protocol”) is just as legitimate, needs no change to the packet, and is what a software decoder typically wants; get\_unit\_id() and get\_agent\_id() exist for exactly that.

If a future block wants to filter or bin by identity, the packet already carries what it needs. Only the consumer is missing.

### 38.2.1 Precedence decides how many protocols exist

This is not a reporting preference. **Precedence determines whether protocol is a global namespace or a per-unit one, and that changes the ceiling by a factor of 256** – 16 protocol identities against  $256 \times 16 = 4096$ , with the full message space scaling by the same  $256 \times$  (see the table below).

protocol is 4 bits. Read protocol-major – protocol / packet\_type / unit – and those 4 bits are a **global** namespace: 16 protocols for the entire SoC, for all time, 5 already spent. Every new one must be centrally allocated and must never collide with anything, anywhere. It is a scarce, coordinated resource and it runs out.

Read unit-major – unit / protocol / type – and protocol is **scoped by unit\_id**. Unit 9's protocol 3 and unit 12's protocol 3 are unrelated. Nothing central allocates them; each unit owns its own 16.

|                              | protocol-major                      | unit-major                              |
|------------------------------|-------------------------------------|-----------------------------------------|
| protocol namespace           | global                              | per unit_id                             |
| distinct protocol identities | 16                                  | 256 x 16 = <b>4096</b>                  |
| full message space           | 16 x 16 x 256 = 65,536              | 256 x 16 x 16 x 256 = <b>16,777,216</b> |
| allocating a new protocol    | central registry, collides SoC-wide | local to the unit, collides with nobody |

For any practical purpose the unit-major space is unbounded – you exhaust unit\_id long before you exhaust protocols, and unit\_id is the field you were going to assign per-block anyway.

**This is the same mechanism as event\_code, applied one level up.** event\_code is already scoped per {protocol, packet\_type} rather than globally, which is why adding events never competes with existing assignments. Scoping protocol under unit\_id extends that property to protocols themselves. The packet already works this way one level down; the only question is how far up you take it.

**What it costs.** A protocol-major packet is self-describing: any decoder can read protocol and know what it is holding, with no context. A unit-major packet is not – protocol = 3 means nothing until you know the unit, so every consumer needs the unit-to-protocol map, and that map becomes a real artifact the project must maintain and version. You also give up the free global query: “every AXI error in the SoC” is one mask protocol-major, and a map lookup per unit otherwise.

So it is a genuine trade: **a hard 16-protocol ceiling with zero bookkeeping, against an effectively unbounded space that requires a maintained map.** Small or single-protocol projects should take the ceiling. Large multi-team SoCs, where blocks arrive with their own event vocabularies and no one wants a central protocol registry, should take the map.

### 38.2.2 Fix the choice per project

Whichever you pick, **pick once and hold it for the life of the project.** A different project may reasonably pick the other; two consumers inside the same project may not – with unit-major especially, since a consumer that assumes global protocol will silently mis-decode every packet from a unit that reused a value.

Precedence also leaks into everything downstream: decoder grouping, coverage report row order, the register map a filter is programmed through, dashboards, and the shorthand people use in bug reports. Half a project sorting one way and half the other makes all of those incomparable.

Nothing in the RTL enforces either choice – the packet is a flat tuple and no hardware indexes on identity – so this is a convention, and conventions need a home. Record it where the project records its other integration decisions, alongside the UNIT\_ID and AGENT\_ID assignments, since

those are the same conversation. If the project is unit-major, the unit-to-protocol map lives there too.

38.2.3 Coordinate A – classification: what happened

protocol (4b, 16 slots)    AXI / AXIS / APB / ARB / CORE  
  packet\_type (4b)        error / timeout / completion / threshold /  
perf / debug / ...  
    event\_code (8b)    ARB\_ERR\_STARVATION,  
AXI\_PERFWIN\_BP\_CYCLES, ...

Each level narrows the one above, and **each level is independently useful**. A consumer that only understands packet\_type can still sort errors from performance samples in a stream containing protocols it has never heard of. One that understands protocol too can route AXI traffic to an AXI decoder and pass the rest through untouched. Only the innermost level requires knowing the block.

This is what lets the filter work in stages (drop a whole type, then individual codes within a type you are keeping, then restrict to an address window) and what lets the histogram bin on the whole tuple without knowing what any of it means.

38.2.4 Healthy classes vs fault classes

The packet types split into two groups, and the split changes how you cover them.

**Healthy classes – completion, addr\_match, perf/perfwin/perfhist.** These arise from *correct* operation. Every transaction completes; every access matches a configured watch range; every window closes with utilization buckets. Drive normal traffic and they appear. A capture that tallies them is measuring a working system.

**Fault classes – error, timeout, threshold.** These arise only from a *misbehaving* slave or an *illegal* access, and in a correct system they **never occur** – the fault tally reads **zero**, and that zero is the pass condition, the “nothing went wrong” signal. You cannot exercise them with healthy traffic; you must **explicitly inject the fault**:

| Fault packet | Inject by                                                                                                                                 |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| timeout      | a slave that does not respond (hold R/B past the timeout window)                                                                          |
| threshold    | a slave that is slow (latency past LATENCY_THRESH, under the timeout)                                                                     |
| error        | a slave that returns SLVERR/DECERR, or an access outside an enabled address-range allowlist (see <a href="#">axi_monitor_addr_check</a> ) |

**An error, by definition, hangs the system.** A transaction that errors or is never answered does not retire, so injecting an error *wedges* the traffic that provoked it – that stall is the fault, not a test artifact to engineer around. The monitor’s job, and its whole value, is to **emit the fault packet as the hang happens** so a downstream capture records what stalled and why, instead of the system simply going dark. Consequently, fault coverage belongs in a dedicated fault-injection test that deliberately misbehaves the slaves – kept separate from the healthy-traffic capture, whose job is to assert the fault tally stays zero.

### 38.2.5 Coordinate B – identity: who it happened to

```
unit_id (8b) which subsystem
 agent_id (16b) which instance within it
 channel_id (9b) which channel, or which AXI transaction ID
```

Note this nests the same way the classification does, and to the same depth – three levels, each narrowing the one above. That symmetry is why either can serve as the major key without the other becoming awkward.

UNIT\_ID and AGENT\_ID are elaboration parameters on the monitor, so identity is assigned structurally at integration time, not discovered at runtime. Two instances of the same wrapper differ only by parameter. A consumer can aggregate at any level – all errors in a subsystem, or one channel of one instance – without the producers knowing which grouping anyone intends.

### 38.2.6 Topology – how packets reach the capture point

monbus\_arbiter takes CLIENTS monbus inputs and produces **one monbus output of exactly the same shape**. That is the whole trick: an arbiter’s output is a valid arbiter input, so aggregation nests to any depth without a special “root” or “leaf” module.

STREAM builds a real three-level tree this way – one arbiter per level, each merging the level below:

| Level  | Module                    | CLIENTS          |
|--------|---------------------------|------------------|
| leaf   | scheduler_group           | 2                |
| middle | scheduler_group_ar<br>ray | NUM_CHANNELS + 1 |
| root   | stream_core               | 3                |

Each level collapses its children onto one stream, and the packet is unchanged by the trip – agent\_id still says which leaf produced it. Only the root attaches to a monbus\_\*\_group and off-chip transport. Adding a channel changes one CLIENTS parameter; it does not change the packet, the transport, or any consumer.

Note that the timestamp rides beside the packet through every level, and monbus\_arbiter carries the 192-bit (packet, timestamp) pair atomically at each hop – so depth in the tree does not risk pairing a packet with the wrong time.



**The shape of the tree is arbitrary, and it is expected to change.** Nothing in the packet, the transport or any consumer encodes how many levels there are, which leaf hangs off which branch, or how wide each merge is. It is an integration choice: re-parent a block, add a level, collapse two levels into one wider CLIENTS, and every downstream stage behaves identically. That freedom is the point of the input/output symmetry – the tree can be re-drawn to match floorplan or timing without touching a line of monitor logic.

Two consequences follow, and both are contracts rather than advice:

- **Arrival order across producers means nothing.** Each level merges with `arbiter_round_robin` in ACK mode, so the interleaving you observe is a product of arbitration phase, backpressure and tree shape – all three of which can change. **Order packets by their timestamp, never by the order they came out of the pipe.** A consumer that infers causality from arrival order is reading an artifact of the topology it was captured on, and will silently disagree with itself when the tree is re-drawn.
- **Per-producer order IS preserved.** A single producer’s packets stay in the order it emitted them, at every level, because arbitration reorders *between* clients and never *within* one. So “this agent’s events, in sequence” is reliable; “these two agents’ events, interleaved” is not.

What the tree does guarantee is fairness and no loss: round-robin prevents a chatty leaf from starving a quiet one, and the grant-hold contract (the grant retires only on `grant && valid && ready`) means backpressure delays packets rather than dropping them. Depth costs latency, never data.

---

### 38.3 Adaptability: what is deliberately left open

---

The format is locked, which is what makes bit-exact tooling possible. It is not, however, full:

| Space         | Used                          | Free                                 |
|---------------|-------------------------------|--------------------------------------|
| protocol      | 5 (AXI, AXIS, APB, ARB, CORE) | 11 slots                             |
| packet_type   | 13                            | 0xA, 0xB, 0xC reserved               |
| event_code    | per {protocol, packet_type}   | 8 bits <i>per pair</i>               |
| reserved bits | none                          | [123:109], 15 bits of forward-compat |

| Space      | Used    | Free                                     |
|------------|---------|------------------------------------------|
| event_data | payload | slack<br>64 bits, fits a full<br>address |

The event\_code row is the important one. Because the code is scoped by the {protocol, packet\_type} pair rather than global, a new protocol gets a fresh 256-value space for each packet type it uses. Extension does not compete with existing assignments, so nobody has to coordinate a global registry.

PROTOCOL\_CORE exists as the general-purpose slot for blocks that are not a bus protocol – reach for it before spending one of the 11 free protocol values. Spend a protocol slot only when the block is a *family* worth separating in filters and coverage reports.

The 15 reserved bits are genuine slack, not padding to a round number: they sit between packet\_type and protocol, so a future field can be added without moving either the type tag at the top or the 64-bit payload at the bottom, and without changing the width. Existing decoders that mask on the documented fields keep working.

## 38.4 Functional Description

| DETECT<br>CAPTURE                                                                                                                                                                                                                                                                                                                      | SHAPE                                                                                                                                                                                               | TRANSPORT                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| -----<br>-----                                                                                                                                                                                                                                                                                                                         | -----                                                                                                                                                                                               | -----                                                             |
| protocol wrapper<br>monbus_*_group<br>axi4_master_rd_mon<br>error FIFO<br>axi5_slave_wr_mon<br>-> AXI read<br>axil4_*_mon<br>(IRQ)<br>apb4_monitor<br>write FIFO<br>apb5_monitor<br>-> AXI write<br><br>(bulk trace)<br>custom producer<br>arbiter_rr_pwm_monbus<br>monbus_compressor<br>arbiter_wrr_pwm_monbus<br>(optional, in-line) | axi_monitor_base<br><br>trans_mgr (CAM)<br><br>timer / timeout<br><br>reporter_error<br><br>reporter_timeout<br><br>reporter_compl<br><br>reporter_threshold<br><br>reporter_perf<br>reporter_debug | monbus_arbiter<br><br>(N:1 merge,<br><br>packet+ts<br><br>atomic) |

your block

```
monbus_pkt_tally
 > axi_monitor_filtered (counting only)
 (staged drop filter)
```

Four stages, and you can stop after any of them.

### 38.4.11. Detect

A **protocol wrapper** pairs one protocol block with the shared monitor core.

axi4\_master\_rd\_mon instantiates axi4\_master\_rd and axi\_monitor\_filtered; the wrapper's whole job is to present protocol signals to a core that does not know what AXI is. These live with their protocols (rtl/amba/axi4/, axi5/, axil4/, apb/, apb5/), not in rtl/amba/monitor/.

For AXI-shaped traffic the core tracks outstanding transactions in monitor\_trans\_cam, a multi-port ID CAM with payload storage, managed by axi\_monitor\_trans\_mgr. That table is what makes completion, timeout and latency reporting possible: an event is attributed to the transaction that caused it.

**A custom block skips all of this.** See “Instrumenting something that is not a bus protocol” below.

### 38.4.22. Shape

axi\_monitor\_base decides what is worth reporting, through six reporters that each own one packet type:

| Reporter                       | Emits             | Answers                                                                                                               |
|--------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------|
| axi_monitor_reporter_error     | PktTypeError      | a protocol or response error occurred                                                                                 |
| axi_monitor_reporter_timeout   | PktTypeTimeout    | a transaction outlived its budget                                                                                     |
| axi_monitor_reporter_compl     | PktTypeCompletion | a transaction finished (and how long it took)                                                                         |
| axi_monitor_reporter_threshold | PktTypeThreshold  | a watermark was crossed                                                                                               |
| axi_monitor_reporter_perf      | PktTypePerf only  | lifetime completion/error count rollups. <b>Not</b> PerfWin/PerfHist – nothing packetizes the window buckets onto the |

| Reporter                       | Emits        | Answers                                                                           |
|--------------------------------|--------------|-----------------------------------------------------------------------------------|
|                                |              | MonBus yet (see Performance Monitoring below); they are readable as counters only |
| axi_monitor_reporter_<br>debug | PktTypeDebug | trace points                                                                      |

Each has an `ENABLE_*_LOGIC` parameter that removes its detection cone at elaboration. This matters: an unused reporter is not gated at runtime, it is **not built**. `ENABLE_TIMEOUT_LOGIC=0` also drops the `axi_monitor_timeout` instance outright. Size the monitor to what you will actually read.

### 38.4.33. Filter

Three mechanisms drop packets *at the source*, before one ever consumes transport bandwidth. Note that these are not the Level 1/2/3 numbers used inside `axi_monitor_filtered.sv` – that module numbers its own two active stages 1 and 3, with 2 reserved. Here they are grouped by what they do:

1. **cfg\_axi\_pkt\_mask** – drop whole packet types (16 bits, one per type).
2. **per-type event masks** – `cfg_axi_error_mask`, `cfg_axi_timeout_mask`, `cfg_axi_compl_mask`, `cfg_axi_thresh_mask`, `cfg_axi_perf_mask`, `cfg_axi_addr_mask`, `cfg_axi_debug_mask`. Drop individual event codes within a type you are otherwise keeping.
3. **address-range selection** – `axi_monitor_addr_check` (and `apb_monitor_addr_check` for APB) restrict reporting to address windows.

Filtering at the source rather than at the consumer is deliberate. The monitor bus is a shared resource and the failure mode is congestion: enable everything at once and the interesting packets queue behind the boring ones.

### 38.4.44. Transport and capture

`monbus_arbiter` merges N producer streams into one, carrying packet and timestamp atomically. The `monbus_group` family then provides two independent drains from one `monbus_group_core`:

- **Error / interrupt path** – a FIFO drained over an AXI4-shaped *slave read* interface, 192 bits per record. An IRQ handler walks records as 3 x 64-bit beats; `arlen` may fetch several records per burst. This is the “wake the CPU, something is wrong” path.

- **Master-write path** – a beat-granular FIFO drained over an AXI4-shaped *master write* interface with watermark and timeout flush, bursting as far as FIFO contents, MAX\_BURST\_BEATS, the 4 KB boundary and the address-window wrap allow. This is the bulk-trace path.

Four wrappers exist for the four combinations of interface width on those two ports: `monbus_axi4_axi4_group`, `monbus_axi4_axil4_group`, `monbus_axil4_axi4_group`, `monbus_axil4_axil4_group`. They are wrappers only – the logic is in `monbus_group_core`.

## 38.5 Choosing a capture strategy

This is the decision that actually shapes a deployment, and there are three answers.

|               | Bulk trace                                | Compressed trace                      | Counting                       |
|---------------|-------------------------------------------|---------------------------------------|--------------------------------|
| Module        | <code>monbus_group_core</code> write path | + <code>monbus_compressor</code>      | <code>monbus_pkt_tally</code>  |
| Off-chip cost | 24 B per record                           | ~8 B per record on template hits      | zero until readback            |
| Run length    | bounded by log SRAM depth                 | bounded, but several times longer     | <b>unbounded</b>               |
| Keeps         | every packet, exactly                     | every packet, exactly                 | counts per message identity    |
| Loses         | nothing                                   | nothing                               | ordering, timestamps, payloads |
| Use when      | debugging a specific failure              | long capture, still need every packet | “did this ever happen?”        |

### 38.5.1 Bulk trace

Raw mode emits complete 24-byte (3-beat) records. Simple, exact, and the run length is whatever your log SRAM holds.

### 38.5.2 Compressed trace – `monbus_compressor`

Opt in per wrapper via `USE_COMPRESSION`. It sits in front of the master writer and turns (packet, timestamp) records into 64-bit self-tagged slots:

| Tag | Format     | Beats | When                           |
|-----|------------|-------|--------------------------------|
| 0x0 | RAW escape | 3     | no template match              |
| 0x1 | T1-A       | 1     | template hit, small payload    |
| 0x2 | T1-B       | 1     | template hit, big delta_ts     |
| 0x3 | T1-C       | 1     | template hit, event_data delta |

A 32-entry true-LRU CAM (`monbus_cam`) holds the templates, keyed on message identity. Each entry stores **its own** last timestamp, so `delta_ts` is measured per template – interleaved producers do not force each other into the raw escape. Throughput is one record per cycle on template hits, one per three cycles on escapes.

The format is a locked spec, and the acceptance criterion is unusually strong: the hardware slot stream is **bit-exact** to what the Python Encoder in `bin/TBClasses/monbus/monbus_compressor.py` produces from the same record sequence. Per-tier hit counters (`tier1_a`, `tier1_b`, `tier1_c`, `tier0_escape`) let you measure the compression you are actually getting rather than assuming it.

`monbus_halfbeat_packer` sits downstream and packs two 30-bit half-slots into one 64-bit beat where the format allows.

### 38.5.3 Counting – `monbus_pkt_tally`

The one that changes what is *possible* rather than what is efficient.

It counts accepted packets into an SRAM histogram addressed by `{protocol, packet_type, event_code}`, fronted by a 32-entry LRU write-combining cache so back-to-back hits on hot bins never reach the SRAM. One readback sweep dumps the matrix.

**A counter absorbs any arrival rate.** The trace paths are bounded by log SRAM depth; the tally is not bounded at all. A coverage run can span millions of cycles. This is the silicon twin of the simulation-side packet-type coverage matrix (`bin/monbus_coverage_report`, `TBClasses.monbus.parse`) – a bin count greater than zero means *this message was observed on hardware*.

You give up ordering, timestamps and payloads. If the question is “which of these 200 events ever fired in a week of running”, that is the right trade.

## 38.6 Instrumenting something that is not a bus protocol

The system was built for AXI and then generalised, and the arbiters are the proof that the generalisation is real. `PROTOCOL_ARB` is a first-class protocol with its own event codes:

| Code | Event                                                                      |
|------|----------------------------------------------------------------------------|
| 0x0  | ARB_ERR_STARVATION – a client never got service                            |
| 0x1  | ARB_ERR_ACK_TIMEOUT                                                        |
| 0x2  | ARB_ERR_PROTOCOL_VIOLATION                                                 |
| 0x3  | ARB_ERR_CREDIT_VIOLATION                                                   |
| 0x4  | ARB_ERR_FAIRNESS_VIOLATION – weighted round-robin is not honouring weights |
| 0x5  | ARB_ERR_WEIGHT_UNDERFLOW                                                   |
| 0x6  | ARB_ERR_CONCURRENT_GRANTS                                                  |
| 0x7  | ARB_ERR_INVALID_GRANT_ID                                                   |
| 0x8  | ARB_ERR_ORPHAN_ACK                                                         |
| 0x9  | ARB_ERR_GRANT_OVERLAP                                                      |
| 0xA  | ARB_ERR_MASK_ERROR                                                         |
| 0xB  | ARB_ERR_STATE_MACHINE                                                      |
| 0xC  | ARB_ERR_CONFIGURATION                                                      |

None of these mean anything to AXI. They ride the identical packet, the identical arbiter, the identical filter, and land in the identical histogram bin. `arbiter_monbus_common` carries the shared production logic; `arbiter_rr_pwm_monbus` and `arbiter_wrr_pwm_monbus` are the instrumented round-robin and weighted round-robin arbiters.

### To instrument your own block:

1. Pick a protocol value. `PROTOCOL_CORE` (4'h4) exists for exactly this – blocks that are not a bus. Add a new `protocol_type_t` only if your block is a family worth separating; there are 16 slots and 5 are used.
2. Allocate `event_code` values in a package next to `monitor_arbiter_pkg.sv`, and document them. The code is 8 bits per {protocol, packet\_type} pair, so the space is not tight.
3. Emit a `monitor_packet_t`. Fill `unit_id` / `agent_id` so the packet identifies your instance, and use the 64 bits of `event_data` for whatever the event needs.

4. Feed `monbus_arbiter`. From that point on you inherit filtering, both transports, compression and the coverage histogram without writing any of it.

Step 4 is the payoff. The reason to use this system rather than adding your own debug registers is that everything after emission already exists.

---

## 38.7 Performance monitoring

---

Two distinct perf paths exist, and only one currently reaches the bus:

- **PktTypePerf rollup (implemented).** `ENABLE_PERF_LOGIC` (legacy alias `ENABLE_PERF_PACKETS`) builds `axi_monitor_reporter_perf`, which rolls up lifetime completion/error counts and emits `PktTypePerf` (`AXI_PERF_COMPLETED_COUNT = 0x7`, `AXI_PERF_ERROR_COUNT = 0x8`). This is the perf packet you see on the MonBus today. It advances only while the output is idle and loses the reporter mux to completion/threshold, so it surfaces in the gaps between traffic rather than under sustained load.
- **PktTypePerfWin / PktTypePerfHist window+histogram (RFC Stage B/F – not yet emitted).** The window state machine below exists (perfmon Stage A in `axi_monitor_base`) and maintains its bucket counters, but the code that *packetizes* those counters onto the MonBus is not wired yet. **No module currently emits `PktTypePerfWin` or `PktTypePerfHist`** – the buckets are readable only as counters, so a MonBus tally can never bin them. The description below is the intended shape once Stage B/F lands.

The window state machine buckets every cycle into one of four states and (once Stage B/F lands) reports at window close:

| Code | AXI_PERFWIN_* | Meaning                         |
|------|---------------|---------------------------------|
| 0x1  | PROD_CYCLES   | valid && ready – productive     |
| 0x2  | BP_CYCLES     | valid && !ready – backpressured |
| 0x3  | STARV_CYCLES  | !valid && ready – starved       |
| 0x4  | IDLE_CYCLES   | !valid && !ready – idle         |



Those four partition the window, which is what makes them useful: a link at 40% utilization is a different problem depending on whether the other 60% is backpressure (the consumer is slow) or starvation (the producer is). Alongside them: BEAT\_COUNT, BYTE\_COUNT (beats x 1<<axsize, masked by strb), BURST\_COUNT, window start and end timestamps, and per-channel splits (CHAN\_PROD, CHAN\_STARV, CHAN\_BP).

PktTypePerfHist reports histogram buckets instead of totals: event\_code[7:4] selects the histogram, event\_code[3:0] the bucket (0..15, log2 cycle thresholds). Use it for latency distributions, where a mean hides the tail that actually matters.

## 38.8 Configuration cautions

Two that have bitten before:

- **Do not enable every packet type at once.** cfg\_compl\_enable together with cfg\_perf\_enable is the documented congestion case – completion packets are frequent and perf packets are bursty, and together they crowd out errors. Use separate test configurations.
- **The transaction table has a saturation-recovery contract.** Command-originated entries are capped below MAX\_TRANSACTIONS so a saturated table can always drain; cmd\_entry\_reserve() in monitor\_common\_pkg is the single source of truth, and both axi\_monitor\_trans\_mgr and axi\_monitor\_base derive from it. Tables smaller than 16 take reserve 0 and trade the guarantee for capacity. This was a comment-encoded invariant once, and that is how a saturation wedge shipped.

## 38.9 Related Modules

| For                                   | Read                                  |
|---------------------------------------|---------------------------------------|
| Packet fields, event-code assignments | <a href="#">monitor_package_spec</a>  |
| The monitor core                      | <a href="#">axi_monitor_base</a>      |
| Filtering                             | <a href="#">axi_monitor_filtered</a>  |
| Capture back end                      | <a href="#">monbus_group_core</a>     |
| Compression                           | <a href="#">monbus_compressor</a>     |
| Counting                              | <a href="#">monbus_pkt_tally</a>      |
| Arbiter instrumentation               | <a href="#">arbiter_monbus_common</a> |
| A protocol wrapper                    | <a href="#">axi4_master_rd_mon</a>    |

## 38.10 Navigation

- [← Back to Shared Infrastructure Index](#)
- [← Back to rtl-amba Index](#)
- [← Back to Main Documentation Index](#)

## 39 monitor\_trans\_cam

**Module:** monitor\_trans\_cam.sv **Location:** rtl/amba/monitor/ **Category:** AXI Monitor Infrastructure **Status:** Production Ready

### 39.1 Overview

monitor\_trans\_cam is a **multi-port ID CAM with opaque payload storage** used by the axi\_monitor\_trans\_mgr to track outstanding AXI transactions by ID. It replaces the inline parallel-match logic that previously lived in axi\_monitor\_trans\_mgr (2026-04-23 revision), moving the keying state and payload storage into a reusable shared module without changing the synthesis shape that closes 100 MHz on the xc7a100t-1 Artix-7.

Don't confuse it with monbus\_cam — the two solve different problems:

| Feature       | monbus_cam                          | monitor_trans_cam                                     |
|---------------|-------------------------------------|-------------------------------------------------------|
| Use case      | Caching (compressor template index) | Associative storage (AXI ID tracking)                 |
| Eviction      | True LRU                            | None (explicit alloc / free)                          |
| Ports         | 1 lookup                            | 3 lookups + alloc priority encoder                    |
| Payload       | 64 b                                | Parameterizable (typically \$bits(bus_transaction_t)) |
| Default depth | 32                                  | 16                                                    |
| Match output  | hit + idx + old_data                | Three one-hot vectors                                 |

### 39.1.1 Key Features

- 3 independent ID lookup ports per cycle (addr / data / resp)
  - One-hot match vectors (\*\_match\_oh) plus a lowest-index “first” variant (data\_match\_first\_oh; provided by the CAM but left unconnected by the current axi\_monitor\_trans\_mgr, which resolves multi-match cases itself, oldest-first)
  - Free-slot vector (free\_oh) and priority-encoded **3-way mutex alloc** (addr\_alloc\_oh / data\_alloc\_oh / resp\_alloc\_oh)
  - Per-slot write port (one-hot enable, separate next-state inputs per slot)
  - Per-slot read port (registered current state) exposed for the caller
  - (\* keep = "true" \*) attributes on match/free vectors so synth can't fuse them into downstream update cones (preserves the trans\_mgr WNS fix)
  - Simulation-only protocol assertions on the caller
- 

## 39.2 Parameters

| Parameter     | Default | Notes                                                                          |
|---------------|---------|--------------------------------------------------------------------------------|
| DEPTH         | 16      | Maximum outstanding transactions. Asserts in TBs require $\geq 4$ .            |
| ID_WIDTH      | 8       | AXI ID width. Typically 4–8 for AXI4.                                          |
| PAYLOAD_WIDTH | 128     | Caller's opaque payload. trans_mgr uses \$bits(bus_transaction_t) (~285 bits). |

The trans\_mgr instantiates with PAYLOAD\_WIDTH = \$bits(bus\_transaction\_t), storing the full struct (including the redundant id field — the CAM's entry\_id and payload.id are written atomically so they always agree).

---

## 39.3 Ports

The full interface, straight from the RTL:

```
module monitor_trans_cam #(
 parameter int DEPTH = 16,
```

```

parameter int ID_WIDTH = 8,
parameter int PAYLOAD_WIDTH = 128
) (
 input logic clk,
 input logic rst_n,

 // Synchronous clear: on the next edge, invalidate every entry
 // (empty the CAM) so the host can rebase the transaction table
 // without a full rst_n. Takes priority over per-slot entry_we.
 input logic clear,

 // ---- Lookup ports (combinational) ----
 input logic [ID_WIDTH-1:0] lookup_addr_id,
 input logic [ID_WIDTH-1:0] lookup_data_id,
 input logic [ID_WIDTH-1:0] lookup_resp_id,
 output logic [DEPTH-1:0] addr_match_oh,
 output logic [DEPTH-1:0] data_match_oh,
 output logic [DEPTH-1:0] resp_match_oh,
 output logic [DEPTH-1:0] data_match_first_oh,
 // lowest-index match for
data port

 // ---- Free + 3-way alloc priority encoder ----
 output logic [DEPTH-1:0] free_oh,
 input logic addr_wants_alloc,
 input logic data_wants_alloc,
 input logic resp_wants_alloc,
 // Per-phase allocation masks -- TIE ALL ONES for unrestricted
behaviour.
 // Every alloc loop gates on these; leaving them unconnected
floats them
 // (X in simulation) or ties them to 0 (synthesis), and allocation
then
 // never fires at all.
 input logic [DEPTH-1:0] addr_alloc_mask,
 input logic [DEPTH-1:0] data_alloc_mask,
 input logic [DEPTH-1:0] resp_alloc_mask,
 output logic [DEPTH-1:0] addr_alloc_oh,
 output logic [DEPTH-1:0] data_alloc_oh,
 output logic [DEPTH-1:0] resp_alloc_oh,

 // ---- Per-slot write port ----
 input logic [DEPTH-1:0] entry_we,
 input logic [DEPTH-1:0] entry_valid_next,
 input logic [ID_WIDTH-1:0] entry_id_next
[DEPTH],
 input logic [PAYLOAD_WIDTH-1:0] entry_payload_next

```

```

[DEPTH],

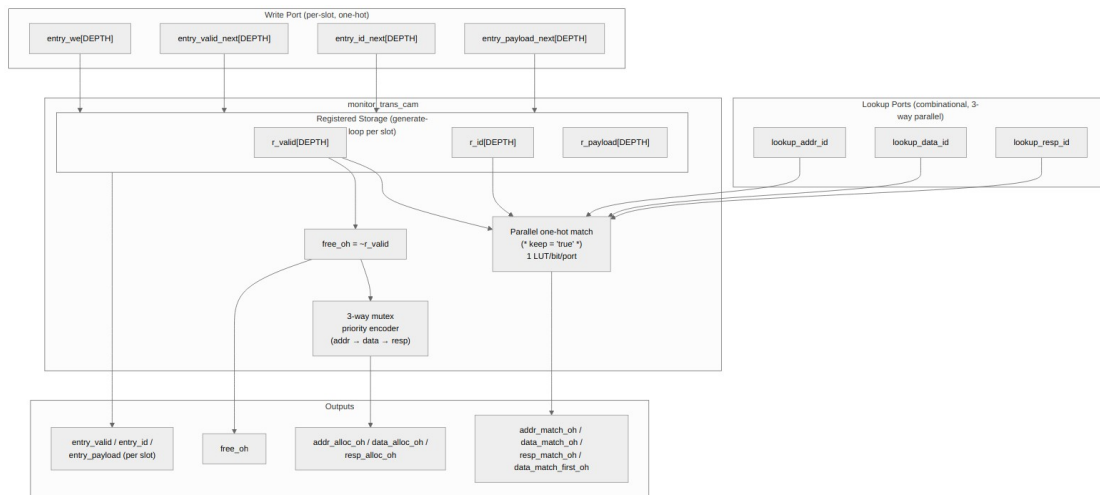
 // ---- Per-slot read port (registered) ----
 output logic [DEPTH-1:0] entry_valid,
 output logic [ID_WIDTH-1:0] entry_id
[DEPTH],
 output logic [PAYLOAD_WIDTH-1:0] entry_payload
[DEPTH]
);

```

---

## 39.4 Functional Description

### 39.4.1 Architecture



*monitor\_trans\_cam block diagram*

Source: monitor\_trans\_cam.mmd

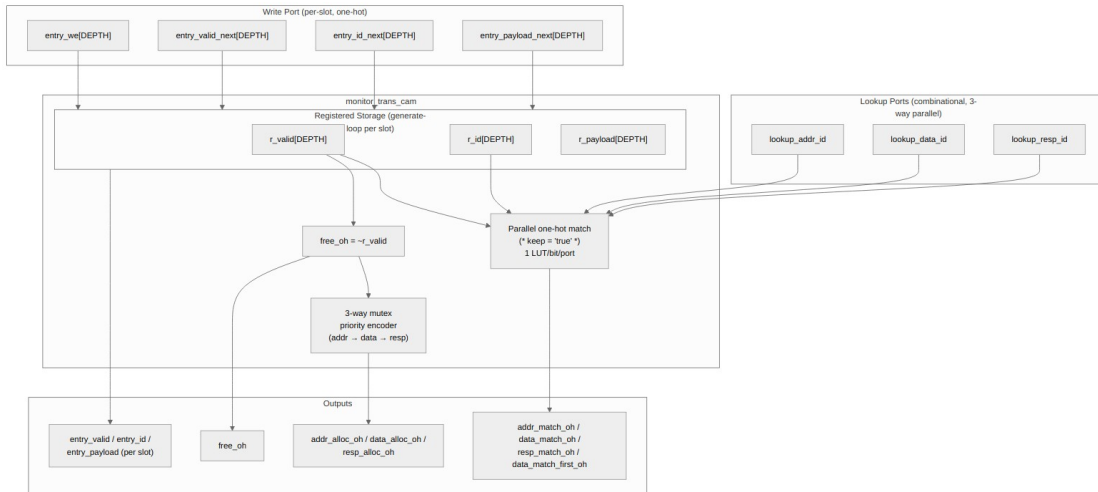


Diagram 12

### 39.4.2 Why Three Lookup Ports?

The AXI monitor needs to match an arriving transaction beat against the outstanding transaction table along three independent channels in the same cycle:

| Lookup port    | Purpose                                                                                                   |
|----------------|-----------------------------------------------------------------------------------------------------------|
| lookup_addr_id | Matches an arriving AW/AR beat to an in-flight transaction (or detects “this is a brand-new transaction”) |
| lookup_data_id | Matches an arriving R beat (read) or detects orphan-data-without-AW (write, AXI-Lite only)                |
| lookup_resp_id | Matches an arriving B beat (write) to an in-flight transaction                                            |

All three lookups go in parallel. The CAM’s internal storage feeds 3 one-hot match vectors, each a 1-LUT-per-bit valid && (id == lookup\_id) test. With DEPTH=16 that’s 16 small independent cones per port, no chained dependencies.

The (\* keep = "true" \*) attribute on the match vectors prevents Vivado from fusing them into the downstream axi\_monitor\_trans\_mgr update logic — without that, the 16 entries’ update cones would fold together into one shared wide function, producing ~12 LUT levels and failing 100 MHz on the -1 part. With the attribute, each entry’s update stays an independent small cone.

### 39.4.3 The “First-Match” Variant

For **AXI4 writes**, the data channel has no WID. The `trans_mgr` matches an arriving W beat to an outstanding write transaction by state predicate (`valid && state ∈ {ADDR_PHASE, DATA_PHASE} && cmd_received && !data_completed`), NOT by id. That state predicate is computed *outside* the CAM (it requires fields the CAM doesn’t see, like `state` and `cmd_received`), and — because several entries can satisfy it — is resolved to the **oldest** candidate via the `trans_mgr`’s rank-based `pick_oldest()` selector, matching AXI4’s write-data ordering rule.

The CAM also provides `data_match_first_oh`, the lowest-index bit of `data_match_oh` (the id-based match). The current `axi_monitor_trans_mgr` leaves this output **unconnected**: lowest-index is not issue order once slots are recycled, so for reads the `trans_mgr` feeds the raw `data_match_oh` candidates through `pick_oldest()` instead, and for writes it uses its own state-predicate vector as above. The port remains for callers that need a deterministic lowest-index pick. The two paths share the rest of the CAM machinery (free, alloc, write port) so the cost is minimal.

### 39.4.4 3-Way Mutex Alloc Priority Encoder

When a new transaction arrives that doesn’t match an existing entry, the `trans_mgr` needs to allocate a free slot for it. Up to three independent “want to alloc” requests can fire in the same cycle (`addr`, `data`, `resp`), and each needs a *different* slot — the priority encoder ensures no two phases ever claim the same free slot.

The encoder is purely combinational:

```
remaining_free = free_oh

if addr_wants_alloc:
 addr_alloc_oh = lowest-set-bit(remaining_free & addr_alloc_mask)
 remaining_free &= ~addr_alloc_oh

if data_wants_alloc:
 data_alloc_oh = lowest-set-bit(remaining_free & data_alloc_mask)
 remaining_free &= ~data_alloc_oh

if resp_wants_alloc:
 resp_alloc_oh = lowest-set-bit(remaining_free & resp_alloc_mask)
 remaining_free &= ~resp_alloc_oh
```

The three outputs are guaranteed to be disjoint (mutex assertion enforces this in simulation). If `wants_alloc` is asserted with no free slot, the corresponding `*_alloc_oh` is '0 — the caller need not gate its `wants_alloc` requests on free-slot availability (the `trans_mgr` gates them on match suppression and its command-entry cap instead).

Priority order is `addr → data → resp`. AXI4 writes can have all three firing in the same cycle on a tight pipeline, and the order matches the phase order so the slot indices in the table grow in a reasonable chronological order.

### 39.4.5 Per-Slot Storage and Write Port

The CAM owns the registered storage:

```
logic r_valid [DEPTH];
logic [ID_WIDTH-1:0] r_id [DEPTH];
logic [PAYLOAD_WIDTH-1:0] r_payload [DEPTH];
```

The write port is one-hot per slot:

```
// Per slot i: if (entry_we[i]) then on next clock edge,
// r_valid[i] <= entry_valid_next[i];
// r_id[i] <= entry_id_next[i];
// r_payload[i] <= entry_payload_next[i];
```

To clear a slot, drive `entry_we[i]=1` with `entry_valid_next[i]=0`. The payload and id get written too (with whatever the caller drives) but the valid bit determines whether subsequent lookups see the entry.

The per-slot updates live in a generate loop of independent `always_ff` blocks, one per slot. Each block depends only on local one-hot bits and shared inputs — synth cannot fuse them across slots. The result is DEPTH parallel small update cones rather than one giant shared one.

---

## 39.5 Timing Characteristics

---

This module is **sequential**: it contains clocked logic (via `always_ff` or the repository's `ALWAYS_FF_RST` macro) and therefore holds state. Outputs driven from those blocks are registered and appear one clock after the inputs that produced them.

Per-path cycle counts are not enumerated here; read the block that drives the signal you care about. No synthesis frequency or area figures are quoted – none have been measured against a target device.

Timing closure is therefore a question of the surrounding logic's slack, not of this module's cycle count. No synthesis figures are quoted; none have been measured.

---

## 39.6 Usage Examples

---

### 39.6.1 Caller Pattern (`axi_monitor_trans_mgr`)

The `trans_mgr` uses the CAM's outputs to compute per-slot next-state in a generate-loop `always_comb`, then drives the write port:

```
For each slot i (combinational):
 next_payload = current_payload (default: hold)
```



```

next_we = 0

if addr_alloc_oh[i]:
 # New transaction starting in addr phase
 next_payload = initialize from cmd_*
 next_we = 1
 next_id = cmd_id

elif addr_update_oh[i] && cmd_handshake:
 # Existing entry getting addr fields updated
 next_payload.cmd_received = 1
 next_payload.addr_timestamp = timestamp
 next_we = 1

... data and resp phases similar ...

if can_cleanup[i]:
 next_payload.valid = 0
 next_we = 1

if event_reported_flag[i] && !current_payload.event_reported:
 next_payload.event_reported = 1
 next_we = 1

```

Drive entry\_we[i], entry\_valid\_next[i], entry\_payload\_next[i] for the CAM.

The CAM's entry\_valid output mirrors the payload's .valid field — the trans\_mgr writes them in sync atomically. Lookups use entry\_valid (the CAM's view) so the match logic stays tight.

## 39.7 Design Notes

### 39.7.1 Synthesis Notes

Key design choices to preserve the trans\_mgr's 2026-04-23 WNS fix:

| Construct                                  | Rationale                                                                          |
|--------------------------------------------|------------------------------------------------------------------------------------|
| (* keep = "true" *) on *_match_oh, free_oh | Anchors the match vectors so Vivado doesn't fuse them into downstream update cones |
| Per-slot generate-loop always_ff           | DEPTH independent small update cones instead of one shared cone                    |

| Construct                                        | Rationale                                                                                              |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Combinational lookup + commit-in-one-cycle write | Caller can drive lookup → update without an extra pipeline stage                                       |
| Position-IS-rank NOT used                        | This CAM has no LRU — slots are positionally assigned by the caller; the index has no recency semantic |

The combinational depth from input port to CAM output is:

lookup\_id → match\_oh (per-slot compare; the module exports the one-hot vector, and the consumer ORs it if it needs a hit flag)

first\_oh (priority encoder)

That's ~3–4 LUT levels for DEPTH=16 — comfortably under the 100 MHz budget on the -1 part.

## 39.8 Related Modules

| Module                | Role                                                                                 |
|-----------------------|--------------------------------------------------------------------------------------|
| axi_monitor_trans_mgr | Sole consumer — uses the CAM for ID tracking and per-slot storage                    |
| monbus_cam            | Sister CAM for the bulk-trace compressor (LRU, single port, different design intent) |
| bin/TBClasses/shared/ | Test framework utilities                                                             |

## 39.9 Testing

val/amba/test\_monitor\_trans\_cam.py runs 17 sub-tests covering:

| # | Sub-test                                                    |
|---|-------------------------------------------------------------|
| 1 | Reset state — all entries invalid, free_oh = all-1s         |
| 2 | Single write + lookup — write slot 0 with id=K, lookup hits |
| 3 | Independent writes — each slot writable in isolation        |

| #  | Sub-test                                                              |
|----|-----------------------------------------------------------------------|
| 4  | Three-port lookup — distinct addr/data/resp ids resolve independently |
| 5  | data_match_first_oh — duplicate ids: lowest-index wins                |
| 6  | free_oh tracks valid bit per slot                                     |
| 7  | Clear via we=1 + valid_next=0                                         |
| 8  | Alloc one phase — addr_wants_alloc → lowest-index free slot           |
| 9  | Alloc mutex — all three phases want → three distinct slots            |
| 10 | Alloc partial wants — only some phases want, others zero              |
| 11 | Alloc when full — no free slots → all *_alloc_oh = 0                  |
| 12 | Payload preserved across idle cycles                                  |
| 13 | Concurrent writes — entry_we to multiple slots in one cycle           |
| 14 | Lookup miss — id absent from CAM → match_oh = 0                       |
| 15 | Alloc priority cascade — pre-occupy slots, watch alloc skip them      |
| 16 | ID collision tolerance — two valid slots with same id                 |
| 17 | Random stress — random write/lookup/alloc mix vs Python model         |

The Python golden model mirrors the RTL state and combinational outputs exactly. At FULL REG\_LEVEL the random stress runs 5000 ops per config, and every cycle cross-checks all 9 combinational outputs (3 \*\_match\_oh, first\_oh, free\_oh, 3 \*\_alloc\_oh, entry\_valid) against the model.

```
pytest val/amba/test_monitor_trans_cam.py -v
```

REG\_LEVEL parameter sweep: - **GATE**: 1 config (8/64/16) - **FUNC**: 2 configs (+ 4/32/8) - **FULL**: 7 configs (depth 4/8/16/32, id 4/6/8, payload 32/64/128/256)

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## 39.10 Navigation

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